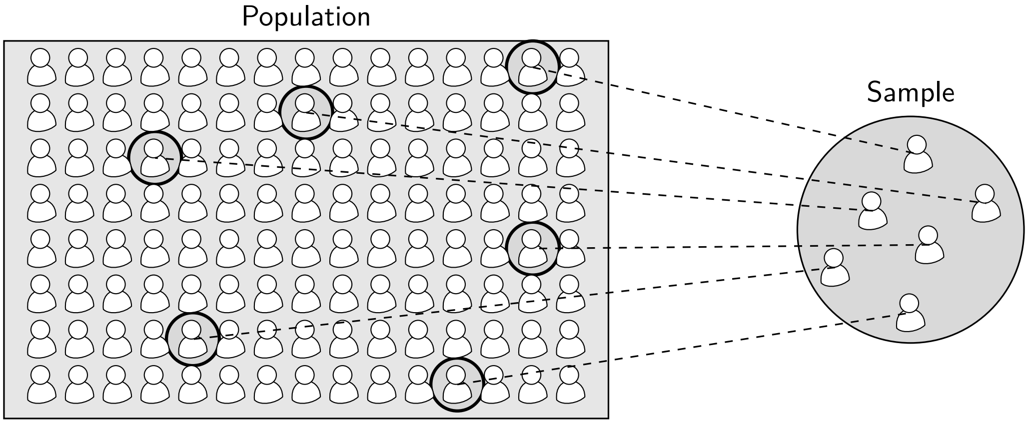
Contents

Introduction

Statistics is the science of collecting, organising, presenting, and analysing data. It plays a vital role in helping us understand the world and enables informed decision-making based on available information. Governments, organisations, businesses, and individuals across a wide range of fields rely on statistics and statisticians.



The *Advanced Higher Statistics* course aims to equip learners with a sound foundation in statistical knowledge. This textbook has been designed to provide key facts, examples, and exercises to support progression through the course. It covers all major stages of statistical investigations, including:

- Collecting data using appropriate *sampling techniques*.
- Exploring data using *descriptive statistics*, such as:
  - Visualising data with charts and graphs
  - Calculating sample statistics
- Applying *statistical inference* to evaluate claims about population parameters, including:
  - An introduction to *probability theory*, which underpins statistical inference
  - Performing hypothesis tests
  - Constructing confidence intervals

By the end of the course, students should have the knowledge and confidence to collect their own data and carry out independent statistical investigations.

1

Exploratory Data Analysis

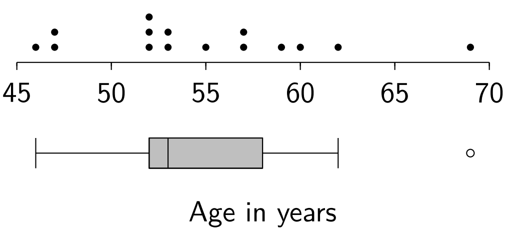
Data that has been collected but not yet processed is referred to as *raw data*. Before conducting any statistical analysis, a statistician will typically want to get a quick sense of the data, checking for potential issues and identifying key features. Common first steps are therefore often to: *organise* the data into suitable lists or tables; *visualise* the data using plots and graphs; and *measure* key properties of the data.

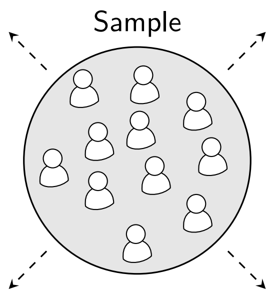
This process is called *exploratory data analysis*. Also sometimes referred to as *descriptive statistics*, it helps us understand the data before applying more formal *inferential statistics* methods.

Patient data table (first five entries only)

age (yrs)	height (cm)	NHS Health Board
57	182.2	Fife
43	168.2	Tayside
48	179.4	Tayside
64	162.6	Forth Valley
52	171.5	Fife

Dot plot and box plot (age, NHS Fife patients)

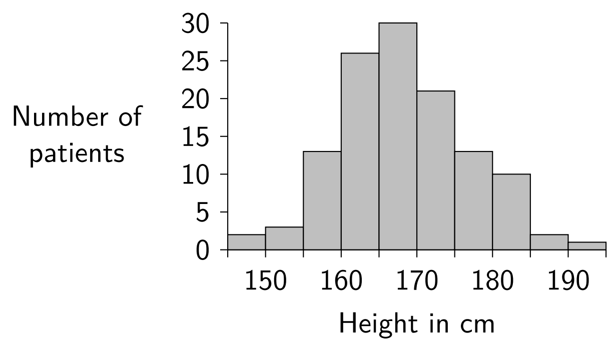




Summary (height in cm, all patients)

sample size	$n = 120$
mean	$\bar{x} = 168.5$
standard deviation	$s = 8.26$
median	median $= 168.9$
interquartile range	IQR $= 10.3$
minimum value	min $= 149.0$
maximum value	max $= 190.1$

Histogram (height, all patients)

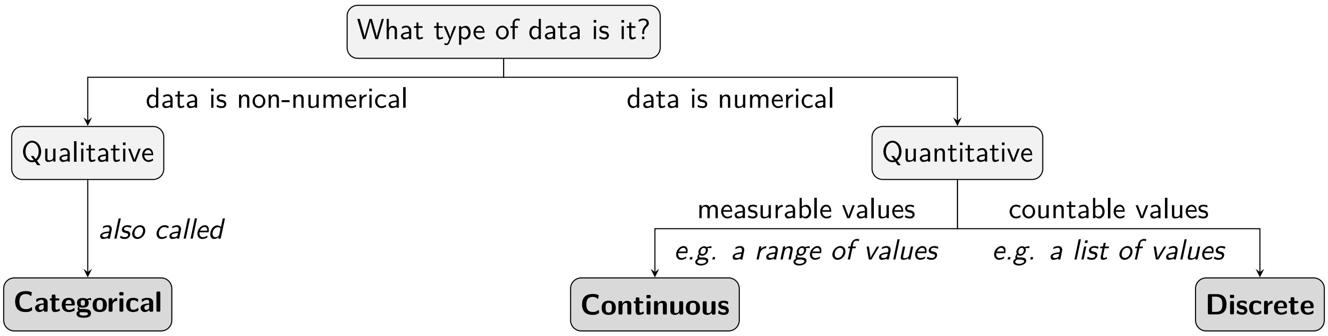


1.1 Types of Data

Knowing what *type* of data has been collected helps determine what kind of graphs and statistics may be appropriate.

Data that is numerical is called *quantitative*, of which there are two types. **Discrete** data consists of values that can be counted, such as the number of siblings someone has. **Continuous** data can take *any* value within a given range, though in practice values are often rounded due to measurement precision, such as the mass of a mouse.

Qualitative data (also called **categorical**) is *non-numerical*, such as whether a squirrel is red or grey.



For example, the table below shows data relating to some football players:

Position	Team	Height (m)	Goals	Assists
Midfielder	Blackburn Rovers	1.78	11	14
Defender	Sheffield United	1.93	3	5
Forward	St Mirren	1.75	23	7
Forward	Celtic	1.78	19	12
Categorical		**Continuous**	**Discrete**	

Exercise 1.1

1. A survey is carried out on the pets owned by a class. State whether each would result in **quantitative** or **qualitative** data.

   (a) How many pets each pupil owns.
   (b) What kind of animal the pets are.
   (c) The pet's age.
   (d) The name of the pet.
   (e) The mass of the pet.
   (f) Whether the pet is a mammal.

2. For each of the following, state whether the quantitative data collected would be **discrete** or **continuous**.

   (a) The number of people in a room.
   (b) The mass of a book.
   (c) The number of pages in a book.
   (d) The number of songs in an album.
   (e) The volume of water in a bottle.
   (f) Time taken to complete a puzzle.

3. A sample of members of a gym were surveyed and the first few results are shown in the table below.

Activity level	Smoker	Height (inches)	Mass (pounds)	Pulse (bpm)
Moderate	No	66	140	64
Moderate	No	72	145	58
A lot	Yes	73.5	160	62
Slight	Yes	73	190	66

   State the type of data that was collected for each variable.

1.2 Displaying Data

Before analysing a data set, it is common practice to *display* the data in a simple form so that an initial impression can be formed. Some key *figures* for the course are shown below, using a sample of 10 sunflower heights measured by a gardener 8 weeks after germination, measured to the nearest centimetre:

34 35 38 41 46 47 47 51 62 73

Figure 1: Dot Plot

Dot plots display individual data values, stacking repeated values vertically for clarity. They are best suited to *discrete* data

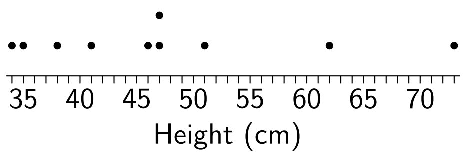


Figure 2: Stem-and-Leaf Diagram

Stem-and-leaf diagrams group data while preserving the original values. Leaves should be in order, and a *key* and the sample size (n) included.

3	4 5 8
4	1 6 7 7
5	1
6	2
7	3

$n = 10$

3 | 5 represents 35 cm

Figure 3: Box Plot

Box plots offer a visual representation of a *five-number summary*, consisting of the minimum value, lower quartile LQ, median, upper quartile UQ, and maximum value.

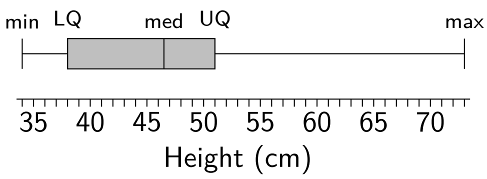
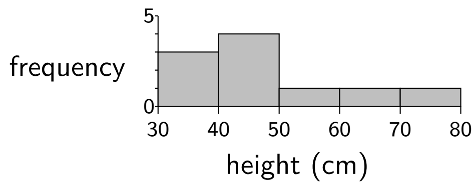


Figure 4: Histogram

Histograms display continuous data grouped into intervals. Unlike bar charts, the bars are adjacent (no gaps). When interval widths are equal, the height of each bar represents frequency.



Exercise 1.2

1. The ages (in years) of competitors in a junior golf tournament are below. Draw a dot plot to represent this data.

   16 14 19 13 14 17 16 19 16 17 12 16 14 19

2. The price of an ice cream is recorded at 12 stalls (in £). Represent the data using a stem-and-leaf diagram.

   1.50 2.20 2.50 2.10 1.80 1.40 2.00 1.90 3.60 0.90 1.70 1.50

3. The masses (in kg) of the luggage of 10 passengers on a plane to Barra were recorded, and a five-number summary of the data obtained using `computer output` from statistical software. Construct a box plot.

```
min  LQ  median  mean  UQ  max  n
10   14   16.5   16.6  19   22  10
```

4. The *frequency table* below shows the masses (in grams) of 40 medium-sized eggs, grouped into equal-width intervals and counted. Construct a histogram to display the data.

Interval	52-53.99	54-55.99	56-57.99	58-59.99	60-61.99	62-63.99
Frequency	2	4	11	17	5	1

1.3 Tables, Figures and Output

There are many ways in which data can be presented, extending beyond those covered on the previous page. In statistical reports they are often clearly labelled as **tables** or **figures**, and numbered so they can be more easily referred to.

Table 1: Contingency table of mode of transport.

		Transport			
		walk	bus	car	bike
Year Group	3rd	12	15	3	4
	2nd	7	13	8	1
	1st	4	14	11	2

Figure 1: Bar chart of mode of transport.

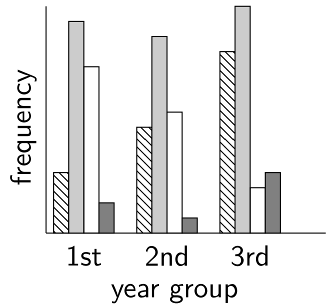
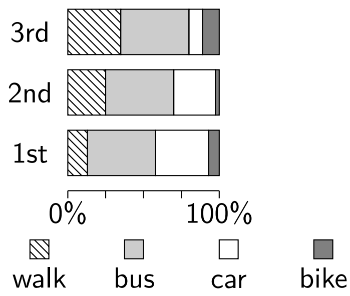


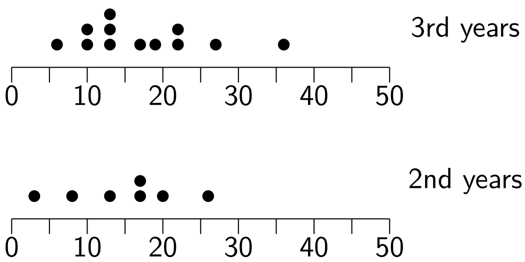
Figure 2: Chart of transport percentages.



Not all graphs encountered in real life will be up to scratch; some graphs may be unclear, accidentally misleading or even intentionally misleading. Being alert to potential flaws in tables and figures and identifying possible improvements is an important skill.

The widespread availability of computing power means that specialist software is often used by statisticians to avoid the need for time-consuming calculations where possible. Whilst this course requires no use of statistical software, examples of `computer output` will be displayed from which the required information must be extracted.

Figure 3: Dot plots of time (mins) to walk to school for 2nd and 3rd years.



Output 1: Summary data for walk times (mins) for 2nd and 3rd years.

```
data: walk time

3rd yr: n=12    mean=17.3   sd=8.42

2nd yr: n=****  mean=14.9   sd=7.65

alternative hypothesis: means are not equal

t=0.619, df=17, p-value=0.544
```

Whilst this textbook aims to enable familiarity with common figures, tables and computer output, the best way to practise obtaining information from them is to *read a wide range of relevant statistical articles and reports*.

Exercise 1.3

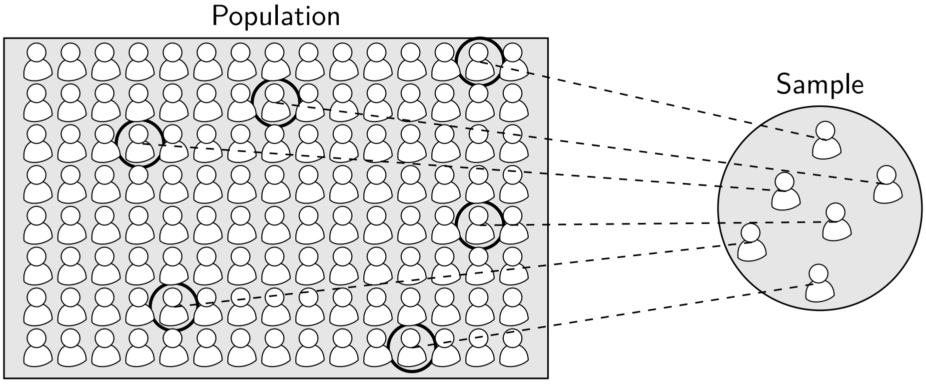
1. Use **Table 1** to calculate the proportion of pupils in the sample that walk to school, as a:

   (a) Fraction. (b) Decimal. (c) Percentage.

2. Suggest two possible improvements to **Figure 1**.
3. Using **Figure 2 only**, compare the proportions of pupils who take the bus between each year group.
4. From **Figure 3**, state the approximate *shortest* and *longest* walking times amongst 2nd and 3rd year pupils.
5. One piece of information from **Output 1** has been replaced by ****. State the missing value.

1.4 Population Parameters and Sample Statistics

In a statistical study, the term **population** refers to the entire set of objects of interest, whilst a **sample** is a subset chosen from the population. Understanding the difference is fundamental to statistics.



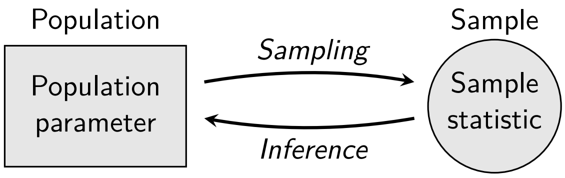
If data were available for an entire *population*, then various **population parameters** could be calculated *exactly*, such as the *population mean*. Since this is rarely possible in practice, instead *samples* are taken from populations. Measures calculated from a sample are called **sample statistics**, such as the *sample mean*.

The distinction between population parameters and sample statistics is often reflected in the notation used. Letters from the Greek alphabet are commonly used to represent population parameters. For example:

Population Parameters	**Sample Statistics**
Population mean μ *("mu")*	Sample mean $\bar{x}$ *("x bar")*
Population standard deviation σ *("sigma")*	Sample standard deviation s

Sample statistics vary depending on the sample chosen and can be viewed as *estimates* of the true values of their corresponding population parameters, which are treated as fixed (but usually unknown).

Statistical inference is the process of using *sample statistics* to assess claims about *population parameters*. This requires learning about the inherent *random variation* that arises when selecting samples from a population. Statistical inference, and *sampling techniques*, will be introduced in later chapters.



The purpose of discussing sampling at this point is to caution against making *definitive* statements based solely on *sample statistics*, when it is the (usually unknown) *population parameters* that are of primary interest.

Exercise 1.4

For each, state using words and notation which sample statistic or population parameter has been obtained. For example: "Sample mean ($\bar{x}$)." or "Population standard deviation (σ)."

1. Whilst studying house price valuations on a street, the valuations are found of every house on the street, and the mean valuation is calculated.
2. A bus company in Scotland checks the mileage for some of its buses when they come in for a service, and the mean mileage of these buses is calculated.
3. The iron levels are checked for a sample of patients at a hospital, and the standard deviation of these is calculated.
4. Whilst researching the capacities of football stadiums in the Scottish Football League, the standard deviation is calculated using a full list of all the stadiums used in the Scottish Football League.
5. For a study on lunchtime spending in Scottish schools, the mean lunchtime spend is calculated from surveying every pupil studying at a large secondary school in Edinburgh.

1.5 Measures of Location and Spread

Summary statistics are numerical measures calculated from sample data, such as the mean or standard deviation. These provide a concise way to describe key features of the data. For example, for the *raw data set* of 0, 2, 2, 6, 12, 14, 16, 18, 18, 18, 20, the table below shows a selection of summary statistics.

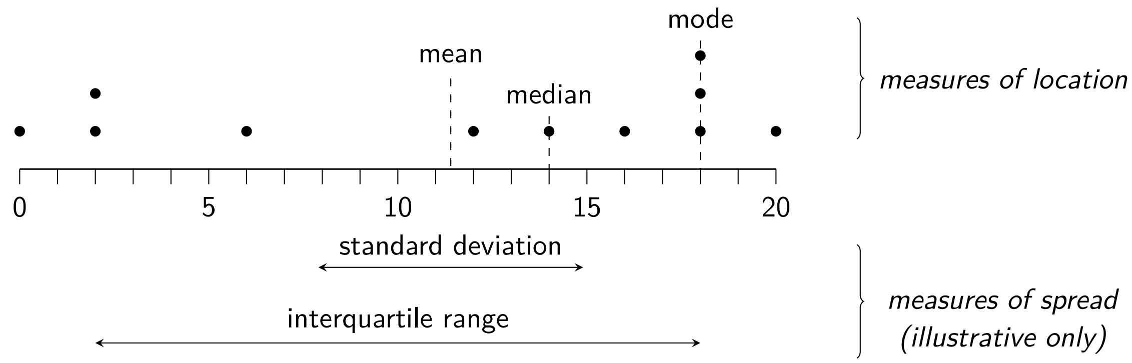
n	mean	sd	sum	sum of squares	min	LQ	median	UQ	max	range	IQR	mode
11	11.45	7.50	126	2012	0	2	14	18	20	20	16	18

There are two main types of summary statistic: those that describe the *central tendency* of data (or its **location**); and those that describe the *variability* of data (or its **spread**). It is common to measure both the location *and* spread, since two data sets can have the same average but very different variability.

Measures of Location

The three main measures of location (or *averages*) may already be familiar. To avoid ambiguity, statisticians will typically avoid referring to an *'average'*, and will instead specify *which* average they are referring to:

- **Mode** - the most frequently occurring value.
- **Median** - the middle value after the data has been put into numerical order.
- **Mean** - the sum of the values divided by the number of values.



Measures of Spread

The measure of spread which is usually encountered first in a maths classroom is the *range*, defined as the difference between the largest and smallest values. The range, however, is less informative than other measures as it uses only the two most extreme values. There are three main measures of spread commonly used, two of which are likely to already be familiar:

- **Interquartile Range** - the difference between the upper quartile and the lower quartile.
- **Variance** - the *mean of the squared differences* between each value and the mean.
- **Standard deviation** - the square root of the variance.

The Sample Median and the Sample Interquartile Range

A lay person's interpretation of what an 'average' seeks to describe is the *typical* value for a data set. However, consider the hourly rate of pay of six people working in a store, with one manager and five shop workers:

£10 £10 £10 £10 £14 £51

The mean of these values is £17.50, yet this figure does not seem to represent a *typical* hourly rate of pay of people working in the store. This example may seem contrived, yet this exact issue is why countries' average salaries are not measured using the mean. This is because the *mean* takes into account *all values* in any data set, and so is influenced by *extreme* values.

The median (Q_2) and interquartile range (IQR) are often chosen to measure location and spread when the data is particularly *skewed* (see page 15), as they are less influenced by *extreme values* than the mean and standard deviation. The *median* represents the *middle* of a data set: a value which approximately 50% of the data lies below, and approximately 50% of the data lies above. Where a data set has an *odd* number of values, the median is simply the middle value. Where a data set has an *even* number of values, the median is the mean of the *two* middle values. The median is used to describe the *location* of a population.

median = 9

Data set A ($n = 11$): 4 6 6 7 9 9 11 12 13 16 18

median = 12

Data set B ($n = 10$): 5 7 7 9 11 | 13 13 15 17 17

The *lower quartile* (LQ or Q_1) and the *upper quartile* (UQ or Q_3) split an ordered data set into four approximately equal parts, along with the median. From the lowest value to LQ contains approximately 25% of the data, from LQ to the median contains approximately the next 25%, and so on.

ignore middle value

LQ = 6 ↓ UQ = 13

Data set A ($n = 11$): 4 6 6 7 9 9 11 12 13 16 18

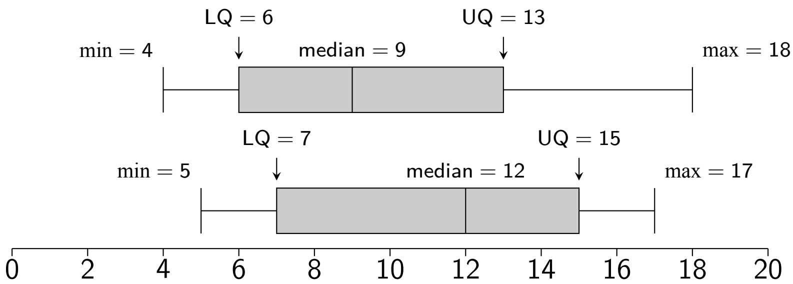
LQ = 7 UQ = 15

Data set B ($n = 10$): 5 7 7 9 11 | 13 13 15 17 17

The *interquartile range*, calculated as IQR = UQ − LQ, measures the range of the approximate 'middle 50%' of a data set, ignoring the remaining lowest and highest quarters. On a box plot (sometimes called a *box-and-whisker* plot), the interquartile range defines the *width of the box*, ignoring the 'whiskers' at either side. The larger a data set's interquartile range is, the more spread it suggests the data may be.

Data set A:
IQR = 13 − 6 = 7

Data set B:
IQR = 15 − 7 = 8



The box plots above suggest that population B may have a higher median value than population A, and that population B's distribution may be slightly more varied, since its sample gave a higher interquartile range.

The Sample Mean and the Sample Standard Deviation

The mean (*'add up all the values then divide by how many there are'*) is what most people are typically thinking of when they talk about *the average*. It has the property of taking into account *every value* in a data set, which is often desirable except when *extreme values* are present, as discussed on the previous page.

If data for an entire *population* of values $x_1, x_2, \dots, x_N$ were available, where N is the population size, then the **population mean** (μ) could be calculated. This can be expressed concisely using summation notation, $\sum$ ("Sigma"), where $\sum x_i$ denotes the sum of all values from x_1 to x_N.

$$\mu = \frac{x_1 + x_2 + \cdots + x_N}{N} = \frac{\sum x_i}{N}$$

Since in practice a population is rarely available, instead it is common to calculate the **sample mean** ($\bar{x}$), using a sample of data from a population. Given a sample of data $x_1, x_2, \dots, x_n$, where n is the sample size, the sample mean can be calculated as, in its most commonly used form:

$$\bar{x} = \frac{\sum x_i}{n}$$

The sample mean is an *unbiased estimator* of the population mean, meaning that, on average, it gives the correct value across repeated sampling.

Similarly, if data for an entire *population* of values $x_1, x_2, \dots, x_N$ were available, then the **population variance** (σ^2) could be obtained. More useful for its mathematical properties than as a value to *interpret*, it is the *mean of the squared differences* between each value x_i and the population mean, μ. Its units are the *square* of the units of the original data, and so for data measured in grams, units of variance are (somewhat unintuitively) grams^2. It can be calculated, again using an abbreviated form of summation, as:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

Replicating the same formula using sample data gives a *biased* estimate for variance. This is because the formula requires the population mean μ be replaced with the *sample mean* $\bar{x}$ as its estimate, which will tend to minimize the sum of square deviations and so tend to lead to an *underestimate* of the true population variance. *Bessel's correction* is used to produce a formula for the **sample variance** (s^2), which is an *unbiased* estimator for the population variance, taking a denominator of $n - 1$. The below two forms are equivalent.

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} \quad \text{or} \quad s^2 = \frac{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}{n-1}$$

The *standard deviation* is the *square root* of the variance, which brings the units it takes back to match those of the original data (e.g. grams). The formula for the **sample standard deviation** (s) appears on **page 4** of the Data Booklet. It is given in terms of the *sum of squares for* x, notated as S_{xx}, which is the numerator of the right-hand formula above. A formula for the sample standard deviation is therefore:

$$s = \sqrt{\frac{S_{xx}}{n-1}} \quad \text{where} \quad S_{xx} = \sum x_i^2 - \frac{(\sum x_i)^2}{n}$$

In practice, only the *sample mean* and *sample standard deviation* (and sample variance) will need to be calculated in this course, and both can be performed by a graphical calculator directly from a sample of values with no need for manual calculation. However, an insight into their formulae is essential in particular for anyone who wishes to take statistics further beyond this course.

For example, calculating the mean and standard deviation sample data set: 4, 6, 7, 9, 10

$$\sum x_i^2 = 282,\ \sum x_i = 36,\ n = 5 \qquad \bar{x} = \frac{36}{5} = 7.2 \qquad s = \sqrt{\frac{282 - \frac{36^2}{5}}{4}} = 2.387 \text{ (to 3dp)}$$

Exercise 1.5

1. Summary statistics from sample data are given below. Use appropriate formulae to calculate the mean and standard deviation for each sample. Note that summations such as $\sum x_i$ are often simplified to $\sum x$.

   (a) $\sum x = 359,\ \sum x^2 = 15119,\ n = 10$ (b) $S_{xx} = 1037.75,\ \sum x = 994,\ n = 16$

2. Sets of sample data are shown below. Using a graphical calculator if available, and formulae otherwise, find the mean and standard deviation for each.

   (a) 32 25 29 37 16 39 23 32

   (b)
   17 63 34 23 40 57 72 70 43
   56 29 37 41 50 13 24 30 58

3. The frequency tables below represent sample data. Calculate the mean for each data set.

   (a)

Value	6	7	8	9	10
Frequency	1	5	4	3	2

   (b)

Value	0	1	2	3	4	5	6
Frequency	3	7	5	0	4	1	2

4. Sets of sample data are shown below. Using a graphical calculator, or by first putting the data set in order, find the median and interquartile range for each data set.

   (a) 32 25 29 37 16 39 23 32

   (b)
   17 63 34 23 40 57 72 70 43
   56 29 37 41 50 13 24 30 58

5. Random samples of data from populations A and B are shown below.

Sample A	4	7	9	10	10	14	15	20
Sample B	9	10	10	11	12	12	17	

   (a) Find the mean and standard deviation for sample A.
   (b) Calculate the variance for sample A.
   (c) Calculate the median and interquartile range for sample B.
   (d) Suggest two further statistics that could be calculated to make a fair comparison between the location and spread of the two populations.

6. Eight physics students sit a test in September and a test in November, each out of ten. Each student calculates the *difference* in their marks between the two tests (November – September). The results are shown in the table below, with the differences for the final four students yet to be calculated:

Student	A	B	C	D	E	F	G	H
September	6	9	4	10	8	5	7	8
November	8	8	6	10	9	3	7	7
Difference	2	-1	2	0	?	?	?	?

   (a) Calculate the remaining four differences.
   (b) Find the *median of the differences* and the *interquartile range of the differences*.
   (c) Find the *mean of the differences* and the *standard deviation of the differences* (writing the answers as $\bar{x}_d$ and s_d, instead of $\bar{x}$ and s respectively, for clarity).

1.6 Outliers and Fences

Sometimes a data set may contain values that seem *extremely high or low* in comparison to the rest of the values. It is important to be wary of such values and the potentially significant impact they could have on any statistical analysis performed. There is a range of objective approaches that can be used to determine whether one or more points are extreme enough to be classified as **outliers**. In this course, *fences* will be used.

Identifying Outliers

$$\text{Lower Fence} = Q_1 - 1.5 \times \text{IQR}.$$

$$\text{Upper Fence} = Q_3 + 1.5 \times \text{IQR}.$$

Any values *below the lower fence* or *above the upper fence* are **outliers**.

If an outlier is identified in a data set, an investigation should be conducted to determine the cause. If it is concluded that the outlier is a rogue point, such as an error in measurement, it can be removed from the data set and summary statistics re-calculated. However, if an outlier is a valid point in the data set then it should not be removed.

Returning to the sunflower heights from page 5, the data can now be checked for possible outliers:

Raw data

Heights (cm)									
34	35	38	41	46	47	47	51	62	73

Summary data

n	min	Q1	median	Q3	max
10	34	38	46.5	51	73

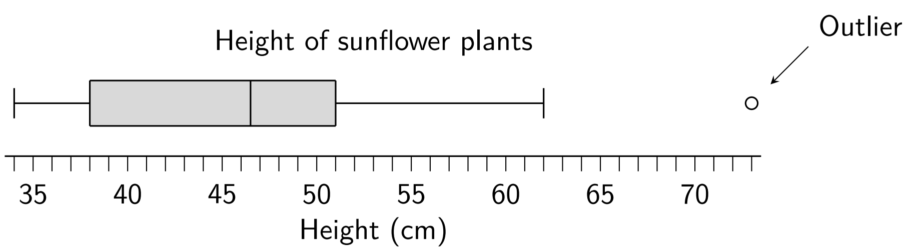
$$\text{IQR} = Q_3 - Q_1 = 51 - 38 = 13$$

$$\text{Lower fence} = Q_1 - 1.5 \times \text{IQR} = 38 - 1.5 \times 13 = 18.5$$

$$\text{Upper fence} = Q_3 + 1.5 \times \text{IQR} = 51 + 1.5 \times 13 = 70.5$$

Since $73 > 70.5$, it is an outlier.

On a box plot, outliers are indicated separately as shown below, with the "whiskers" extend only as far as the largest and smallest values that are *not* outliers. However the quartiles remain as those calculated using the *full* set of values, including any outliers. *Plotting outliers separately in this way is to highlight their presence, and is not a judgement on whether those values are rogue or valid.*



The outlier in this data set could be valid if, for instance, the sunflower plant is in a particular sunny spot in the garden. However, if the height was measured incorrectly then it can be removed from the data set.

Example

Problem: A vet conducts a survey, asking 10 randomly chosen cat owners registered with his practice to share the mass of their adult cat, in kilograms. The results are:

4.6 3.6 4.1 4.5 3.7 3.9 4.0 0.6 3.9 4.8

(a) Show that the data set contains an outlier.

(b) Suggest a possible cause for this outlier which would mean it should be removed.

Solution:

(a) $Q_1 = 3.7$, $Q_3 = 4.5$, $\text{IQR} = 4.5 - 3.7 = 0.8$.

Lower fence $= 3.7 - 1.5 \times 0.8 = 2.5$

Upper fence $= 4.5 + 1.5 \times 0.8 = 5.7$

Since $0.6 < 2.5$, the value of 0.6 is an outlier.

(b) The owner could have mistakenly measured the mass of their cat in stones instead of kilograms.

Exercise 1.6

1. Strawberry tarts are sold every day in a bakery over the summer months. The shop manager monitors the sales and takes a sample of the number of strawberry tarts sold on 10 randomly selected days over one month.

   23 31 45 32 35 19 24 25 30 38

   Calculate the upper and lower fences for this data set and identify any possible outliers.

2. The number of hours spent studying each week by a randomly selected group of 24 students is given below.

10	18	15	12	17	27	4	14	14	14	17	15
15	12	17	17	13	15	15	16	16	9	11	15

   (a) Identify any possible outliers.
   (b) Draw a box plot to display this data.

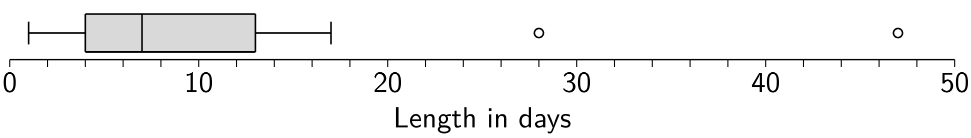
3. A call centre records the times (in minutes) a sample of customers have to wait before being connected to an agent.

   55 20 31 12 18 32 28 16 14 33

   Show that the data set contains at least one outlier.

4. A hospital trust compiled data relating to the length of stay of patients in a particular hospital. A random sample of 18 patients gave the following summary data, measured in days.

n	mean	sd	min	Q1	median	Q3	max
18	10.78	11.17	1	4	7	13	47

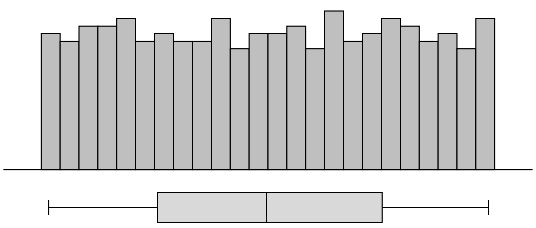
   

   (a) Show that the two values indicated by circles on the box plot are outliers.
   (b) Suggest a cause for these outliers that would make it valid to remove them from further analysis.

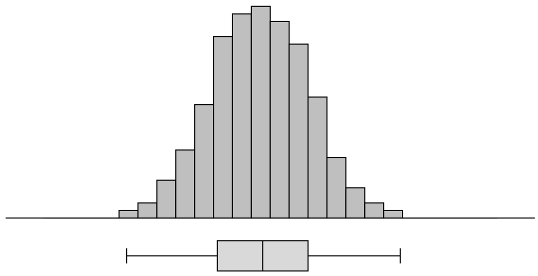
1.7 Describing the Shapes of Distributions

Summary statistics alone are not sufficient to describe the full complexity of the way data is distributed. It is also important to consider and be able to describe the **shape** of the distribution, visualised most easily using a graph. Whilst real-life data rarely follows a precise mathematical model *exactly*, a number of shapes are commonly observed. The following data was generated by *simulating random sampling* from populations with the described distributions.

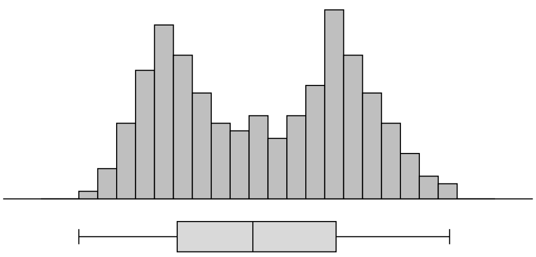
uniformly distributed (n = 103)



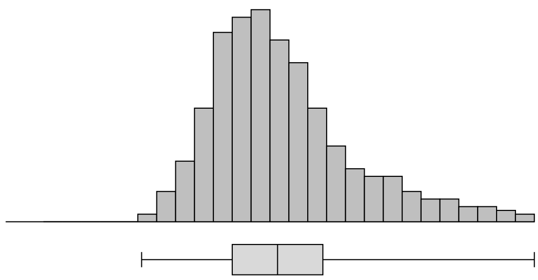
normally distributed (bell-shaped distribution) (n = 215)



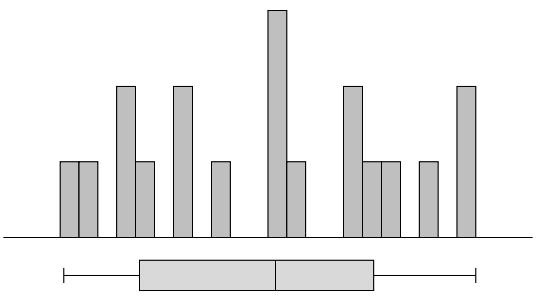
bimodal (n = 87)



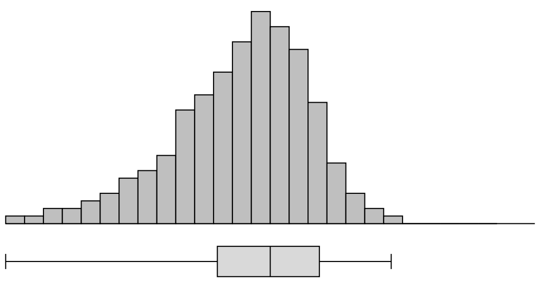
positively skewed (and so not symmetrical) (n = 134)



difficult to infer distributions from small samples (n = 19)



negatively skewed (and so not symmetrical) (n = 97)



Note that the histograms above were made by selecting some of the most *well-behaved* random samples. Even when data is drawn from a particular distribution, histograms of this sample data may not appear perfectly shaped, especially for smaller sample sizes. Graphs of sample data do not **show** what the shape, location or spread of distribution of the underlying population is; instead, they merely suggest what may or may not be plausible. Any overly definitive conclusion from a visual representation of sample data should be avoided.

As well as considering whether the **underlying population distribution** of data appears to be *normal*, *skewed*, *uniform* or *bimodal*, the appropriateness of some statistical inference techniques will depend on whether the distribution is *symmetrical*, or whether two sets of data follow the *same distribution*, with matching shapes and spread. Whilst the following example references techniques that will be introduced later in the course, it is intended to give an idea of the importance of studying distributions in selecting appropriate methods for statistical inference.

Example

Problem: An environmental scientist collected water samples from a number of rivers in a region to explore whether the concentration of a harmful chemical in the water has changed since farming practices used locally have changed. They are considering the following choices of hypothesis test:

- A *t-test*, the most appropriate choice if the data follows a normal distribution.
- A *Wilcoxon Signed Rank test*, appropriate if the data is distributed symmetrically.

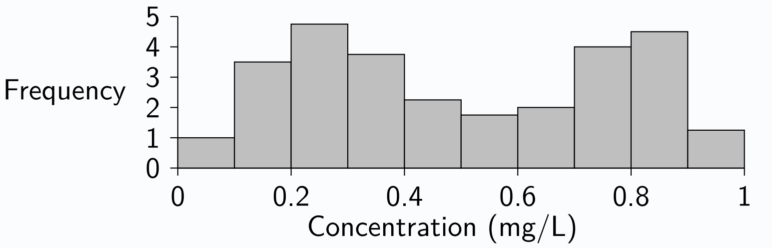


Figure 1: Concentration of chemical in river sites

(a) Use **Figure 1** to make a comment on the distribution of chemical concentrations.

(b) Comment on the appropriateness of each choice of hypothesis test suggested for this study.

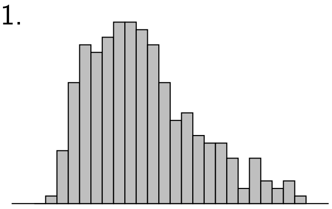
Solution:

(a) Figure 1 suggests that the population distribution of concentrations of the chemical in river sites may be bimodal.

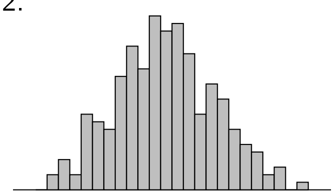
(b) Since the histogram suggests the population distribution of chemical concentration may be symmetrical but not normal, a Wilcoxon Signed Rank test would be appropriate whilst a t-test would not.

Exercise 1.7

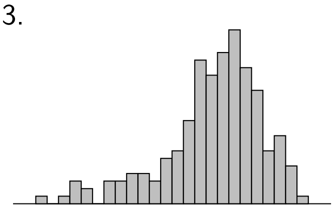
Comment on each population distribution using the simplified histograms, which show data from samples.

1.


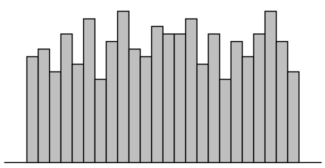
Hazelnut mass (grams)

2.


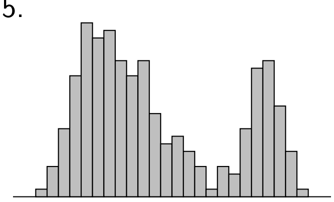
Tree height (metres)

3.


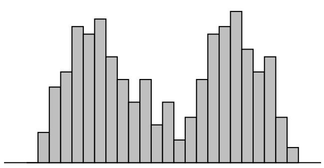
Number of patients

4.


Waiting time (minutes)

5.


Number of successful passes

6.


Distance to city centre (km)

1.8 Comparing Distributions

When comparing two or more samples of data, a good starting point is to create suitable plots and obtain summary statistics. Differences and similarities should be noted, considering:

- **Location** (i.e. means, medians, or clear visual differences in locations)
- **Spread** (i.e. standard deviations, interquartile ranges, or clear visual differences in spread)
- **Shapes** (i.e. symmetrical, skewed, bimodal, bell-shaped curve)

At this *exploratory* stage, it is again important to avoid making any *conclusions* relating to the populations from which the sample data was drawn.

Example

Problem: Within the sport of professional cycling, it is common for riders to specialise in certain types of cycling. *Sprinters* aim to win races that finish on flat roads at high speed, whilst *climbers* aim to win races with an uphill finish. A researcher studying the characteristics of elite-level cyclists randomly selects samples of 30 *sprinters* and 30 *climbers*, recording the heights of each. The results are shown summarised in **Table 1**.

Table 1

variable	n	Min	Q1	Median	Q3	Max	IQR
sprinters	30	165	174	179	184	200	10
climbers	30	158	168	174	177	195	9

Box plots for the two groups of cyclists are shown below in **Figure 1**, and dot plots in **Figure 2**.

Figure 1

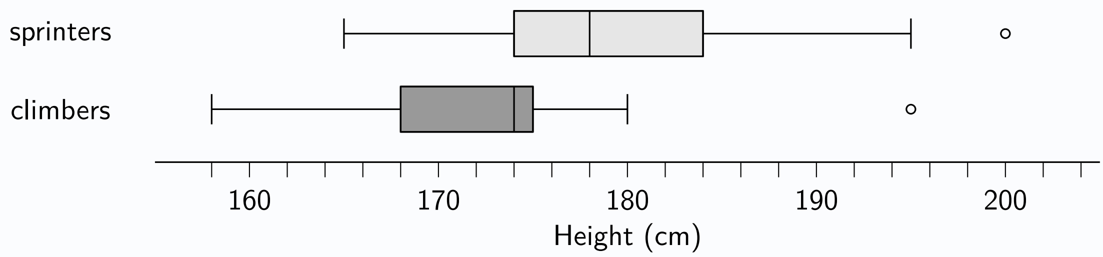
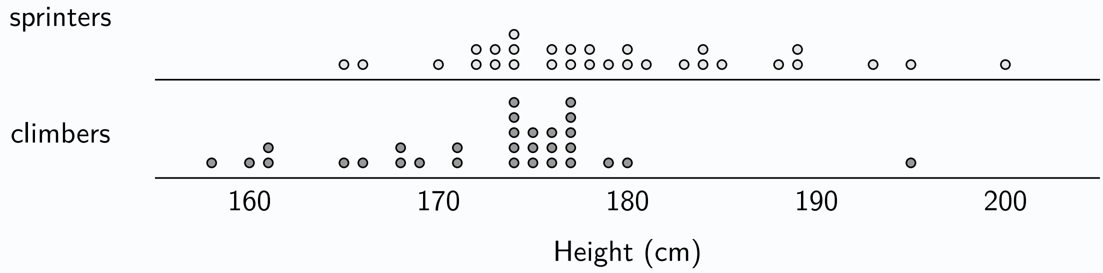


Figure 2



(a) Use **Figure 1** to make two comments comparing the heights of sprinters and climbers.

(b) Use **Figure 2** to comment on the shape of the distribution of the heights of climbers.

Solution:

(a) The sample median height of the sprinters was greater than that of the climbers.

The larger interquartile range suggests that the population of sprinters' heights is more varied.

(b) The dot plot suggests the underlying distribution of climbers' heights is negatively skewed.

Exercise 1.8

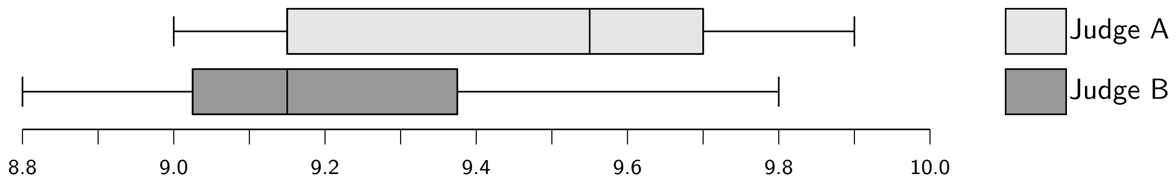
1. Two models of coffee machines produced by the brand *DOLCAFÉ* are supposed to produce a white coffee with the same volume of liquid each time, but natural variation means the actual amount dispensed is not quite the same every time. The volumes dispensed, in ml, are recorded for random samples from each model.

Model A	295	298	299	301	303	304	309
Model B	282	294	300	302	306	312	320

   (a) Determine the sample mean and standard deviation for each model.

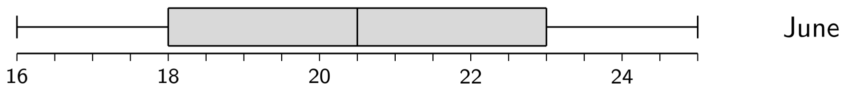
   (b) Make two comments comparing the amount of coffee dispensed by the two models.

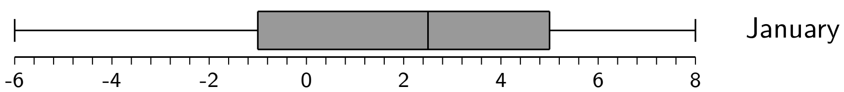
2. The scores awarded by two judges in a diving competition are recorded and displayed in box plots, shown below.



   Using the box plots, make two comments comparing the location and spread of the judges' scoring.

3. The temperature at 2pm was noted for 14 randomly selected days in January and June at a local tourist spot and box plots drawn to display the results.





   (a) Using the box plots, make two comments comparing temperatures in January and June.

   (b) Explain why the box plots may be misleading.

4. The manager of a lemonade bottling plant is interested in comparing the performances of two production lines, one of which has only recently been installed. For each line she selects 10 one-hour periods at random and records the number of crates completed in each hour. The table below gives the results:

Production Line	Number of crates completed per hour									
New	78	87	79	82	87	81	85	80	82	83
Old	74	77	78	70	87	83	76	78	81	76

   (a) Draw dot plots to represent the data.

   (b) Find the mean and standard deviation for each sample.

   (c) Make two comments comparing location and spread for the two lemonade production lines.

Review Exercise

1. A music fan wishes to investigate the characteristics of songs that are popular on a well-known music streaming app. She is able to obtain data on the thousand most-streamed songs of all time, a small selection of which are displayed in the table below.

Track	Artist	Released	Length (m:s)	No. of Streams
Creep	Radiohead	1992	3:59	1271293243
Blank Space	Taylor Swift	2014	3:52	1355959075
Something In The Way	Nirvana	1991	20:37	368646862
Nonsense	Sabrina Carpenter	2022	2:43	342897938
Drivers License	Olivia Rodrigo	2021	4:02	1858144199
Iris	The Goo Goo Dolls	1998	4:50	1284942608

   (a) State which of the five variables contain:

      i. Categorical data. ii. Discrete data. iii. Continuous data.

   (b) If it was suspected that *Something In The Way* is an outlier in terms of length, state what could be calculated in order to determine whether it is in fact an outlier.

2. A sports clothing store is seeking to better understand its sales of running shoes. For a random sample of twelve sales of pairs of running shoes, the sizes are recorded as follows:

   4 7 9 11 6 9 8 4 10 9 6 5

   (a) Construct a dot plot to display the data.

   (b) Find the median and interquartile range for the sizes of running shoes sold.

   (c) Find the mean and standard deviation for the sizes of running shoes sold.

   (d) State the type of data recorded.

3. The back-to-back stem-and-leaf diagram below shows the heights of trees in two areas of woodland.

Woodland A		Woodland B
5	2	1 3 3 8
7 4	3	0 2 4 7
9 7 7 2	4	2 5
7 6 3	5	

   (a) Suggest one improvement needed for the diagram.

   (b) Without performing any calculations, make one comment comparing the heights of trees in the two areas.

4. The petal lengths (in centimetres) for samples of flowers from two species of iris flowers is recorded: *iris setosa* and *iris versicolor*. The data is summarised in the table below using statistical software:

species	n	min	Q1	Q2	Q3	max
iris setosa	50	1.00	1.40	1.50	1.58	1.90
iris versicolor	50	3.00	4.00	4.35	4.60	5.10

   (a) For the species *iris setosa*, determine whether the data set contains any outliers.

   (b) Make two comments comparing the petal lengths of the two species of iris.

2

An Introduction to Probability Theory

The purpose of statistics is to make sense of the world by analysing data. In practice, however, it is only possible to obtain data from a random sample of the full population of interest, introducing uncertainty that must be accounted for. For this reason, developing an understanding of statistical analysis begins with learning about probability theory.

People commonly talk about *chance* in everyday language. For example, 'St Mirren are unlikely to win this weekend', or 'It is probably going to rain today'. However, neither of these comments seek to assign a numerical value to quantify those chances. In this chapter, probability as a *numerical measure* will be studied.

Key definitions

In probability theory, an experiment for which all possible *outcomes* can be defined, and yet the actual outcome of any given *trial* cannot be known in advance, is called a *random experiment*. The set of all possible outcomes is called the *sample space* of the random experiment. Probability is assigned to individual outcomes or to sets of outcomes, called *events* (subsets of the sample space). Probability can be interpreted as the **long-run relative frequency** of an outcome or event occurring, and values range from 0 (*impossible*) to 1 (*certain*).

For example, in the context of rolling a fair, cubical die:

- the procedure of rolling the die and observing the result is a **random experiment**.
- each roll of the die may be referred to as a **trial**.
- the **sample space** consists of a list of all of the possible **outcomes**: $\{1, 2, 3, 4, 5, 6\}$
- the **probability** of a 5 is $\frac{1}{6}$ since *in the long-run* it can be expected to occur $\frac{1}{6}$ of the time.
- a possible **event** of interest may be that the outcome is *prime*: $\{2, 3, 5\}$

Probability notation

The probability that a fair cubical die lands on 5 is *one-in-six*, or $\frac{1}{6}$. Using probability notation:

$$\text{P}(\text{rolling a 5}) = \frac{1}{6}$$

2.1 Measuring Probability

Experimental Probability

The probability of an event can be estimated through repeated trials - this is referred to as the *experimental probability* of the event. For example, if a motorist passes a set of traffic lights every morning and wishes to know the probability of having to wait at a red light on any given morning, they could record the *proportion* of mornings on which they had to wait at a red light over a period of time. If, over the course of 200 mornings, they have to wait at a red light 114 times:

$$\text{P(red light)} \approx \frac{114}{200} = 0.57$$

Theoretical Probability

When each outcome in the sample space can be reasonably assumed to be **equally likely**, then the probability of outcomes or events can be determined. This is called *theoretical probability*.

Equally likely outcomes typically arise from the symmetry of a physical object (such as a coin or die) or a reasonable assumption. The probability of any *individual outcome* in a sample space containing N equally likely outcomes is $\frac{1}{N}$. For example, if a playing card is picked at random from a standard pack of 52 cards, containing no jokers, then the probability of picking the Queen of Hearts is $\frac{1}{52}$.

An **event** is a set of one or more outcomes from the sample space. The probability of an event is given by:

Probability of an Event

$$\text{P(event)} = \frac{\text{number of outcomes from the sample space in the event}}{\text{total number of outcomes in the sample space}}$$

Example

Problem: A bag contains 13 red counters, 5 blue counters and 4 green counters. A counter is chosen at random. Find the probability that the counter is green.

Solution:

$$\text{P(green)} = \frac{4}{22} = \frac{2}{11}$$

When considering two events in probability theory, *"and"* means that *both* events are true, and *"or"* means that *at least one* of the events is true (including the possibility that they both occur, where applicable).

Example

Problem: A car dealership has 80 cars in its showroom. They have a mix of electric, hybrid, petrol and diesel cars, both new and second-hand. The table below shows the numbers of each type.

	electric	hybrid	petrol	diesel
new	15	17	28	3
second-hand	1	7	4	5

A car is selected at random. Determine the probability that the car selected is a second-hand hybrid car.

Solution:

$$\text{P(second-hand and hybrid)} = \frac{7}{80} = 0.0875$$

In this textbook, probabilities will be given as fractions or *rounded to four decimal places* for consistency.

Exercise 2.1

1. A letter of the English alphabet is chosen at random. State the probability that the letter chosen is:

   (a) A vowel. (b) A consonant. (c) In the word *'statistics'*.

2. A bag contains four blue, five green and six red counters. Find the probability that a randomly chosen counter is:

   (a) Blue. (b) Either green or red. (c) Not red.

3. A box of chocolates contains chocolates of three different flavours. Five are caramel, four are orange and three are strawberry. A chocolate is chosen at random. State the value of:

   (a) P(orange). (b) P(orange and strawberry). (c) P(not caramel)

4. The UK contains 76 cities, of which 55 are in England, 8 in Scotland, 7 in Wales and 6 in Northern Ireland. If a city in the UK is selected at random, find the probability that the chosen city is:

   (a) In Scotland. (b) In Wales or Northern Ireland. (c) Not in England.

5. A board game uses green counters numbered from 1 to 8, blue counters numbered from 1 to 6 and red counters numbered from 1 to 4. A counter is chosen at random. Determine the probability that the counter selected is:

   (a) The blue numbered 5. (b) Numbered 5. (c) Blue or numbered 5.

6. A restaurant grades the fruit contained in a delivery as good or bad. The contents of the previous delivery are shown in the table below.

	Apples	Tangerines	Kiwi
Good	27	19	26
Bad	3	1	4

   Letting G represent that the event the fruit chosen is *good*, and A the event that it is an *apple*, find:

   (a) P(G) (b) P$(G$ and $A)$ (c) P(Not $A)$

7. A hospital contains a number of doctors, nurses, allied health professionals and support staff, who are each assigned to work on one of Wards 1 to 3. The breakdown of this is shown in the table below.

	Doctor	Nurse	Allied	Support
Ward 1	3	7	5	8
Ward 2	4	10	2	9
Ward 3	2	8	1	5

   If a member of staff working at the hospital is chosen at random, find the probability that they are:

   (a) A nurse. (b) Not on Ward 2. (c) On Ward 2 but not a doctor

2.2 Sample Spaces as Lists and Grids

It can help to write out sample spaces. Some sample spaces can be easily described using a simple list, but more organisation may be needed for more complicated sample spaces, such as approaching the list in a structured manner or using a grid. Sample spaces are particularly useful when each outcome is equally likely, as this allows probabilities to be calculated directly by counting outcomes.

If two coins are tossed, it may be tempting to suggest that the probability of both landing tails is one-in-three, since the experiment can result in either no coins landing tails, only one coin landing tails or two coins landing tails. However a list of the four possible *equally likely* outcomes in the sample space shows that this is not the case:

Not equally likely outcomes: $\{\text{no tails}, \text{one tail}, \textbf{two tails}\}$

These three outcomes are *not* equally likely, so the probability is *not one-in-three*, or: $\text{P}(\text{both tails}) \neq \frac{1}{3}$

Equally likely outcomes: $\{\text{HH}, \text{HT}, \text{TH}, \textbf{TT}\}$

These four outcomes *are* equally likely, so the probability is *one-in-four*, or: $\text{P}(\text{both tails}) = \frac{1}{4}$.

Example

Problem: Three coins are tossed at the same time. Find the probability that exactly two of the coins land showing tails.

Solution: *A list of equally likely outcomes will help here. Note there are $2 \times 2 \times 2 = 8$ outcomes.*

$$HHH \quad HHT \quad HTH \quad \underbrace{HTT} \quad THH \quad \underbrace{THT} \quad \underbrace{TTH} \quad TTT$$

$$\text{P}(\text{exactly two tails}) = \frac{3}{8}$$

Example

Problem: Two fair dice are rolled. Determine the probability that the sum obtained is at least nine.

Solution: *A grid of outcomes will help here. Note there are $6 \times 6 = 36$ outcomes.*

first die

second die

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$$\text{P}(\text{sum} \geqslant 9) = \frac{10}{36} = \frac{5}{18}$$

Exercise 2.2

For most of the following questions, it is recommended to write out a full list of all the possible equally likely outcomes. There may be times when a full list would be excessive in length, in which case careful multiplication can reveal the *number* of outcomes, and only the *relevant* ones may be needed to be listed.

1. A coin is tossed and a cubical, fair die numbered 1 to 6 is rolled. Write down:
   (a) A list of all outcomes.
   (b) P(even and tails)

2. Each morning, a teacher picks at random whether to start the day with tea, coffee or apple juice, and also picks at random which one of muesli, granola, toast or porridge to eat for breakfast. By first listing all the possible ways the teacher may start their day, determine the probability that on any particular morning the teacher has:
   (a) Tea and toast.
   (b) muesli but not coffee.
   (c) Neither porridge nor tea.

3. A coin is tossed three times. Find the probability that it shows:
   (a) Tails on two consecutive tosses.
   (b) Tails at least twice.

4. A random number is chosen from 1 to 140, inclusive. By first listing all relevant results, determine the probability of the number chosen having digits that sum to 3.

5. Two fair, cubical dice are rolled. Find the probability that the product of the numbers obtained is less than 5.

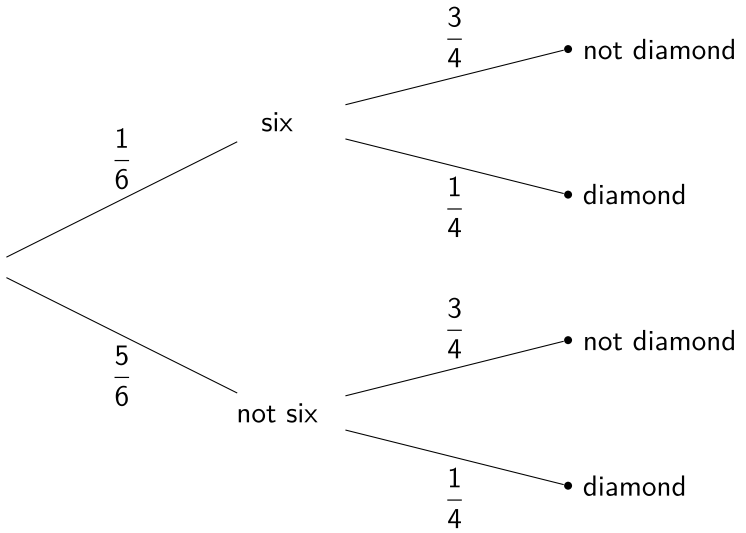
6. A tetrahedral (four-sided) die numbered 1 to 4 and a cubical die numbered 1 to 6 are rolled. Find the probability that at least one of the numbers obtained is 2 or less.

7. Five coins are placed in a bag, with values of 1p, 2p, 5p, 10p and 20p. Two coins are chosen at random, one after another, with the second chosen without replacement. Calculate:
   (a) P(the 20p is selected first)
   (b) P(the 20p is selected as one of the two coins)
   (c) P(total of the two coins $\geqslant$ 15p)
   (d) P(total of the two coins $<$ 12p)

8. Two friends, Alex and Charlie, are part of a class which contains eight pupils in total. One pupil from the class is randomly chosen to hand out mini-whiteboards, and a different pupil is chosen to hand out pens.
   (a) Determine the number of possible ways the two tasks can be assigned to two pupils.
   (b) Hence, find the probability that:
      i. Alex is chosen to hand out the mini-whiteboards, and Charlie is chosen to hand out the pens.
      ii. Neither Alex nor Charlie are chosen for either task.
      iii. At least one of Alex and Charlie are chosen to perform a task.

2.3 Tree Diagrams

Tree diagrams can be created to map out the possible outcomes of a random experiment, and they are especially useful when considering two or more events. For example, consider a game involving rolling a cubical die and picking a playing card at random from a standard set, with an interest in the chance of the die landing on a *six* and the card picked being a *diamond*:



$$P(\text{six } \textbf{and} \text{ diamond}) = \frac{1}{6} \times \frac{1}{4} = \frac{1}{24}$$

Tree diagrams should be complete with each *branch* labelled with a *probability*, and each *node* labelled with an *event*, as above. Since each split into *sibling branches* should show all the possible things that can happen next, the probabilities on sibling branches should add to 1.

$$\text{i.e. } \frac{1}{6} + \frac{5}{6} = 1 \quad \text{and} \quad \frac{3}{4} + \frac{1}{4} = 1$$

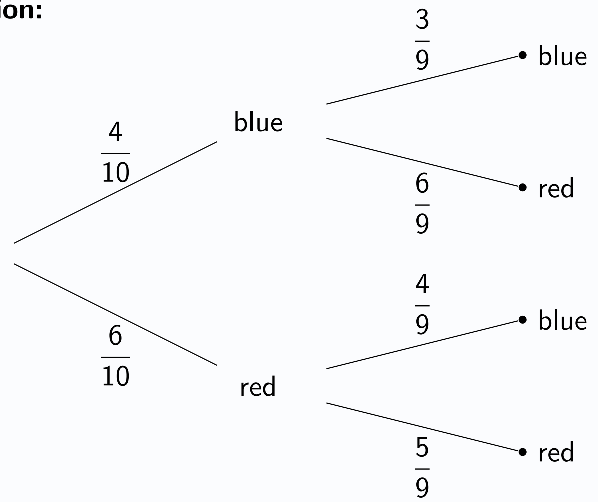
Multiplying the probabilities along a path of branches gives the probability of all those events occurring together.

Note that in the tree diagram above the probability of the card being a diamond does not change depending on whether the roll of a die gives a six or not; this is because the roll of a die and the drawing of a card are **independent**. In the following example the probabilities on the second stage of the tree diagram do change depending on the result of the first stage, since the first and second draws of a cube are **not independent**.

Example

Problem: From a bag containing four blue and six red cubes one is drawn at random, followed by another without replacing the first. Find the probability that two red cubes are drawn.

Solution:



$$P(\text{both red}) = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$$

Exercise 2.3

Construct a tree diagram for each question.

1. A counter is selected at random from a bag containing four green counters and five yellow counters, and then a second counter is selected from the bag without replacing the first counter that was taken.
   (a) Construct a tree diagram to represent the possible results.
   (b) Find the probability that both counters selected are green.
   (c) Find the probability that at least one of the counters selected is green.

2. A game involves a player randomly selecting one card then a second card from a set of six blue cards and four red cards, with replacement. Calculate the probability of a player selecting two cards of the same colour.

3. In Scotland, 41% of the population have the blood type *O positive (O+)*. A doctor in Scotland randomly selects two of their patients and asks for them to take part in a blood screening trial, as part of which their blood type will be checked. Calculate the probability that neither of the two patients has blood type *O+*.

4. A component used in wind turbines is tested at two points in the production process, and marked as either *sufficient quality* or *insufficient quality* on the first testing point and then *pass* or *fail* at the second testing point. A component that is not of sufficient quality at the first testing point will continue to be developed and may still pass at the second testing point. Three-quarters of the components are marked as *sufficient quality* initially, of which nine-tenths go on to pass. Of those initially marked as *insufficient quality*, only three-fifths go on to pass.
   (a) Construct a tree diagram to represent this information.
   (b) Determine the probability that a component selected at random passes at the second testing point.

5. A bus for a particular route that is scheduled to run once per day is cancelled on 3% of days. On the days when it is running, it is running late 12% of the time.
   (a) Construct a tree diagram to represent this information.
   (b) Calculate the probability that the bus is running on time on any given day (i.e. not cancelled and not late).

6. A pencil case contains pencils of three different types: four HB pencils, two 2B pencils and one 2H pencil. Two pencils are selected at random, without replacement.
   (a) Construct a tree diagram to represent the types of the two pencils selected.
   (b) Find the probability that both pencils selected are of the same type.
   (c) Determine the probability that the two pencils selected both are of different types.
   (d) Find the probability that neither pencil selected is an HB pencil.

7. A motorist passes through three sets of traffic lights on their way to work each day. If the probability of having to stop at a red light for each set of lights is 0.6, 0.2 and 0.3 respectively then find the probability that:
   (a) The motorist does not have to stop.
   (b) The motorist has to stop at least once.
   (c) The motorist has to stop at exactly two of the sets of lights.

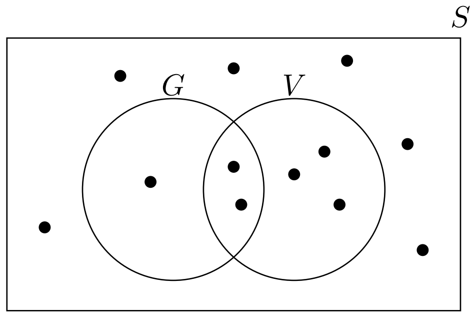
2.4 Venn Diagrams and Set Notation

Venn diagrams are used to show which elements of a set belong to one or more subsets. They can offer an alternative way of representing information about a sample space that might otherwise be shown in a table. For instance, the *contingency table* below shows the twelve dishes from a restaurant's main course menu, broken down by whether or not each is *gluten-free* (G) and whether or not each is *vegan* (V). Beside it is a Venn diagram showing the same information.

	vegan	not vegan
gluten-free	2	1
not gluten-free	3	6

Contingency table

Where a table is used to show categorical data in terms of frequencies in statistics, it is often referred to as a **contingency table**.



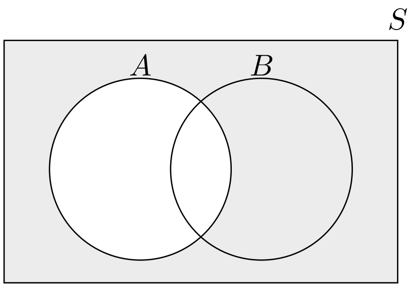
Venn diagram

Complements, Intersections and Unions

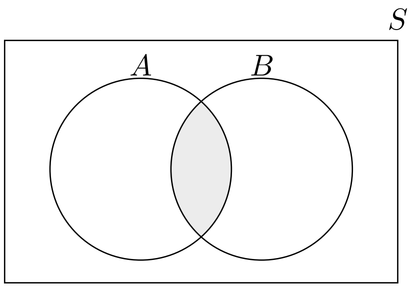
So far, *descriptive* words have been used to refer to different sets of outcomes. Using *set notation*:

Description	Set notation	Name
not A	$\overline{A}$ or A'	Complement of A
A *and* B	$A \cap B$	Intersection of A and B
A *or* B	$A \cup B$	Union of A and B

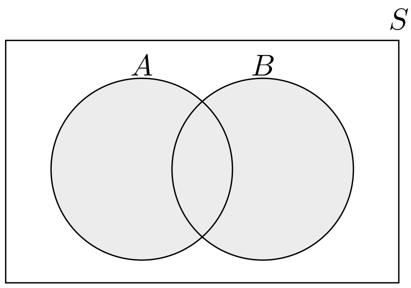
Venn diagrams help illustrate these. Note that *capital letters* are commonly used to represent events.



$\overline{A}$ or A'



$A \cap B$



$A \cup B$

Example

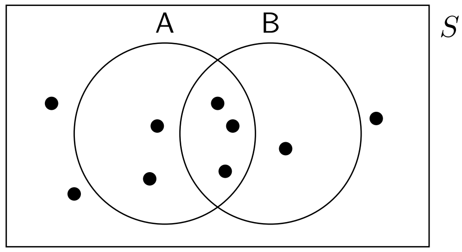
Problem: A dish is picked at random from the twelve on the main course menu shown in the Venn diagram above. State the value of $\mathrm{P}(G \cap V)$.

Solution:
$\mathrm{P}(G \cap V) = \dfrac{2}{12} = \dfrac{1}{6}$

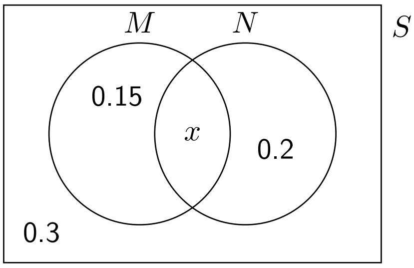
Exercise 2.4

1. One of the nine elements shown on the Venn diagram below is selected at random. State:

(a) P(A)
(b) P(B)
(c) P($\overline{A}$)
(d) P($\overline{B}$)
(e) P($A \cap B$)
(f) P($A \cup B$)
(g) P($\overline{A} \cap B$)
(h) P($A \cap \overline{B}$)

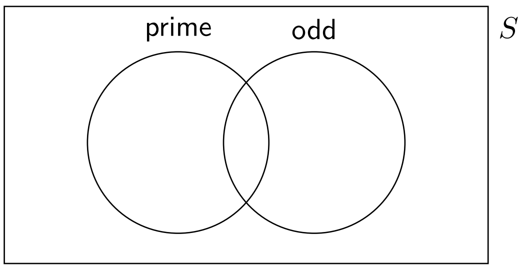


2. The Venn diagram below, containing events M and N, shows the *probabilities* associated with each subset.



(a) State the value of x.
(b) State:
  i. P(N).
  ii. P($\overline{M}$).
  iii. P($M \cup N$).

3. Copy the Venn diagram below, writing each of the integers from 1 to 16 inclusive in an appropriate position. Then, given that an integer from 1 to 16 inclusive is selected at random, state the value of:



(a) P(prime)
(b) P(odd)
(c) P($\overline{\text{prime}}$)
(d) P($\overline{\text{odd}}$)
(e) P(prime ∩ odd)
(f) P(prime ∪ odd)
(g) P($\overline{\text{prime}}$ ∩ odd)
(h) P(prime ∩ $\overline{\text{odd}}$)

4. 100 pupils are asked whether or not they love cats, and whether or not they love dogs, with the results shown in the table below. One pupil is to be selected at random. Let event C represent "a pupil who loves cats is selected", and event D represent "a pupil who loves dogs is selected". By first representing this information on a Venn diagram, find:

(a) P($C \cap D$)
(b) P($C \cup D$)
(c) P($\overline{C} \cap \overline{D}$)
(d) P($\overline{C} \cup \overline{D}$)
(e) P($C \cap \overline{D}$)
(f) P($\overline{C} \cap D$)
(g) P($C \cup \overline{D}$)
(h) P($\overline{C} \cup D$)

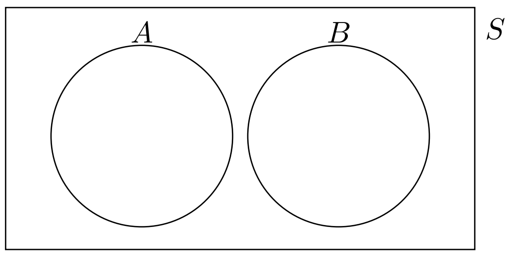
	love cats	do not love cats
love dogs	34	12
do not love dogs	45	9

5. Amongst a class of 24 pupils, seven are studying biology *but not chemistry*, five are studying chemistry *but not biology*, and six are studying neither biology nor chemistry. Find the probability that a pupil chosen at random is studying biology.

6. Of the twenty air pollution monitoring stations in an urban area, nine are showing high levels of particulate matter (PM) in the air and eight are showing high levels of nitrogen dioxide (NO_2) in the air, whilst five are not showing high levels of either pollutant. Determine the probability that one of the twenty stations chosen at random is showing high levels of PM *and* NO_2.

2.5 The Addition Rule for the Union

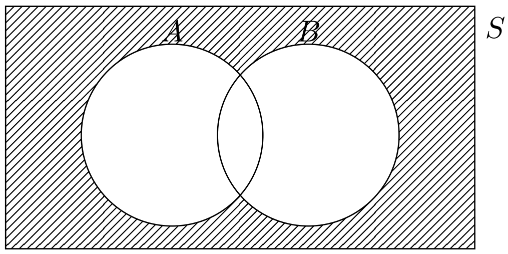
If two sets A and B are such that *no outcomes are common to both A and B*, then they are said to be **mutually exclusive**. This is sometimes visualised on a diagram by two *disjoint* circles, that do not intersect. In terms of probability theory, this means that the probability of the event 'A *and* B' is 0 (impossible).



For Mutually Exclusive events A and B:

$$\mathsf{P}(A \cap B) = 0$$

If two sets A and B are such that *every outcome in the sample space lies in at least one of A or B*, then they are said to be **exhaustive**. This can be visualised on a Venn diagram by crossing out the area of the sample space outside of the two circles. This means that the probability of the event 'A *or* B' is 1 (certain).



For Exhaustive events A and B:

$$\mathsf{P}(A \cup B) = 1$$

In addition to the special cases above (when events are mutually exclusive or exhaustive), the following rules apply to *all* events A, B:

The Complement of A:

$$\mathsf{P}(A) + \mathsf{P}(\overline{A}) = 1$$

The Addition Rule for events A and B:

$$\mathsf{P}(A \cup B) = \mathsf{P}(A) + \mathsf{P}(B) - \mathsf{P}(A \cap B)$$

It should be noted that there are generally numerous different ways to approach probability problems, including using a Venn diagram, a tree diagram or formulae. Whilst the problems on the next page have been designed to allow practice using formulae, a flexible approach may help.

Example

Problem: Given events F and G such that $\mathsf{P}(\overline{F}) = 0.3$, $\mathsf{P}(G) = 0.4$ and $\mathsf{P}(F \cap G) = 0.1$ show that F and G are exhaustive.

Solution:

$$\begin{aligned} \mathsf{P}(F \cup G) &= \mathsf{P}(F) + \mathsf{P}(G) - \mathsf{P}(F \cap G) \\ &= 0.7 + 0.4 - 0.1 \\ &= 1 \quad \therefore \text{ events } F \text{ and } G \text{ are exhaustive} \end{aligned}$$

Exercise 2.5

1. Given events X and Y such that $\mathsf{P}(X \cap Y) = 0.3$, $\mathsf{P}(X) = 0.4$ and $\mathsf{P}(Y) = 0.5$, find $\mathsf{P}(X \cup Y)$.

2. If $\mathsf{P}(\overline{A}) = \frac{1}{5}$, $\mathsf{P}(B) = \frac{2}{3}$ and $\mathsf{P}(A \cap B) = \frac{1}{2}$, determine the value of $\mathsf{P}(A \cup B)$.

3. Calculate the value of $\mathsf{P}(E \cup F)$ given $\mathsf{P}(E) = 0.32$, $\mathsf{P}(F) = 0.29$ and that events E and F are *mutually exclusive*.

4. Given mutually exclusive events C and D such that $\mathsf{P}(\overline{C}) = \frac{5}{7}$ and $\mathsf{P}(D) = \frac{1}{2}$, find $\mathsf{P}(C \cup D)$.

5. Calculate the value of $\mathsf{P}(A \cap B)$ if $\mathsf{P}(A) = 0.591$, $\mathsf{P}(B) = 0.472$ and $\mathsf{P}(A \cup B) = 0.9$.

6. If $\mathsf{P}(\overline{X}) = \frac{17}{80}$, $\mathsf{P}(\overline{Y}) = \frac{29}{40}$ and $\mathsf{P}(X \cup Y) = \frac{19}{20}$, determine $\mathsf{P}(X \cap Y)$.

7. Given *exhaustive* events V and W such that $\mathsf{P}(V) = 0.4$ and $\mathsf{P}(\overline{W}) = 0.2$, find $\mathsf{P}(V \cap W)$.

8. Given events H and J such that $\mathsf{P}(H) = \frac{3}{8}$, $\mathsf{P}(H \cup J) = \frac{7}{9}$ and $\mathsf{P}(H \cap J) = \frac{1}{4}$, find $\mathsf{P}(\overline{J})$.

9. If X and Y are mutually exclusive, $\mathsf{P}(X) = 0.3$ and $\mathsf{P}(\overline{Y}) = 0.4$, calculate $\mathsf{P}(X \cup Y)$.

10. Events A and B are exhaustive. If $\mathsf{P}(A) = \frac{1}{3}$ and $\mathsf{P}(B \cap A) = \frac{1}{6}$, calculate $\mathsf{P}(\overline{B})$.

11. In a fish and chip shop, the probability that a customer's order includes both fish and chips is 0.7, whilst the probability that an order will include at least one of fish or chips is 0.95. If the probability that an order contains chips is 0.8, calculate the probability that an order contains fish.

12. If, for exhaustive events M and N, $\mathsf{P}(M \cap N) = \frac{1}{4}$ and $\mathsf{P}(M) = \mathsf{P}(N)$, calculate $\mathsf{P}(\overline{N})$.

13. Given events R and Q such that $\mathsf{P}(\overline{Q}) = 0.11$, $\mathsf{P}(R \cap Q) = 0.75$ and $\mathsf{P}(R \cup Q) = 0.95$, find $\mathsf{P}(\overline{R})$.

14. For each pair of events, determine whether they are exhaustive or not:

    (a) X and Y given $\mathsf{P}(X) = 0.4$, $\mathsf{P}(Y) = 0.7$ and $\mathsf{P}(X \cap Y) = 0.2$.

    (b) R and T given $\mathsf{P}(R) = \frac{4}{5}$, $\mathsf{P}(\overline{T}) = \frac{1}{10}$ and $\mathsf{P}(T \cap R) = \frac{7}{10}$.

    (c) E and F given $\mathsf{P}(E) = 0.31$, $\mathsf{P}(F) = 0.82$ and $\mathsf{P}(E \cap F) = 0.28$.

15. For each pair of events, determine whether they are mutually exclusive or not:

    (a) W and V given $\mathsf{P}(W) = 0.35$, $\mathsf{P}(V) = 0.45$ and $\mathsf{P}(W \cup V) = 0.5$.

    (b) X and Y given $\mathsf{P}(X) = \frac{1}{3}$, $\mathsf{P}(\overline{Y}) = \frac{4}{5}$ and $\mathsf{P}(X \cup Y) = \frac{8}{15}$.

    (c) A and B given $\mathsf{P}(\overline{A}) = 0.5$, $\mathsf{P}(\overline{B}) = 0.8$ and $\mathsf{P}(\overline{B} \cap \overline{A}) = 0.3$.

Review Exercise

1. A box contains five red balls numbered from 1 to 5, and four yellow balls numbered from 1 to 4. A ball is selected at random from the box. Let *yellow* be the event "the ball selected is yellow".

   (a) State P($\overline{\text{yellow}}$ ∪ 1).

   The ball previously selected is returned to the box. Two balls are then randomly taken from the box, without replacement.

   (b) Determine the probability that neither of the two balls taken from the box are yellow.

2. The table below shows the pupil numbers within a school corridor during a particular period, broken down by subject and course level.

	National 5	Higher
English	21	15
History	17	11

   A pupil is selected at random to run an errand. State:

   (a) P(English ∩ National 5)

   (b) P(History ∪ Higher)

3. For the exhaustive events E and F, such that P$(F \cap E) = 0.7$ and P$(\overline{E}) = 0.2$, determine P(F).

4. A parking ticket machine in a remote location in the Cairngorms National Park takes payment by contactless card reader, which connects to the banking network via a 3G phone signal. On some days, the connection fails and the machine cannot take payment. When the humidity is high, the connection will fail on 18% of days. When the humidity is not high, the connection will fail on 7% of days. Humidity is high in the area on 34% of days. Determine the probability on a randomly selected day of the year that the parking machine can take payment.

5. A board game begins with an *defending* player rolling a tetrahedral die with sides numbered from 1 to 4, and a *attacking* player rolling a cubical die with sides numbered from 1 to 6. The player who obtains the highest number earns a point, with a tie leading to the dice being rolled again.

   (a) Find the probability that the first point is earned, by either player, without the dice having to be rolled again following a tie.

   (b) Find the probability that the attacking player earns the first point.

   If the attacking player wins three points in a row, they are then awarded a *bonus roll* in which they roll *both* dice and gain a bonus point if the product of the numbers obtained is greater than 16.

   (c) Determine the probability that a bonus roll results in a bonus point.

6. The 100 members of an orchestra are shown in the table below, broken down by the section they are in and whether they can attend the upcoming concert.

	String	Woodwind	Brass	Percussion
attend	56	12	10	4
not attend	8	4	5	1

   A member of the orchestra is selected at random. Let B be the event the member selected is in the *brass* section and A that they can *attend* the upcoming concert. State P$(B \cup \overline{A})$.

7. Exhaustive events A, B and C are such that P$(A) = 0.25$, P$(A \cap B) = 0.1$, P$(\overline{B}) = 0.3$ and P$(A \cup C) = 0.6$. Events A and C are mutually exclusive. Determine the value of P$(B \cap C)$.

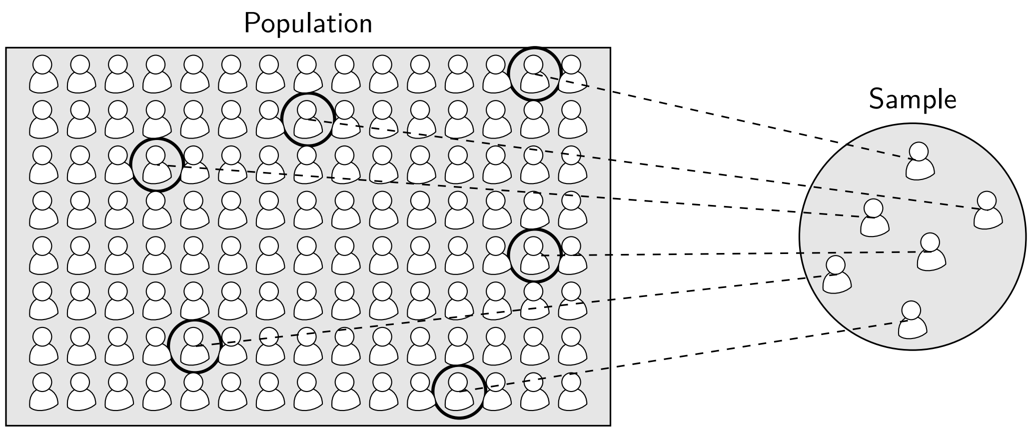
3

Sampling Theory

"Garbage In, Garbage Out" is a phrase used in computer science that applies equally well to statistics. It is important not to assume that the task of a statistician begins with raw data; the manner in which data is collected is of fundamental importance. Statistical inferences based on data that has not been collected with an appropriate *sampling method* should not be trusted.

Populations and Samples

To recap from chapter 1, the term **population** refers to the entire set of objects in which we are interested. A **sample** is a subset of the population.



A **census** is used to collect information from the entire population and a **sample survey** is used to collect data on a sample.

It is usually very difficult to carry out a census as it is time consuming, expensive, and requires an accurate and complete list of every member of the population. In some cases, investigating an entire population could destroy the entire population, such as the burn time of candles. It is therefore usually preferable that a sample survey is carried out and, if representative of the population, it should give an accurate indication of the characteristics of the population.

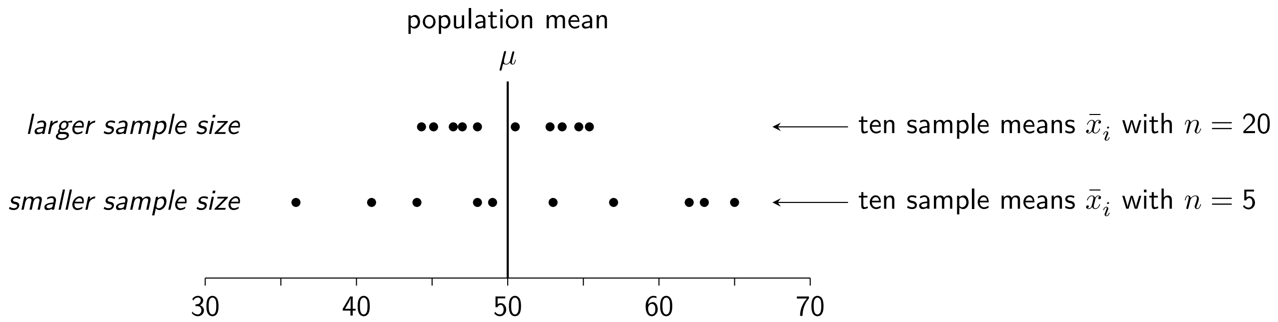
3.1 Sampling and Bias

Samples are used to infer information about the populations from which they are taken, and sample data collected using higher quality sampling techniques should tend to lead to more reliable inferences. There are several key considerations and pitfalls to be aware of whenever the collection of sample data is being planned.

Sample Size

Parameters, such as the population mean, are fixed but usually unknown. If samples are repeatedly taken from a population and their means are obtained, the values of those sample means will sometimes be higher than that of the true population mean and sometimes lower. This comes from *random sampling error* which, despite its name, is a natural effect due to chance variation.

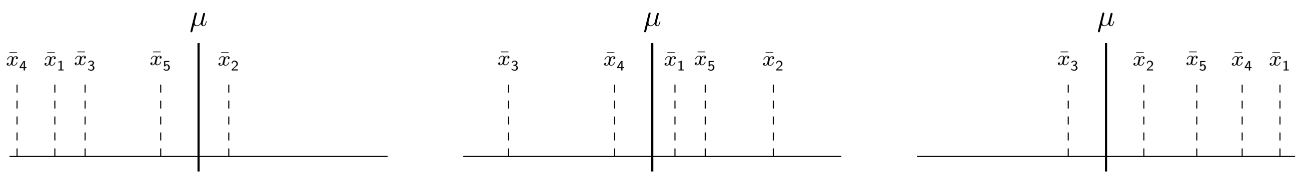
The figure below shows two sets of sample means for random samples taken from the same distribution with a population mean of $\mu = 50$. For one set a sample size of $n = 5$ was used, with $n = 20$ for the other.



It can be seen that larger sample sizes tend to produce sample statistics that better represent the population parameters they are used to estimate. In other words, as sample size increases, the variability of sample statistics decreases. Larger samples are therefore generally preferred, and statistical studies conducted using only small samples of data should typically be treated with more caution than those based on larger samples. When small samples are used, the reasons are usually one or more of cost, time, and availability of data.

Bias

Poor sampling technique may lead to samples that are not *representative* of the population from which they are drawn. This often results in *bias* being introduced: a tendency for sample statistics calculated from these samples to overestimate, or underestimate, the population parameter they aim to represent. The diagrams below illustrate the effects various sampling techniques may have on sample means obtained from repeated sampling; the techniques used for the left and right diagrams are causing bias, whilst the sample means obtained for the middle diagram are *unbiased*.



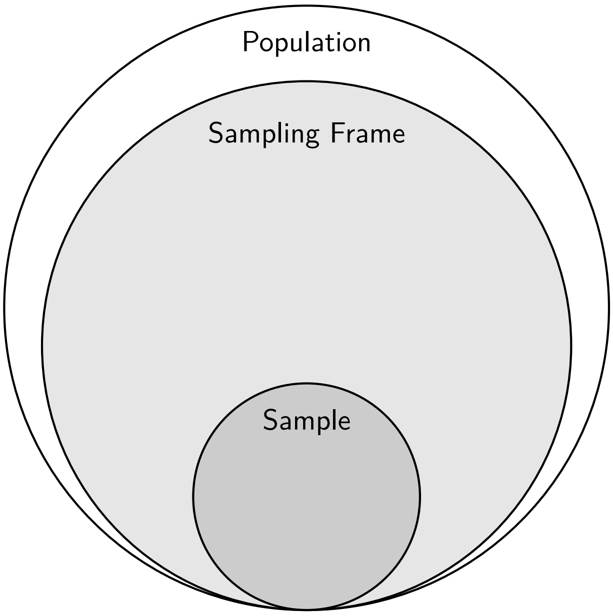
biased (tending to underestimate) *unbiased* (accurate estimation) *biased* (tending to overestimate)

It is very difficult to identify that bias has been introduced, and equally difficult to correct it in practice. The only reliable way to avoid introducing bias is through application of *appropriate random sampling techniques*.

Sampling Frames

Real-life sampling situations for any kind of study are likely to quickly reveal that it is not quite as straightforward as *"just choosing at random"*. Several key elements must be identified and problems considered.

For example, consider the context of wishing to use registration time one morning to survey a sample of pupils at a large secondary school for their views on the school cafeteria.



The **target population** is the *group being studied.*

e.g. all of the pupils who attend a school

The **sampling frame** is the subset of the target population of size N from which a sample is actually drawn, typically in the form of a list (often ordered).

e.g. a numbered list of the school roll

Sampling frames consist of **sampling units** which may be selected to form a sample. These may be people, buildings, square grids on a map, and so on.

e.g. those pupils listed on the school roll

The **sample** consists of the subset of the sampling frame selected to be studied, with size n.

e.g. those pupils selected to be surveyed

The **sampling method** is the way in which sampling units are selected from the sampling frame to create the sample. Sampling methods should feature a *random process* to reduce potential bias.

Care must be taken to ensure that sampling frames are representative of the target population. If the sampling frame does not fully represent the target population, some individuals will have no chance of being selected. For example, if walking habits are being studied, a sampling frame containing only residents of Glasgow city centre is unlikely to be *representative* of an intended target population of people living in Scotland.

The impact on a study of an ill-defined target population, an unrepresentative sampling frame or a poor sampling technique is likely to lead to data which is *biased* and *not representative* of the population, and to *unreliable conclusions*.

Exercise 3.1

1. A maths teacher conducts a survey of pupils' confidence in the subject by asking the first twenty pupils who enter the maths corridor at the start of first period. Suggest one issue with the **sampling frame** and one issue with the **sampling method**.

2. A researcher wants to study the shopping habits of those who use his local supermarket. To collect data, he plans to stand at the exit of the self-checkout area one Sunday morning and ask shoppers if he can take a photo of their receipt. To save time, he will pick only those he feels are most likely to agree to this.
   (a) Identify: (i) The target population. (ii) The sampling frame.
   (b) Suggest two problems with the **sampling frame**, and explain the impact this may have on their data.
   (c) Comment on the suitability of the **sampling method** the researcher plans to use.

3. A large field is full of many wildflowers. A botanist wishes to conduct a survey to assess the health and variety of species of the wildflowers. A random number generator will be used to help obtain a sample.
   (a) Identify one problem with a proposal to use individual flowers as sampling units.
   (b) Suggest an alternative sampling unit for the botanist to use instead.
   (c) Explain the purpose of a random number generator in producing a sample.

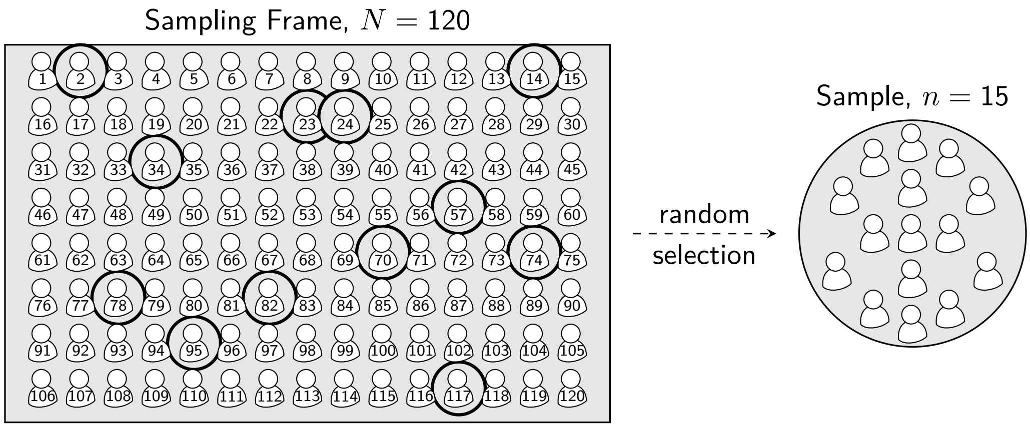
3.2 Six Sampling Methods

In this course, knowledge is required of *four random sampling methods* and *two non-random sampling methods*.

Simple Random Sampling (Random)

With *simple random sampling* each unit of the sampling frame is *equally likely* to be selected, such as by drawing names out of a hat. In general, a simple random sample of size n can be selected from a population of size N by numbering the units in the sampling frame from 1 to N, then selecting n unique random numbers using a random number generator. Those units in the sampling frame corresponding to those numbers generated are then selected to form the sample.

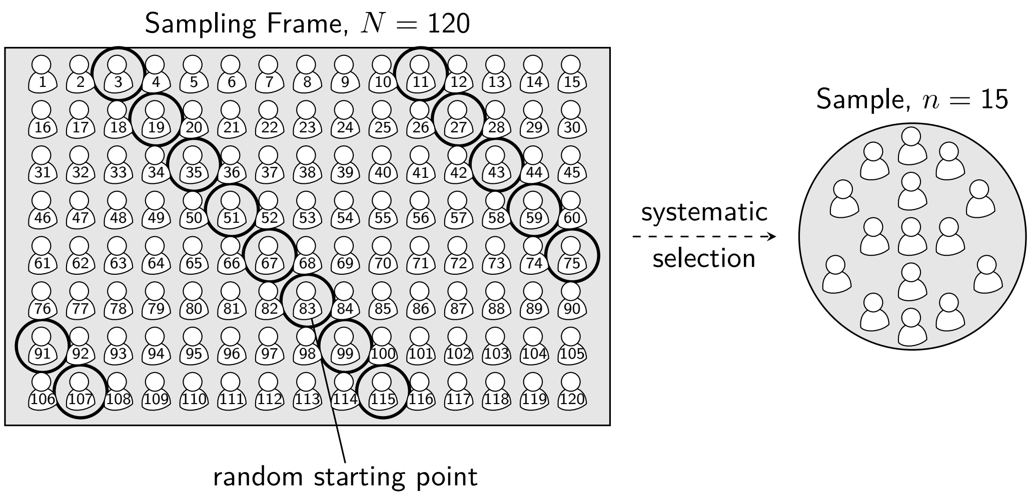
For example, a simple random sample of 15 pupils from a year group of 120 can be made by numbering a list of pupils from the year group from 1 to 120, then selecting 15 random numbers using a random number generator.



Systematic Sampling (Random)

Systematic sampling involves using an orderly *pattern* to select units, starting from a randomly selected point. In general, for a numbered sampling frame of size N, a random number generator may select a number from 1 to N, with the corresponding sampling unit selected. After that initial selection, *every subsequent* $\left(\frac{N}{n}\right)$ th *unit is selected*, looping back to the start of the sampling frame as necessary to complete the sample of size n.

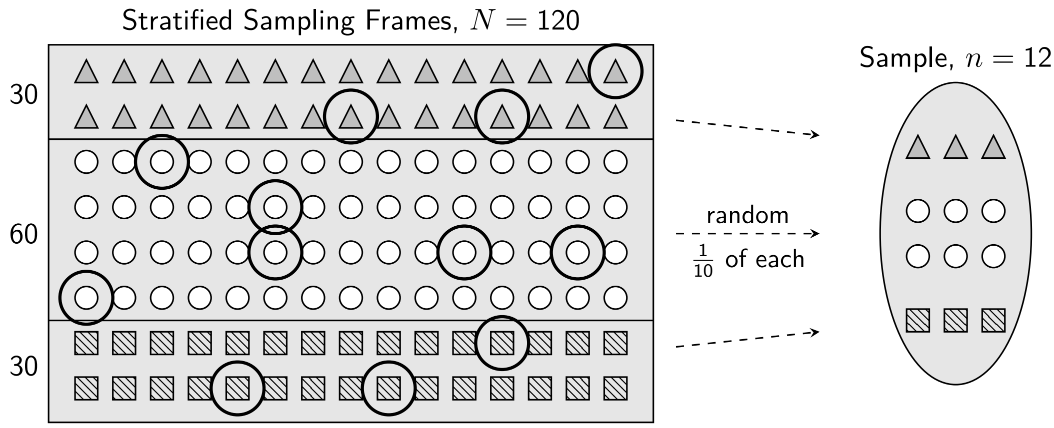
For example, a systematic sample of 15 pupils from a year group of 120 can be made by first selecting a random starting point, such as the 83rd pupil, in an alphabetised list of pupils in the year group. From there, every $\frac{120}{15} = 8$th pupil from the list is chosen, looping back round at the end of the list until 15 pupils have been chosen.



Stratified Sampling (Random)

If a target population contains distinct *strata*, where each stratum may have its own distinct characteristics, *stratified sampling* ensures that each stratum is *proportionally represented*. In general, this involves obtaining an ordered sampling frame for *each stratum* and using the proportion of each stratum in the population to determine how many to sample from each, using simple random sampling. Finally, these samples from the strata are combined.

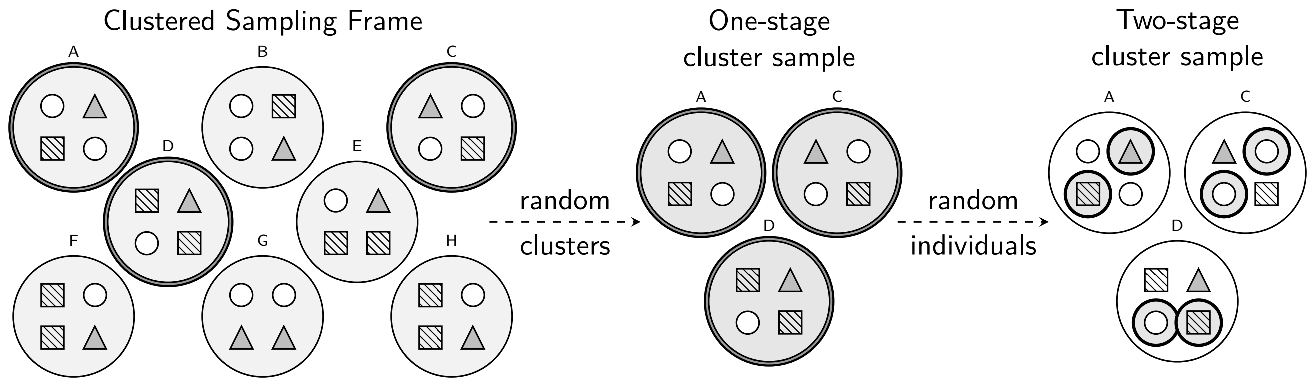
For example, a hospital wishing to survey its staff about the food in the staff canteen may split the $N = 120$ staff into management, doctors/nurses and support staff. For a sample of size $n = 12$, covering $\frac{1}{10}$th of the sampling frame, simple random sampling is used to select $\frac{1}{10}$ of the staff *from each group*. These are then collected to form the stratified sample.



Cluster Sampling (Random)

If a target population is split into clusters, and each cluster can be considered a small representation of the population, containing a similar mix of characteristics, *cluster sampling* may be useful. For a *one-stage* cluster sample, one or more clusters are randomly selected, from a sampling frame of clusters, and then *all individuals* within those clusters form the sample. For a *two-stage* cluster sample, an additional level of sampling takes place to only sample *some* of the individuals within each cluster.

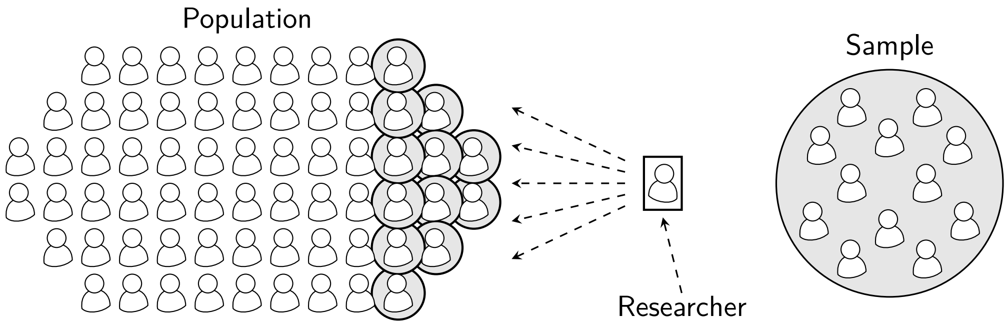
For example, a hospital wishing to survey its staff about the food in the staff canteen may recognise that staff are split into hospital wards A to H. Several wards are randomly selected (A, C and D) and each member of staff working there surveyed. For a two-stage sampling method, simple random sampling could be used to select only *some* of the ward staff from the selected clusters to sample rather than everyone.



The following sampling methods are *non-random*. Random sampling methods are generally preferred as they reduce bias and improve representativeness, whereas non-random methods should be used with caution.

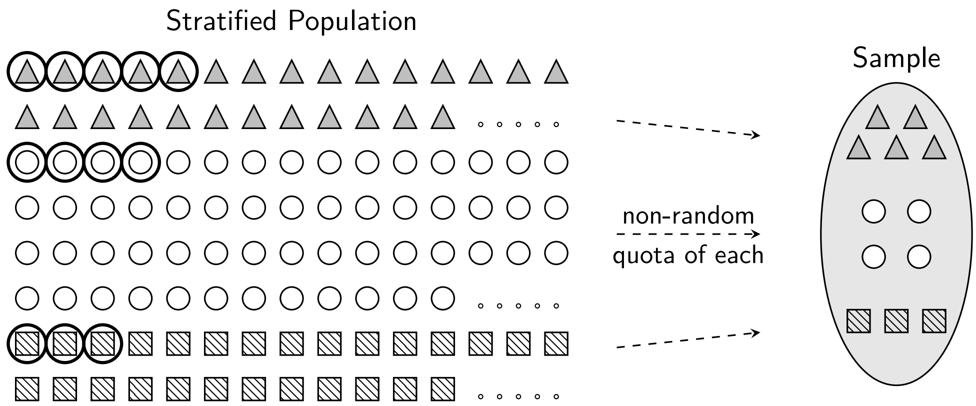
Convenience Sampling (Non-random)

Taking a sample of a target population from an easily obtainable group, without any random process taking place, is called *convenience sampling*. For example, a teacher wishes to investigate the heights of pupils in a particular year group and uses pupils within their class in that year group as the sample.



Quota Sampling (Non-random)

Like stratified sampling, *quota sampling* considers subgroups within the population. However, it does not use a random process, and number sampled from each stratum often comes from a set *quota* rather than being proportional to their size.



For example, interviewers for a marketing company at a mall speak to people who pass, choosing for themselves who to interview from their allocated strata, with no random process used to avoid potential bias.

Advantages and Disadvantages

Advantages and disadvantages of a sampling method for most contexts will typically relate to:

- The time and cost involved in planning and carrying it out (i.e. quick or slow, cheap or expensive).
- The ease with which it can be administered (such as whether access to the sampling frame(s) is straightforward).
- How reliably it may produce a sample which is representative of the population (such as stratified to minimise chances of a subsection of the population being overlooked in a sample).

The method of collecting data may also impact the quality of the data obtained. For example, *emailing* selected individuals a questionnaire may introduce bias by excluding those who don't use email and allowing *self-selection*. Allowing *self-selection*, such as inviting customers to give a rating out of five for a purchase, may also introduce bias which is challenging to account for, as individuals with strong opinions are more likely to respond.

Above all, the greatest concern for the purposes of statistical inference is whether or not a sampling technique applied has used a *random process* to minimise the risk of bias.

Exercise 3.2

1. For each, state the sampling method that is being described.
   (a) The population is made up of groups, each of which contains broadly similar characteristics to each other, and the population. One or more of these groups are randomly selected, and everyone in those selected groups is chosen to make up the sample.
   (b) It is recognised that the population contains a number of subgroups, each of which may have their own distinct characteristics. Individuals are randomly selected from each subgroup, such that the proportions of each subgroup in the sample match those proportions of the population.
   (c) Once an initial random individual from the population has been selected, subsequent individuals are selected at regular intervals until a sample has been obtained.
   (d) Individuals are selected from the population at random, with each having an equal chance of being selected.
   (e) It is recognised that the population contains a number of subgroups, each of which may have their own distinct characteristics. Ten individuals from each subgroup are selected, without the use of a random process.
   (f) The individuals most readily available are selected.

2. The operators of a cruise ship wish to survey the condition of the fittings in the cabins on board by carefully examining a sample of rooms. Of the 360 cabins, 24 are classed as *Deluxe*, 72 are *Premium* and the remaining are *Standard*.
   (a) Explain why the ship's operators might wish to use stratified sampling for their survey.
   (b) Describe the steps involved in sampling 30 of the ship's cabins.

3. A marine biologist is interested in the health of the coral reefs on the West Coast of Scotland. On his next scuba dive near the coastal town in which he lives, he takes some time to photograph the animals and plants on and around the coral reef surface. Based on the survey, he concludes that the health of coral reefs on the West Coast of Scotland has deteriorated.
   (a) State the type of sampling method used by the marine biologist.
   (b) Suggest one reason why the conclusion he reached may not be reliable.

4. The owners of a nationwide chain of petrol stations wish to measure the accuracy of fuel delivery by its pumps. They own 126 petrol stations, each containing typically 8 pumps, and they wish to sample approximately 40 pumps.
   (a) State one advantage of using cluster sampling for the survey.
   (b) Describe the steps involved in creating a one-stage cluster sample of 40 pumps.

5. A music lover has a collection of vinyl records, arranged along a single shelf across one wall. Concerned that some of the actual records may be missing from their sleeves, she checks the one on the far left of the shelf and then every 10th record until she reaches the end of the shelf.
   (a) State the sampling method which most closely matches the method used.
   (b) Identify one way in which the steps taken do not match the ideal process of the sampling method stated in part (a).

6. A school wishes to survey the views of pupils, parents and staff on their opinions of a proposed new school emblem design. Wanting to ensure each group is included, during the annual Sports Day questionnaires are given to twelve of the pupils and twelve of the parents standing in the queue to register, and to the eight members of staff helping run the event.
   (a) State the sampling method used.
   (b) State one disadvantage of this sampling method and describe the impact this may have on the data obtained.

Review Exercise

1. The owners of a chain of nine restaurants, located in various cities around Scotland, wish to interview a sample of the employees working in those restaurants.
   (a) One of the owners suggests that he asks to interview any employees that he sees when he dines at one of the restaurants next week.
      i. State the type of sampling method that he is suggesting the owners use.
      ii. State one disadvantage of this sampling method.
   (b) Another owner proposes instead that they could type the name of each of the 150 employees into a spreadsheet and use a random cell selection tool to pick those to be interviewed.
      i. State the type of sampling method that she is proposing.
      ii. Describe how the owners could instead use systematic sampling to select 30 employees.
   (c) It is pointed out by the HR Director that the employees can be broken down into three categories, each of which may have their own distinct views on what it is like to work in the restaurant chain:
      - Managers and supervisors.
      - Kitchen staff.
      - Waiting and bar staff.

      Each of the restaurants contains a similar mix of staff types.
      i. Describe how a one-stage cluster sample could be conducted.
      ii. Suggest one advantage and one disadvantage of using this method, compared to using systematic sampling.

2. A company employs the following personnel:

Management	Sales staff	Shop floor staff
100	400	4500

   Explain how a stratified sample of 150 employees could be obtained, ensuring that each type of employee is proportionally represented.

3. A nutritionist is studying the calorific contents of pizzas sold in the UK, having read that the salt content is much higher than it is in Italy. They obtain a list of the takeaways which sell pizza in their local area by searching on social media, and randomly select twelve for their study. Suggest two reasons why the sampling frame may not be sufficiently representative of the target population.

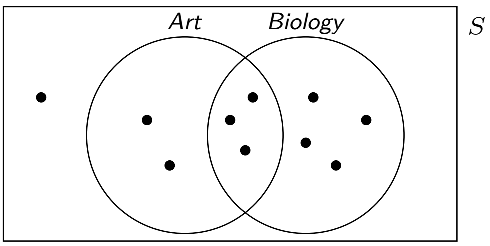
4. A researcher wishes to interview a random sample of pupils taking Higher Mathematics to investigate the proportion of them that are interested in pursuing a career in a mathematical field.
   (a) Suggest a reason why taking a simple random sample of all pupils taking Higher Mathematics in Scotland would not be feasible.
   (b) Explain how a two-stage cluster sample could be taken.

5. A wildflower meadow contains many different species of wildflowers growing together. A botanist wishes to assess the health of the wildflowers in the meadow by studying a sample of the flowers. They have access to a map of the meadow, broken down into boxes with grid references such as A7, and B2. Explain how the botanist could use simple random sampling to obtain a sample.

4

Further Probability Theory

Suppose ten pupils in a class are asked whether they are taking Art, and whether they are taking Biology, and the results are displayed in both a Venn diagram and a table.



	Biology	$\overline{\text{Biology}}$
Art	3	2
$\overline{\text{Art}}$	4	1

From both the table and the Venn diagram, it can be seen that, of the ten pupils, five are taking Art. Therefore, the probability of randomly selecting a pupil from the class and that pupil being found to be taking Art can be stated:

$$\mathrm{P}(\text{Art}) = \frac{5}{10} = \frac{1}{2}$$

Now suppose a pupil is chosen at random and they reveal that they are taking Biology. The probability that the pupil is also taking Art is no longer $\frac{1}{2}$, since only three of the seven pupils taking Biology are taking Art. The probability now being sought is the probability that a pupil is taking Art *given that they are taking Biology*. This is called a *conditional probability* and is written as:

$$\mathrm{P}(\text{Art} \mid \text{Biology}) = \frac{3}{7}$$

"The probability of Art given Biology..."

Conditional probability is fundamental to many statistical analysis techniques introduced later in the course.

4.1 Conditional Probability from Tables and Diagrams

Example

Problem: A medical trial for a drug treating a minor illness has patients given either the drug or a placebo, with a test after one month of treatment determining whether or not each patient has fully recovered.

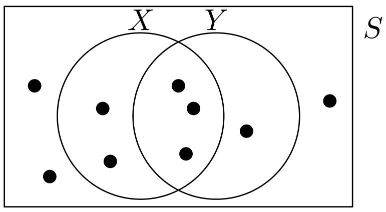
	drug	placebo
fully recovered	42	18
not fully recovered	8	12

A patient is selected at random to be interviewed. State P(fully recovered | placebo).

Solution: $\text{P(fully recovered} \mid \text{placebo)} = \dfrac{18}{30} = \dfrac{3}{5}$

Exercise 4.1

1. The Venn diagram below shows equally likely outcomes in a sample space and events X and Y. State:



(a) $\text{P}(X)$
(b) $\text{P}(X|Y)$
(c) $\text{P}(X \cap Y)$
(d) $\text{P}(X \cup Y)$
(e) $\text{P}(Y)$
(f) $\text{P}(Y|X)$
(g) $\text{P}(Y \cap X)$
(h) $\text{P}(Y \cup X)$

2. A basketball coach records the number of 2-pointer, 3-pointer and free-throw scoring attempts over the course of a season, broken down by whether those attempts scored or missed. An attempt is chosen at random to be analysed. Find:

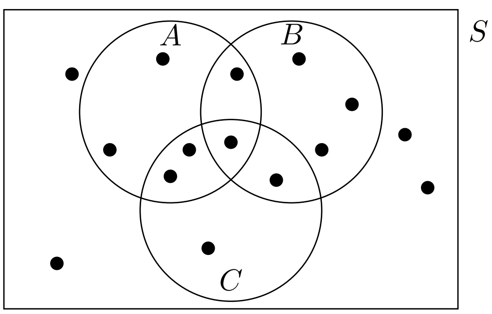
(a) P(scored)
(b) P(scored | 3-pointer)
(c) P(3-pointer)
(d) P(3-pointer | scored)

	2-pointer	3-pointer	free throw
scored	72	30	44
missed	60	52	12

3. On mornings when it is raining, the probability that a red squirrel will leave its drey is 0.4. On mornings when it is not raining, the probability that it will leave its drey is 0.9. It is rainy on 30% of mornings. State the value of:

(a) P(rainy)
(b) P(not rainy)
(c) P(leaves | rainy)
(d) P(does not leave | rainy)
(e) P(leaves | not rainy)
(f) P(does not leave | not rainy)

4. The Venn diagram below shows equally likely outcomes in a sample space and events A, B and C. Determine:



(a) $\text{P}(A)$
(b) $\text{P}(B)$
(c) $\text{P}(C)$
(d) $\text{P}(A \cap B)$
(e) $\text{P}(B \cup C)$
(f) $\text{P}(A \cap B \cap C)$
(g) $\text{P}(A|C)$
(h) $\text{P}(C|B)$
(i) $\text{P}(\overline{B}|C)$
(j) $\text{P}(C|\overline{A})$
(k) $\text{P}((A \cap B)|C)$

4.2 The Conditional Probability Formula

Where a conditional probability cannot be intuitively stated, the formula for conditional probability may be used, shown below. It can be obtained by rearranging the *multiplication rule*.

Conditional Probability:

$$\mathsf{P}(A|B) = \frac{\mathsf{P}(A \cap B)}{\mathsf{P}(B)}$$

Similarly, the complement rule applies under conditioning: $\mathsf{P}(\overline{A}|B) = 1 - \mathsf{P}(A|B)$.

Example

Problem: Given events V and W such that $\mathsf{P}(V) = 0.7236$ and $\mathsf{P}(V \cap W) = 0.1046$, find $\mathsf{P}(W|V)$.

Solution: $\mathsf{P}(W|V) = \dfrac{\mathsf{P}(W \cap V)}{\mathsf{P}(V)} = \dfrac{0.1046}{0.7236} = 0.1446$

Exercise 4.2

1. Find, by first writing out the conditional probability formula for these events:
   (a) $\mathsf{P}(A|B)$ given $\mathsf{P}(B) = 0.8$ and $\mathsf{P}(A \cap B) = 0.16$.
   (b) $\mathsf{P}(\text{red} \mid \text{blue})$ given $\mathsf{P}(\text{red} \cap \text{blue}) = 0.1526$, $\mathsf{P}(\text{blue}) = 0.3147$ and $\mathsf{P}(\text{red}) = 0.2142$.
   (c) $\mathsf{P}(X|Y)$ given $\mathsf{P}(Y) = \frac{1}{2}$ and $\mathsf{P}(Y \cap X) = \frac{1}{3}$.
   (d) $\mathsf{P}(M|N)$ given $\mathsf{P}(\overline{N}) = 0.6137$, $\mathsf{P}(N \cap M) = 0.1249$ and $\mathsf{P}(M \cup N) = 0.4511$.
   (e) Using (a) to (d), state:

      i. $\mathsf{P}(\overline{A}|B)$ ii. $\mathsf{P}(\overline{\text{red}} \mid \text{blue})$ iii. $\mathsf{P}(\overline{X}|Y)$ iv. $\mathsf{P}(\overline{M}|N)$

2. Given events V and W such that $\mathsf{P}(V \cap W) = 0.28$ and $\mathsf{P}(W) = 0.4$, find $\mathsf{P}(\overline{V}|W)$.

3. Numbered cards of different colours are selected at random from a pack during a game such that the probability of a selected card being *yellow* is 20%, and the probability of it being *yellow and a five* is 4%. Find the probability that a selected card is a five *given that it is a yellow*.

4. A football team knows from looking at their past results that they concede at least two goals in $\frac{2}{5}$ of all games. In $\frac{1}{4}$ of all of their games they concede at least two goals and lose the game. Find the probability that, given that they concede at least two goals in a game, they lose.

5. Given events M and N such that $\mathsf{P}(M) = 0.6$, $\mathsf{P}(N) = 0.4$ and $\mathsf{P}(M \cup N) = 0.9$, find $\mathsf{P}(M|N)$.

6. For events V and W, $\mathsf{P}(V) = 0.6$, $\mathsf{P}(V|W) = 0.8$ and $\mathsf{P}(V \cap W) = 0.5$. Calculate $\mathsf{P}(V \cup W)$.

4.3 Independent Events

The probability of tossing a coin and it landing tails is $\frac{1}{2}$. The probability of it landing tails *given that it is a weekday*, notated as P(tails | weekday), is *still* $\frac{1}{2}$. Since knowing whether it is a weekday does not change the probability of the coin showing tails, the two events are said to be *independent*.

A and B are independent if:

$$\mathrm{P}(A|B) \quad = \quad \mathrm{P}(A)$$

The probability of a randomly selected card from a standard pack of playing cards being a Diamond is $\frac{1}{4}$. The probability the card is a Diamond *given that we discover the card picked is red*, notated as P(Diamond | red), is $\frac{1}{2}$. Since knowing that the card is red changes the probability of it being a Diamond, the events are *not independent*.

A and B are not independent if:

$$\mathrm{P}(A|B) \quad \neq \quad \mathrm{P}(A)$$

Example

Problem: Given $\mathrm{P}(D) = 0.25$, $\mathrm{P}(E) = 0.6$ and $\mathrm{P}(D \cap E) = 0.15$, show that E and D are independent.

Solution:

$$\mathrm{P}(D|E) = \frac{\mathrm{P}(D \cap E)}{\mathrm{P}(E)} = \frac{0.15}{0.6} = 0.25$$

Since $\mathrm{P}(D|E) = \mathrm{P}(D)$, events D and E are independent.

Exercise 4.3

1. For each pair of events X and Y, state whether you would expect them to be independent or not.

   (a) A person is selected at random in a survey:
   - X: They were born in January.
   - Y: They work in an office.

   (b) A mechanic is inspecting a car's tyre treads:
   - X: The front tyres have low tread.
   - Y: The rear tyres have low tread.

2. For each pair of events, determine whether or not they are independent.

   (a) A and B given:
   - $\mathrm{P}(A) = 0.3$
   - $\mathrm{P}(B) = 0.8$
   - $\mathrm{P}(A \cap B) = 0.16$

   (b) X and Y given:
   - $\mathrm{P}(X) = \frac{3}{4}$
   - $\mathrm{P}(Y) = \frac{7}{12}$
   - $\mathrm{P}(X \cap Y) = \frac{7}{16}$

   (c) M and N given:
   - $\mathrm{P}(M \cup N) = 0.67$
   - $\mathrm{P}(\overline{M}) = 0.55$
   - $\mathrm{P}(M \cap N) = 0.18$

3. A music tutor taught 40 students last year, each learning one of the violin, trumpet or saxophone, and each had lessons either midweek or at the weekend. A student is chosen at random to take a survey to give feedback. Let V represent that the student chosen took violin lessons, T that they took trumpet lessons and W that they had lessons at the weekend.

	violin	trumpet	saxophone
midweek	12	11	9
weekend	3	4	1

   (a) Show that V and W are independent.

   (b) Show that T and W are not independent.

4.4 The Multiplication Rule for the Intersection

The conditional probability rule can be rearranged to allow the **intersection** of two events to be calculated using a conditional probability. For independent events, since $\mathrm{P}(B|A) = \mathrm{P}(B)$, this simplifies further.

Multiplication rule for any A and B:

$$\mathrm{P}(A \cap B) \;=\; \mathrm{P}(A) \times \mathrm{P}(B|A)$$

Multiplication rule for Independent A and B:

$$\mathrm{P}(A \cap B) \;=\; \mathrm{P}(A) \times \mathrm{P}(B)$$

Example

Problem: Studies of a species of plant in the UK reveal that 8% of the plants have a particular genetic mutation. Amongst those plants with the genetic mutation, 35% produce yellow flowers. If a plant of this species in the UK is selected at random and observed, determine the probability that it will both have the genetic mutation and produce yellow flowers.

Solution:
$\mathrm{P}(\text{mutation} \cap \text{yellow}) = \mathrm{P}(\text{mutation}) \times \mathrm{P}(\text{yellow} \mid \text{mutation}) = 0.08 \times 0.35 = 0.028$

Exercise 4.4

1. Calculate each probability by first *writing out the formula required*:

   (a) $\mathrm{P}(X \cap Y)$ given:
   - $\mathrm{P}(X) = 0.4$
   - $\mathrm{P}(Y) = 0.26$
   - $\mathrm{P}(Y|X) = 0.18$

   (b) $\mathrm{P}(\text{blue} \cap \text{yellow})$ given:
   - $\mathrm{P}(\overline{\text{blue}}) = \frac{2}{3}$
   - $\mathrm{P}(\text{yellow}) = \frac{1}{2}$
   - $\mathrm{P}(\text{blue} \mid \text{yellow}) = \frac{3}{4}$

   (c) $\mathrm{P}(E \cap F)$ given:
   - E and F are independent
   - $\mathrm{P}(E) = 0.82$
   - $\mathrm{P}(\overline{F}) = 0.7$

2. Events A and B are such that $\mathrm{P}(A) = 0.4$, $\mathrm{P}(B) = 0.7$ and $\mathrm{P}(A|B) = 0.56$. Determine the value of:

   (a) $\mathrm{P}(A \cap B)$ (b) $\mathrm{P}(A \cup B)$ (c) $\mathrm{P}(B|A)$

3. Independent events X and Y are such that $\mathrm{P}(X) = 0.4$ and $\mathrm{P}(Y) = 0.8$. Determine the value of $\mathrm{P}(X \cup Y)$.

4. A large box in a maths classroom contains a number of calculators, with 85% of them made by Arizona Instruments and the rest made by Calculo. 70% of the calculators overall are working, whilst 80% of those made by Calculo are working. Determine the probability that a calculator selected at random from the box is a working Calculo calculator.

5. As part of an analysis of a large company's vulnerability to a cyber attack, it is found that 16% of company-issued employee laptops have out-of-date antivirus protection, whilst 24% are only protected by a weak password. Of those with out-of-date antivirus protection, 60% are only protected by a weak password. If a laptop is selected at random, determine the probability that it both has out-of-date antivirus protection and is only protected by a weak password.

6. Given:
   - $\mathrm{P}(\overline{\text{pink}}) = 0.5$
   - $\mathrm{P}(\overline{\text{pink}} \mid \overline{\text{blue}}) = 0.4$
   - $\mathrm{P}(\overline{\text{pink}} \cap \overline{\text{blue}}) = 0.32$

   Find the value of $\mathrm{P}(\text{blue} \cap \text{pink})$.

4.5 Bayes Theorem and Reversing the Condition

In 1763 the Reverend Thomas Bayes, also a statistician, published *"An Essay towards solving a Problem in the Doctrine of Chances"*. In it, he stated a formula that has come to be known as *Bayes' Theorem*, which can be applied in a particular, specialised way to enable existing knowledge to be updated as new information is collected. In turn, this has resulted in an approach towards statistical inference that has been growing since the mid-20th century called *Bayesian statistics*, substantially made more feasible by modern computing power.

Although knowledge of Bayes' Theorem is required in the Advanced Higher Statistics course, Bayesian *statistics* is *not* part of this course, and so will not be covered here. If you would like to find out more about Bayesian statistics, and whether it may be relevant to you in the future, it is recommended that you discuss this with your teacher at a later point in the course. University modules introducing Bayesian statistics are typically available as part of an undergraduate degree in Mathematics or Statistics.

Bayes' Theorem can be derived directly from the formulae already introduced in this chapter. Note that:

$$\mathsf{P}(A \cap B) = \mathsf{P}(A|B) \times \mathsf{P}(B)$$
$$\mathsf{P}(B \cap A) = \mathsf{P}(B|A) \times \mathsf{P}(A)$$

Since $\mathsf{P}(A \cap B) = \mathsf{P}(B \cap A)$:

$$\mathsf{P}(A|B) \times \mathsf{P}(B) = \mathsf{P}(B|A) \times \mathsf{P}(A)$$

Dividing both sides by $\mathsf{P}(B)$ gives:

Bayes' Theorem:

$$\mathsf{P}(A|B) = \frac{\mathsf{P}(B|A) \times \mathsf{P}(A)}{\mathsf{P}(B)}$$

Bayes' Theorem provides a link between a conditional probability and its *reverse*. Using it to find a conditional probability from its reverse is sometimes referred to as *reversing the condition*.

Example

Problem: Given events V and W such that $\mathsf{P}(V|W) = 0.3$, $\mathsf{P}(W) = 0.8$ and $\mathsf{P}(\overline{V}) = 0.1$, find $\mathsf{P}(W|V)$.

Solution: $\mathsf{P}(W|V) = \dfrac{\mathsf{P}(V|W) \times \mathsf{P}(W)}{\mathsf{P}(V)} = \dfrac{0.3 \times 0.8}{0.9} = \dfrac{4}{15}$

Careful reading of text may be required to identify when a probability provided is a *conditional* probability.

Example

Problem: In a neighbourhood, 14% of homes have a garage, whilst 62% of homes have a garden. Of those homes with a garage, 87% have a garden. If a home chosen at random has a garden, determine the probability that it has a garage.

Solution: $\mathsf{P}(\text{garage} \mid \text{garden}) = \dfrac{\mathsf{P}(\text{garden} \mid \text{garage}) \times \mathsf{P}(\text{garage})}{\mathsf{P}(\text{garden})} = \dfrac{0.87 \times 0.14}{0.62} = 0.1965$

Exercise 4.5

1. For each, first write out the formula for Bayes' Theorem explicitly:

   (a) Events X and Y are such that:
   - $\mathsf{P}(X) = 0.7$
   - $\mathsf{P}(Y) = 0.6$
   - $\mathsf{P}(Y|X) = 0.5$

   Calculate $\mathsf{P}(X|Y)$.

   (b) For events V and W:
   - $\mathsf{P}(W) = \frac{4}{5}$
   - $\mathsf{P}(\overline{V}) = \frac{1}{3}$
   - $\mathsf{P}(W|V) = \frac{3}{4}$

   Determine $\mathsf{P}(V|W)$.

2. A bag contains numbered counters of various colours, a quarter of which are red. Three-eighths of the counters are numbered 1, of which two-fifths are red. If a counter is selected at random and it is red, determine the probability that it is numbered 1.

3. Research has shown that 1.7% of people possess a specific gene. A prototype test for the presence of the gene in a person is being developed, and it shows as *positive* for the gene for 7% of all people tested. It is later established that the probability a test shows as positive for a person known to possess the gene is 0.9. If a person is selected at random to take the test, and it shows positive, determine the probability that they actually do possess the gene.

4. Given that $\mathsf{P}(M) = \frac{2}{3}$, $\mathsf{P}(N) = \frac{3}{4}$ and $\mathsf{P}(N|M) = \frac{3}{5}$:

   (a) Calculate $\mathsf{P}(M|N)$.

   (b) Calculate $\mathsf{P}(\overline{M}|N)$.

5. An MOT test is an annual examination of cars and other vehicles required to check they are roadworthy. 83% of the vehicles a local garage performs MOT tests on are cars, whilst the rest are vans. Considering only the first MOT test performed on each vehicle, before any repairs are carried out:
   - 15% of all tests result in a fail
   - Cars account for 86% of all failed tests.

   Determine the proportion of cars that *pass* their first MOT.

6. A region has access to local radio stations, *Pacific 525*, as well as a number of national stations. When a resident of the region directs their car radio to automatically find a station, the probability that it will connect to *Pacific 525* is 0.64, otherwise it will instead connect to a national station. 72% of the time, when their car automatically connects there will be music playing. When their car radio connects to *Pacific 525* the probability music will be playing is 0.84. If the car radio connects to a station and music is playing, determine the probability that the station is *Pacific 525*.

7. Events E and F are such that:
   - $\mathsf{P}(\overline{F}|E) = 0.23$.
   - $\mathsf{P}(F) = 0.87$.
   - $\mathsf{P}(\overline{E}) = 0.09$.

   Calculate $\mathsf{P}(\overline{E}|F)$.

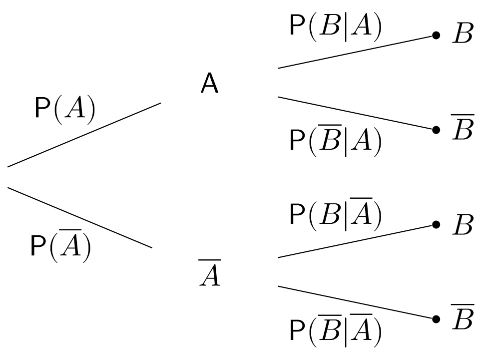
4.6 Total Probability and Tree Diagrams

The *law of total probability* for X is:

$$\mathsf{P}(X) = \sum_{i=1}^{n} \mathsf{P}(Y_i) \times \mathsf{P}(X|Y_i) \qquad \text{(provided } Y_i \text{ are mutually exclusive and exhaustive)}$$

It may help to recognise the right hand side of the equation as containing the *multiplication rule for the intersection*: the total probability of X is obtained by adding together the probabilities of all (disjoint) intersections in which X occurs. In other words, the total probability of an event can be found by considering all the different ways in which it can occur, and adding their probabilities.

The tree diagram below shows the appropriate notation for each stage. The first stage represents event A, and the second stage represents event B.



$$\mathsf{P}(A) \times \mathsf{P}(B|A) = \mathsf{P}(A \cap B) \quad *$$
$$\mathsf{P}(A) \times \mathsf{P}(\overline{B}|A) = \mathsf{P}(A \cap \overline{B})$$
$$\mathsf{P}(\overline{A}) \times \mathsf{P}(B|\overline{A}) = \mathsf{P}(\overline{A} \cap B) \quad *$$
$$\mathsf{P}(\overline{A}) \times \mathsf{P}(\overline{B}|\overline{A}) = \mathsf{P}(\overline{A} \cap \overline{B})$$

The *total probability* of B, $\mathsf{P}(B)$, can be calculated as:

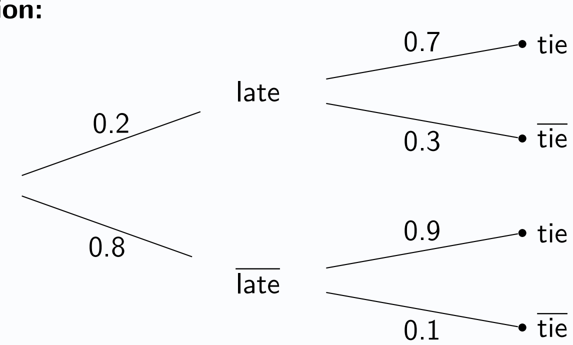
$$\begin{aligned} \mathsf{P}(B) &= \mathsf{P}(A \cap B) + \mathsf{P}(\overline{A} \cap B) \\ &= \mathsf{P}(A) \times \mathsf{P}(B|A) + \mathsf{P}(\overline{A}) \times \mathsf{P}(B|\overline{A}) \end{aligned}$$

The advantage of the use of tree diagrams, for questions where they are suitable, is that they can offer a more intuitive and visual approach to tackling problems than solely using formulae. If they are used, they must be annotated carefully with events and probabilities, and accurate probability notation must still be used when giving answers.

Example

Problem: A teacher observes that one of their pupils is late to school 20% of the time, and when this happens they have their tie with them 70% of the time. When they are on time they have their tie 90% of the time. Construct a tree diagram and calculate the probability that, on a randomly chosen day, they have no tie.

Solution:



$$\mathsf{P}(\text{late} \cap \overline{\text{tie}}) = 0.06 \quad *$$
$$\mathsf{P}(\overline{\text{late}} \cap \overline{\text{tie}}) = 0.08 \quad *$$

$$\mathsf{P}(\overline{\text{tie}}) = 0.06 + 0.08 = 0.14$$

Exercise 4.6

Construct a tree diagram for each question.

1. A UK-based car company makes a special edition model with a colour choice of blue or yellow. It sells cars both within the UK as well as exporting to Europe.

   - 72% of the special edition models manufactured by the company are blue.
   - The rest are yellow.
   - 84% of the blue cars are sold within the UK and 95% of the yellow cars are sold within the UK.

   Calculate the probability that a randomly selected special edition model is sold within the UK.

2. A large number of trees are planted as part of a tree-planting project.

   - Three-quarters of trees are oak.
   - The rest are alder trees.
   - The probability an oak tree failing is 0.356.
   - The probability of failure for an alder tree is 0.234.

   Calculate the probability that a tree randomly selected for observation survives.

3. The star striker for a football team has a probability of 0.7 of scoring in any given game. When they do, the probability that the team goes on to win is 0.8, whilst when they don't the probability of a win drops to 0.45. Calculate the probability that the team wins a randomly chosen game.

4. When Betty hears a song on the radio:

   - The probability that it is a song she knows is 0.8.
   - When she knows it, the probability that it is a song she likes is 0.6.
   - When she doesn't know it, the probability she will like it is 0.3.

   Determine the proportion of songs Betty hears on the radio that she likes.

5. An obstacle course consists of three parts, with each part needed to be completed successfully for a competitor to be allowed to move on to the next. 30% of competitors fail to complete the first part, and of those who make it onto the second part only 40% manage to make it onto the third part. Of those that do make it on to the third part, however, 65% successfully complete it. Calculate the probability that a competitor reaches the third part.

6. As part of a study examining the survival rates of hedgehogs over winter hibernation periods, a number of hedgehogs are captured, weighed, tagged and then released. Depending on their weight they are classed as having a "good weight", an "average weight" or being "underweight", with 18%, 76% and 6% observed respectively in each category. From the study it is found that the probability of survival for each category is 91%, 72% and 35% respectively. Assuming that the distribution of hedgehog weights and probabilities of survival can be taken as representative of hedgehog populations in general, find the proportion of hedgehogs that do not survive winter hibernation.

7. The probability of being dealt a pair (such as two fives) in an opening hand in a game of poker is 0.06. If this happens, a particular player knows that based on their game history they have a probability of 0.83 of winning the hand, whilst otherwise they have a probability of 0.36. Calculate the probability that they win a hand.

4.7 Bayes Theorem and Tree Diagrams

In this chapter, Bayes' Theorem has already been used to calculate the *reverse* of a conditional probability, for example calculating $\mathsf{P}(B|A)$ when $\mathsf{P}(A|B)$ is known. However, this process can be made more intuitive and more readily extended to more complex problems by using tree diagrams combined with the formula for conditional probability:

$$\mathsf{P}(A|B) = \frac{\mathsf{P}(A \cap B)}{\mathsf{P}(B)}$$

A key misunderstanding often made is that a high value for $\mathsf{P}(B|A)$ must lead to a similarly high value for $\mathsf{P}(A|B)$. The following example demonstrates that this is not always the case, and gives an insight into a common problem that must be considered whenever medical testing is being undertaken, especially for rare conditions or diseases:

Example

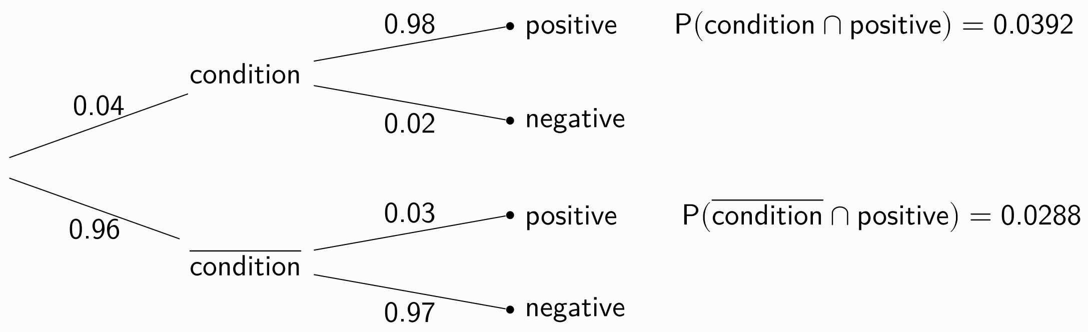
Problem: A medical test for a condition has a *sensitivity* of 98%, meaning that when the patient *has the condition*, the test will correctly return a *positive* result with a probability of 0.98. This means there is a *false negative* rate of 2%. The test has a *specificity* of 97%, meaning that when the patient *does not* have the condition, the test will correctly return a *negative* result with a probability of 0.97. This means there is a *false positive* rate of 3%. The *prevalence* in the general population is 4%, meaning only 4% of people have the condition.

If a person is selected at random from the general population to be tested for the condition:

(a) Calculate the probability that the test comes back *positive*.

(b) Given that the test comes back positive, determine the probability that they do actually *have* the condition.

Solution:



(a) $\mathsf{P}(\text{positive}) = 0.0392 + 0.0288 = 0.068$

(b) $\mathsf{P}(\text{condition} \mid \text{positive}) = \dfrac{\mathsf{P}(\text{condition} \ \cap \ \text{positive})}{\mathsf{P}(\text{positive})} = \dfrac{0.0392}{0.068} = 0.5765$

So even though the test is highly accurate, only about 57.65% of people who test positive actually have the condition.

It may be surprising that a test which appears so effective can lead to such a high proportion of false positives, and calculations such as this should be considered when healthcare professionals establish testing procedures.

Exercise 4.7

Construct a tree diagram for each question.

1. A company sends customers' orders by either regular or express delivery.
   - 35% of orders are sent by express delivery.
   - Of those sent by express delivery, 98% arrive undamaged.
   - 91% of the orders sent by regular deliveries arrive undamaged.

   Customers are requested to submit a form online to confirm whether their order arrived damaged or undamaged.

   (a) Determine the probability that a form received by the company states that an order arrived damaged.

   (b) Given that an order arrived damaged, calculate the probability that the order was sent by express delivery.

2. Anti-virus software is being developed to detect whether attachments sent by email are malicious or safe, and flag those that are malicious. In its current state it can correctly flag that an attachment is malicious 92% of the time. It incorrectly flags safe attachments 2% of the time. A company is interested in using the software, and 1.2% of all attachments emailed to their employees are actually malicious.

   (a) Determine the proportion of attachments that are flagged as malicious.

   (b) Given that an attachment is flagged, calculate the probability that it is malicious.

3. A pupil sitting a multiple choice test is able to answer $\frac{3}{4}$ of the questions and guesses the rest. They know from trying practice tests that the probability that they are correct in the questions they are able to answer is 0.9, whilst they have a one-in-five chance of getting a correct answer by guessing. Determine the probability that a question which they get wrong was one that they guessed.

4. A marketing campaign for a new product is estimated to have been seen by three times as many people through its social media advert as by its promotional email. The probability that seeing the social media advert leads to a person visiting their website is 0.037, whilst for the promotional email that figure drops to 0.021.

   (a) Determine the probability that someone who sees the marketing campaign in either form visits the website.

   (b) Determine the probability that a visit to the website due to the marketing campaign came from that person seeing the social media advert.

5. A sports team is given an extra rest day half of the time after winning a game, a quarter of the time after drawing a game and only a tenth of the time when they lose a game. They win 55% of their games, draw 25% of their games and lose the rest. Given that they are given an extra rest day, find the probability that they lost the last game.

6. All precision-engineered parts at a small factory undergo two inspections as part of quality control. During Inspection A, 8% of parts are flagged for a more thorough inspection during Inspection B. Of those that are more thoroughly inspected during Inspection B, 15% are scrapped due to flaws in the manufacturing process. 3% of those that had not been flagged initially are also then scrapped during Inspection B as flaws are spotted that were initially missed. Finally, of those parts that have not been scrapped, 1% are set aside for stress-testing of the final products and never leave the factory. Given that a part doesn't leave the factory, find the probability that it was due to a flaw.

7. For medical tests for conditions that are known to have the given specificity and sensitivity, as well as the prevalence of the condition, calculate the probability that a randomly tested person has the condition, given that they test positive:

   (a) Sensitivity = 94%, Specificity = 89%, Prevalence = 9%

   (b) Sensitivity = 98%, Specificity = 99%, Prevalence = 1%

Review Exercise

1. Events E and F are such that:
   - P$(F \cap E) = 0.7$
   - P$(\overline{E}) = 0.2$
   - E and F are exhaustive.

   Calculate P$(E|F)$.

2. A second-hand online computing retailer has 40 laptops in stock, broken down by whether they use Windows or Mac, and the grades A, B and C depending on the condition they are in:

	A	B	C
Windows	7	18	5
Mac	4	5	1

   A laptop is selected at random.
   W is the event that the laptop selected uses Windows.
   A is the event that the laptop selected is in grade A condition.
   B is the event that the laptop selected is in grade B condition.

   (a) Find:
      i. P$(W \cup A)$.
      ii. P$(\overline{W} \cap B)$.

   80% of the Windows laptops and 90% of the Mac laptops come with the original power supply.

   (b) Calculate the probability that a randomly selected laptop comes with its original power supply.
   (c) Determine the probability that a randomly selected laptop is a Mac laptop, given that it does not come with its original power supply.

3. At the start of each round of a board game played by four players, a random process determines who gets to go 1st, who goes 2nd who goes 3rd and who goes 4th. Sam knows that when she plays against her two friends, she wins the round two-thirds of the time when she goes 1st, half of the time when she goes 2nd and only a fifth of the time when she goes either 3rd or 4th.

   (a) Assuming that each round is independent of the result of previous rounds, calculate the proportion of rounds that Sam wins.
   (b) Given that Sam lost the last round, calculate the probability that she went 1st.

4. For the events M and B: P$(M \cap B) = \frac{3}{25}$, P$(M) = \frac{1}{5}$ and P$(\overline{B}) = \frac{2}{5}$. Show that events M and B are independent.

5. As part of a machine learning project, a program is created that tries to correctly identify whether it is being shown a picture of a dog, a cat or a human. It has been tested and found to correctly identify a human with the probability 0.8, correctly identify a dog with the probability 0.6 and correctly identify a cat with the probability 0.55. When subsequently put to use by the public, the program is shown a picture of a human 60% of the time, a dog 25% of the time and a cat 15% of the time. Assuming the probabilities of successful detection are the same when used by the public as when under testing, calculate:

   (a) the proportion of times that the program is incorrect
   (b) the probability that the program was shown a cat, given that it is incorrect.

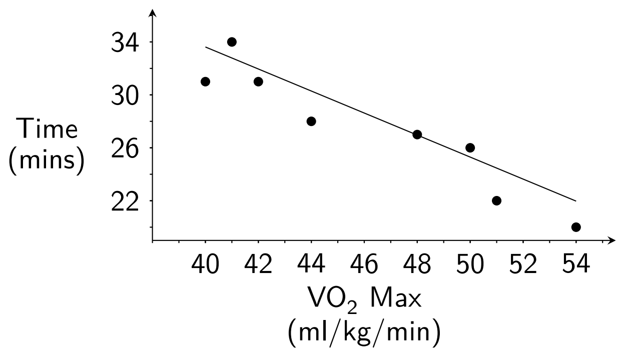
5

An Introduction to Linear Regression

A common aim when working with data is to investigate possible relationships between two numerical variables. For example, suppose a sports scientist wishes to explore the relationship between runners' VO_2 Max, which is a measure of aerobic fitness, and their 5km running times. From a sample of eight runners, they may record the values of the two *variables* for each individual, presenting the results in a table as follows:

Runner	A	B	C	D	E	F	G	H
VO_2 Max (ml/kg/min)	50	41	54	42	44	48	40	51
5km time (mins)	26	34	20	31	28	27	31	22

Before applying any statistical analysis techniques to this **bivariate data**, the sports scientist constructs a **scatterplot** to visualise the relationship between them, looking for any apparent patterns or problems. In this example, VO_2 Max is treated as the **predictor** variable (plotted on the x-axis) and 5km running time is treated as the **response** variable (plotted along the y-axis).



In general, the predictor variable is chosen as the one which may help explain or predict changes in the response variable, and is typically used to calculate *estimates* of the response variable. Here, this suggests that VO_2 Max may be used to estimate 5km running time for other runners whose times are not yet known.

Although the sample size is very small, the scatterplot seems to suggest that there is a *negative linear relationship* between a runner's VO_2 Max and their 5km running time, as higher VO_2 Max values are associated with lower running times. To illustrate this, a **line of best fit** (or *trend line*) has been added to the scatterplot.

This chapter will introduce the use of a *correlation coefficient* to evaluate the strength of any linear relationship between numerical variables and the construction of a *linear regression model* to describe this relationship and make predictions.

5.1 Scatterplots and Linear Correlation

When deciding whether a linear relationship exists between two numerical variables, a scatterplot should always be drawn before any analysis is undertaken using the bivariate data collected. This is the first step in determining the level of ***linear relationship*** *(or* linear correlation*) between the variables. Whilst other kinds of relationship may exist between two numerical variables, in this course the focus is on the existence and strength of* linear* relationships. Consider the following:

Figure 1

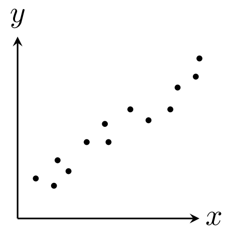


Figure 2

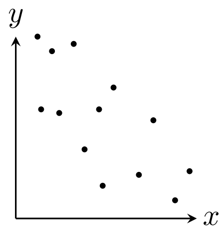


Figure 1 shows a **strong positive linear correlation** between x and y, where both variables tend to increase together.

Figure 2 shows a **weak negative linear correlation** between x and y, where, as one variable increases, the other tends to decrease.

Figure 3

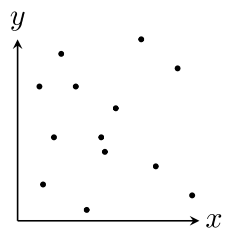
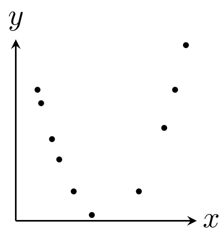


Figure 4



In **Figure 3**, as one variable increases, there appears to be no clear pattern as to how the other variable behaves. There appears to be **no linear relationship** between x and y.

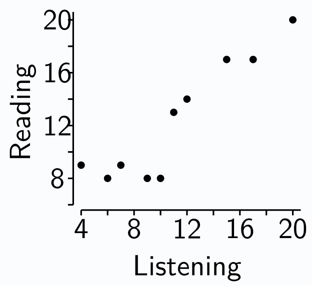
Figure 4 is also an example of **no linear relationship** between x and y, although there appears to be some non-linear (possibly quadratic) association between the variables.

Note that **correlation does not imply causation**. A linear relationship can exist between variables without meaning that one causes the other; there may be other variables influencing both. Seemingly linear relationships can also appear by chance - this is referred to as *spurious correlation*.

Example

Problem: A French class takes two tests; *listening* and *reading*. The results of the 10 pupils are shown in the table and scatterplot below, along with some summary statistics. Comment on the scatterplot.

Listening	9	7	11	20	4	6	17	12	10	15
Reading	8	9	13	20	9	8	17	14	8	17



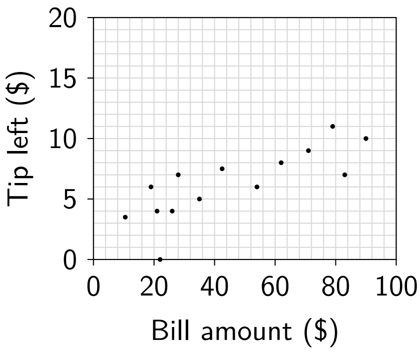
Solution: The scatterplot appears to show a strong positive linear relationship between pupils' listening and reading scores.

Note that there is no universally agreed scale for describing correlations as 'strong', 'moderate' or 'weak'. However, it is still recommended for comments to include some description of the strength of a linear relationship.

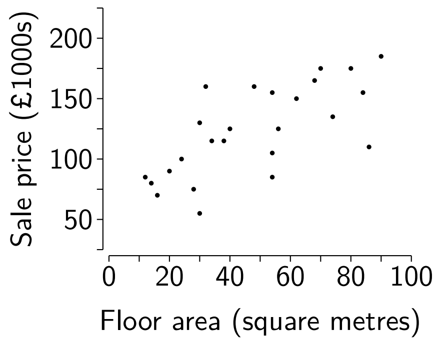
Exercise 5.1

1. For each, describe the relationship shown in the scatterplot.

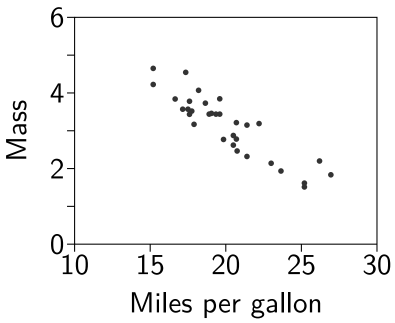
(a) Data is collected by a diner in the USA on the *bill amount* and *tip left* by its guests.

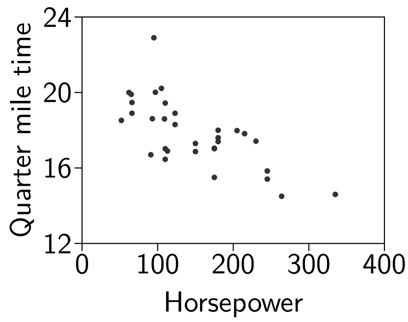


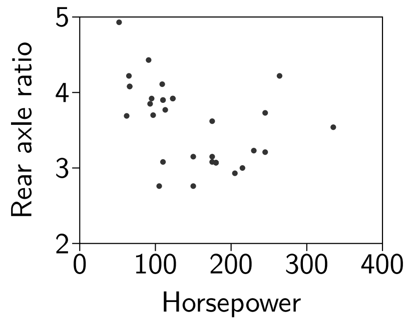
(b) The *sale price* and *floor area* are recorded for a number of 1-bedroom city centre apartments.



2. Information on the designs of 32 different models of cars from the 1970s was collected by a motoring magazine. Data collected included *miles per gallon*, *mass* (in thousands of pounds), *quarter mile time* (in seconds), *rear axle ratio* and *horsepower*. Three scatterplots are shown below to explore possible relationships between the variables.

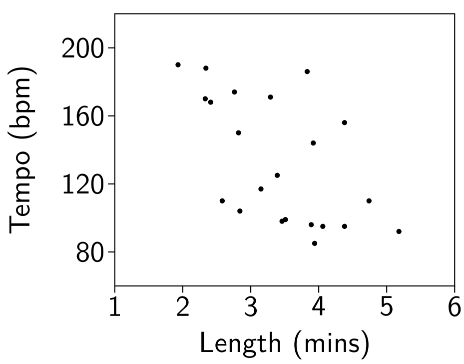






Comment on each scatterplot.

3. A statistician is keen to learn what can be learned by studying data on songs provided by a music streaming platform. For a random sample of pop songs on the platform, their *length* (x, in minutes) and *tempo* (y, in bpm) are recorded, and a scatterplot is constructed. Some summary statistics were also calculated.



$\Sigma x = 75.13$

$\Sigma y = 2919$

$\Sigma x^2 = 272.1$

$\Sigma y^2 = 416279$

$\Sigma xy = 9583.94$

$n = 22$

(a) Comment on the scatterplot.

(b) Use the summary statistics to calculate the mean song length and the mean tempo for the sample.

5.2 Pearson's Product Moment Correlation Coefficient

Instead of relying on vague terms such as 'strong', 'moderate' and 'weak', the strength of the *linear relationship* between two variables can be measured and better communicated using a **correlation coefficient**. Whilst there are a number of different correlation coefficients used in statistics, the only one used in this course is *Pearson's Product Moment Correlation Coefficient (PMCC)*, which can now simply be referred to as the *correlation coefficient*. Given a sample of bivariate data, the *sample correlation coefficient*, r, can be calculated as:

Pearson's Correlation Coefficient

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

The formulae for the *"Three Ss"* and the *"Five Sigmas"* are given on **page 4** in the Data Booklet:

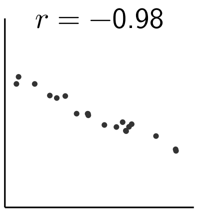
Three Ss:

$$S_{xx} = \Sigma(x-\bar{x})^2 = \Sigma x^2 - \frac{(\Sigma x)^2}{n}$$
$$S_{yy} = \Sigma(y-\bar{y})^2 = \Sigma y^2 - \frac{(\Sigma y)^2}{n}$$
$$S_{xy} = \Sigma(x-\bar{x})(y-\bar{y}) = \Sigma xy - \frac{\Sigma x \Sigma y}{n}$$

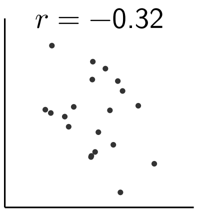
Five Sigmas:

Σx
Σy
Σx^2
Σy^2
Σxy

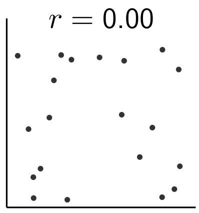
For all sets of data, $-1 \leqslant r \leqslant 1$. A correlation coefficient of **zero** suggests there is **no linear relationship** between the two variables, whilst any non-zero correlation coefficient suggests the presence of a possible linear relationship. The figures below show scatterplots of sample data along with their sample correlation coefficients, r, and an interpretation.



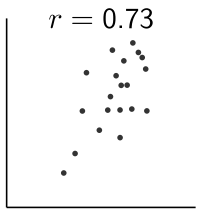
very strong negative linear relationship



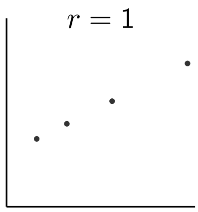
weak negative linear relationship



no linear relationship

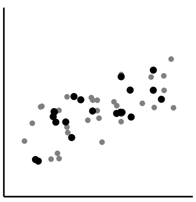


strong positive linear relationship



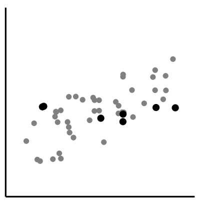
perfect positive linear relationship

As the correlation coefficient, r, is calculated for a *sample*, it is a **sample statistic**. It gives an estimate for the *population correlation coefficient*, ρ (the Greek letter, *"rho"*). For this reason, a non-zero value of r calculated from a sample only *suggests* a linear relationship exists between the variables.



$\rho = 0.66$
$r = 0.71$
$n = 20$

This **sample** has resulted in a value for r which approximates well the true **population** correlation coefficient, ρ.



$\rho = 0.66$
$r = -0.18$
$n = 7$

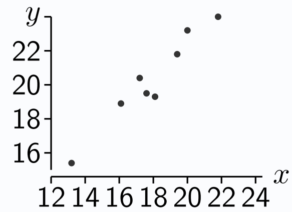
This **sample** has resulted in a value for r which is far from the true **population** correlation coefficient, ρ.

Example

Problem: A biologist records the tail length (x) and total height (y) (in cm) for a random sample of 8 red squirrels. A scatterplot and summary statistics for the data are produced and shown below.

$\Sigma x = 143.4 \quad \Sigma y = 163.3 \quad \Sigma xy = 2978.5$

$\Sigma x^2 = 2618.3 \quad \Sigma y^2 = 3391.8 \quad n = 8$



Calculate the correlation coefficient and interpret its value.

Solution:

$$S_{xx} = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 2618.3 - \frac{143.4^2}{8} = 47.82$$

$$S_{yy} = \Sigma y^2 - \frac{(\Sigma y)^2}{n} = 3391.8 - \frac{163.3^2}{8} = 58.43$$

$$S_{xy} = \Sigma xy - \frac{\Sigma x \Sigma y}{n} = 2978.5 - \frac{143.4 \times 163.3}{8} = 51.39$$

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} = \frac{51.39}{\sqrt{47.82 \times 58.43}} = 0.9354$$

This suggests a strong positive linear correlation between the length of squirrels' tails and their heights.

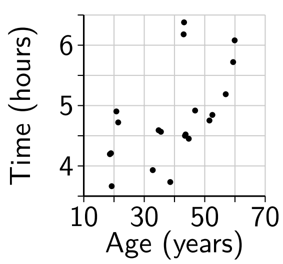
Exercise 5.2

1. Students' average homework mark and their test score are recorded. Calculate the correlation coefficient.

Homework (average, out of 20)	17.1	11.2	19.6	15.4	19.8	8.1
Test (%)	81	46	94	83	94	64

2. For each, calculate the correlation coefficient and interpret its value.

(a) For 20 first-time marathon runners, their age (x, in years) and finishing time (y, in hours) is recorded.



$\Sigma x = 785.9$

$\Sigma y = 96$

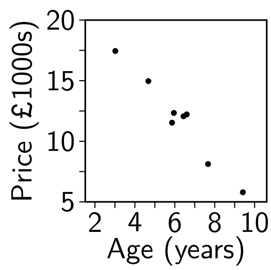
$\Sigma x^2 = 34481$

$\Sigma y^2 = 472.6$

$\Sigma xy = 3888.7$

$n = 20$

(b) The age (x, in years) and selling price (y, £1000s) of 8 used cars is recorded.



$\Sigma x = 49.63$

$\Sigma y = 94.48$

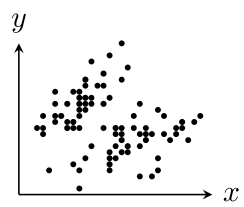
$\Sigma x^2 = 333.1$

$\Sigma y^2 = 1208.0$

$\Sigma xy = 538.87$

$n = 8$

3. A *sepal* is part of a flower. The lengths (x, in cm) and widths (y, in cm) are measured for a random sample of 100 *iris* flowers, of two different varieties. A scatterplot is plotted, and a correlation coefficient of -0.206 is obtained. The researcher concludes that this proves that there is a weak negative linear relationship.



(a) Without considering the scatterplot, explain why the researcher should not have used the word "proves" in their conclusion.

(b) Comment on what might be observed by inspecting the scatterplot, and explain what the researcher should do with the data as a consequence.

(c) Suggest two possible improvements to the design of the scatterplot.

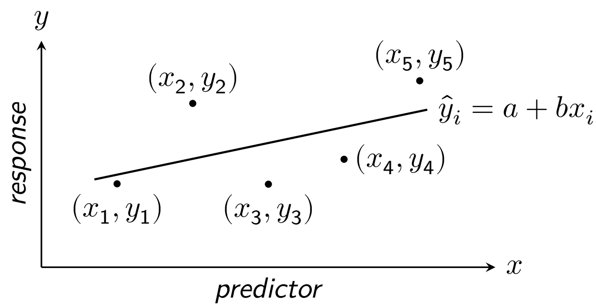
5.3 Least Squares Linear Regression

A y *on* x regression model is one constructed with the aim of producing a trend line which can be used to predict unknown values of y based on known values of x. For a sample of bivariate data $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$, this *regression line* has an equation of the form $\hat{y}_i = a + bx_i$, where:

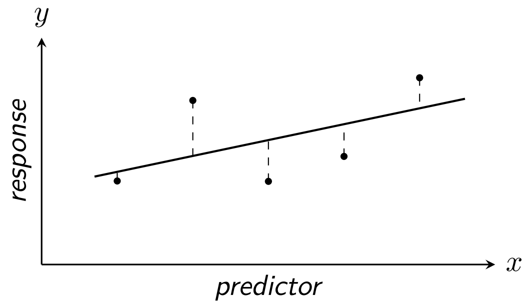
$\hat{y}_i$ (*'y hat'*) is the *best estimate* of y_i given x_i.

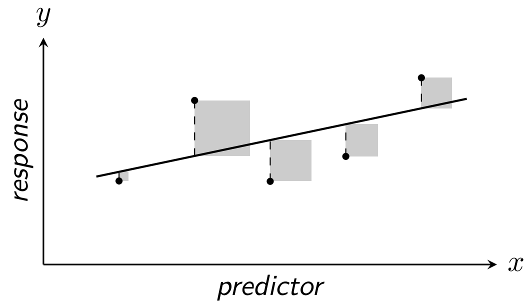
Sample statistic a represents the line's **intercept**, and is an estimate for the true *intercept parameter* α (*"alpha"*).

Sample statistic b represents the line's **slope**, and is an estimate for the true *slope parameter* β (*"beta"*).



To determine the equation of this regression line, the method of *least squares regression* is used. Since the goal is to produce the *best estimates* for y_i, vertical distances between observed values and the line are considered. The *least squares* regression line is that which minimises the sum of the squares of these vertical distances.





Since regression lines are generally calculated from a random *sample* of bivariate data points, the values for the population parameters α and β have to be estimated, with the sample statistics a and b respectively calculated as *estimators*:

Estimation of Slope and Intercept Parameters

$$\hat{\beta} = b = \frac{S_{xy}}{S_{xx}} \qquad \text{and} \qquad \hat{\alpha} = a = \bar{y} - b\bar{x}$$

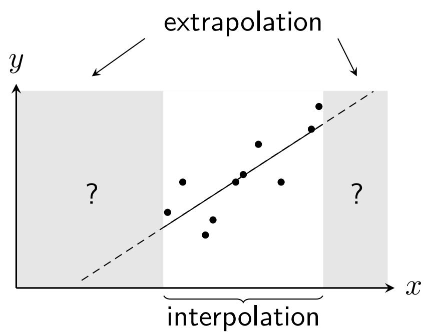
Note that the calculation for a implies that the regression line always passes through the point $(\bar{x}, \bar{y})$.

Once values for a and b have been calculated, and so the least squares regression line of y on x obtained as $\hat{y}_i = a + bx_i$, unknown values of y_i may be estimated for given values of x_i. This equation (and the line it defines) is *not* suitable for predicting x from y; this process will be covered in a later chapter.

Whether a linear relationship will continue to hold for x values lower or higher than any observed within the range of the data can be unpredictable, and should not be assumed.

Interpolation, prediction within the range of the control data, is generally reliable if the correlation is high.

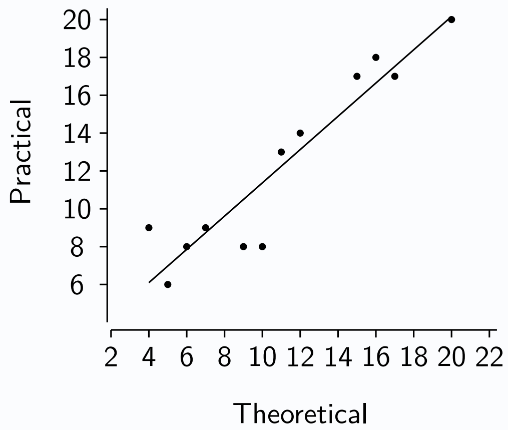
However, **extrapolation**, prediction outwith the range of the data, should be avoided as unreliable and unsupported.



Example

Problem: A biology class takes two tests, one theoretical and one practical. The results of the 12 pupils are shown in the table and scatterplot below, along with some summary statistics.

Theoretical test result (x)	5	9	7	11	20	4	6	17	12	10	15	16
Practical test result (y)	6	8	9	13	20	9	8	17	14	8	17	18



$$\Sigma x = 132, \quad \Sigma y = 147,$$

$$\Sigma x^2 = 1742, \quad \Sigma y^2 = 2057, \quad \Sigma xy = 1872$$

(a) Calculate the equation of the least squares regression line of y on x.

(b) Calculate a predicted mark for the practical test for a pupil who missed it, having scored 8 in the theoretical test.

(c) Comment on the reliability of this prediction.

Solution:

(a)

$$S_{xx} = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 1742 - \frac{132^2}{12} = 290$$

$$S_{xy} = \Sigma xy - \frac{\Sigma x \Sigma y}{n} = 1872 - \frac{132 \times 147}{12} = 255$$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{255}{290} = 0.879$$

$$a = \bar{y} - b\bar{x} = \frac{147}{12} - 0.879 \times \frac{132}{12} = 2.578$$

$$\hat{y}_i = 2.578 + 0.879 x_i$$

(b) $\hat{y} = 2.578 + 0.879 \times 8 = 9.61$

(c) This is an example of interpolation as the value of x used is within the range of values of the other students, and consequently it can be considered a robust prediction, especially given the strong linear relationship suggested by the scatterplot.

Another way of writing the equation found above is: practical $= 2.578 + 0.879 \times$ theoretical. This suggests that each increase of 1 mark in the theory test is associated with an increase of 0.879 marks in the practical.

Scatterplots should always be constructed and closely scrutinised before further analysing bivariate data.

By inspecting a scatterplot, it may become apparent, for instance, that the data set is separated into *distinct groups*, which ought to be considered separately, including calculating a correlation coefficient and constructing a regression line for each group.

Outliers can have a large influence on both the correlation coefficient and the regression line, so they should always be (visually) identified and considered, removing them only if doing so can be justified.

Exercise 5.3

1. A sample of bivariate data collected by a researcher is shown in the table below.

x	38	27	19	45	8	23	11	34
y	34	53	42	13	19	30	29	45

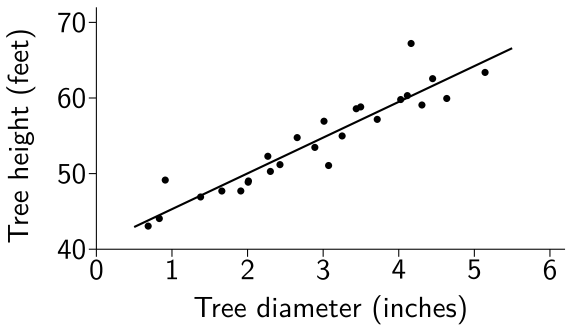
The researcher constructed a scatterplot, from which they determined that a least squares regression line would not be appropriate. State what feature they are likely to have observed on the scatterplot.

2. A bivariate sample has summary statistics:

$$\sum x = 117 \qquad \sum x^2 = 1869 \qquad \sum y = 660 \qquad \sum y^2 = 54638 \qquad \sum xy = 9519 \qquad n = 8$$

It is determined from inspecting a scatterplot that there appears to be a linear relationship between the variables. Obtain the equation of the least squares regression line of y on x.

3. 26 randomly selected trees in a forest have their diameter at a set distance from the ground measured, in inches, and the height of each tree also measured, in feet. The equation of the least squares regression line for y on x is obtained, and shown on the scatterplot below.



Summary statistics:

$\sum x = 74.76$

$\sum x^2 = 253.76$

$\sum y = 1408.3$

$\sum y^2 = 77277$

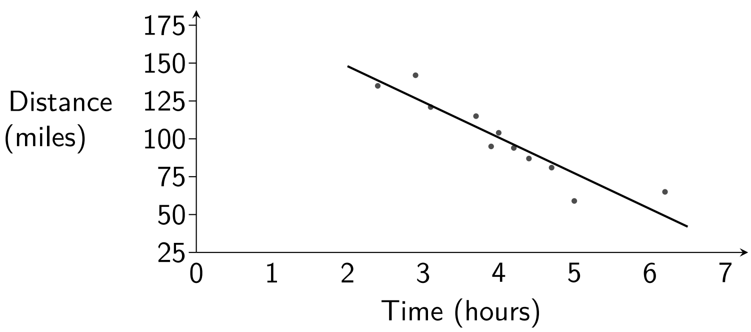
$\sum xy = 4233.1$

$n = 26$

(a) Determine the equation of the least squares regression line of y on x.

(b) Calculate an estimate for the height of a tree with a diameter of 3.9 inches.

4. A running website randomly selected eleven amateur runners who recently completed a marathon and asked them to complete a survey which included their finishing time in the marathon and information about their training routine. The scatterplot below shows, for each runner, their finishing time (x, in hours) and the total distance they ran in the final four weeks of their training (y, in miles).



Summary statistics:

$\sum x = 44.5$

$\sum x^2 = 191.21$

$\sum y = 1098$

$\sum y^2 = 116768$

$\sum xy = 4179.2$

$n = 11$

A least squares regression line for y on x is shown on the graph.

(a) Obtain the equation of the least squares regression line.

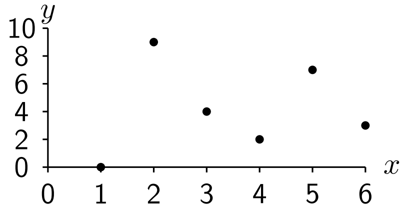
(b) Calculate the correlation coefficient, and comment on its value.

The website wishes to allow users the ability to input the number of miles in their training plan for the four weeks leading up to the race, and receive in response an estimate for their finishing time.

(c) Explain why the equation provided above is unsuitable for this purpose.

5. A researcher laid a transect line from the base of a tree and placed a quadrat at one metre intervals along the transect line. The number of seeds dropped by the tree contained in each quadrat was recorded.

Distance from base of tree (x)	1	2	3	4	5	6
No. of seeds found (y)	0	9	4	2	7	3



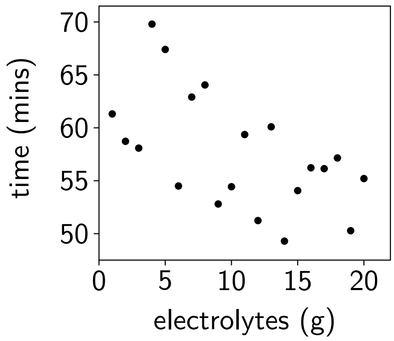
(a) Calculate the correlation coefficient.
(b) Explain why the researcher may have decided not to obtain a regression line.

6. *Electrolyte powders* are often used to help improve hydration during exercise. A keen cyclist records the time (y minutes) it takes them to complete a familiar route during a series of rides over the summer, randomly choosing the mass (x grams) of electrolyte powder to add to their water bottle each time.

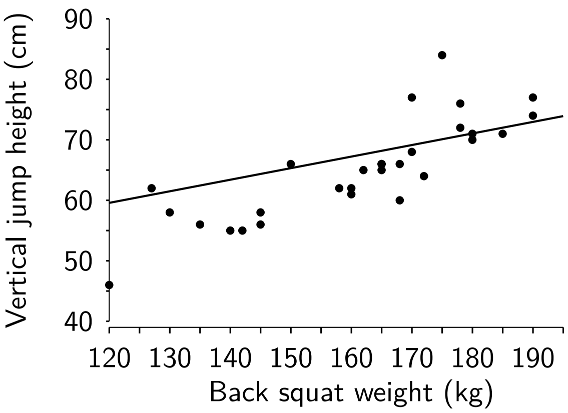
Summary statistics for the rides are below:

S_{xx}	S_{yy}	S_{xy}	n
665	567.2	-346.4	20

Determine the equation of the y on x regression line, and hence estimate the route time for a ride for which 5g of electrolyte powder was used.



7. A strength and conditioning coach wants to increase the vertical jump height performance in their trainees and considers whether back squat weight has an impact on vertical jump height performance. The summary statistics below are taken from the back squat weight (x in kg) and vertical jump height (y in cm) achieved by a random sample of 29 of the coach's trainees.



$$\Sigma x = 4673$$
$$\Sigma x^2 = 763101$$
$$\Sigma xy = 308020$$
$$\Sigma y = 1889$$
$$\Sigma y^2 = 124945$$
$$n = 29$$

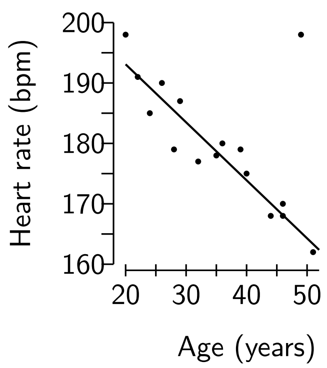
(a) Use the scatterplot to comment on the relationship between vertical jump height and back squat weight achieved by the trainees.
(b) Calculate the product moment correlation coefficient for this data set.
(c) The coach wishes to use the data to predict vertical jump height.
  i. Find the equation of the least squares regression line of vertical jump height on back squat weight.
  ii. Estimate the vertical jump height of a trainee who back squats 165kg and comment on the reliability of this estimate.

Based on the correlation coefficient the coach advises that increasing their back squat weight will increase their vertical jump height.

(d) Give a reason why this advice is not necessarily correct.

Review Exercise

1. As part of a study on intensive exercise, a sports scientist recorded the peak heart rates (y) of a random selection of sixteen volunteers of different ages (x) who took regular exercise. The linear regression equation was calculated for the data shown in the scatter diagram and found to be $y = 203.9 - 0.67x$.



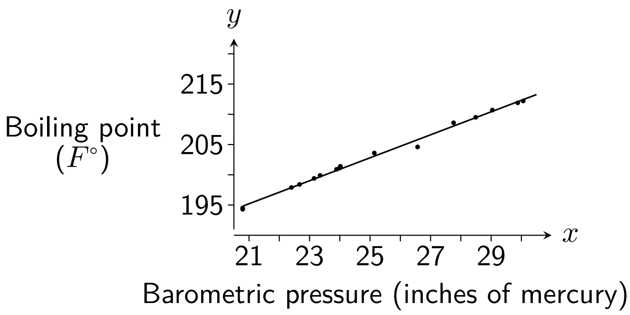
However, after considering the scatter diagram for the data, it was realised that one piece of data has been misrecorded, and this volunteer's data (spotted as a possible outlier) was removed from the data set.

(a) State the approximate age of the volunteer whose data was removed.

(b) Calculate the new equation of the least squares regression line using the revised summary data:
$\Sigma x = 518, \Sigma y = 2687, \Sigma x^2 = 19196, \Sigma y^2 = 482691, \Sigma xy = 91534$

(c) Comment on the difference this makes to the prediction for the average peak heart rate of a 45 year old volunteer.

2. James Forbes was a Scottish physicist who, amongst other things, was interested in finding a way to estimate altitude from measurements of the boiling temperature of water. Some of his observations on boiling point (y, in F°) and barometric pressure (x, in inches of mercury) are summarised below and shown in the scatterplot.



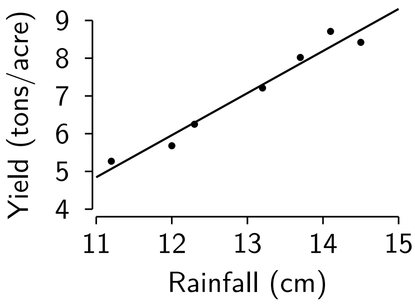
Summary data for sample:

$S_{xx} = 145.94$
$S_{yy} = 530.78$
$S_{xy} = 277.54$
$n = 17$

The least squares regression line of $y = 155.29 + 1.9018x$ was fitted. Show the calculations required to obtain the values of the sample intercept and sample slope parameter.

3. The yield of a particular crop on a farm is thought to depend principally on the amount of rainfall in the growing season. The values of the yield, y, in tons per acre, and the rainfall, x, in centimetres, for seven successive years are given in the table and displayed in the scatterplot.

Rainfall (x)	12.3	13.7	14.5	11.2	13.2	14.1	12.0
Yield (y)	6.25	8.02	8.42	5.27	7.21	8.71	5.68



(a) Comment on what the scatterplot shows.

(b) Calculate the product moment correlation coefficient for this data set.

(c) Find the equation of the least squares regression line of y on x.

(d) Comment on the appropriateness of using the regression line found to
estimate the rainfall in a year if the yield for the year is known.

6 Random Variables

A random variable is a function that assigns a numerical value to each outcome of a random experiment, used to *model* the behaviour of random processes. Random variables are notated using *capital letters*, such as X and Y, whilst the *specific values* that they can take are notated with corresponding small letters such as x and y. Random variables can be used to model either *discrete* or *continuous* quantitative variables.

Discrete Random Variables

If a random variable takes *discrete* numerical values, and a probability is assigned to each, then it is a **discrete random variable**. Discrete values are those that can be listed (either finitely or countably infinitely). Some examples of discrete random variables are:

Random variable (e.g. X)	**Discrete values** (e.g. x)
The outcome of a roll of a cubical die	$1, 2, 3, 4, 5, 6$
The number of goals scored in a game of football	$0, 1, 2, 3, 4, 5, 6, 7...$
The value of a randomly picked UK coin, in pounds	$0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2$

The probability that a discrete random variable X takes the value x is notated as:

$$\mathsf{P}(X = x)$$

For example, if random variable X is used to represent the outcome of the roll of a fair, six-sided die, then:

$$\mathsf{P}(X = 5) = \frac{1}{6}$$

6.1 Probability Distributions for Discrete Random Variables

A *probability function* assigns probabilities to each possible value of a discrete random variable, and completely describes its behaviour. There are a number of ways of presenting the *probability distribution* created. For example, consider a discrete random variable X that takes the value 3 with probability 0.2, the value 4 with probability 0.5 and the value 5 with probability 0.3. Below, the probability distribution for X has been *tabulated* and *graphed*.

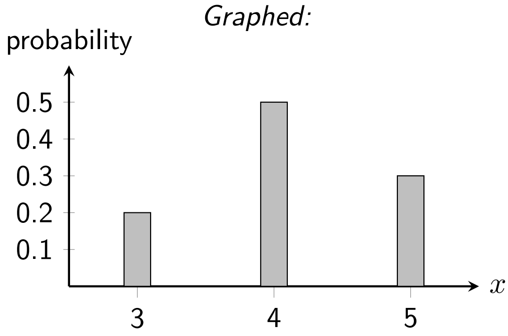
Tabulated:

x	3	4	5
$\mathsf{P}(X = x)$	0.2	0.5	0.3

A discrete probability distribution must define *mutually exclusive* and *exhaustive* outcomes, and so *the sum of all probabilities must equal 1.*

$$0 \leqslant \mathsf{P}(X = x_i) \leqslant 1 \text{ for all } i \quad \text{and} \quad \sum \mathsf{P}(X = x_i) = 1$$

Graphed:



From both the table and the graph, it can be seen that, for example, $\mathsf{P}(X = 3) = 0.2$.

Probabilities such as $\mathsf{P}(X \leqslant 4)$ can be calculated by adding the probability of each outcome (since they are mutually exclusive), or by subtracting from 1 (since they are exhaustive).

$$\mathsf{P}(X \leqslant 4) = \mathsf{P}(X = 3) + \mathsf{P}(X = 4) = 0.2 + 0.5 = 0.7$$

or

$$\mathsf{P}(X \leqslant 4) = 1 - \mathsf{P}(X = 5) = 1 - 0.3 = 0.7$$

Example

Problem: The probability distribution table for the discrete random variable D is shown below.

d	6	7	8	9	10
$\mathsf{P}(D = d)$	0.35	0.3	c	0.2	0.1

(a) Calculate the value of c.

(b) Calculate $\mathsf{P}(D \leqslant 7)$.

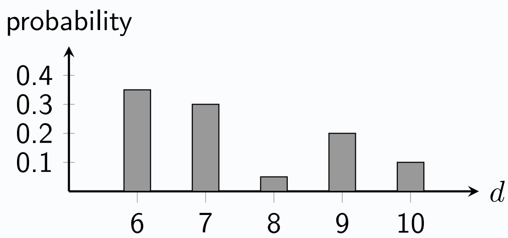
(c) Calculate $\mathsf{P}(D > 7)$.

Solution:
(a) $c = 1 - (0.35 + 0.3 + 0.2 + 0.1) = 0.05$

d	6	7	8	9	10
$\mathsf{P}(D = d)$	0.35	0.3	0.05	0.2	0.1

(b) $$\begin{aligned} \mathsf{P}(D \leqslant 7) &= \mathsf{P}(D = 6) + \mathsf{P}(D = 7) \\ &= 0.35 + 0.3 \\ &= 0.65 \end{aligned}$$

(c) $$\begin{aligned} \mathsf{P}(D > 7) &= 1 - \mathsf{P}(D \leqslant 7) \\ &= 1 - 0.65 \\ &= 0.35 \end{aligned}$$



Exercise 6.1

1. Explain why each of the following tables do not represent a tabulated probability distribution.

(a)

a	7	8	9
$\mathsf{P}(A=a)$	0.6	0.3	0.2

(b)

b	2	5	10
$\mathsf{P}(B=b)$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{2}{5}$

(c)

c	red	blue	green
$\mathsf{P}(C=c)$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{2}$

(d)

d	2	4	6
$\mathsf{P}(D=d)$	0.9	−0.3	0.4

2. Each table below represents the probability distribution of a discrete random variable. Determine the value of k for each.

(a)

q	1	2	3	4
$\mathsf{P}(Q=q)$	0.3	0.2	k	0.1

(b)

r	0	1	2
$\mathsf{P}(R=r)$	0.4	$2k$	k

(c)

s	5	10	15
$\mathsf{P}(S=s)$	$\frac{1}{10}$	$\frac{7}{10}$	k

(d)

t	1	2	3	4
$\mathsf{P}(T=t)$	$2k$	$4k$	$6k$	$8k$

3. The discrete random variable X has probability distribution:

x	1	2	3	4	5	6	7	8
$\mathsf{P}(X=x)$	0.15	0.2	0.1	0.05	c	0.1	0.08	0.02

Calculate:

(a) The value of c.
(b) $\mathsf{P}(X=4)$.
(c) $\mathsf{P}(X<4)$.
(d) $\mathsf{P}(X\leqslant 4)$.
(e) $\mathsf{P}(X>4)$.
(f) $\mathsf{P}(X\geqslant 4)$.
(g) $\mathsf{P}(2<X<5)$.
(h) $\mathsf{P}(2\leqslant X\leqslant 5)$.

4. A bag contains nine counters numbered 1 to 9. Two are selected at random, without replacement. Tabulate the probability distribution for the random variable E, representing the number of even numbers selected.

5. The discrete random variable X has probability distribution given by the function $\mathsf{P}(X=x)=k(x+1)$, where k is a constant and X takes values $x=0,1,2,3$.

   (a) Tabulate the probability distribution for X.

   (b) Find $\mathsf{P}(X\geqslant 2)$.

6. The discrete random variable X has probability distribution given by the function $\mathsf{P}(X=x)=kx^2$, where k is a constant and X takes values $x=1,2,3,4$.

   (a) Tabulate the probability distribution for X.

   (b) Find $\mathsf{P}(X<4)$.

7. Tabulate the probability distribution of each of the following discrete random variables:

   (a) F, the number of fives obtained when two cubical dice are rolled.

   (b) T, the number of tails obtained when three coins are tossed.

   (c) S, the sum obtained when two cubical dice are rolled.

6.2 The Expectation of a Discrete Random Variable

Whilst a single roll of a die is equally likely to land on any of the numbers from 1 to 6, if it is rolled many times and the *mean* of the observed values is calculated, it is very likely to be around 3.5. Letting random variable X represent the result of one roll of the die, the *long-run mean* of repeated observations of X will tend towards 3.5. This can be referred to as the *mean*, μ, of X, the **expectation** of X, or $\mathsf{E}(X)$.

The expectation μ of discrete random variable X:

$$\mathsf{E}(X) \quad = \quad \sum x_i \mathsf{P}(X = x_i)$$

The calculated mean of a finite number of observations of a random variable is likely to not be perfectly equal to the true *expected value*. However, the mean of repeated observations will tend towards the expected value as the number of observations increases. This is an illustration of the *law of large numbers*.

Example

Problem: Given the probability distribution of random variable X tabulated below, calculate $\mathsf{E}(X)$.

x	3	4	5	6
$\mathsf{P}(X = x)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{8}$

Solution:

$$\mathsf{E}(X) = 3 \times \frac{1}{8} + 4 \times \frac{1}{4} + 5 \times \frac{1}{2} + 6 \times \frac{1}{8} = \frac{37}{8} = 4.625$$

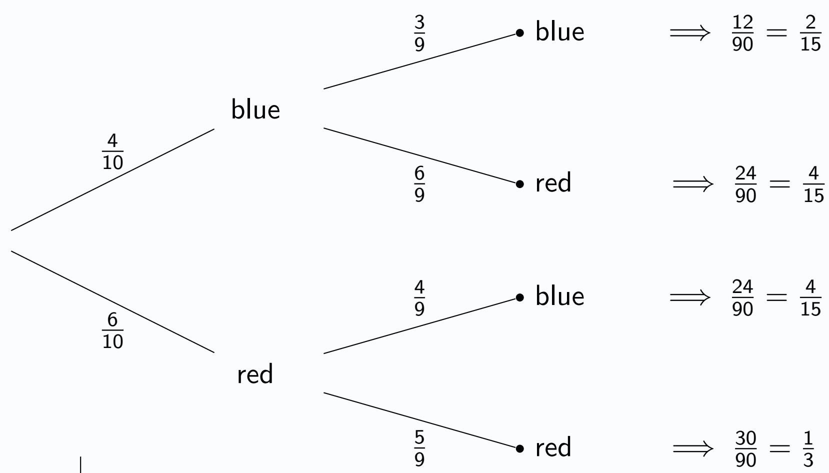
Note that the expected value is not necessarily a value that the random variable can actually take.

Problems may require a probability distribution to first be tabulated, such as by using a tree diagram.

Example

Problem: A bag contains 4 blue and 6 red cubes. One is drawn at random, followed by another without replacing the first. Let discrete random variable X represent the number of red cubes drawn. Find $\mathsf{E}(X)$.

Solution:



x	0	1	2
$\mathsf{P}(X = x)$	$\frac{2}{15}$	$\frac{8}{15}$	$\frac{1}{3}$

$$\mathsf{E}(X) = 0 \times \frac{2}{15} + 1 \times \frac{8}{15} + 2 \times \frac{1}{3} = \frac{6}{5} = 1.2$$

Exercise 6.2

1. Calculate the expected value for each of the probability distributions below.

(a)

x	2	4	6	8
$\mathsf{P}(X=x)$	0.25	0.4	0.2	0.15

(b)

y	1	2	3	4	5
$\mathsf{P}(Y=y)$	$\frac{1}{20}$	$\frac{7}{20}$	$\frac{3}{20}$	k	$\frac{1}{20}$

2. The random variable X represents the result of a roll of a fair, octahedral die with sides numbered 1 to 8.
   (a) Tabulate the probability distribution of X.
   (b) State $\mathsf{P}(4 \leqslant X \leqslant 6)$.
   (c) Calculate $\mathsf{E}(X)$.

3. A discrete random variable X can assume values 10 and 20 only. Given $\mathsf{E}(X) = 16$, tabulate the probability distribution of X.

4. A box contains 3 red marbles and 5 green marbles. Two marbles are taken at random without replacement, and X is the number of green marbles obtained. Find the expectation of X.

5. A box contains 3 red marbles and 5 green marbles. Two marbles are taken at random with replacement, and Y is the number of green marbles obtained. Find the expectation of Y.

6. In the UKMT Senior Maths Challenge, 4 points are gained for a question answered correctly and 1 point is lost for a question answered incorrectly. A candidate guesses two questions, picking at random from the 5 choices for each question. Let random variable Y represent the points gained from those two questions.
   (a) Tabulate the probability distribution of Y.
   (b) Calculate $\mathsf{E}(Y)$.

7. A bag contains three £1 coins and two £2 coins. Two coins are taken at random, without replacement. Let discrete random variable M represent the amount of money that is taken from the bag, in pounds.
   (a) Tabulate the probability distribution of M.
   (b) Calculate μ, the mean value of M.

8. A game costs £4 to enter and consists of tossing a coin three times.
   - £8 is paid out if the coin comes up tails three times.
   - £5 is paid out if the coin comes up tails exactly twice.
   - £2 is paid out if the coin only comes up tails once.
   - Nothing is paid out if the coin shows heads three times.

   Let the random variable W represent the profit, in pounds, that a player makes if the game is played once.
   (a) Tabulate the probability distribution for W.
   (b) Calculate $\mathsf{E}(W)$.

9. The probability distribution of the discrete random variable X is tabulated below.

x	1	3	5	8	10
$\mathsf{P}(X=x)$	0.1	a	0.3	b	0.2

   Given that $\mathsf{E}(X) = 6.3$, find the values of a and b.

6.3 The Law of Expectation

If discrete random variable X takes values $x_1, x_2, \ldots, x_n$ with probabilities $\mathsf{P}(X = x_i)$, then random variable $Y = 2X$ takes values $2x_1, 2x_2, \ldots, 2x_n$ with probabilities $\mathsf{P}(X = x_i)$. For example, let X represent the roll of a die:

x	1	2	3	4	5	6
$\mathsf{P}(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

y	2	4	6	8	10	12
$\mathsf{P}(Y = y)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$\mathsf{E}(X) = 1 \times \tfrac{1}{6} + \ldots + 6 \times \tfrac{1}{6} = 3.5 \qquad \mathsf{E}(2X) = \mathsf{E}(Y) = 2 \times \tfrac{1}{6} + \ldots + 12 \times \tfrac{1}{6} = 7$$

(Note that the distribution of $2X$ is *not* that of the sum of two *independent* dice rolls.)

It can be seen here that $\mathsf{E}(2X) = 2\mathsf{E}(X)$. The expectation of a random variable that is a *linear transformation* of another random variable can be calculated using following law:

For random variable X with constants a and b:

$$\mathsf{E}(aX \pm b) = a\mathsf{E}(X) \pm b$$

This shows that expectation is *linear*, distributing over addition and multiplication by constants.

Proof:

$$\begin{aligned}
\mathsf{E}(aX + b) &= \sum (ax_i + b)\mathsf{P}(X = x_i) \\
&= \sum (ax_i)\mathsf{P}(X = x_i) + \sum b\mathsf{P}(X = x_i) \\
&= a\sum x_i\mathsf{P}(X = x_i) + b\sum \mathsf{P}(X = x_i) \\
&= a\mathsf{E}(X) + b
\end{aligned}$$

Example

Problem: Let X be the random variable such that $\mathsf{E}(X) = 3$. Calculate $\mathsf{E}(4X - 1)$.

Solution:

$$\begin{aligned}
\mathsf{E}(4X - 1) &= 4\mathsf{E}(X) - 1 \\
&= 4 \times 3 - 1 \\
&= 11
\end{aligned}$$

Given the value $\mathsf{E}(aX \pm b)$, the expectation of X may be determined.

Example

Problem: Given that $\mathsf{E}(10 - 2Y) = 50$, determine the value of $\mathsf{E}(Y)$.

Solution:

$$\begin{aligned}
\mathsf{E}(10 - 2Y) &= 50 \\
\mathsf{E}(-2Y + 10) &= 50 \\
-2\mathsf{E}(Y) + 10 &= 50 \\
-2\mathsf{E}(Y) &= 40 \\
\mathsf{E}(Y) &= -20
\end{aligned}$$

Exercise 6.3

1. Calculate:

   (a) $\mathsf{E}(X+4)$ given $\mathsf{E}(X)=3$ (b) $\mathsf{E}(Y-3)$ given $\mathsf{E}(Y)=0.4$ (c) $\mathsf{E}(2+Z)$ given $\mathsf{E}(Z)=-5$

2. Find:

   (a) $\mathsf{E}(3X)$ given $\mathsf{E}(X)=2.5$ (b) $\mathsf{E}(-\frac{1}{2}Y)$ given $\mathsf{E}(Y)=8$ (c) $\mathsf{E}(-W)$ given $\mathsf{E}(W)=-\frac{2}{3}$

3. Determine the value of:

   (a) $\mathsf{E}(2X-3)$ given $\mathsf{E}(X)=-4$ (b) $\mathsf{E}(-\frac{1}{3}Y+2)$ given $\mathsf{E}(Y)=12$(c) $\mathsf{E}(3-4C)$ given $\mathsf{E}(C)=\frac{1}{2}$

4. Determine:

   (a) $\mathsf{E}(X)$ given $\mathsf{E}(4X+3)=15$ (b) $\mathsf{E}(Y)$ given $\mathsf{E}(4-Y)=2.4$ (c) $\mathsf{E}(Q)$ given $\mathsf{E}(5+\frac{2}{3}Q)=-1$

5. The probability distribution of X is tabulated as follows:

x	4	5	6	7
$\mathsf{P}(X=x)$	0.3	0.5	k	0.05

   Determine the values of:

   (a) k (b) $\mathsf{P}(4<X\leqslant 6)$ (c) $\mathsf{E}(X)$ (d) $\mathsf{E}(4X+1)$

6. In a board game a fair, tetrahedral die with faces numbered from 1 to 4 is thrown, and the result is multiplied by 3 to give the number of places a player can move. Let random variable M represent the number of places a player can move. Determine the mean, μ, of M.

7. Random variables X and Y are defined such that the mean of X is $\frac{1}{4}$ and $Y=6X+4$. Determine the mean of Y.

8. A betting game involves a player rolling a fair, octagonal die with faces numbered from 1 to 8. Let X represent the result of a roll of the die.

   (a) Determine $\mathsf{E}(X)$.

   Each game costs £10 to enter, with the number obtained on a single roll of the die being multiplied by 2 to give an amount in pounds that the player receives. Let Y represent the *profit* a player makes from playing the game a single time.

   (b) State an equation linking Y and X.

   (c) Determine the value of $\mathsf{E}(Y)$.

   (d) Explain what this suggests will happen if a player plays the game repeatedly.

9. Random variables X and Y are defined such that the mean of Y is -5 and $Y=2X-10$. Determine the mean of X.

6.4 The Variance of a Random Variable

The *variance* of the values observed from repeated rolls of a fair cubical die will tend towards a fixed value in the long-run. Letting the random variable X represent the result of a roll of a die, the **variance**, σ^2, of X is $\frac{35}{12}$, or $\mathsf{V}(X) = \frac{35}{12}$.

The variance σ^2 of random variable X:

$$\mathsf{V}(X) \quad = \quad \mathsf{E}(X^2) \; - \; (\mathsf{E}(X))^2$$

As a reminder, the variance measures how spread out the values of a random variable are about the mean.

Note that $\mathsf{E}\left(X^2\right)$ is not equal to $(\mathsf{E}(X))^2$.

Proof:

$$\begin{aligned}
\mathsf{V}(X) &= \mathsf{E}[(X-\mu)^2] \\
&= \mathsf{E}[(X-\mu)(X-\mu)] \\
&= \mathsf{E}[X^2 - 2\mu X + \mu^2] \\
&= \mathsf{E}(X^2) - 2\mu\mathsf{E}(X) + \mu^2 \\
&= \mathsf{E}(X^2) - 2(\mathsf{E}(X))^2 + (\mathsf{E}(X))^2 \\
&= \mathsf{E}(X^2) - (\mathsf{E}(X))^2
\end{aligned}$$

Knowing the *expectation* and *variance* of random variables help statisticians better understand the *long-run* behaviour of repeated observations:

- $\mathsf{E}(X) = \mu_X$, the *long-run mean* of the distribution of X.
- $\mathsf{V}(X) = \sigma_X^2$, the *variance* of the distribution of X.

The standard deviation, σ, of a random variable X, notated $\mathsf{SD}(X)$, is the square root of the variance:

$$\mathsf{SD}(X) = \sqrt{\mathsf{V}(X)} \qquad \text{and} \qquad \mathsf{SD}(X), \mathsf{V}(X) \geqslant 0$$

Example

Problem: The probability distribution of random variable X is given below:

x	0	1	2	3
$\mathsf{P}(X = x)$	0.3	0.4	0.1	0.2

Calculate $\mathsf{E}(X)$ and $\mathsf{V}(X)$

Solution: *(Remember that $\mathsf{E}(X^2)$ is not the variance.)*

$$\mathsf{E}(X) = 0 \times 0.3 + 1 \times 0.4 + 2 \times 0.1 + 3 \times 0.2 = 1.2$$

$$\mathsf{E}(X^2) = 0^2 \times 0.3 + 1^2 \times 0.4 + 2^2 \times 0.1 + 3^2 \times 0.2 = 2.6$$

$$\begin{aligned}
\mathsf{V}(X) &= \mathsf{E}(X^2) - (\mathsf{E}(X))^2 \\
&= 2.6 - 1.2^2 \\
&= 1.16
\end{aligned}$$

Exercise 6.4

1. Calculate the mean and variance of each of the following probability distributions.

(a)

x	1	2	5	10
$\mathsf{P}(X=x)$	0.4	0.3	0.2	0.1

(b)

y	0	1	2	3	4
$\mathsf{P}(Y=y)$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{4}$	k	$\frac{2}{5}$

2. The random variable X represents the result of a roll of a fair, tetrahedral die with sides numbered 1 to 4.
   (a) Tabulate the probability distribution of X.
   (b) Find $\mathsf{E}(X)$.
   (c) Calculate $\mathsf{V}(X)$.

3. A box contains 6 pink marbles and 4 orange marbles. Two marbles are taken at random without replacement, and X is the number of pink marbles obtained. Find the expectation and variance of X.

4. A box contains 2 red marbles and 6 green marbles. Two marbles are selected at random with replacement, and Y is the number of green marbles obtained. Find $\mathsf{E}(Y)$ and $\mathsf{V}(Y)$.

5. A discrete random variable X can assume values 4 and 12 only. Given $\mathsf{E}(X) = 10$, determine the variance of X.

6. A multiple-choice quiz gives four possible answers for each question, with 4 marks awarded for a correct answer and 1 mark taken away for an incorrect answer. Determine the expected gain in marks when the answer to a single question is guessed, and the variance in the number of marks gained.

7. A game costs £10 to enter, and the player rolls two fair, cubical dice together. If both dice show a six the player wins £50, a single six means the player gets £10 back and no sixes means the player gets nothing. Determine the mean and variance in the amount of profit for playing the game once.

8. A fair coin is tossed three times. Let random variable X represent the number of tails shown.
   (a) Tabulate the probability distribution of X.
   (b) Find $\mathsf{E}(X)$ and $\mathsf{V}(X)$.

9. The random variable X takes values 7, 8 and 9. Given $\mathsf{P}(X = 8) = p$ and $\mathsf{P}(X = 7) = \mathsf{P}(X = 9)$:
   (a) Determine $\mathsf{E}(X)$.
   (b) Given $\mathsf{V}(X) = 0.8$, determine the value of p.

10. Random variable Y takes the value 1 with probability 0.9 and otherwise takes the value k. Determine the value of k if $\mathsf{V}(Y) = 0$.

6.5 The Law of Variance

Since variance is based on *squared* differences from the mean, $\mathsf{V}(2X) \neq 2\mathsf{V}(X)$.

For random variable X with constants a and b:

$$\mathsf{V}(aX \pm b) = a^2\mathsf{V}(X)$$

This shows that multiplying a random variable by a constant changes its variance by the square of that constant, while adding a constant does not affect the variance. In contrast to expectation, variance is *not linear* and does not distribute over addition or multiplication by constants.

Proof:

$$\begin{aligned}
\mathsf{V}(aX+b) &= \mathsf{E}\big[(aX+b-\mathsf{E}(aX+b))^2\big] \\
&= \mathsf{E}\big[(aX+b-a\mathsf{E}(X)-b)^2\big] \\
&= \mathsf{E}\big[(aX-a\mathsf{E}(X))^2\big] \\
&= \mathsf{E}\big[(a(X-\mathsf{E}(X)))^2\big] \\
&= \mathsf{E}[a^2(X-\mathsf{E}(X))^2] \\
&= a^2\mathsf{E}[(X-\mathsf{E}(X))^2] \\
&= a^2\mathsf{V}(X)
\end{aligned}$$

Example

Problem: Let X be the random variable such that $\mathsf{E}(X) = -7$ and $\mathsf{V}(X) = 4$. Calculate $\mathsf{V}(2-3X)$.

Solution: *(It helps to rearrange into the form $aX \pm b$.)*

$$\begin{aligned}
\mathsf{V}(2-3X) &= \mathsf{V}(-3X+2) \\
&= (-3)^2\mathsf{V}(X) \\
&= 9 \times 4 \\
&= 36
\end{aligned}$$

Note that any laws relating to the *standard deviation* of a linear transformation ($\mathsf{SD}(aX \pm b)$) cannot be expressed as concisely, so such problems must be tackled by *first finding the variance* ($\mathsf{V}(aX \pm b)$).

Example

Problem: Random variable Y has mean 1.5 and standard deviation 0.8.
Determine the standard deviation of $10Y + 25$.

Solution:

$$\begin{aligned}
\mathsf{V}(10Y+25) &= 10^2\mathsf{V}(Y) \\
&= 100 \times 0.8^2 \\
&= 64 \\
\mathsf{SD}(10Y+25) &= 8
\end{aligned}$$

Exercise 6.5

1. Given E$(X) = 6$ and V$(X) = 8$, determine the values of:

   (a) E$(2X)$ (b) V$(2X)$ (c) SD$(2X)$

2. Given E$(X) = -2$ and V$(X) = 3$, determine the values of:

   (a) E$(X + 5)$ (b) V$(X + 5)$ (c) SD$(X + 5)$

3. Given E$(X) = 1.25$ and SD$(X) = 0.64$, determine the values of:

   (a) E$(3X - 2)$ (b) V$(3X - 2)$ (c) SD$(3X - 2)$

4. Given E$(X) = 500$ and V$(X) = 400$, determine the values of:

   (a) E$(0.1X + 10)$ (b) V$(0.1X + 10)$ (c) SD$(0.1X + 10)$

5. The discrete random variable X has the following probability distribution.

x	1	2	4	8
P$(X = x)$	0.1	0.5	0.3	0.1

   (a) Calculate E(X) and V(X). (b) Find E$(2X + 1)$. (c) Find V$(2X + 1)$.

6. Given E$(Y) = -5$ and V$(Y) = 3$, determine the values of:

   (a) E$(-3Y - 2)$ (b) V$(-3Y - 2)$ (c) SD$(-3Y - 2)$

7. Random variable X has a mean of 2.4 and a standard deviation of 0.8. Find:

   (a) E$(1 - 10X)$ (b) V$(1 - 10X)$ (c) SD$(1 - 10X)$

8. Given random variable X such that E$(4X - 1) = 23$ and V$(4X - 1) = 80$, determine the values of E(X) and V(X).

9. Given random variable Y such that E$(3Y + 12) = 0$ and V$(3Y + 12) = 81$, determine the mean and standard deviation of Y.

10. Given random variables X and Z such that E$(Z) = 0$, V$(Z) = 1$ and $X = a + bZ$:

    (a) State SD(Z).

    (b) Find E(X).

    (c) Determine the values of V(X) and SD(X).

6.6 Combining Random Variables

As well as linear transformations of a single random variable, new variables can be created by adding (or subtracting) random variables. Given random variables X and Y, for example, $W = X + Y$ is a new random variable formed from them. When studying the long-run behaviour of combined random variables, by considering their expectation and variance, it is important to consider whether the variables are *independent*. The laws relating to **variance** below *only* apply when the variables are **independent**.

For random variables X and Y:

For all X and Y:		For independent X and Y:
$\mathsf{E}(X+Y) = \mathsf{E}(X) + \mathsf{E}(Y)$		$\mathsf{V}(X+Y) = \mathsf{V}(X) + \mathsf{V}(Y)$
$\mathsf{E}(X-Y) = \mathsf{E}(X) - \mathsf{E}(Y)$	and	$\mathsf{V}(X-Y) = \mathsf{V}(X) + \mathsf{V}(Y)$
$\mathsf{E}(aX \pm bY) = a\mathsf{E}(X) \pm b\mathsf{E}(Y)$		$\mathsf{V}(aX \pm bY) = a^2\mathsf{V}(X) + b^2\mathsf{V}(Y)$

If X and Y are not independent variables then determining $\mathsf{V}(X \pm Y)$ is beyond the scope of the course.

Example

Problem: Two independent random variables F and G have means of 15 and 12 and standard deviations of 3 and 4 respectively. Calculate $\mathsf{E}(3G - 2F)$ and $\mathsf{V}(F - G)$.

Solution: $(\mathsf{E}(F) = 15, \mathsf{E}(G) = 12, \mathsf{SD}(F) = 3, \mathsf{SD}(G) = 4)$

$$\begin{aligned} \mathsf{E}(3G - 2F) &= 3\mathsf{E}(G) - 2\mathsf{E}(F) \\ &= 3 \times 12 - 2 \times 15 \\ &= 36 - 30 \\ &= 6 \end{aligned}$$

$$\begin{aligned} \mathsf{V}(F) &= 3^2 = 9 \\ \mathsf{V}(G) &= 4^2 = 16 \\ \mathsf{V}(F - G) &= \mathsf{V}(F) + \mathsf{V}(G) \\ &= 9 + 16 \\ &= 25 \end{aligned}$$

Random variables **independently and identically distributed (i.i.d)** with the same distribution as X are denoted as $X_1, X_2, X_3, \ldots$

Example

Problem: Books in the fiction section of a library have a mean thickness of 3.1cm with standard deviation 0.2cm. If three books are randomly selected from the fiction section and stacked together, determine the mean and standard deviation of the total thickness of the three books, stating one assumption required.

Solution: Let X represent the thickness of a book, so $\mathsf{E}(X) = 3.1$ and $\mathsf{V}(X) = 0.2^2 = 0.04$

Total thickness $T = X_1 + X_2 + X_3$. Assuming that the thickness of each book is independent:

$\mathsf{E}(T) = \mathsf{E}(X_1 + X_2 + X_3) = \mathsf{E}(X_1) + \mathsf{E}(X_2) + \mathsf{E}(X_3) = 3.1 + 3.1 + 3.1 = 9.3$

$\mathsf{V}(T) = \mathsf{V}(X_1 + X_2 + X_3) = \mathsf{V}(X_1) + \mathsf{V}(X_2) + \mathsf{V}(X_3) = 0.04 + 0.04 + 0.04 = 0.12$

$\mathsf{SD}(T) = \sqrt{\mathsf{V}(T)} = \sqrt{0.12} = 0.346$

Hence the mean total thickness of the three books is 9.3cm, and the standard deviation is 0.346cm.

Exercise 6.6

1. Random variables J and K are defined such that E$(J) = 3$ and E$(K) = -2$.
   (a) State the information required to be known about J and K in order to determine whether the laws introduced on the previous page may be applied.
   (b) Assuming that J and K are such that those laws may be applied, find:

      i. E$(J + K)$ ii. E$(J - K)$ iii. E$(2K - 3J)$

   (c) Assuming that J and K are such that those laws may be applied, find:

      i. V$(J + K)$ ii. V$(J - K)$ iii. V$(2K - 3J)$

2. Independent random variables X and Y are defined such that E$(X) = 5$, V$(X) = 6$, E$(Y) = 3$ and V$(Y) = 8$. Determine the values of:

   (a) E$(2X + 6Y)$
   (b) V$(2X + 6Y)$
   (c) E$(4X + 5Y)$
   (d) V$(4X + 5Y)$
   (e) E$(\frac{1}{2}Y - 2X)$
   (f) V$(\frac{1}{2}Y - 2X)$

3. Random variables F and G have means of 3 and 2 respectively, and standard deviations of 5 and 6 respectively. State the information that is missing which means that it is not currently possible to determine the values of E$(2F + G)$ and V$(2F + G)$.

4. Independent random variables A and B have means of 1.5 and 1.2 respectively, and standard deviations of 1 and 0.8 respectively. Determine the values of E$(A - B)$ and V$(A - B)$.

5. Random variable X is defined such that E$(X) = 10$ and V$(X) = 3$. Random variables $\{X_1, X_2, X_3 \ldots\}$ represent independent observations of X. Determine the mean and variance of $X_1 + X_2 + X_3 + X_4$, representing the sum of four such observations of X.

6. A spreadsheet contains a large amount of cells, each containing numbers such that the mean of the numbers is 12 and the standard deviation is 1.2. Five numbers from the spreadsheet are selected at random, with replacement. Determine the mean and standard deviation of the total of the five numbers.

7. It is known that the mean volume of coffee poured per cup in a staffroom for teachers is 250ml, with a standard deviation of 10ml. Determine the mean and standard deviation of the total volume of coffee used for six cups, stating an assumption required.

8. Let random variable X represent the number obtained when a fair, cubical die numbered 1 to 6 is thrown, and random variable Y the number obtained when a fair, tetrahedral die numbered 1 to 4 is thrown.
   (a) Find E(X) and E(Y).
   (b) Find V(X) and V(Y).
   (c) Obtain the mean and variance of the result when both dice are thrown and the scores are added together.

9. The mean length of a car is 4.5 metres, with a standard deviation of 30 centimetres. Drivers arriving to queue for a ferry are told to line up one in front of another leaving as small a gap as possible between cars. When they do this, the mean gap they leave is 40 centimetres, with a standard deviation of 10 centimetres. Stating two assumptions required, determine the mean and standard deviation of the total length of a line of eight cars asked to line up in this way.

Review Exercise

1. A trial consists of tossing two unbiased coins. Let the random variable X represent the number of heads obtained.

   (a) Tabulate the probability distribution of X after one trial.

   (b) Calculate:

      i. E(X).

      ii. V(X).

      iii. SD(X).

2. Two independent random variables P and Q have means 5 and 10 and standard deviations 2 and 3 respectively. Find the mean and variance of each of the random variables:

   (a) $4P - 8$.

   (b) $3Q - P$.

3. A box contains 4 blue marbles and 5 red marbles. Three marbles are taken from the box without replacement. Let the random variable X represent the number of red marbles taken.

   (a) Tabulate the probability distribution of X.

   (b) Calculate the mean and variance of X.

4. Given that E$(2X + 6) = 16$ and V$(2X + 6) = 12$:

   (a) Calculate E(X).

   (b) Calculate V(X).

5. Two independent random variables A and B have means 53 and 21 and standard deviations 6 and 5 respectively. Find the mean and standard deviations of each of the following random variables:

   (a) $5B + 6$.

   (b) $2A - 3B$.

6. It is known that items of hand luggage carried by passengers on an airline have mean mass of 5.6kg and a standard deviation of 2.0kg, and that their hold luggage has a mean mass of 22.0kg and a standard deviation 5.0kg.

   (a) Calculate the mean and standard deviation of the total luggage mass for one passenger, stating an assumption you have made.

   The passengers themselves have masses with mean 75kg and standard deviation 12kg. A small plane carries 10 passengers.

   (b) Calculate the mean and standard deviation of the total load, including passengers and their luggage.

7
Discrete Distributions

"*All models are wrong, but some models are useful.*"

Models, Conditions and Assumptions

The above quote, attributed to the statistician George Box, highlights that no mathematical or statistical *model* can perfectly describe the behaviour of something in the real world. Nevertheless, if a model can get *close enough* then results obtained using it can be sufficiently accurate and useful.

In this chapter, three *discrete probability distributions* that can model real-life data will be introduced:

- The Discrete Uniform Distribution
- The Binomial Distribution
- The Poisson Distribution

Studying a problem in context and identifying an appropriate model to use is an important skill for anyone working with data. The validity of the use of a model is based on a number of *conditions* being met. When using a model it is vital to consider the conditions required for its use, and specifically whether they are each satisfied. In this chapter, and subsequent chapters introducing more probability distributions and hypothesis testing, the *conditions for the valid use* of models and tests are clearly indicated as they are introduced.

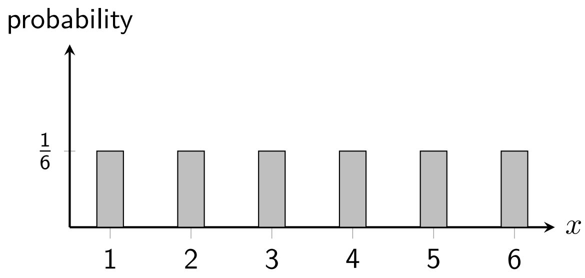
Sometimes it will be clear that a particular condition has been met, but there will be times when it is unclear. Whenever it is not clear that a condition has been met, statisticians must ask whether they can reasonably make an *assumption* that the condition is satisfied. It should always be recognised when an assumption is required, and it may be necessary to *justify* the use of an assumption in a context.

7.1 The Discrete Uniform Distribution

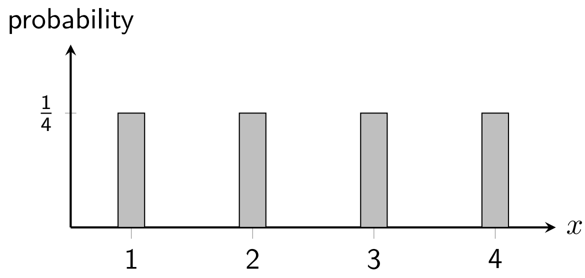
Consider the discrete random variables X and Y, representing the scores obtained from a cubical die numbered $\{1,2,3,4,5,6\}$ and a tetrahedral die numbered $\{1,2,3,4\}$ respectively. Their probability distributions have been tabulated and graphed below, and the expectation and variance calculated for each:

x	1	2	3	4	5	6
$\mathsf{P}(X=x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

y	1	2	3	4
$\mathsf{P}(Y=y)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$



$\mathsf{E}(X)=3.5$ and $\mathsf{V}(X)=\frac{35}{12}$



$\mathsf{E}(Y)=2.5$ and $\mathsf{V}(Y)=\frac{5}{4}$

Whilst X and Y do not follow the same distribution, they share key characteristics, namely that they both take a *finite number of integer values*, $\{1,2,3,\ldots,k\}$, and that *each of these values is equally likely*. It is these *shared characteristics* that define the *discrete uniform distribution*.

The Discrete Uniform Distribution:

- Notated $X\sim\mathsf{U}(k)$, defined by the single parameter k.
- X takes integer values from 1 to k: $\{1,2,3,\ldots,k\}$.
- Each outcome is *equally likely*, with probability $\dfrac{1}{k}$.

Parameters define the precise form of a distribution within a *family of distributions*. Here, we can say that X *"is distributed uniformly"* with parameter $k=6$ (with values from 1 to 6). Using $\sim$, or *"tilde"*, to mean *"is distributed..."*, such statements can be written more efficiently:

$$X\sim\mathsf{U}(6) \quad \text{and} \quad Y\sim\mathsf{U}(4)$$

The Data Booklet (**page 4**) gives the formulae to calculate *expectation* and *variance* for the discrete uniform distribution. For $X\sim\mathsf{U}(k)$:

$$\mathsf{E}(X)=\frac{k+1}{2} \quad \text{and} \quad \mathsf{V}(X)=\frac{k^2-1}{12}$$

The previously stated values for the expectation and variance of X and Y can be quickly obtained using these formulae. It also provides the *probability function* (**pf**) for the distribution $\mathsf{U}(k)$ as $\frac{1}{k}$. Probability functions show how to calculate the probability of any given outcome for a discrete random variable. Here, the probability depends *only on the parameter* k. For example, given $X\sim\mathsf{U}(5)$:

$$\mathsf{P}(X=1)=\frac{1}{5}, \quad \mathsf{P}(X=2)=\frac{1}{5}, \quad \mathsf{P}(X=3)=\frac{1}{5}, \quad \text{and so on...}$$

Example

Problem: Given $T \sim \mathsf{U}(10)$ and $W = 3T + 1$, calculate $\mathsf{E}(W)$ and $\mathsf{V}(W)$.

Solution:

(a)
$$\begin{aligned} \mathsf{E}(T) &= \frac{10+1}{2} = 5.5 \\ \mathsf{E}(W) &= \mathsf{E}(3T+1) \\ &= 3\mathsf{E}(T) + 1 \\ &= 3 \times 5.5 + 1 \\ &= 17.5 \end{aligned}$$

(b)
$$\begin{aligned} \mathsf{V}(T) &= \frac{10^2 - 1}{12} = 8.25 \\ \mathsf{V}(W) &= \mathsf{V}(3T+1) \\ &= 3^2\mathsf{V}(T) \\ &= 9 \times 8.25 \\ &= 74.25 \end{aligned}$$

Exercise 7.1

1. Explain why, for each, random variable X does **not** follow a $\mathsf{U}(k)$ distribution.
   (a) Let X represent the total score obtained from rolling two fair six-sided dice.
   (b) Let X represent the day of the month on which a person was born.
   (c) Let X represent a single-digit prime number picked at random.
   (d) Let X represent the number of 6s obtained when a die is rolled ten times.

2. Let $X \sim \mathsf{U}(5)$. Calculate the mean and variance of X.

3. Let $X \sim \mathsf{U}(15)$. Calculate the mean and variance of X.

4. A discrete random variable is uniformly distributed such that $\mathsf{P}(X = x) = \frac{1}{k}$, with $k = 8$.
   (a) Calculate $\mathsf{E}(X)$ and $\mathsf{V}(X)$.
   (b) Evaluate $\mathsf{E}(2X + 5)$ and $\mathsf{V}(2X + 5)$.

5. A regular, unbiased icosahedral die is rolled. Let the random variable X be the score on the topmost face.
   (a) State the distribution of X.
   (b) Determine the values of $\mathsf{E}(X)$ and $\mathsf{V}(X)$.
   (c) Evaluate $\mathsf{E}(1 - 3X)$ and $\mathsf{V}(1 - 3X)$.

6. Two independent random variables X and Y are defined such that $X \sim \mathsf{U}(3)$ and $Y \sim \mathsf{U}(5)$.
   (a) Calculate $\mathsf{E}(X - Y)$.
   (b) Calculate $\mathsf{V}(X - Y)$.

7. Given $X \sim \mathsf{U}(k)$ such that $\mathsf{E}(X) = 5.5$:
   (a) Determine the value of k.
   (b) State $\mathsf{P}(2 \leqslant X < 6)$.
   (c) Find the standard deviation of X.

8. Given $Y \sim \mathsf{U}(k)$ such that the standard deviation of Y is $\frac{\sqrt{21}}{2}$:
   (a) Determine the value of k.
   (b) Find $\mathsf{E}(1 - 2Y)$.

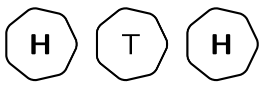
9. Discrete, independent random variables X and Y are uniformly distributed such that:
   - $\mathsf{P}(X = 1) < \mathsf{P}(Y = 1)$
   - $\mathsf{E}(X - Y) = 2$ and $\mathsf{V}(X - Y) = 14$

   Determine the distributions of X and Y.

7.2 The Binomial Distribution

A *Bernoulli trial* is an experiment in which each trial has only two possible outcomes, such as a coin landing on either *heads* or *tails*, or asking a *yes or no* question in a survey. These responses can be more generally referred to as *successes* or *failures*. The *binomial distribution* models the **number of successes** obtained from a *fixed number* of *independent* Bernoulli trials.

A simple example is tossing a coin three times and counting the number of *heads* obtained:



two successes (two heads showing)

If X represents the number of heads shown then it is a *discrete random variable* with outcomes $\{0, 1, 2, 3\}$.

The Binomial Distribution:

- Notated $X \sim \mathrm{B}(n, p)$, defined by the two parameters n and p.
- X represents the **number of successes** in a **fixed number of trials**, n, and takes integer values $\{0, 1, 2, \ldots, n\}$.
- Each trial has only **two possible outcomes**: success or failure.
- Each trial is independent, with a **constant probability of success**, p.

For the coin tosses above (assuming the coin is fair):

- There are a fixed number of trials ($n = 3$).
- Each trial results in one of two outcomes: success (heads) or failure (tails).
- The outcome of each trial is independent (since one coin toss does not affect another).
- The probability of success (heads) for each trial is constant ($p = 0.5$).
- The number of successes (heads) is being counted.

If X is the number of heads obtained from three coin tosses, then X is *binomially distributed* with $X \sim \mathrm{B}(3, 0.5)$.

Exercise 7.2

For each, if X is a binomially distributed random variable then state its distribution, such as $X \sim \mathrm{B}\left(8, \frac{2}{3}\right)$. Otherwise, explain clearly which condition(s) for a binomial distribution are **not** satisfied.

1. A student spends one minute rolling a die. Let X represent the number of times the die showed a six.

2. A student rolls a die 20 times. Let X represent the number of times the die showed a six.

3. A bag contains 30 red counters and 30 blue counters. A student selects a counter from the bag at random, notes the colour, then **replaces it** into the bag. In total, they do this 12 times. Let random variable X represent the number of times that the counter selected is red.

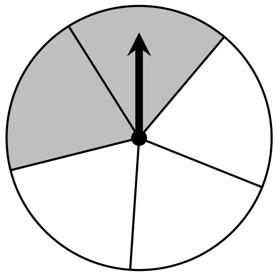
4. A bag contains 30 red counters and 30 blue counters. A student selects a counter from the bag at random, notes the colour, then **does not replace it** into the bag. In total, they do this 12 times. Let random variable X represent the number of times that the counter selected is red.

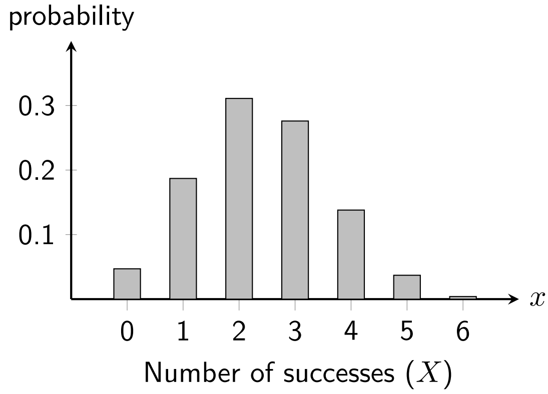
5. Bob takes twelve observations from a $\mathrm{U}(10)$ distribution. Each time the outcome is at least 8, he writes down the value observed. Let X represent the values he writes down.

7.3 The Binomial Probability Function

Unlike the discrete uniform distribution, the binomial distribution has a probability function that is less obvious. It is therefore useful to understand where the formula comes from.

Consider a spinning wheel that lands on a shaded sector with probability $\frac{2}{5}$, or 0.4. Let random variable X represent the number of times the wheel will land on the shaded sector in 6 trials, with possible values from 0 to 6. Here X is *binomially distributed*, with $n = 6$ (number of trials) and $p = 0.4$ (probability of success), or: $X \sim \mathsf{B}(6, 0.4)$





One way of obtaining *exactly two successes* is for the first two spins to be successes and the remaining four to be failures. Since each trial is independent:

$$\mathsf{P}(\text{success, success, fail, fail, fail, fail}) = 0.4 \times 0.4 \times 0.6 \times 0.6 \times 0.6 \times 0.6 = 0.4^2 \times 0.6^4 = 0.0207$$

However, this is only one of many possible arrangements that contain exactly two successes. For example, the successes could occur on spins 1 and 3, spins 2 and 5, and so on. Since each arrangement contains two successes and four failures, each has probability 0.0207. The number of ways of choosing the x trials on which success occurs from the n trials is given by the binomial coefficient, ${}^n\mathsf{C}_x$. For 6 trials with exactly 2 successes, the number of possible arrangements is ${}^6\mathsf{C}_2 = 15$.

$$\mathsf{P}(X = 2) = {}^6\mathsf{C}_2 \times 0.4^2 \times 0.6^4 = 0.3110$$

The same reasoning applies generally. To obtain exactly x successes from n independent trials, there are ${}^n\mathsf{C}_x$ possible arrangements of the successes, each with probability $p^x \times (1-p)^{n-x}$. The probability function for the binomial distribution $X \sim \mathsf{B}(n, p)$ is given on **page 4** of the Data Booklet as:

$$\mathsf{P}(X = x) = {}^n\mathsf{C}_x \times p^x \times (1-p)^{n-x}$$

Exercise 7.3

1. Given that $X \sim \mathsf{B}(10, 0.25)$, calculate $\mathsf{P}(X = 3)$.

2. Given that $X \sim \mathsf{B}(8, 0.6)$, calculate $\mathsf{P}(X = 5)$.

3. Given that $Y \sim \mathsf{B}(13, 0.4)$, calculate $\mathsf{P}(Y \leqslant 2)$. (*Hint:* use $\mathsf{P}(Y = 0)$, $\mathsf{P}(Y = 1)$ and $\mathsf{P}(Y = 2)$).

4. 5% of bluebells have white flowers. The remainder have blue flowers. Determine the probability that a random sample of twelve bluebell plants includes **none** with white flowers.

5. In Bob's class of 20 pupils, their teacher assigns each pupil a number from 1 to 20 and, each day, uses a random number generator to select which pupil should be responsible for handing out rulers. Calculate the probability that Bob is selected **at least once** in one school week.

7.4 The Binomial Distribution using Cumulative Probability Tables

In practice, it will seldom be needed to calculate binomial probabilities directly using its probability function. Graphical calculators can provide both *individual* ($\text{P}(X = x)$) and *cumulative* ($\text{P}(X \leqslant x)$) probabilities. On **page 7**, the Data Booklet provides **cumulative probability tables**, giving **"less than or equal to"** probabilities.

It is therefore important to be able to write any given probability that uses an inequality in terms of a *less than or equal to* probability. Consider discrete random variable X taking values $x = \{0, 1, 2, 3, 4, 5, 6\}$:

$\text{P}(X < 5) = \text{P}(X \leqslant 4)$	$\{0, 1, 2, 3, 4, 5, 6\}$
$\text{P}(X > 5) = 1 - \text{P}(X \leqslant 5)$	$\{0, 1, 2, 3, 4, 5, 6\}$
$\text{P}(X = 5) = \text{P}(X \leqslant 5) - \text{P}(X \leqslant 4)$	$\{0, 1, 2, 3, 4, 5, 6\}$
$\text{P}(1 \leqslant X \leqslant 5) = \text{P}(X \leqslant 5) - \text{P}(X \leqslant 0)$	$\{0, 1, 2, 3, 4, 5, 6\}$
$\text{P}(1 < X < 5) = \text{P}(X \leqslant 4) - \text{P}(X \leqslant 1)$	$\{0, 1, 2, 3, 4, 5, 6\}$

Example

Problem: It is estimated that around 10% of people are left-handed. Taking this to be accurate, calculate the probability that, in a random sample of 12 people, at least two will be left-handed.

Solution: Let random variable X represent the number of left-handed people out of the sample of 12.

$X \sim \text{B}(12, 0.1)$

$\text{P}(X \geqslant 2) = 1 - \text{P}(X \leqslant 1) = 1 - 0.6590 = 0.3410$

Exercise 7.4

1. Given that $X \sim \text{B}(10, 0.3)$, calculate:

   (a) $\text{P}(X \leqslant 6)$ (b) $\text{P}(X = 6)$ (c) $\text{P}(X < 6)$ (d) $\text{P}(X > 6)$ (e) $\text{P}(X \geqslant 6)$

2. Given that $X \sim \text{B}(6, 0.15)$, calculate:

   (a) $\text{P}(X \leqslant 2)$ (b) $\text{P}(X \geqslant 2)$ (c) $\text{P}(1 \leqslant X < 3)$

3. Given that $X \sim \text{B}(14, 0.45)$, calculate:

   (a) $\text{P}(X < 8)$ (b) $\text{P}(X > 12)$ (c) $\text{P}(3 < X \leqslant 6)$

4. A *biased* coin shows heads with probability 0.35. Calculate the probability that it shows heads fewer than twice in twelve tosses.

5. If random variable X represents the number of *successes* in ten independent trials, each with probability of success 0.75, state:

   (a) The distribution of X.
   (b) The distribution of Y, where Y represents the number of *failures*.
   (c) Express $\text{P}(X > 7)$ in terms of Y.
   (d) The value of $\text{P}(X > 7)$, using the Data Booklet.

6. Given that $X \sim \text{B}(8, 0.8)$, calculate:

   (a) $\text{P}(X = 5)$ (b) $\text{P}(X \geqslant 7)$ (c) $\text{P}(3 < X < 6)$

7.5 The Mean and Variance of the Binomial Distribution

The *expectation* of a binomial distribution is fairly intuitive (the number of trials multiplied by the probability of success on each trial), whilst the *variance* is less so. The Data Booklet gives the formulae for the *mean* and *variance* of the binomial distribution on **page 4**. Given random variable $X \sim \mathrm{B}(n, p)$:

$$\mathrm{E}(X) = np \qquad \text{and} \qquad \mathrm{V}(X) = npq \qquad (\text{where } q = 1 - p)$$

Example

Problem: A multiple choice quiz is comprised of eight questions, each with four possible choices. Let random variable X represent the number of correct answers obtained out of eight from a student guessing completely at random. Calculate the mean and variance of X.

Solution: Since the student is guessing randomly, the probability of a correct answer is $p = \frac{1}{4}$.

$X \sim \mathrm{B}(8, 0.25)$

$\mathrm{E}(X) = np = 8 \times 0.25 = 2$

$\mathrm{V}(X) = npq = 8 \times 0.25 \times 0.75 = 1.5$

Exercise 7.5

1. Calculate the mean and variance for each random variable:

   (a) $X \sim \mathrm{B}(4, 0.2)$ (b) $Y \sim \mathrm{B}(6, \frac{1}{3})$ (c) $X \sim \mathrm{B}(4, 0.1)$ (d) $Y \sim \mathrm{B}(30, 0.4)$

2. An octahedral die numbered 1 to 8 is rolled ten times. Let the random variable S represent the number of times the die shows a square number from the ten rolls.

   (a) State the distribution of S.

   (b) Calculate $\mathrm{E}(S)$ and $\mathrm{V}(S)$.

   (c) Determine $\mathrm{P}(S > 5)$.

3. Given that $X \sim \mathrm{B}(6, p)$ such that $\mathrm{E}(X) = 1.2$:

   (a) Determine the value of p.

   (b) Calculate $\mathrm{SD}(X)$.

4. Given that $Y \sim \mathrm{B}(12, p)$ such that $\mathrm{V}(Y) = \frac{8}{3}$ and $p > 0.5$:

   (a) Determine the value of p.

   (b) Calculate $\mathrm{E}(Y)$.

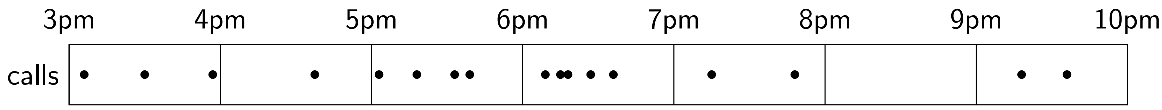
5. Given that $X \sim \mathrm{B}(n, p)$ such that X has mean and variance of 16 and 3.2 respectively:

   (a) Determine the values of n and p.

   (b) Calculate $\mathrm{P}(5 < X < 10)$.

7.6 The Poisson Distribution

Consider a mobile network's customer support department looking at how many call agents to have on duty at any one time during an evening shift. The *mean rate* of calls may be well known, such as 2.5 per hour. However, in any one hour there may be 4 calls, 10 calls, no calls at all, or any other non-negative integer number of calls.

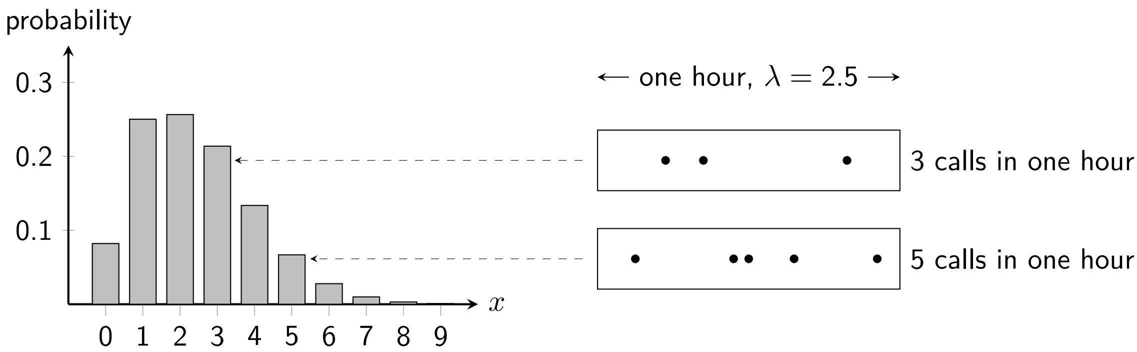


Producing a model for the number of calls in any given hour is clearly of interest to the department, as it allows appropriate staffing levels to be determined. Let the random variable X represent the number of calls in any particular hour. If the conditions for a Poisson model are satisfied, then X follows a **Poisson distribution** with parameter $\lambda = 2.5$.

The Poisson Distribution:

- Notated $X \sim \text{Po}(\lambda)$, defined by one parameter λ *("lambda")*.
- X represents the **number of events occurring** in a set amount of time or space, with events occurring randomly throughout the interval.
- λ represents the **constant mean rate** of events per unit of time or space.
- Each event must be **independent** and **occur singly**.

Assume that the mean rate of 2.5 phone calls per hour is *constant* across the opening hours of the customer support department, and that there is no major network-wide event causing calls to become *dependent* on one another. The Poisson distribution can then be used to calculate, for example, the probability that there are exactly 4 calls in any one hour. Note that the graph below is truncated, and the Poisson distribution has non-zero probabilities for any non-negative integer $\{0, 1, 2, 3, \dots\}$.



As with the binomial distribution, individual Poisson probabilities can be calculated using its *probability function* (**pf**), provided on **page 4** of the Data Booklet. The derivation of this formula relies on mathematics beyond the scope of this course.

Given random variable $X \sim \text{Po}(\lambda)$:

$$\text{P}(X = x) = \frac{e^{-\lambda} \times \lambda^x}{x!}$$

For example $\text{P}(X = 4)$ in the probability distribution above (where $\lambda = 2.5$) can be calculated as:

$$\text{P}(X = 4) = \frac{e^{-2.5} \times 2.5^4}{4!} = 0.1336$$

Example

Problem: During a spell of rain, drops of rain fall on a lawn at a constant mean rate of 6 drops per square metre per second. Calculate the probability that exactly 10 drops fall in one square metre of the lawn in a second.

Solution:
Let random variable X represent the number of drops of rain. Assuming a Poisson process applies:

$X \sim \mathrm{Po}(6)$

$$\mathrm{P}(X = 10) = \frac{e^{-6} \times 6^{10}}{10!} = 0.0413$$

As with the binomial distribution, in general it is much faster to use the **cumulative probability tables** from **page 10** of the Data Booklet, or a graphical calculator, to calculate Poisson probabilities. Unlike the binomial distribution, there is no fixed upper limit to the number of events that can occur, and so (for example) $\mathrm{P}(X \geqslant 5)$ must be calculated as $1 - \mathrm{P}(X \leqslant 4)$.

Exercise 7.6

1. For each, explain why random variable X may **not** follow a Poisson distribution, with reference to the conditions required for a Poisson model.
   (a) Let X represent the number of people arriving at a restaurant each hour.
   (b) Let X represent the amount of rain falling in one hour.
   (c) Let X represent the number of vehicles passing set of traffic lights each minute.
   (d) Let X represent the number of buses arriving at a bus station in an hour.
   (e) Let X represent the number of times a football fan checks their phone during a match.

2. Given random variable X such that $X \sim \mathrm{Po}(2.5)$, use the **probability function** to calculate:

   (a) $\mathrm{P}(X = 4)$. (b) $\mathrm{P}(X = 0)$. (c) $\mathrm{P}(X < 2)$. (d) $\mathrm{P}(X \geqslant 2)$.

3. Let random variable X represent the number of goals in a game of Sunday League football, known to follow a Poisson distribution with a mean of 4 goals.
   (a) State the distribution of X.
   (b) Use the probability function to calculate the probability that a game finishes goalless.

4. The random variable X follows a Poisson distribution with mean 6. Use the **cumulative probability tables** to calculate:

   (a) $\mathrm{P}(X < 4)$. (b) $\mathrm{P}(X > 5)$. (c) $\mathrm{P}(X \leqslant 2)$. (d) $\mathrm{P}(X \geqslant 7)$.

5. The random variable Y follows a Poisson distribution with mean 1.5. Use the tables to calculate:

   (a) $\mathrm{P}(Y < 1)$. (b) $\mathrm{P}(3 < Y < 6)$. (c) $\mathrm{P}(1 \leqslant Y \leqslant 6)$. (d) $\mathrm{P}(Y \geqslant 3)$.

6. Whilst watching a section of the night sky, the number of visible shooting stars is known to follow a Poisson distribution with a mean rate of one per hour. Let random variable X represent the number of shooting stars seen in an hour.
   (a) State the distribution of X.
   (b) Calculate the probability of more than three shooting stars being seen in any given hour.

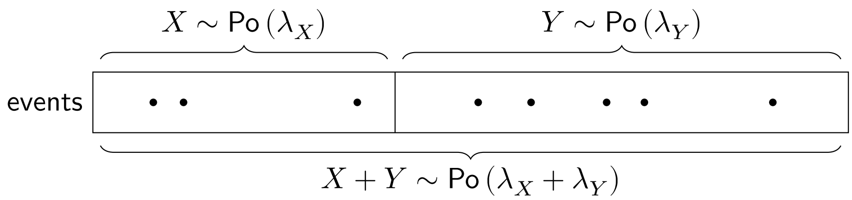
7.7 Combining Poisson Random Variables

The mean and variance of a Poisson distribution are stated on **page 4** of the Data Booklet. Given $X \sim \mathsf{Po}(\lambda)$:

$$\mathsf{E}(X) = \mathsf{V}(X) = \lambda \qquad \text{and} \qquad \mathsf{SD}(X) = \sqrt{\lambda}$$

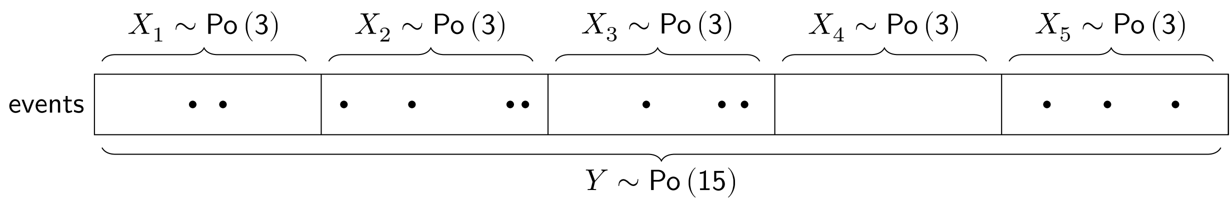
A key feature of the Poisson distribution is that **the mean and variance are equal** This can help to identify whether a Poisson model is appropriate when analysing data.

Given two **independent** random variables X and Y such that $X \sim \mathsf{Po}(\lambda_X)$ and $Y \sim \mathsf{Po}(\lambda_Y)$, the *sum* of X and Y is also a Poisson random variable, for which $\lambda = \lambda_X + \lambda_Y$.



The same result can be used to extend, or shorten, the time period of interest for a Poisson distribution, provided there is a *constant mean rate* throughout the entire time period. For example, consider a call centre for which the number of calls per hour, X, can be modelled using $X \sim \mathsf{Po}(3)$.

For a five-hour shift at the call centre, the number of calls may be modelled as $Y \sim \mathsf{Po}(15)$ since $\lambda = 3 \times 5 = 15$.



This assumes that the number of calls received in each time period is independent.

Example

Problem: Betty has observed that the number of work emails she receives on weekdays between 10am and 11am follows a Poisson distribution with a mean rate of 7 emails. The number of personal emails she receives in this time is also Poisson distributed with a mean rate of 3 emails. Calculate the probability that she receives fewer than 8 emails in total on a weekday between 10am and 11am.

Solution:
Let random variable X represent the number of work emails, and Y the number of personal emails.

$\lambda_X = 7$ and $X \sim \mathsf{Po}(7)$

$\lambda_Y = 3$ and $Y \sim \mathsf{Po}(3)$

Let random variable T represent the total number of emails in an hour:

$\lambda_X + \lambda_Y = 7 + 3 = 10$, so $T \sim \mathsf{Po}(10)$

$\mathsf{P}(T < 8) = \mathsf{P}(T \leqslant 7) = 0.2202$

So there is a roughly 22% probability that Betty receives fewer than 8 emails in total between 10am and 11am, assuming that each event (an email arriving) is independent. This would not be the case if some emails she receives are duplicated between her work and email accounts, such as subscriptions to services emailing all the addresses a customer has registered with them.

Exercise 7.7

1. Given independent random variables $X \sim \text{Po}(1.6)$ and $Y \sim \text{Po}(0.4)$, calculate $\text{P}(X + Y \leqslant 4)$.

2. Given $F \sim \text{Po}(3.3)$ and $G \sim \text{Po}(1.2)$, with F and G independent, calculate $\text{P}(F + G \geqslant 2)$.

3. The number of wickets taken in a morning session of country cricket in a league in Scotland follows a Poisson distribution with a mean of 3.5. In an afternoon session, the mean is 2.5.

   (a) Calculate the probability that fewer than 4 wickets are taken in a **morning** session.

   (b) Find the probability that exactly 8 wickets are taken in a full day, comprising of a morning session and an afternoon session.

   (c) State an assumption required for the calculation in part (b).

4. For a restaurant's Glasgow branch, the number of bottles of champagne sold on a Saturday night follows a Poisson distribution with a mean rate of 3 per night. The number sold for their Edinburgh branch is Poisson distributed with a mean of 5 per night. Stating an assumption required, calculate the probability at least 10 but fewer than 15 bottles of champagne are sold on a Saturday night between the two branches.

5. Shooting stars can be seen in a section of night sky at a mean rate of 1 per hour, and the number visible in an hour is Poisson distributed. Calculate the probability of watching the night sky for two hours and not seeing a single shooting star.

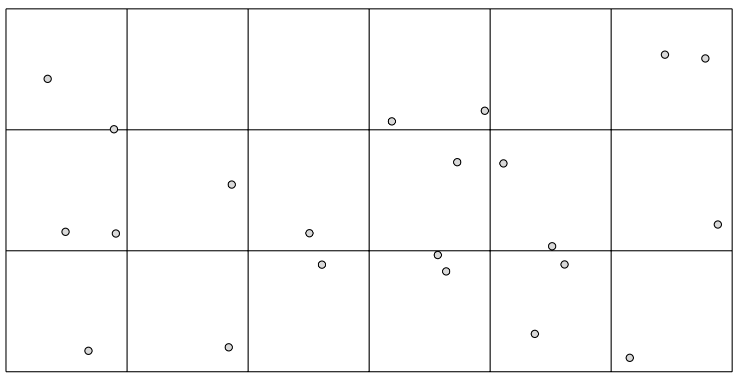
6. A city is experiencing problems with the number of pieces of chewing gum stuck to pavements. A worker from the council's environmental health department notes that in the city's main pedestrian square, the mean number of pieces of chewing gum per square paving stone was 1.5.

   It may be assumed that the number of pieces on a paving stone, X, follows a Poisson distribution.

   (a) State the distribution of X.

   (b) Use the Data Booklet to calculate the probability that a randomly selected paving stone contains at least one piece of chewing gum.

   The worker counts the number of pieces of chewing gum on a grid of 18 paving stones. The diagram below is for illustration, and does not show what the actual number of pieces the worker observed.



   (c) Calculate the probability that a selection of 18 paving stones contains exactly 30 pieces of chewing gum in total, stating an assumption required.

7. A data storage company using a large number of hard drives has seen that the number of drives failing per day follows a Poisson distribution with a constant mean rate of 1.5 per day. The company has staff monitoring the servers that contain the drives 24 hours per day, working in three separate 8-hour shifts each day. Calculate the probability of more than 2 drives failing during one shift.

Review Exercise

1. Given that $X \sim \mathrm{U}(8)$, calculate:
   (a) $\mathrm{P}(X < 6)$.
   (b) The mean and variance of X.

2. Given that $X \sim \mathrm{B}(12, 0.35)$:
   (a) Find $\mathrm{P}(X = 3)$.
   (b) Find $\mathrm{P}(5 < X < 9)$.

3. X and Y are independent random variables such that $X \sim \mathrm{Po}(1.7)$ and $Y \sim \mathrm{Po}(6.8)$, find:
   (a) $\mathrm{P}(X + Y = 7)$.
   (b) $\mathrm{P}(X + Y \geqslant 9)$.

4. The random variable X follows a Poisson distribution with standard deviation 2. Find $\mathrm{P}(X \leqslant 3)$.

5. A gardener plants seeds in batches of 20 and knows that only 85% of seeds actually germinate.
   (a) Find the chance of more than 15 seeds in the batch germinating.
   (b) Find the mean and standard deviation of the number of seeds that germinate.
   (c) State one assumption required for the valid use of the probability distribution used.

6. A restaurant has two food mixers, A and B. The number of times per week that mixer A breaks down has a Poisson distribution with mean 0.4, while independently the number of times that mixer B breaks down has a Poisson distribution with mean 0.1. Calculate the probability that in the next three weeks:
   (a) Mixer A will not break down at all.
   (b) There will be a total of 2 breakdowns.

7. A restaurant menu contains 20 dishes, consisting of starters, main courses and desserts. The table below breaks down the number of dishes for each course and by whether or not they are vegetarian:

	Starter	Main	Dessert
Vegetarian	2	2	6
Not Vegetarian	4	6	0

   Each week, the owner chooses one dish at random to check that the quality is up to standard. Each dish is equally likely to be selected, and a dish being chosen one week does not change the probability that it will be chosen in the following weeks.
   (a) Given that a dish selected to be checked is vegetarian, state the probability that it is a main course.
   Let random variable X represent the number of dishes checked over an 8 week period that are desserts.
   (b) State the distribution of X.
   (c) Calculate the probability that more than half of the dishes checked in the 8 week period are dessert dishes.

8. For each random variable, calculate the probability of the value of an observation being greater than one standard deviation above the mean.
   (a) $X \sim \mathrm{U}(20)$.
   (b) $Y \sim \mathrm{Po}(6)$.
   (c) $Q \sim \mathrm{B}(16, 0.2)$.

8

Continuous Distributions

Data that results from measuring is often continuous, such as the height of a person, the mass of a piece of fruit or the volume of tea in a cup. *Continuous* means that if there were no limits to the precision of a measuring device, there would be infinitely many possible values in any given interval. A **continuous random variable** can take any value within a continuous range of real numbers, rather than a list of separate values. For example, the time taken to complete a task could be *any positive real number*, which can be written as $\{x \in \mathbb{R} : x > 0\}$.

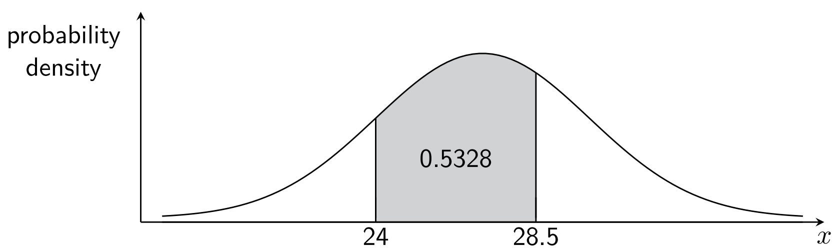
Probability and Continuous Distributions

Since a continuous random variable can take infinitely many values in any interval, probabilities are assigned to ranges of values rather than individual values. Consequently, for any specific value x, $\mathrm{P}(X = x) = 0$. For example, let random variable X represent the time taken to complete a task, measured in seconds. Then it is possible that:

$$\mathrm{P}(X = 26) = 0 \quad \textit{but} \quad \mathrm{P}(24 < X < 28.5) = 0.5328$$

This also means that for continuous random variables: $\mathrm{P}(X \leqslant a) = \mathrm{P}(X < a)$.

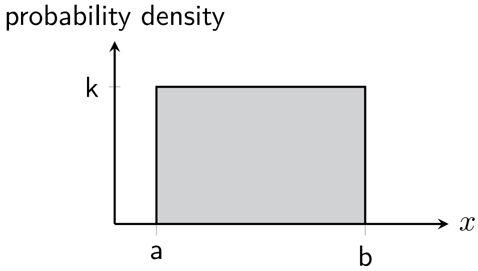
Probabilities are calculated not from the *height* of the graphed probability distributions as with discrete random variables, but instead from the **area under the curve** within the required interval. In this course, graphical calculators and tables of values mean that calculus will not be required to calculate these areas through integration.



The height of the curve is defined by a continuous distribution's *probability density function* (**pdf**). Greater height indicates greater relative likelihood, but probabilities are determined by area rather than height. Since the total probability must be 1, the **total area under the curve is always 1**.

8.1 The Continuous Uniform Distribution

If X can take **any real value** in the interval $[a, b]$ and intervals of equal length have **equal probability**, then X follows a *continuous uniform distribution* with *two* parameters: a and b. This should not be confused with the *discrete* uniform distribution, which takes a *single* parameter only. Since graphing this distribution produces a rectangle, it is often also called a *rectangular distribution*.



The Continuous Uniform Distribution:

- Notated $X \sim \mathrm{U}(a, b)$.
- Defined by two parameters, a and b.
- Takes **continuous** values from a to b.
- Constant probability density of k.

The probability for any range of values can be calculated simply as the area of the rectangle enclosed. The value of its continuous probability density k can be calculated using the formula given on **page 4** of the Data Booklet, derived from the area under any "curve" for any continuous distribution being **equal to 1**. The formulae for the **expectation** and **variance** for the continuous uniform distribution are also given for $X \sim \mathrm{U}(a, b)$:

$$k = \frac{1}{b-a} \qquad \mathrm{E}(X) = \frac{a+b}{2} \qquad \mathrm{V}(X) = \frac{(b-a)^2}{12}$$

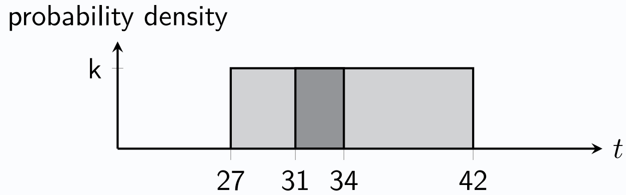
Example

Problem: Given random variable T such that $T \sim \mathrm{U}(27, 42)$, calculate $\mathrm{P}(31 < T < 34)$.

Solution: *(A sketch can help here.)*

$k = \frac{1}{b-a} = \frac{1}{42-27} = \frac{1}{15}$

$\mathrm{P}(31 < T < 34) = 3 \times \frac{1}{15} = \frac{3}{15} = \frac{1}{5}$



Exercise 8.1

1. Explain why, for each, random variable X does **not** follow a $\mathrm{U}(a, b)$ distribution.
   (a) Let X represent the height of an adult.
   (b) Let X represent the score from a fair six-sided die.
   (c) Let X represent the lifetime of a light bulb.
   (d) Let X represent the arrival time of a randomly selected pupil, between 8am and 9am.

2. The continuous random variable X is uniformly distributed over the interval $[6, 15]$. Find:
   (a) $\mathrm{P}(X < 12)$.
   (b) $\mathrm{P}(X > 8.1)$.
   (c) $\mathrm{P}(9.3 < X < 10.2)$.

3. The continuous random variable X is uniformly distributed over the interval $[24, 36]$. Find:
   (a) $\mathsf{P}(X \geqslant 32)$.
   (b) $\mathsf{P}(X = 29.5)$.
   (c) $\mathsf{P}(26.1 \leqslant X \leqslant 34)$.

4. The continuous random variable Y is uniformly distributed over the interval $[1.6, 4]$. Find:
   (a) $\mathsf{P}(Y < 2)$.
   (b) $\mathsf{P}(2 < Y < 5)$.
   (c) $\mathsf{P}(Y + 1 > 4)$.

5. The continuous random variable X is uniformly distributed over the interval $[-4, 6]$.
   (a) Evaluate the mean and standard deviation of X.
   (b) Calculate $\mathsf{E}(3X - 1)$ and $\mathsf{V}(5 - 2X)$.
   (c) Find $\mathsf{P}(X \leqslant 2.4)$.
   (d) Find $\mathsf{P}(-3 < X - 5 < 3)$.

6. The amount of time, in minutes, that a person must wait for a bus is uniformly distributed between 0 and 15 minutes, inclusive.
   (a) Find the probability that a person waits for less than 12.5 minutes.
   (b) Find the average amount of time that a person waits.
   (c) Find the standard deviation of the waiting time.

7. A manufacturer produces sweets of length L millimetres, where L has a continuous uniform distribution with range $[15, 30]$.
   (a) Find the probability that a randomly selected sweet has a length greater than 24 millimetres.

   These sweets are randomly packed into bags containing 20 sweets.

   (b) Find the probability that a randomly selected bag will contain at least eight sweets of length greater than 24 millimetres.

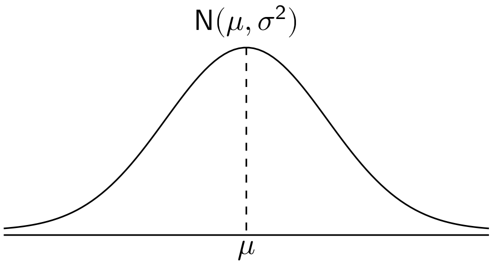
8. Given continuous random variable X such that $X \sim \mathsf{U}(1, 15)$:
   (a) Find μ, the mean of X.
   (b) Find σ, the standard deviation of X.
   (c) Determine $\mathsf{P}(X > \mu)$.
   (d) Find the probability that an observation from X falls within one standard deviation of the mean.

9. Given that continuous random variable X is uniformly distributed with a mean of 8 and a maximum value of 14:
   (a) State the distribution of X.
   (b) Find the standard deviation of X.
   (c) Determine $\mathsf{P}(X < 7)$.

10. Given that continuous random variable X is uniformly distributed such that $\mathsf{E}(X) = 19$ and $\mathsf{SD}(X) = \sqrt{3}$:
    (a) Find the distribution of X.
    (b) Determine $\mathsf{P}(\mathsf{E}(X) - \mathsf{SD}(X) < X < \mathsf{E}(X) + \mathsf{SD}(X))$.

8.2 The Normal Distribution

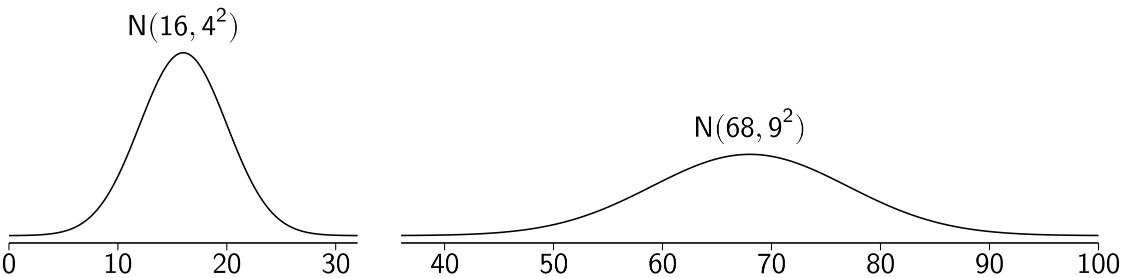
The *normal distribution*, also sometimes called the *Gaussian Distribution*, is one of the most important probability distributions in statistics. One of the reasons for this is that many continuous variables, from the mass of an animal to the time taken to complete a task, can be suitably modelled by it.



pdf: $\dfrac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

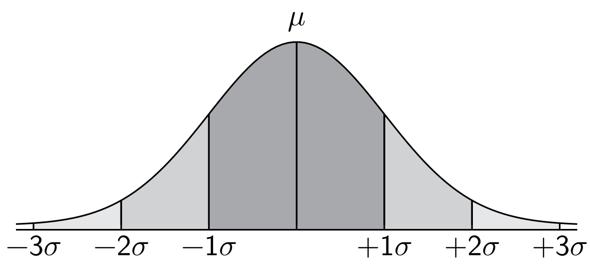
The Normal Distribution:

- Notated $X \sim \mathrm{N}(\mu, \sigma^2)$.
- Defined by two parameters, μ and σ.
- Often described as a *"bell-shaped curve"*.
- μ (the **mean**) determines the *location* of the curve.
- σ (the **standard deviation**) determines the *spread* of the curve.
- The curve is *symmetrical about the mean*.



The shape of the curve illustrates that values closer to the mean are more likely to be observed, whilst those further from the mean are less likely. Whilst *any real value* is possible under the model, for **every normal distribution**:

- 68.27% of observed values lie within **one** standard deviation of the mean.
- 95.45% of observed values lie within **two** standard deviations of the mean.
- 99.73% of observed values lie within **three** standard deviations of the mean.



Exercise 8.2

For each, sketch the distribution, labelling the mean and two standard deviations either side of the mean.

1. (a) $X \sim \mathrm{N}(60, 16)$ (b) $X \sim \mathrm{N}(8, 1.3^2)$ (c) $X \sim \mathrm{N}(243, 9)$ (d) $X \sim \mathrm{N}(15, 1.44)$

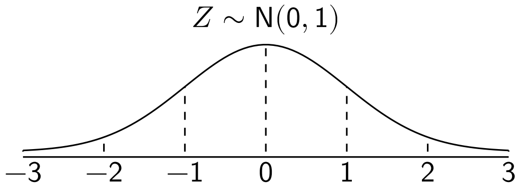
2. The mass of a variety of pumpkin is normally distributed with mean 7kg and standard deviation 2kg.

3. The time taken to polish a spoon is normally distributed with $\mu = 8$ and $\sigma = 1.5$ (measured in seconds).

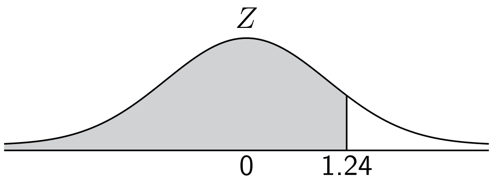
4. The depth of water in a city's puddles after heavy rain is normally distributed with mean 32mm and variance 81mm^2.

8.3 The Standard Normal Distribution *Z*

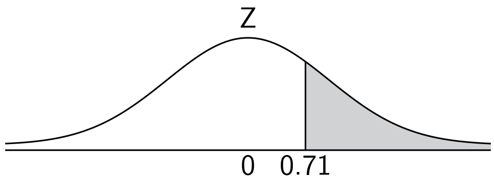
Determining probabilities for the normal distribution by directly calculating an area under the curve requires integration, and is not required in this course (or generally by statisticians at all). Instead, a useful random variable Z is specifically defined as normally distributed, with $\mu = 0$ and $\sigma^2 = 1$ (i.e. mean 0 and variance 1):



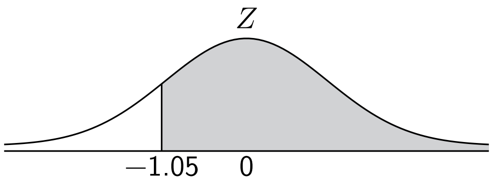
Z is called the **standard normal distribution**, and it is used to calculate probabilities for **any** normal distribution. Probabilities for the Z distribution can be obtained using the **cumulative probability table** on **page 11** of the Data Booklet, or directly from a graphical calculator. Since the cumulative table only gives values for *positive* z, probabilities involving negative z-values must be found using the *symmetry* of the normal distribution.



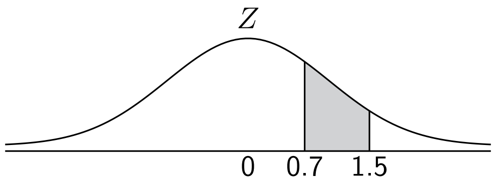
$$\mathrm{P}(Z < 1.24) = 0.8925$$



$$\begin{aligned}\mathrm{P}(Z > 0.71) &= 1 - \mathrm{P}(Z < 0.71)\\ &= 1 - 0.7611\\ &= 0.2389\end{aligned}$$



$$\begin{aligned}\mathrm{P}(Z > -1.05) &= \mathrm{P}(Z < 1.05)\\ &= 0.8531\end{aligned}$$



$$\begin{aligned}\mathrm{P}(0.7 < Z < 1.5) &= \mathrm{P}(Z < 1.5) - \mathrm{P}(Z < 0.7)\\ &= 0.9332 - 0.7580\\ &= 0.1752\end{aligned}$$

Exercise 8.3

1. Find:

(a) $\mathrm{P}(Z = 1.72)$ (b) $\mathrm{P}(Z < 1.72)$ (c) $\mathrm{P}(Z \leqslant 1.72)$ (d) $\mathrm{P}(Z < 0)$
(e) $\mathrm{P}(Z < 0.35)$ (f) $\mathrm{P}(Z < 2.03)$ (g) $\mathrm{P}(Z > 2.03)$ (h) $\mathrm{P}(Z > 1.19)$
(i) $\mathrm{P}(Z < -1.19)$ (j) $\mathrm{P}(Z > -0.56)$ (k) $\mathrm{P}(Z > -1.44)$ (l) $\mathrm{P}(Z < -0.57)$

2. Find:

(a) $\mathrm{P}(1.3 < Z < 2)$ (b) $\mathrm{P}(0.48 < Z < 1.93)$ (c) $\mathrm{P}(-2.3 < Z < -1.7)$
(d) $\mathrm{P}(-1.54 < Z < -0.76)$ (e) $\mathrm{P}(-0.7 < Z < 1.3)$ (f) $\mathrm{P}(-1 < Z < 1)$
(g) $\mathrm{P}(-2 < Z < 2)$ (h) $\mathrm{P}(-1.96 < Z < 1.96)$ (i) $\mathrm{P}(-1.64 < Z < 1.64)$

8.4 The Z-Transformation

A linear transformation of a normally distributed random variable will result in *another* normally distributed random variable, with the laws of expectation and variance applying as usual. In this way, any normally distributed variable can be related to the *standard normal distribution*, Z. Consider a random variable X with mean μ and variance σ^2:

$$X \sim \mathsf{N}(\mu, \sigma^2)$$

Multiplying Z by σ and adding μ results in a normally distributed random variable with mean μ and variance σ^2:

$$\begin{aligned} \mathsf{E}(\sigma Z + \mu) &= \sigma \mathsf{E}(Z) + \mu & \qquad \mathsf{V}(\sigma Z + \mu) &= \sigma^2 \mathsf{V}(Z) \\ &= \sigma \times 0 + \mu & &= \sigma^2 \times 1 \\ &= \mu & &= \sigma^2 \end{aligned}$$

Hence $X = \sigma Z + \mu$, which can be rearranged to obtain the **z-transformation** for any normal random variable $X \sim \mathsf{N}\left(\mu, \sigma^2\right)$:

The z-transformation:

$$Z = \frac{X - \mu}{\sigma}$$

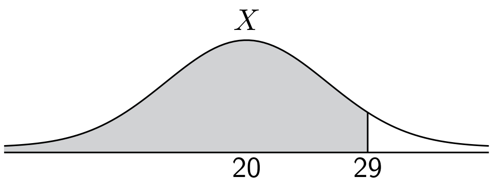
A useful way of thinking about the result of a z-transformation is that it is a measure of *how many standard deviations a value is from the mean*, which then can be used to calculate a probability.

For example, consider normally distributed random variable X with a mean of 20 and standard deviation of 6. The probability that a random observation of X results in a value less than 29 can be sketched on the distribution of X, and compared to the *transformed value* of 1.5 on the distribution of Z. Note that 29 is 1.5 standard deviations greater than the mean of 20.

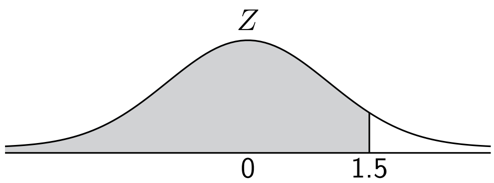
$$z = \frac{x - \mu}{\sigma} = \frac{29 - 20}{6} = 1.5$$

Hence, a probability statement about X can be rewritten as a probability statement about Z:

$$\mathsf{P}(X < 29) = \mathsf{P}\left(Z < \frac{x - \mu}{\sigma}\right) = \mathsf{P}\left(Z < \frac{29 - 20}{6}\right) = \mathsf{P}(Z < 1.5)$$



$\mathsf{P}(X < 29) = 0.9332$



$\mathsf{P}(Z < 1.5) = 0.9332$

Example

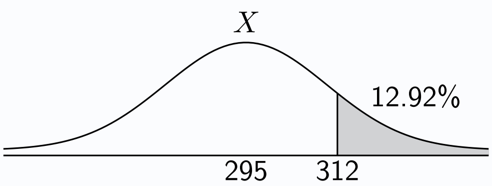
Problem: The mass of an adult red squirrel in an area of woodland is normally distributed with a mean of 295g and a standard deviation of 15g. Calculate the probability that a squirrel chosen at random has a mass greater than 312g.

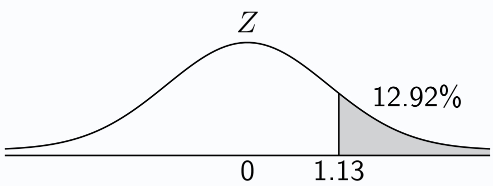
Solution:

Let random variable X represent the mass of a squirrel.

$X \sim \mathrm{N}(295, 15^2)$

$\mathrm{P}(X > 312) = \mathrm{P}\left(Z > \dfrac{312-295}{15}\right) = \mathrm{P}(Z > 1.13) = 0.1292$





Exercise 8.4

1. Given $X \sim \mathrm{N}(24, 9)$, find:

   (a) $\mathrm{P}(X < 26)$ (b) $\mathrm{P}(X \leqslant 21)$ (c) $\mathrm{P}(X > 23)$ (d) $\mathrm{P}(21 < X < 25)$

2. Given that random variable X is normally distributed with mean 80 and standard deviation 4, find:

   (a) $\mathrm{P}(X > 83.1)$ (b) $\mathrm{P}(X < 79.3)$ (c) $\mathrm{P}(X > 72.7)$ (d) $\mathrm{P}(81 < X < 89)$

3. Let random variable X represent the tail length of a certain breed of mouse, which follows a normal distribution with a mean of 67.4mm and a standard deviation of 6mm.

   (a) State the distribution of X.

   (b) Find the probability that a mouse chosen at random will have a tail length:

      i. Less than 70mm. ii. More than 65mm. iii. Between 61mm and 66mm.

4. In a crop of 900 turnips harvested by a farmer, the mass of each is normally distributed with a mean of 150g and a standard deviation of 12g.

   (a) Calculate the probability that a turnip chosen at random will have a mass:

      i. Less than 165.36g. ii. Between 126.48g and 173.52g.

   (b) Estimate the number of turnips of the 900 that have a mass:

      i. Greater than 162g. ii. Less than 155g. iii. Between 140g and 160g.

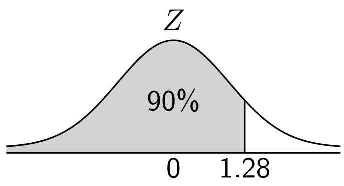
8.5 Working Backwards

"Working backwards" means calculating a value for a normally distributed random variable when a **probability relating to this value is given**. For example, if random variable X is normally distributed with mean 280 and variance 16, it may be desired to calculate the value exceeded by only 10% of the observations. Typical steps are:

- Determine the z-value. $\mathrm{P}(Z < z) = 0.9 \rightarrow z = 1.28$
- Use the z-transformation. $\dfrac{x - 280}{4} = 1.28$

$$x = 1.28 \times 4 + 280$$

- Solve for x. $x = 285.12$



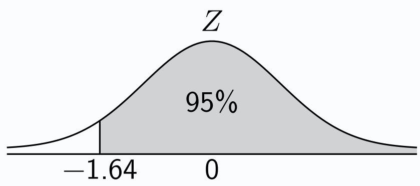
Hence, in the long run only 10% of observations exceed 285.12.

Example

Problem: The time before failure of a model of TV is normally distributed with mean 31 months and standard deviation 4 months. Calculate the minimum warranty length, in months, such that fewer than 5% of TVs will be expected to fail under warranty.

Solution:
Let random variable X represent the time taken for a TV to fail.

$$\begin{aligned} X \sim & \mathrm{N}(31, 4^2) \\ \mathrm{P}(Z < z) = & 0.05 \rightarrow z = -1.64 \\ -1.64 = & \frac{x - 31}{4} \\ x = & 24.44 \end{aligned}$$

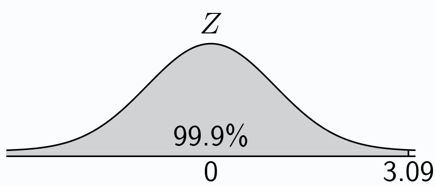


Hence, a 25 month warranty should expect to see less than 5% of TVs fail under warranty.

Example

Problem: The masses of pumpkins grown in a farmer's field are normally distributed, with a mean of 6.8kg. Given that only one in every 1000 pumpkins has a mass greater than 10kg, calculate the standard deviation of the mass of a pumpkin.

Solution:
Let random variable X represent the mass of a pumpkin.



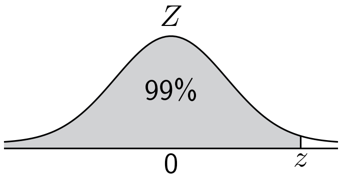
$$\begin{aligned} X \sim & \mathrm{N}(6.8, \sigma^2) \\ \mathrm{P}(Z < z) = & 0.999 \rightarrow z = 3.09 \\ 3.09 = & \frac{10 - 6.8}{\sigma} \\ 3.09\sigma = & 3.2 \\ \sigma = & \frac{3.2}{3.09} \\ \sigma = & 1.04 \end{aligned}$$

Hence, the standard deviation in the mass of a pumpkin is 1.04kg.

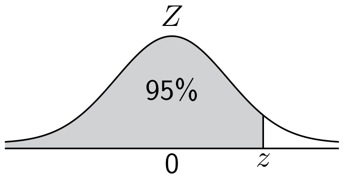
Exercise 8.5

1. Given each sketch of $Z \sim \mathsf{N}(0,1)$ below, use page 12 of the Data Booklet to state the value of z:

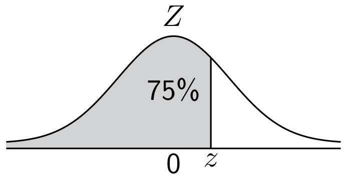
(a) $\mathsf{P}(Z < z) = 0.99$



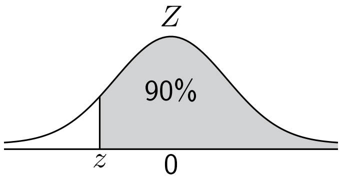
(b) $\mathsf{P}(Z < z) = 0.95$



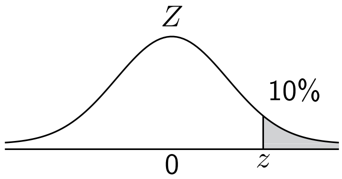
(c) $\mathsf{P}(Z < z) = 0.75$



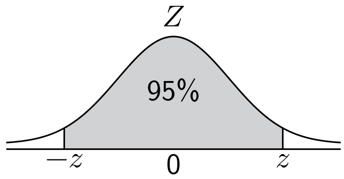
(d) $\mathsf{P}(Z > z) = 0.90$



(e) $\mathsf{P}(Z > z) = 0.1$



(f) $\mathsf{P}(-z < Z < z) = 0.95$



2. Given $Z \sim \mathsf{N}(0,1)$, use a sketch to find the value of z if:

(a) $\mathsf{P}(Z > z) = 0.1$ (b) $\mathsf{P}(Z > z) = 0.05$ (c) $\mathsf{P}(Z > z) = 0.001$ (d) $\mathsf{P}(Z < z) = 0.99$

(e) $\mathsf{P}(Z < z) = 0.995$ (f) $\mathsf{P}(Z < z) = 0.025$ (g) $\mathsf{P}(Z < z) = 0.005$ (h) $\mathsf{P}(Z > z) = 0.9$

(i) $\mathsf{P}(Z > z) = 0.975$ (j) $\mathsf{P}(Z < z) = 0.999$ (k) $\mathsf{P}(Z < z) = 0.791$ (l) $\mathsf{P}(Z > z) = 0.003$

3. Let random variable X represent the total volume of petrol used by a motorist on their daily commute to and from work. The amount of petrol used is normally distributed with a mean of 2.4 litres and standard deviation 0.2 litres.

   (a) State the distribution of X.

   (b) Calculate the probability that less than 2 litres of petrol is used on a daily commute to and from work.

   (c) The motorist wishes to know the amount of petrol only exceeded on 1% of daily commutes. Find this value.

4. The time taken for a particular type of steak to be cooked until it is medium rare is normally distributed with a mean of 8.2 minutes and a standard deviation of 0.3 minutes.

   (a) Find the time needed, in minutes, such that 95% of such steaks will have reached medium rare.

   (b) Determine the range of times, symmetrical around the mean, between which 95% of all such steaks will reach medium rare.

5. In a Maths exam, the scores obtained by candidates are normally distributed with a mean of 57.3. Given that 5% of pupils fail to gain a grade from A to D, and that a score of 41 is required to gain a D, calculate the standard deviation.

6. In a Physics exam, the scores obtained by candidates are normally distributed. Given that 25% of pupils gain a grade A with a mark of 74 required, and 10% of pupils didn't get the 46 marks required for a grade C, calculate the mean and standard deviation.

8.6 Combining Normal Random Variables

Sums and differences of independent and normally distributed random variables are *themselves normally distributed*, and the mean and variance of these may be determined using the laws of expectation and variance introduced in section 6.6. For example, given **independent** normally distributed random variables X and Y with means of μ_X and μ_Y and standard deviations of σ_X and σ_Y respectively:

$$X + Y \sim \mathsf{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2) \quad \text{and} \quad X - Y \sim \mathsf{N}(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

Note that $2X$ is *not* the same thing as the sum of two independent observations of X, which can be expressed as $X_1 + X_2$. They share the same expectation, but different variance. Given $X \sim \mathsf{N}(\mu, \sigma^2)$:

$$\begin{aligned} \mathsf{E}(X_1 + X_2) &= \mathsf{E}(X_1) + \mathsf{E}(X_2) \\ &= 2\mathsf{E}(X) \\ \mathsf{V}(X_1 + X_2) &= \mathsf{V}(X_1) + \mathsf{V}(X_2) \\ &= 2\mathsf{V}(X) \end{aligned}$$

$$\begin{aligned} \mathsf{E}(2X) &= 2\mathsf{E}(X) \\ \mathsf{V}(2X) &= 2^2 \times \mathsf{V}(X) \\ &= 4\mathsf{V}(X) \end{aligned}$$

The sum of n independent observations of X is $X_1 + X_2 + \cdots + X_n$.

$$\begin{aligned} \mathsf{E}(X_1 + X_2 + \ldots + X_n) &= \mathsf{E}(X_1) + \mathsf{E}(X_2) + \cdots + \mathsf{E}(X_n) = n\mathsf{E}(X) \\ \mathsf{V}(X_1 + X_2 + \ldots + X_n) &= \mathsf{V}(X_1) + \mathsf{V}(X_2) + \cdots + \mathsf{V}(X_n) = n\mathsf{V}(X) \end{aligned}$$

When combining normal random variables, it can be helpful to define a new random variable, such as T for *total*, D for *difference* and so on.

Example

Problem: The actual volume of paint contained within tins marked as containing 25 litres is normally distributed with a mean of 25.2 litres and a standard deviation of 0.3 litres. Calculate the probability that the total volume in 8 tins is less than 200 litres, stating an assumption required

Solution:

Let random variable $X \sim \mathsf{N}(25.2, 0.3^2)$ represent the volume of paint in a tin.

Let T represent the total volume of paint in the eight tins.

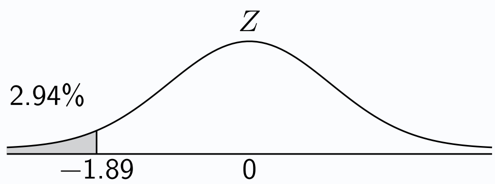
Assuming the volume of paint in each tin is independent.

$T = X_1 + X_2 + \cdots + X_8$

$\mathsf{E}(T) = 8 \times \mathsf{E}(X) = 8 \times 25.2 = 201.6$

$\mathsf{V}(T) = 8 \times \mathsf{V}(X) = 8 \times 0.3^2 = 0.72$

$T \sim \mathsf{N}(201.6, 0.72)$



$$\mathsf{P}(T < 200) = \mathsf{P}\left(Z < \frac{200 - 201.6}{\sqrt{0.72}}\right) = \mathsf{P}(Z < -1.89) = 0.0294$$

Exercise 8.6

1. The time taken for a chef to chop an onion is normally distributed with mean 19 seconds and standard deviation 3 seconds. Assuming the time taken to chop each onion is independent, calculate the probability that it takes over **one minute** in total for the chef to chop three onions.

2. The mass of a library book in a school's library is normally distributed, with a mean of 400g and a standard deviation of 60g. Twenty books are selected at random to be placed on a table as part of a display.

   (a) Calculate the probability of the total mass of the 20 books being less than 8000g.

   (b) State an assumption required, and explain why this assumption may be reasonable.

3. A coffee machine has an option which dispenses a portion of milk. The volume of milk dispensed each time is normally distributed with mean 28ml and variance 3ml^2. Assuming that the volume of each portion is independent, determine the probability that the total volume of four portions exceeds 110ml.

4. The mass of a *house martin*, a species of bird, is normally distributed with mean 18g and variance 2g^2. Five house martins are observed perched together on a telephone line. Stating an assumption required, determine the probability that the total mass of the five birds is greater than 70g.

5. Basketball games are played across four *quarters*, each lasting ten minutes. Extensively collected data shows that the distance run by professional basketball players during a quarter follows a normal distribution with a mean of 1km and a variance of 0.1km^2. Give a reason why this does not necessarily imply that the total distance run by a player during a full game of four quarters can be modelled by a normal distribution with mean 4km and variance 0.4km^2.

6. Bob observes that making his usual portion of mashed potatoes can be broken down into three distinct processes. The time taken to peel the potatoes is normally distributed with a mean of 5 minutes and a standard deviation of 1.2 minutes. The time taken to boil the potatoes is also normally distributed with a mean of 27 minutes and a standard deviation of 3.4 minutes. Finally, the mean time taken to mash and season the potatoes is 2 minutes with a standard deviation of 0.3 minutes.

   (a) Find the distribution of the time taken for Bob to make a portion of mash, stating two assumptions required.

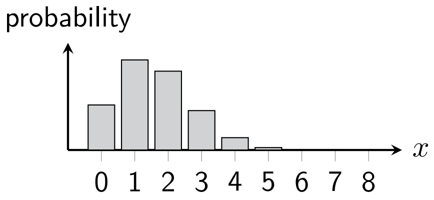
   (b) Calculate the probability that it takes Bob under 35 minutes to make a portion.

7. The bar staff in a hotel ignore legal requirements to carefully measure any alcohol poured, and instead just estimate the amount of champagne to pour into each glass. The amount they pour is normally distributed with a mean of 127ml and a standard deviation of 5ml. Assuming that the amount of champagne poured in each of several glasses is independent, determine the probability that the total volume in five glasses is less than 625ml.

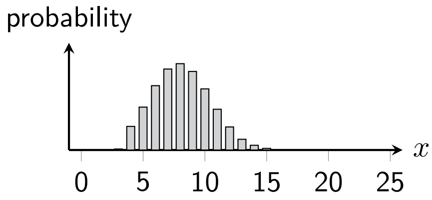
8. The amount of milk a school's Maths department uses per week is normally distributed, with a mean of 3.5 litres and a standard deviation of 0.4 litres. The Business department's milk use is also normally distributed, with a mean of 2.8 litres and a standard deviation of 0.5 litres. For any given week, calculate the probability that the Business department uses more milk than the Maths department, stating any assumptions required.

9. The length of a best-of-five match of tennis is normally distributed with a mean of 165 minutes and a standard deviation of 21 minutes, whilst the length of a best-of-three match has a mean of 95 minutes and a standard deviation of 10 minutes. A tournament has two best-of-five matches and one best-of-three match scheduled to be played on Court 7, beginning at 10am. Find the probability that the three matches will be completed by 4pm, stating two assumptions required.

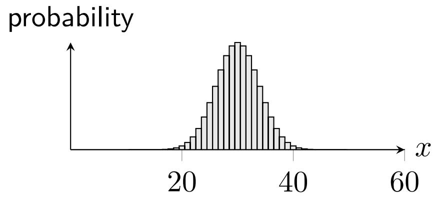
8.7 Normal Approximation to the Binomial Distribution



B$(8, 0.2)$



B$(25, 0.4)$



B$(60, 0.5)$

As the number of trials n grows (and for probabilities of success closer to 0.5), the *discrete* binomial distribution can begin to resemble a *continuous* normal distribution. Given random variable $X \sim \mathsf{B}(n, p)$, a *normal distribution* can approximate the binomial distribution, with *sufficient accuracy*, when:

Conditions for normal approximation to binomial distributions:

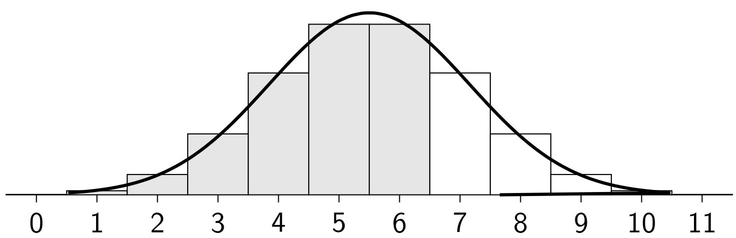
$np > 5$ and $nq > 5$

The mean and variance of the *approximate* normal distribution are, as usual for a binomial distribution, $\mathsf{E}(X) = np$ and $\mathsf{V}(X) = npq$. Using '$\approx$' instead of '$\sim$' is important to show that the normal distribution is only an *approximation* to the actual binomial distribution.

$$\text{If } X \sim \mathsf{B}(n, p) \text{ then } X \approx \mathsf{N}(np, npq)$$

Historically, the ability to approximate the binomial distribution using the normal distribution will have prevented the need for extensive tables of binomial probabilities. Additionally, binomial distribution calculations tend to be more computationally-intensive than normal distributions calculations. Whilst modern computing power may seem to render this argument null, it can still be desirable to speed up computation in return for a trivial loss in accuracy.

Approximating probabilities for a *discrete* distribution using a *continuous* distribution may lead to inaccuracy unless a **continuity correction** is applied. The continuity correction adjusts for the fact that a discrete distribution assigns probability to whole numbers, while a continuous distribution assigns probability over intervals. For example, consider random variable $X \sim \mathsf{B}(11, 0.5)$, with its normal approximation $X \approx \mathsf{N}(5.5, 2.75)$ overlaid and the *discrete* probability $\mathsf{P}(X < 7)$ shaded:



$$\mathsf{P}(X < 7) = \mathsf{P}\left(Z < \frac{6 + 0.5 - 5.5}{\sqrt{2.75}}\right)$$

Discrete	Continuous approximation
$\mathsf{P}(X = 7)$	$\mathsf{P}(6.5 < X < 7.5)$
$\mathsf{P}(X < 7)$	$\mathsf{P}(X < 6.5)$
$\mathsf{P}(X \leqslant 7)$	$\mathsf{P}(X < 7.5)$
$\mathsf{P}(X > 7)$	$\mathsf{P}(X > 7.5)$
$\mathsf{P}(X \geqslant 7)$	$\mathsf{P}(X > 6.5)$

When calculating the discrete probability $\mathsf{P}(X < 7)$, the bar representing the discrete integer 7 should not be included. The continuous probability $\mathsf{P}(X < 6.5)$, making a **continuity correction**, would provide a more accurate estimate of the desired probability. For the discrete probability $\mathsf{P}(X \leqslant 7)$, where the discrete integer value of 7 should be included, the continuous probability $\mathsf{P}(X < 7.5)$ would be more accurate.

Example

Problem: A quiz contains 40 multiple choice questions, with four answers for every question. If a pupil picks answers entirely at random, use a suitable approximation to calculate the probability that they get at least half correct, justifying the suitability of the approximation used.

Solution: *"Suitable approximation" highlights that the normal approximation should be used, even if a graphical calculator can obtain the precise probability.*

Let random variable X represent the number of questions answered correctly.

$X \sim \mathrm{B}(40, 0.25)$

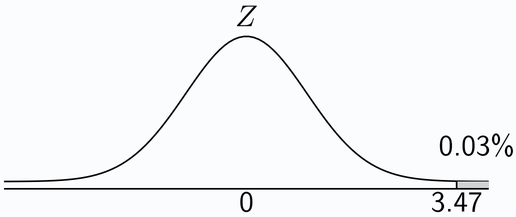
$np = 40 \times 0.25 = 10$, $nq = 40 \times 0.75 = 30$

$np > 5$ and $nq > 5$ therefore a normal approximation is appropriate.

$\mathrm{E}(X) = np = 10$

$\mathrm{V}(X) = npq = 40 \times 0.25 \times 0.75 = 7.5$

$X \approx \mathrm{N}(10, 7.5)$

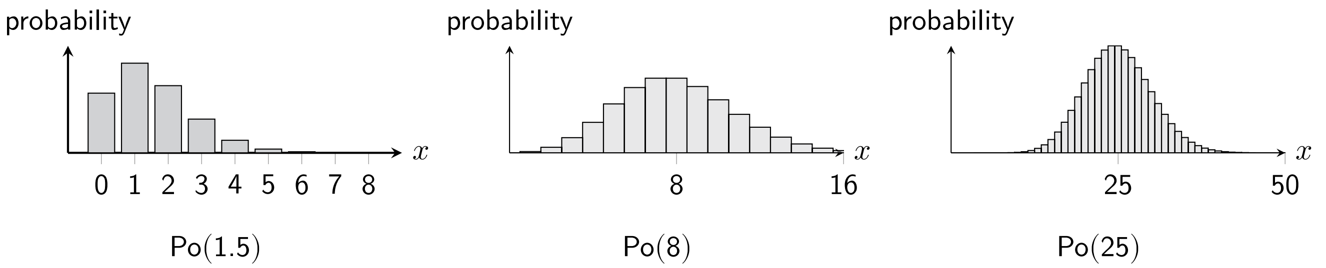


$$\mathrm{P}(X \geqslant 20) = \mathrm{P}\left(Z > \frac{19.5 - 10}{\sqrt{7.5}}\right) = \mathrm{P}(Z > 3.47) = 0.0003$$

Exercise 8.7

1. Given $X \sim \mathrm{B}(40, 0.4)$:
   (a) Show that a normal approximation is appropriate.
   (b) Find:
      i. $\mathrm{P}(X \leqslant 20)$
      ii. $\mathrm{P}(25 \leqslant X \leqslant 28)$
      iii. $\mathrm{P}(X > 31)$
2. Given $Y \sim \mathrm{B}\left(81, \frac{2}{3}\right)$:
   (a) Show that a normal approximation is appropriate.
   (b) Find:
      i. $\mathrm{P}(Y \geqslant 56)$
      ii. $\mathrm{P}(Y = 48)$
      iii. $\mathrm{P}(50 < Y < 60)$
3. To complete the mandatory online training required by a large corporation, 600 employees in an office each try to log on to the corporation's online training portal. The probability of an employee managing to successfully log in on their first attempt is 85%. Let the random variable X represent the number of employees that manage to log in successfully on their first attempt. State the distribution of X and use a suitable approximation to estimate the probability of over 100 of the 600 employees not managing to log in successfully on their first attempt, justifying its use.
4. A farmer believes that 60% of the apples that are brought into her grading shed are classified as Grade A. Assuming she is right, use a suitable approximation to estimate the probability that, in a random sample of 120 apples, there are more than 75 Grade A apples.
5. 200 people are randomly picked as part of a survey and asked whether or not they plan to vote for the party currently in power in an upcoming election. It is generally thought that 54% of people will vote for this party. Using a suitable approximation, calculate the probability that less than 45% of the people sampled indicate that they will vote for the party currently in power, and comment on what this might suggest should it happen.

8.8 Normal Approximation to the Poisson Distribution



As with the binomial distribution, the *discrete* Poisson distribution can also begin to resemble a *continuous* normal distribution; it will do so when it becomes closer to symmetrical as λ increases. Given the random variable $X \sim \text{Po}(\lambda)$, a normal distribution can *approximate* the Poisson distribution with sufficient accuracy when:

Conditions for normal approximation to Poisson distributions:

$\lambda > 10$

The mean and variance of the *approximate* normal distribution are $\text{E}(X) = \text{V}(X) = \lambda$.

$$\text{If } X \sim \text{Po}(\lambda) \text{ then } X \approx \text{N}(\lambda, \lambda)$$

Since again this is a *continuous approximation* to a *discrete distribution*, any probabilities calculated will not be sufficiently accurate unless a **continuity correction** is applied. For a continuous approximation to a discrete distribution taking integer values, as is the case with both such approximations encountered in this chapter, continuity corrections will always require an addition or subtraction of 0.5.

Example

Problem: In a football league, goals occur at a constant mean rate of 2.5 goals per game. A full weekend of fixtures includes 10 games. Using a suitable approximation, and justifying its use, calculate the probability that there will be between 15 and 20 goals in total, inclusive, in a full weekend of fixtures.

Solution:
Let random variable T represent the number of goals across the 10 games, with $\lambda = 2.5 \times 10 = 25$.

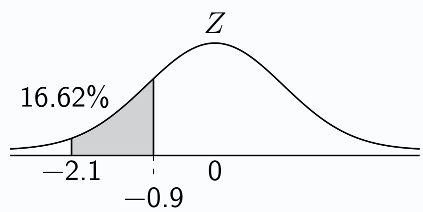
$T \sim \text{Po}(25)$

Since $\lambda > 10$, a normal approximation is appropriate.

$\text{E}(T) = \text{V}(T) = 25$

$T \approx \text{N}(25, 25)$

$$\begin{aligned}
\text{P}(15 \leqslant T \leqslant 20) =& \text{P}\left(\frac{14.5-25}{\sqrt{25}} < Z < \frac{20.5-25}{\sqrt{25}}\right) \\
=& \text{P}(-2.1 < Z < -0.9) \\
=& \text{P}(Z < -0.9) - \text{P}(Z < -2.1) \\
=& 0.1841 - 0.0179 \\
=& 0.1662
\end{aligned}$$



Exercise 8.8

1. Given $X \sim \text{Po}(24)$, use the normal approximation to find:

   (a) $\text{P}(X \leqslant 25)$ (b) $\text{P}(X = 21)$ (c) $\text{P}(X > 23)$

2. Given $X \sim \text{Po}(60)$, use the normal approximation to find:

   (a) $\text{P}(X \geqslant 57)$ (b) $\text{P}(50 < X \leqslant 56)$ (c) $\text{P}(62 < X < 66)$

3. The number of calls received by an office switchboard per hour follows a Poisson distribution with a parameter 30. Using the normal approximation to the Poisson distribution, find the probability that in one hour:

   (a) There are more than 33 calls.
   (b) There are between 25 and 28 calls, inclusive.
   (c) There are exactly 34 calls.

4. In a certain factory the number of incidents occurring in a month follows a Poisson distribution with mean 4. Find the probability that there will be at least 40 incidents during one year.

5. The number of bacteria on a plate viewed under a microscope follows a Poisson distribution with parameter 60.

   (a) Find the probability that there are between 55 and 75 bacteria, inclusive, on a plate.
   (b) A plate is rejected if fewer than 38 bacteria are found. If 2000 plates are viewed, determine the expected number of plates that will be rejected.

6. In an experiment with a radioactive substance, the number of particles reaching a counter over a given period of time follows a Poisson distribution with mean 22. Find the probability that the number of particles reaching the counter over the given period of time is:

   (a) Fewer than 22.
   (b) Between 25 and 30, not inclusive.
   (c) 18 or more.

7. The number of accidents on a certain railway line occurs at an average rate of one every 2 months. Justifying your method, estimate the probability that there are:

   (a) 28 or more accidents in 4 years.
   (b) Fewer than 30 accidents in 5 years.

8. The number of eggs laid by an insect follows a Poisson distribution with parameter 200. Estimate the probability that between 180 and 220 eggs, inclusive, are laid.

Review Exercise

1. Given $X \sim$ U(1,7), find:

   (a) E(X)

   (b) V$(3 - X)$

   (c) P$(X < 3)$

2. Given $X \sim$ N$(20, 6)$, find:

   (a) P$(X < 25)$

   (b) P$(14 < X < 20)$

3. The height of an adult giraffe is normally distributed with a mean of 18.1 feet and a standard deviation of 0.7 feet. Let random variable H represent the height of an adult giraffe, in feet.

   (a) State the distribution of H.

   (b) Find P$(H > 20)$.

   (c) A biologist wishes to know the height only exceeded by 5% of giraffes. Find this height.

4. A guitar player experiences broken strings at a constant mean rate of 2 per month.

   (a) Find the probability that no strings break in one month.

   (b) Use a suitable approximation to estimate the probability that exactly 26 strings break in one year.

5. For continuous random variable $X \sim$ N$\left(\mu, 3^2\right)$, P$(X < 16) = 0.1$. Determine the value of μ for X.

6. A continuous uniform distribution is defined over the interval 10 to 20.

   (a) For a single observation from this distribution, calculate the probability of obtaining a value not greater than 13.

   (b) If a random sample of size 10 is taken from the distribution, calculate the probability of at least 3 of the observations resulting in value not greater than 13.

   (c) If a random sample of size 100 is taken from the distribution, use a suitable approximation to estimate the probability that over 30 of the observations result in a value not greater than 13.

7. A sports scientist is interested in the time taken for 800 metre runners to complete each of the two laps that make up the distance. They observe that the time it takes for an athlete they are studying to complete each lap is normally distributed, with a mean of 53.2 seconds with a standard deviation of 1.3 seconds.

   (a) Find the probability that the athlete completes the two laps in under 1 minute and 44 seconds.

   (b) An important assumption is required to calculate the answer for part (a). State the assumption and explain whether or not it is likely to have been met.

9

The Distribution of the Sample Mean

Previous chapters have investigated probabilities related to *individual* observations from random variables with known distributions. However, statisticians are rarely in the business of making a single observation from a population - instead, they are far more likely to draw a *sample* of observations and, commonly, calculate the *sample mean*. Given a random variable X, the sample mean of observations from X is also a random variable and is denoted $\overline{X}$.

Letting $X_1, X_2, \ldots, X_n$ represent a sample of n *independent and identically distributed* (i.i.d.) observations, where independence may be assumed through the use of *random sampling* methods, the sample mean may be expressed as:

$$\overline{X} = \frac{1}{n}\left(X_1 + X_2 + X_3 + \ldots + X_n\right)$$

The Expectation and Variance of the Sample Mean

The expected value of the sample mean is equal to the true value of the population mean, meaning the sample mean is an *unbiased estimator* of the population mean. That is, if a series of samples are taken from a distribution with population mean μ and the mean is calculated for each sample, in the long-run, the mean of these sample means tends toward μ.

The *standard deviation of the sample mean* $\overline{X}$ is called the **standard error** (SE). For an underlying population with standard deviation σ, the standard error of the sample mean for a sample of size n is $\dfrac{\sigma}{\sqrt{n}}$.

Proof: Given random variable X such that $\mathsf{E}(X) = \mu$ and $\mathsf{V}(X) = \sigma^2$, from which a sample of n independent observations $(X_1, X_2, \ldots, X_n)$ are drawn:

$$\begin{aligned}
\mathsf{E}(\overline{X}) &= \mathsf{E}\left(\frac{1}{n}(X_1 + X_2 + \ldots + X_n)\right) \\
&= \frac{1}{n}(\mathsf{E}(X_1) + \mathsf{E}(X_2) + \ldots + \mathsf{E}(X_n)) \\
&= \frac{1}{n}(n\mu) \\
&= \mu
\end{aligned}$$

$$\begin{aligned}
\mathsf{V}(\overline{X}) &= \mathsf{V}\left(\frac{1}{n}(X_1 + X_2 + \ldots + X_n)\right) \\
&= \frac{1}{n^2}(\mathsf{V}(X_1) + \mathsf{V}(X_2) + \ldots + \mathsf{V}(X_n)) \\
&= \frac{1}{n^2}(n\sigma^2) \\
&= \frac{\sigma^2}{n}
\end{aligned}$$

This shows that, as the sample size n increases, the variability of the sample mean decreases. The result for the variance relies on the observations being independent, which is typically ensured through the use of random sampling.

9.1 The Sample Mean of a Normally Distributed Random Variable

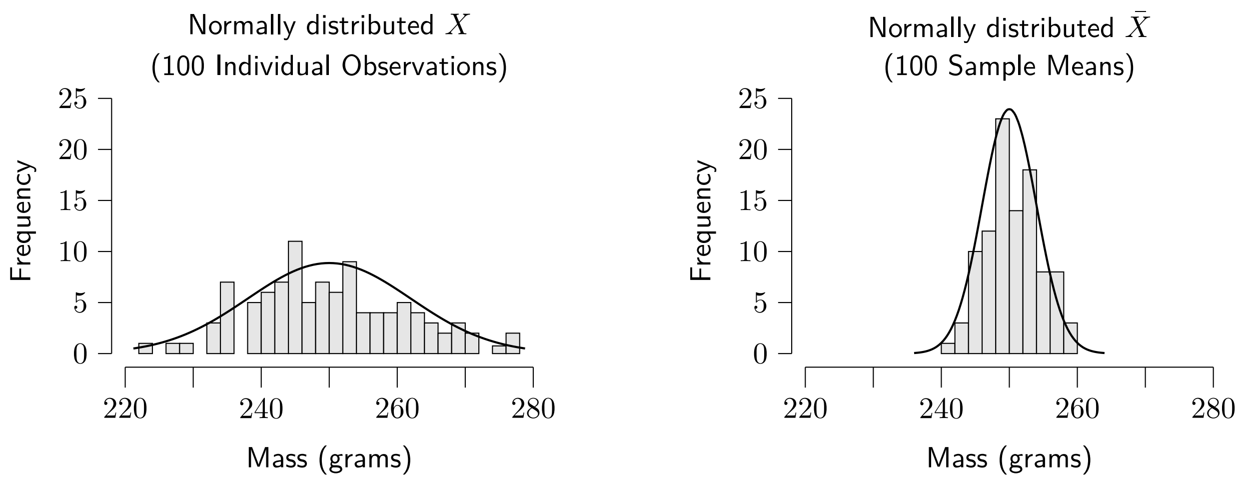
Since the sample mean $\overline{X}$ is a linear combination of independent observations of X, if X is normally distributed then $\overline{X}$ is *also normally distributed*.

Sample mean of normally distributed X:

If $X \sim \mathrm{N}(\mu, \sigma^2)$ then $\overline{X} \sim \mathrm{N}\left(\mu, \dfrac{\sigma^2}{n}\right)$

Due to the lower variance of the sample mean, *extreme results are less likely* when taking the mean of a sample in comparison to a single observation. For example, suppose the mass of adult red squirrels in a location is known to be normally distributed, with mean 250 grams and standard deviation 12 grams. If a single squirrel is randomly selected and its mass measured, it would not be particularly unlikely to encounter a squirrel lighter than 240 grams or heavier than 260 grams. However, if a random sample of three squirrels is taken, each weighed and the mean of the sample calculated, then it would be very unlikely for the sample mean to be lower than 240 grams or heavier than 260 grams.

The histograms below show 100 simulated individual observations of squirrel masses compared to the means of 100 simulated samples, each of size 3. Note that the sample means also show a distribution that looks normal with a mean of 250 grams, but that the spread of the sample means is much less than that of the individual observations. This reflects the fact that the standard deviation of the sample mean (the standard error) is $\dfrac{\sigma}{\sqrt{n}}$, which is smaller than σ.



The z-transformation for *single observations* from $X \sim \mathrm{N}(\mu, \sigma^2)$ is $Z = \dfrac{X - \mu}{\sigma}$.

The z-transformation for the **sample mean** $\overline{X} \sim \mathrm{N}\left(\mu, \dfrac{\sigma^2}{n}\right)$ has the **standard error** as its denominator:

The z-transformation for the sample mean:

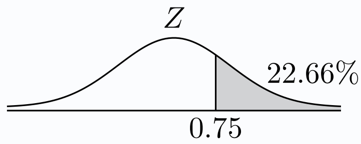
$$Z \quad = \quad \frac{\overline{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

This result holds exactly for normally distributed populations, regardless of sample size.

Example

Problem: The length of a mass-produced brand of fence panels is normally distributed with a mean of 180cm and a standard deviation of 0.4cm. A random sample of nine panels is taken and each is measured. State the distribution of the mean of a random sample of nine panels, and calculate the probability of the sample mean being greater than 180.1cm.

Solution: Let r.v. X represent the length of a fence panel.



$$\overline{X} \sim \mathsf{N}\left(180, \frac{0.4^2}{9}\right)$$

$$\mathsf{P}(\overline{X} > 180.1) = \mathsf{P}\left(Z > \frac{180.1 - 180}{\frac{0.4}{\sqrt{9}}}\right) = \mathsf{P}(Z > 0.75) = 0.2266$$

Exercise 9.1

1. Given random variable $X \sim \mathsf{N}(24, 4^2)$:
   (a) Calculate for a single observation:
      i. $\mathsf{P}(X > 25.2)$
      ii. $\mathsf{P}(X < 23.4)$
   (b) For a sample from X of size 3:
      i. State the distribution of $\overline{X}$
      ii. Calculate $\mathsf{P}(\overline{X} > 25.2)$
      iii. Calculate $\mathsf{P}(\overline{X} < 23.4)$

2. The tail length of a breed of mouse is normally distributed with a mean of 5.2cm and a standard deviation of 0.8cm. A random sample of 12 mice are captured, their tail lengths measured and then released. Calculate the probability that the mean tail length of the sample of mice is less than 5cm.

3. The time taken for a telecommunications company to interview candidates for a sales role is normally distributed with a mean of 28 minutes and a standard deviation of 4 minutes.
   (a) Determine the probability that an interview lasts over half an hour.
   (b) Stating an assumption required, calculate the probability that a sample of five interviews will have a mean length greater than half an hour.

4. The mass of a medium-sized whole chicken sold by a supermarket is normally distributed with a mean of 2.5kg and a standard deviation of 0.15kg. Calculate the probability that the mean mass of a random sample of 9 medium-sized chickens is less than 2.4kg.

5. The amount of food waste discarded by a family per day is normally distributed with a mean of 1.96kg and a standard deviation of 0.4kg.
   (a) Calculate the probability that, on any random day, less than 1kg is discarded.
   (b) Calculate the probability that over the course of a week the mean amount of daily food waste discarded exceeds 2kg.
   (c) Suggest a reason why the calculation in part (b) may not be reliable.

6. The distances recorded in long-jump attempts by a group of junior long-jump specialists at an athletics club are normally distributed with a mean of 6.5m and a standard deviation of 0.2m. The long-jump distances recorded by the junior heptathletes are also normally distributed, with mean 6.3m and standard deviation 0.4m. A coach randomly observes attempts by 4 long-jump specialists and 3 heptathletes. Calculate the probability that the heptathletes' mean distance is greater than that of the long-jump specialists. *(Extension)*

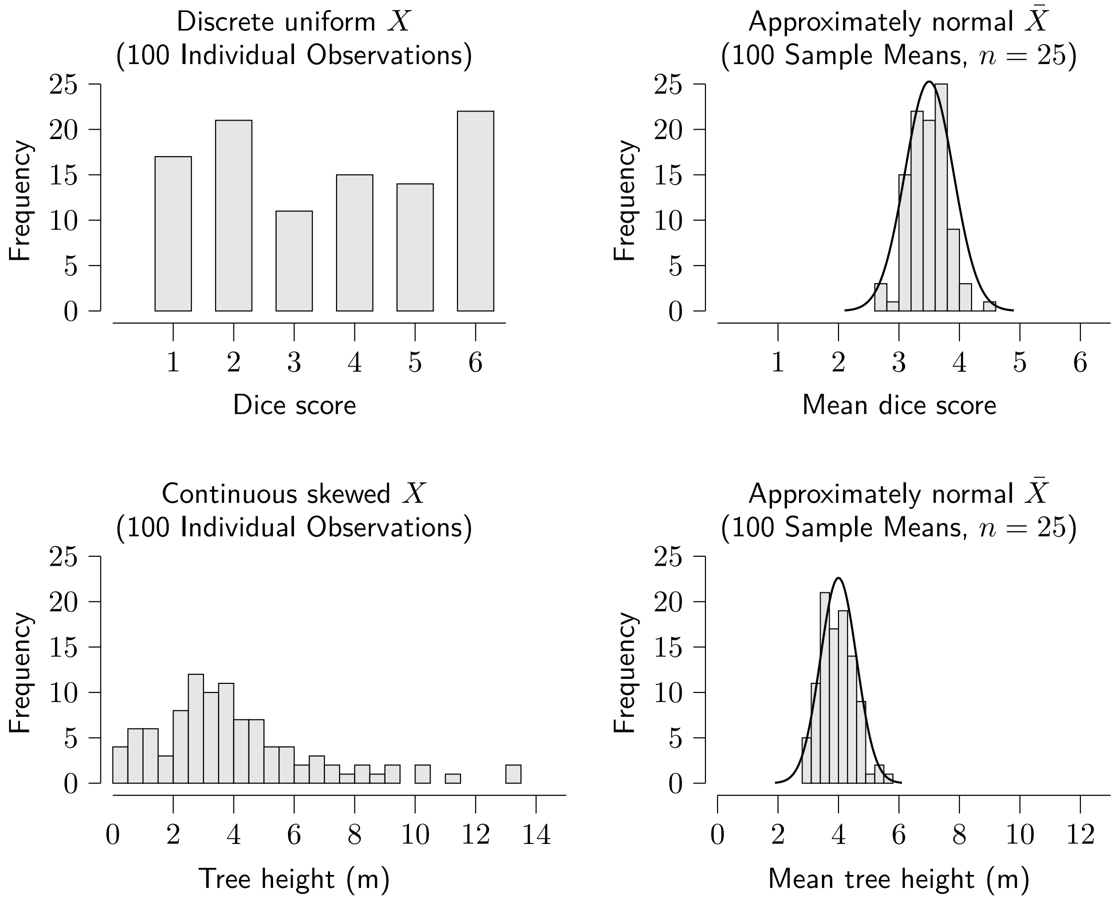
9.2 The Central Limit Theorem

The Central Limit Theorem (CLT) is one of the most important and useful concepts in statistics. It provides insight into the widespread occurrence of the normal distribution and, crucially for statisticians, allows the normal distribution to be used for statistical inference even when the underlying data is *not* normally distributed. The CLT says:

When the underlying population is *not* normally distributed,
the distribution of the **sample mean** is **approximately normal**
for sufficiently large sample sizes.

This approximation relies on independent observations (typically ensured through random sampling) and becomes more accurate as the sample size increases. A *"sufficiently large sample"* in this course is **greater than 20** ($n > 20$); however, some statisticians may use a value of 30, whilst in practice the nature of the underlying distribution may allow for smaller samples to be sufficient, or require larger samples for the distribution of the sample mean to be sufficiently close to normal.

Proof of the CLT is beyond the scope of this course (and this textbook), but statistical software and/or experimentation can help give a sense of what this looks like in practice. The simulations below show random individual observations from distributions on the left, with corresponding distributions of sample means to the right. Every random sample in both cases was of size 25.



The **approximate** distribution of the sample mean can be stated provided the expectation and variance of the underlying distribution are known, or can be determined. The notation '$\approx$' is used to highlight that this is an *approximate* distribution.

The Central Limit Theorem:

If $\mathsf{E}(X) = \mu$ and $\mathsf{V}(X) = \sigma^2$ then $\overline{X} \approx \mathsf{N}\left(\mu, \frac{\sigma^2}{n}\right)$, provided $n > 20$.

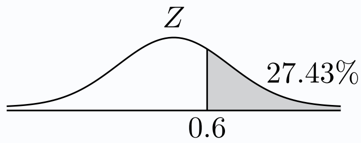
Example

Problem: The mean mass of ceramic bowls produced by a potter is 400g and the standard deviation is 20g, and they can be ordered singly or in bulk. Bulk orders contain 36 bowls and, when such an order is placed, the potter selects the 36 bowls at random to be boxed up and shipped. Letting the random variable X represent the weight of a bowl, and $\overline{X}$ the mean bowl mass in a bulk order:

a. State the distribution of $\overline{X}$.

b. Calculate the probability that the mean bowl mass in a bulk order exceeds 402g.

Solution:



a. $\overline{X} \approx \mathrm{N}\left(400, \frac{20^2}{36}\right)$

b. $\mathrm{P}(\overline{X} > 402) = \mathrm{P}\left(Z > \dfrac{402-400}{\frac{20}{\sqrt{36}}}\right) = \mathrm{P}(Z > 0.6) = 0.2743$

Exercise 9.2

1. For each sample from the random variable given, state the distribution of the sample mean where possible:
   (a) A sample of size 40 from X where $\mathrm{E}(X) = 2.3$ and $\mathrm{SD}(X) = 5$.
   (b) A sample of size 25 from normally distributed Y where $\mathrm{E}(Y) = 83$ and $\mathrm{V}(Y) = 9$.
   (c) A sample of size 7 from normally distributed D where $\mathrm{E}(D) = 126$ and $\mathrm{V}(D) = 10$.
   (d) A sample of size 12 from W where $\mathrm{E}(W) = 14$ and $\mathrm{SD}(W) = 2$.

2. The discrete random variable A has a mean of 18 and a standard deviation of 5. Calculate the probability that a sample of size 30 taken from A results in a mean less than 20.

3. The continuous random variable B has a mean of 248 and a variance of 16. Determine the probability of the mean of a sample of size 24 being greater than 250.

4. The mean height of the sunflowers grown in a field is 1.05m, with a standard deviation of 0.25m. The grower measures the height of a random sample of 40 sunflowers. What is the probability that the mean height of the sample of sunflowers is less than a metre?

5. Eggs classified as *medium* by a farm have a mean mass of 78g and a standard deviation of 5g. Calculate the probability that, for a box of 24 randomly selected eggs, the mean egg mass is recorded as 78g when given to the nearest gram.

6. When *"40 litres"* of petrol is dispensed by a fuel pump design used by a supermarket, the mean actual amount dispensed is 40.1 litres with a standard deviation of 0.35 litres.
   (a) Explain why the known information is not sufficient to calculate the probability that a single *"40 litre"* fill actual gives less than 40 litres.

   As part of the supermarket's monitoring of the accuracy of its pumps, it takes 32 fills, each of size *"40 litres"*, from randomly chosen pumps at its stores nationwide.

   (b) Calculate the probability that the mean obtained from this random sample is less than 40 litres.
   (c) Explain why the Central Limit Theorem was required in order to obtain the answer for part (b).

Review Exercise

1. Given a normally distributed random variable X, with a mean of 15 and a standard deviation of 3, from which a sample of eight independent observations are taken:
   (a) State the distribution of the sample mean, $\overline{X}$.
   (b) Calculate P$(\overline{X} < 17)$.

2. A fair, tetrahedral (four-sided) die numbered 1 to 4 is rolled 25 times, and the score obtained for each roll recorded. Find the probability that the mean score for the 25 rolls is greater than 3.

3. At the start of a marathon, each runner is given a bottle containing 1 litre of water. The mean volume of water consumed by a runner during the marathon is 600 millilitres, whilst the standard deviation is 70 millilitres. At the end of the marathon, 40 runners are selected at random, and the volume of water consumed by each is measured. The mean volume of water consumed by the sample of 40 runners is then calculated.
   (a) In context, explain what the Central Limit Theorem says about the sample mean calculated.
   (b) State the distribution of the sample mean.
   (c) Find the probability that the sample mean calculated is between 610 millilitres and 620 millilitres.

4. The mass of mince pies sold individually at the counter at a bakery is normally distributed, with a mean of 60 grams and a standard deviation of 2 grams.
   (a) A customer asks for a mince pie, and it is selected at random by the bakery staff. Let random variable X represent the mass of the mince pie chosen.
      (i) State the distribution of X
      (ii) Find the probability that the mince pie chosen has a mass of less than 59 grams.
   (b) Another customer asks for eight mince pies, and they are selected at random by the bakery staff. Let $\overline{X}$ represent the mean mass for the sample of eight mince pies chosen.
      (i) State the distribution of $\overline{X}$.
      (ii) Find the probability that the sample mean is less than 59 grams.
   (c) The bakery changes its process, and now when a customer orders a number of mince pies the bakery staff places them on a set of weighing scales, one by one. Explain why this change is likely to mean the distribution of $\overline{X}$ is no longer as it previously was.

5. Let discrete random variable X represent the number of Video Assistant Referee (VAR) reviews during the course of a game of football, known to follow a Poisson distribution with a mean of 4.5 reviews per game.
   (a) State the distribution of X.
   (b) Calculate the probability that a football game has no more than two VAR reviews.

   Thirty games are randomly chosen to be studied in greater detail by data analysts. They count the number of VAR reviews in each game, and calculate the mean number of reviews for this sample of thirty games.
   (c) State the distribution of $\overline{X}$, the calculated sample mean.
   (d) Find the probability that the sample mean calculated is greater than five.

10

An Introduction to Hypothesis Testing

In brief, hypothesis testing is an approach to assessing the statistical evidence for a claim through the analysis of data from a random sample. The claim will typically relate to some aspect of the distribution of a population of interest, often a *parameter* such as the *population mean*, μ, and the foundation of hypothesis testing lies in *probability*.

The Logic of Hypothesis Testing

Consider the following scenarios:

- A coin is tossed 50 times, and it lands showing tails 47 of those times.
- A cubical die is tossed 80 times, and a six is obtained only once.
- A random number generator gives the same number eight times in a row.

In each of these scenarios, there are two possible conclusions:

The coin, die and number generator are all fair, and a very unlikely event happened by random chance...

or

The coin is biased, the die rigged, the random number generator flawed...

An *objective* process is required that allows a decision to be made between these two possibilities. This process, called **hypothesis testing**, will form a substantial part of the remainder of the course.

10.1 The Null Hypothesis and the Alternative Hypothesis

Most statistical studies begin with a claim, which is called the **alternative hypothesis**, or H_1. Initially in this chapter, these will consist of a claim about the value of the *population mean*, μ. A conservative stance in opposition to the alternative hypothesis will be used as the basis for all calculations, assuming that the claim is *not true*. This is called the **null hypothesis**, or H_0. These may be thought of *informally* as:

The null hypothesis	H_0	*"Nothing to see here..."*
The alternative hypothesis	H_1	*"Something is happening here..."*

It is *essential* that both hypotheses are formed *before* observing the data. It should also first be decided whether evidence is being sought for the true value of the population mean being *less than* a specified value, *greater than* a specified value, or simply *different from* a specified value. When the alternative hypothesis is only interested in a *single direction*, the test being performed is described as ***one-tailed***. When the alternative hypothesis is interested in a change in *either direction* from the specified value, it is referred to as a ***two-tailed*** test.

Example

Problem: State the null and alternative hypotheses for each of the following claims.

(a) The mean volume of liquid in espressos sold by a café is less than the 30ml stated on their menu.

(b) The mean mass of red squirrels in a woodland population is different to the previously measured 240g.

Solution:

(a) $H_0 : \mu = 30$ One-tailed.
$H_1 : \mu < 30$

(b) $H_0 : \mu = 240$ Two-tailed.
$H_1 : \mu \neq 240$

Exercise 10.1

For each proposed hypothesis test, state the **null hypothesis** and the **alternative hypothesis**.

1. A telecommunications provider wishes to test whether the mean amount of data used by its customers is more than 8.6 gigabytes per month.

2. A large school plans to test whether the mean amount its teachers spend on photocopying is different from the £12 per year they currently budget for.

3. A farmer wants to test whether the mean mass of the newborn lambs from his flock is less than 4.2 kilograms.

4. A sports statistician is planning a hypothesis test to assess the evidence that the mean actual playing time during a game of football has changed from 58 minutes.

5. A kettle manufacturer wishes to test whether the mean time for its kettles to boil one litre of water is more than the 65 seconds stated on the packaging.

6. A teacher is monitoring the time a sample of teachers from across Scotland spend marking writing reports during a school year, to assess whether there is evidence that an education authority's claim that the mean is 10 hours is valid.

7. A music fan measures the tempo (in beats per minute) of a sample of songs on a streaming platform to test whether a suggested mean of 160 beats per minute is too low.

8. The mean volume of petrol a driver uses driving from Glasgow to Blackburn is 15.7 litres. The driver wishes to test whether the mean volume of petrol used has changed.

Evidence to Suggest

The *true value* of a population parameter cannot be known by studying a sample. This means that, without a census of the entire population, a statistician will never say that they have *proved* or *disproved* a claim, and statistical conclusions are always expressed in terms of evidence, rather than certainty. A suitable conclusion should make a judgement on whether there is **sufficient evidence to suggest** or **insufficient evidence to suggest** a particular claim (represented by H_1) is true.

The Significance Level and the *p*-value

Once the null and alternative hypotheses have been stated, a conservative viewpoint is taken - it is *assumed that the null hypothesis is true*. Only an observed value that is then *very unlikely to have occurred by random chance* will lead to the decision to **reject the null hypothesis** in favour of the alternative hypothesis. The decision to *reject or not reject* is made with respect to the *null hypothesis*, whilst the *evidence (to suggest)*, or lack of, relates to the *alternative hypothesis*.

The probability of observing a result at least as extreme as the one obtained, assuming the null hypothesis is true and in the direction specified by the alternative hypothesis, is called the **p-value**. A decision should be made in advance of collecting the data regarding the **significance level**, α (*"alpha"*). A significance level of 5% ($\alpha = 0.05$) is commonand is often used when no other significance level has been specified. It can be interpreted as the level of 'unlikeliness' that it would take in order to abandon H_0 in favour of H_1.

When the p-value is less than α this is referred to as a *statistically-significant* result, and smaller p-values indicate stronger evidence against the null hypothesis.

The decision rule for the p-value and the significance level α:

If p-value $\leqslant \alpha$ then **reject** H_0 and conclude that
there is *sufficient evidence to suggest* that the alternative hypothesis is true.

If p-value $> \alpha$ then **do not reject** H_0 and conclude that
there is *insufficient evidence to suggest* that the alternative hypothesis is true.

Failing to reject the null hypothesis does not prove that it is true; it simply indicates that there is insufficient evidence against it. To avoid suggesting that evidence will ever be found for the null hypothesis, the phrase 'accept H_0' should **not be used**.

The level of significance influences the long-run risk of making a *Type I Error* - incorrectly rejecting the null hypothesis when it is in fact true. Failing to reject the null hypothesis when it is in fact false is called a *Type II Error*.

	Do Not Reject H_0	Reject H_0
H_0 is true	*Correct decision*	*Type I error*
H_1 is true	*Type II error*	*Correct decision*

When performing *any* significance test the following procedure is generally performed:

- Determine **which hypothesis test** is appropriate for the data and the focus of the study.
- Decide on a suitable **significance level** (α).
- State the **null hypothesis** (H_0) and the **alternative hypothesis** (H_1).
- Calculate the **test statistic** under the assumption that *the null hypothesis is true*.
- Determine the **critical value**, or calculate the p-value.
- Decide whether to **reject or not reject** H_0.
- Write a suitable statistical **conclusion**.

A conclusion should always be written in context, referring to the original claim.

10.2 One-Sample *z*-Test for the Population Mean using a *p*-value

The *one-sample z-test for the population mean* is a hypothesis test used when a sample of data is taken from a population so that a claim about the *true value of the population mean*, μ, can be assessed. Since z-tests rely on assumptions about population parameters and the normal distribution, they are classed as **parametric tests**.

Conditions for the one-sample z-test for the population mean:

- The *sampling distribution of the sample mean* is **normal** (or approximately normal)
- The *population standard deviation*, σ, is **known**.
- The observations are **independent** (typically assured through *random sampling*).

If one or more of these conditions is not known to be satisfied, then it is required to make an **assumption** that it *is* satisfied for the test to be valid. In any statistical study assumptions that have been made should be recognised and consideration should be given as to whether they could be justified.

Note that when the sample size is greater than 20 **the CLT can be invoked**, so the sample mean is at least *approximately* normally distributed regardless of the distribution of the underlying population. Also, for large sample sizes, the sample standard deviation, s, may be used as a reliable estimate of the population standard deviation, σ.

Probabilities relating to observed sample means have already been covered in Chapter 9, and the following examples will demonstrate how this leads to a calculated p-value for a hypothesis test based on a sample mean. Note below that, with a sample mean of 257 observed, and an alternative hypothesis that the true population mean is *less than* 260, the p-value is the probability of observing a sample mean *as extreme or more extreme* than 257. In the context of a claim that $\mu < 260$, the p-value is therefore $\mathrm{P}\left(\bar{X} < 257\right)$.

Example

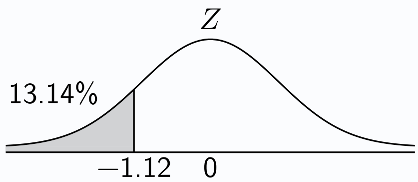
Problem: The designers of an electric car state that the distance that can be covered on a full charge is normally distributed, with a mean of 260 miles and a standard deviation of 10 miles. A car magazine randomly selects 14 of them from showrooms to be tested, with a sample mean of 257 miles obtained. Stating one assumption required, perform a hypothesis test to assess, at the 1% level of significance, the claim by the car magazine that the mean distance is actually less than the designers' claim.

Solution: $\mu = 260$, $\bar{x} = 257$, $n = 14$, $\sigma = 10$

Assume that the standard deviation of the distance covered on a full charge remains 10 miles.

$\mathrm{H}_0: \quad \mu = 260$
$\mathrm{H}_1: \quad \mu < 260$

Under H_0:



$$P(\bar{X} < 257) = P\left(Z < \frac{257 - 260}{\frac{10}{\sqrt{14}}}\right) = P(Z < -1.12) = 0.1314$$

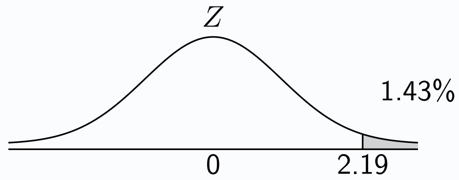
Since $0.1314 > 0.01$, do not reject H_0 at the 1% level of significance. There is insufficient evidence to suggest that the mean distance the car can cover on a full charge is less than the stated 260 miles, and so no evidence to support the car magazine's claim.

Example

Problem: The mass of crisps dispensed into a sharing-sized bag by a machine is normally distributed with a mean of 150g and a standard deviation of 2g. After being serviced, a random sample of 30 bags gives a mean of 150.8g. Test at the 5% level of significance the claim that the machine is now dispensing too much.

Solution: $\mu = 150$, $\bar{x} = 150.8$, $n = 30$, $\sigma = 2$

$H_0: \quad \mu = 150$
$H_1: \quad \mu > 150$



Under H_0:

$$P(\bar{X} > 150.8) = P\left(Z > \frac{150.8 - 150}{\frac{2}{\sqrt{30}}}\right) = P(Z > 2.19) = 0.0143$$

Since $0.0143 < 0.05$, reject H_0 at the 5% level of significance. There is sufficient evidence to suggest that the mean mass of crisps dispensed per bag is more than 150g.

For a *two-tailed* test, the probability of such an extreme result *in either direction* must be considered.

For the example below, this equates to the probability of observing a sample mean with a difference of 1 gram or more from 82. Given a symmetrical distribution of the test statistic, such as with a z-test, a p-value for a two-tailed test is *double* the initial probability obtained by only considering probability in *one* tail.

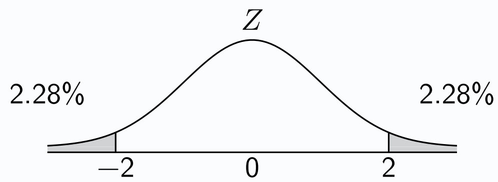
Example

Problem: A tomato grower has been monitoring the mass of the tomatoes he grows, with the mean mass of a tomato shown to be 82 grams. Following a change in the soil composition in which they are grown, the grower wants to see whether the mean mass of their tomatoes has been affected. A sample of 36 tomatoes gives a mean mass of 83 grams and a standard deviation of 3 grams. It is assumed that the sample standard deviation is a reliable estimate of the population standard deviation. Perform a hypothesis test to determine whether there is evidence to suggest that the mean mass of their tomatoes has changed, stating one further assumption required and justifying the validity of the test with reference to the sample size.

Solution: $\mu = 82$, $\bar{x} = 83$, $n = 36$, $\sigma \approx 3$

Assume that the sample of tomatoes was random. Since $n > 20$, the distribution of the sample mean is approximately normally distributed regardless of the underlying distribution of the mass of a tomato.

$H_0: \quad \mu = 82$
$H_1: \quad \mu \neq 82$



Under H_0:

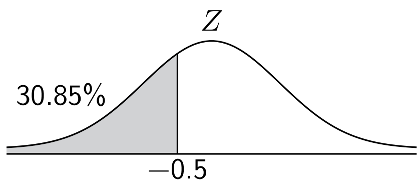
$$P(\bar{X} > 83) = P\left(Z > \frac{83 - 82}{\frac{3}{\sqrt{36}}}\right) = P(Z > 2) = 0.0228$$

p-value $= 2 \times 0.0228 = 0.0456$ and $\alpha = 0.05$

Since $0.0456 < 0.05$, reject H_0 at the 5% level of significance. There is sufficient evidence to suggest that the mean mass of the grower's tomatoes has changed from 82 grams.

Exercise 10.2

1. A practice question is below, followed by a student's **flawed** solution, featuring a number of issues.

> **Practice question:** The mass of dried tea leaves contained in each bag sold by tea company *Brewed Awakenings* is known to be normally distributed with a standard deviation of 0.4 grams, and the mean amount contained in each bag should be 2.5 grams. It is suspected that the company's new machine is putting too little dried tea into each bag. A random sample of 10 bags have their contents measured, and the mean mass obtained from this sample is 2.3 grams.
>
> Perform a z-test to assess whether the new machine is putting too little dried tea into the bags, at the 5% level of significance.
>
> **Student's flawed solution:**
>
> $\text{H}_0:$ mean $= 2.5$
> $\text{H}_1:$ mean < 2.5
>
> 
>
>
> Under H_0:
>
> $\text{P}(\bar{X} < 2.3) = 0.3085$
>
> p-value $= 0.3085$
>
> Since $0.3085 > 0.05$, accept H_0 at the 5% level of significance. There is insufficient evidence to suggest that the mass of dried tea leaves in a bag is less than 2.5 grams.

The hypotheses given by the student lack the required clarity.

(a) State suitable hypotheses which should be used instead.

The p-value of 0.3085 calculated by the student is correct, but the student has not shown sufficient working.

(b) Show a minimum of two additional steps of working, which lead to the value of 0.3085.

Modern statisticians (and Advanced Higher Statistics students) must avoid using the outdated phrase '*accept* H_0', since it suggests the sample provides evidence *for* the null hypothesis.

(c) State an appropriate phrase to use instead of '*accept* H_0'.

The final sentence in the student's conclusion is missing an important word.

(d) State the essential word which is missing.

2. The mass of ground coffee contained in each bag sold by coffee company *Déjà Brew* is known to be normally distributed with a standard deviation of 7 grams, and the mean amount contained in each bag should be 240 grams. It is suspected that the company's new machine is putting too much ground coffee into each bag. A random sample of 18 bags have their contents measured, and the mean mass obtained from this sample is 242 grams.

   Perform a z-test to assess whether the new machine is putting too much ground coffee into the bags, at the 10% level of significance.

3. The mass of cocoa powder contained in each bag sold by cocoa company *Froth It Like It's Hot* is known to be normally distributed with a standard deviation of 11 grams, and the mean amount contained in each bag should be 180 grams. It is suspected that the company's new machine is not putting the right amount of cocoa powder into each bag. A random sample of 15 bags have their contents measured, and the mean mass obtained from this sample is 188 grams.

   Perform a z-test to assess whether the new machine is not putting the right amount of cocoa powder into the bags, at the 1% level of significance.

4. The length of cookery videos uploaded to a video sharing platform has been historically found to be normally distributed, with a mean of 11.7 minutes and a standard deviation of 2.3 minutes. A random sample of 30 videos uploaded this year tagged as 'cookery' on the platform gives a mean of 10.8 minutes. Assuming that the standard deviation of video length remains unchanged, test at the 5% level of significance the claim that the mean length of cookery videos on the platform is now less than 11.7 minutes.

5. The time taken for a teacher to mark an exam paper is normally distributed with a mean of 6.3 minutes and a standard deviation of 0.8 minutes. After more detailed marking instructions are issued, a random sample of 25 papers results in a mean marking time of 6.5 minutes per paper. Assuming that the standard deviation of time taken remains unchanged, assess at the 1% significant level the claim that the mean marking time has increased since the new marking instructions were introduced, using a hypothesis test.

6. The figure often quoted as the mean time spent with the ball being in play during a game of football is 58 minutes, with the standard deviation being 4 minutes, and it is assumed that the time for which the ball is in play is normally distributed. A sports statistician is planning a hypothesis test to assess the evidence that the mean time during which the ball is in play has changed from 58 minutes. A random sample of 15 games gives a mean time in play of 61 minutes.

   A z-test is to be used to assess the data.

   (a) State one assumption required for the valid use of a z-test.

   (b) State suitable hypotheses for the test.

   (c) Calculate the p-value for the test.

   (d) Use the answer from part (c) to conclude the test at the 1% level of significance.

7. A farmer wants to test whether the mean mass of the newborn lambs from his flock is less than 4.2 kilograms. The mass of his newborn lambs is normally distributed, and the standard deviation has been previously established to be 0.3 kilograms. A random sample of 8 newborn lambs are weighed, and the sample mean is found to be 4.0 kilograms.

   (a) State one assumption required for the valid use of a z-test.

   (b) Perform a hypothesis test to assess the evidence at the 1% level of significance that the mean mass of his newborn lambs is less than 4.2 kilograms.

8. Last year, the value of houses in an area were normally distributed, with a mean of £184 000 and a standard deviation of £7000. A random sample of 15 of the houses this year gives a mean value of £187 000. Perform a hypothesis test at the 10% level of significance to assess the claim that the mean house price in the area has increased.

9. The heights of mature oak trees in some woodland are normally distributed with a standard deviation of 1.5 metres, and two years ago the mean height of the trees was found to be 21 metres. A random sample of 30 of these trees taken this year yields a sample mean of 20.4 metres.

   (a) Test at the 10% level of significance the claim that the mean height of the trees has changed.

   (b) Explain why the sample size meant that a parametric test could have been used in part (a) even if it were not known that the heights of the mature oak trees are normally distributed.

10.3 One-Sample *z*-Test for the Population Mean using a Test Statistic

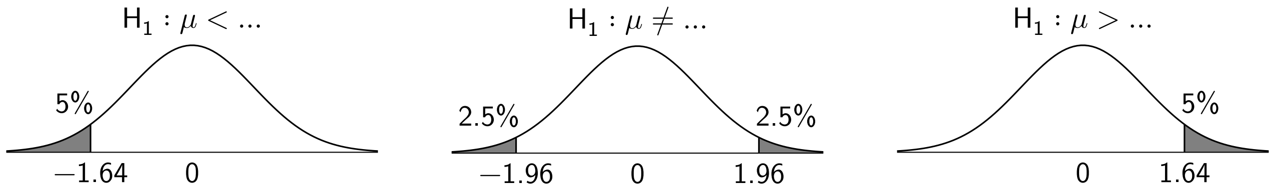
Obtaining a p-value in the previous section required calculating the probability of the observed sample mean, $\bar{x}$, by first using the *z-transformation for a sample mean* to obtain what is called a *z-score*. Instead of concluding a z-test by comparing a p-value to the significance level, an alternative approach is to compare the z-score, referred to as the **test statistic**, to predetermined cut-off values, called **critical values**.

For a one-sample z-test for the population mean:

$$\text{Test statistic: } z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

A *z-test statistic* of 0 occurs when the sample mean of the observed data is equal to the value for μ claimed by the null hypothesis. The further the test statistic is from 0, the more extreme the observed data is.

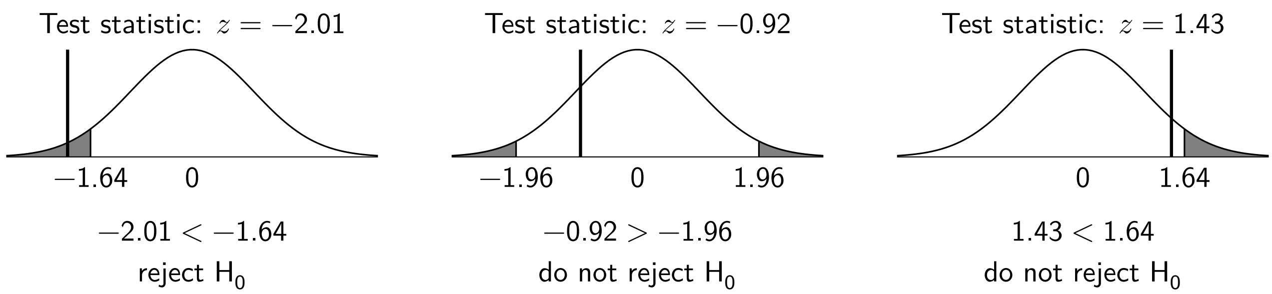
Critical values can be found by first considering *critical regions*. These are the most extreme regions of the sampling distribution of the test statistic; if the calculated test statistic lies within these regions, the null hypothesis is rejected. Critical values are the *boundaries* of these critical regions. Below are examples of critical regions (shown shaded), and critical values (shown below the axis), for a z test statistic with a 5% level of significance.



For a one-tailed test, the most extreme 5% of the distribution in the tail of interest is considered.

For a two-tailed test, the 5% least-likely observed values for the test statistic *in either direction* are considered. The significance level is split between 2.5% in the upper tail and 2.5% in the lower tail. The symmetry of the distribution means that the values to the left of the mean are the *negative* of the equivalent values on the right.

If the test statistic lies **within the critical region** then H_0 is **rejected**, and so there is *sufficient* evidence to suggest that the alternative hypothesis is true. If the test statistic lies **outwith the critical region** then H_0 is **not rejected**, and so there is *insufficient* evidence to suggest that the alternative hypothesis is true.



This approach is equivalent to the p-value method and will always lead to the same conclusion.

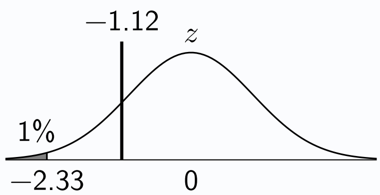
Note that the following examples approach problems already covered in the examples for the previous section, which used a p-value approach, and lead to identical conclusions to the hypothesis tests conducted.

Example

Problem: The designers of an electric car state that the distance that can be covered on a full charge is normally distributed, with a mean of 260 miles and a standard deviation of 10 miles. A car magazine randomly selects 14 of them from showrooms to be tested, with a sample mean of 257 miles obtained. Perform a hypothesis test to assess, at the 1% level of significance, the claim by the car magazine that the mean distance is actually less than the designers' claim.

Solution: $\mu = 260$, $\bar{x} = 257$, $n = 14$, $\sigma = 10$

$H_0: \quad \mu = 260$
$H_1: \quad \mu < 260$



Test statistic: $z = \dfrac{257 - 260}{\dfrac{10}{\sqrt{14}}} = -1.12$

Critical value: $z = -2.33$

Since $-1.12 > -2.33$, do not reject H_0 at the 1% level of significance. There is insufficient evidence to suggest that the mean distance the car can cover on a full charge is less than the stated 260 miles, and so no evidence to support the car magazine's claim.

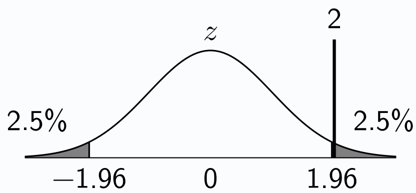
When choosing a critical value for a two-tailed test, in which there are two critical regions, the critical value with the same sign (positive or negative) as the observed test statistic should be used in the conclusion.

Example

Problem: A tomato grower has been monitoring the mass of the tomatoes he grows, with the mean mass of a tomato shown to be 82 grams. Following a change in the soil composition in which they are grown, the grower wants to see whether the mean mass of their tomatoes has been affected. A sample of 36 tomatoes gives a mean mass of 83 grams and a standard deviation of 3 grams. It is assumed that the sample standard deviation is a reliable estimate of the population standard deviation. Perform a hypothesis test to determine whether there is evidence to suggest that the mean mass of their tomatoes has changed, stating one further assumption required and justifying the validity of the test with reference to the sample size.

Solution: $\mu = 82$, $\bar{x} = 83$, $n = 36$, $\sigma \approx 3$

$H_0: \quad \mu = 82$
$H_1: \quad \mu \neq 82$



Test statistic: $z = \dfrac{83 - 82}{\dfrac{3}{\sqrt{36}}} = 2$

Critical value: $z = 1.96$

Since $2 > 1.96$, reject H_0 at the 5% level of significance. There is sufficient evidence to suggest that the mean mass of the grower's tomatoes has changed from 82 grams.

Exercise 10.3

1. A packing process packages ground coffee into bags. The mass of each bag is normally distributed, with standard deviation 15g. To comply with the technical standards stipulated for the process, the mean mass of coffee per bag should be 500g. After a recent adjustment of some of the machinery involved in the process, a random sample of 25 bags is taken and the sample mean is found to be 512g.

   Test the claim that there has been a change in the mean mass of coffee per bag at the 5% level of significance.

2. The lengths of metal bars produced by a particular machine in a factory are normally distributed with mean length 420cm and standard deviation 12cm. The machine is serviced, after which a random sample of 80 bars gives a mean length of 423cm.

   Perform a parametric test to assess whether there is evidence at the 1% level of significance of an increase in the mean length of the bars produced, assuming the standard deviation of bar length remains the same.

3. In 1980 the mass of a herring in the North Sea was found to be normally distributed with a mean of 160g and a standard deviation of 10g. In 2020 a sample of 15 herrings yielded a mean mass of 158g.

   Assuming that the standard deviation is unchanged, test at the 1% level of significance the hypothesis that the mean mass has decreased since 1980.

4. The mass of a pack of salt is normally distributed with a mean of 200g and a standard deviation of 12g. After servicing, it is suspected that the salt dispensing machine is not dispensing the same mean amount of salt. A sample of 23 bags yields a mean mass of 205g per pack.

   Stating two assumptions required, test the hypothesis that the mean amount of salt dispensed by the machine is no longer 200g.

5. Last season, the mean distance a player ran per game by the players of a football team was 7.1km. A random sample, of size 26, of distances run by players so far this season gave a sample mean distance covered of 7.3km and a standard deviation of 0.7km.

   It is assumed that the sample size is sufficiently large that the sample standard deviation is a reliable estimate of the population standard deviation.

   (a) Perform a z-test to assess the evidence, at the 5% level of significance, that the mean distance players are covering in a game this season has changed.

   (b) Explain why the sample size helped enable the use of a z-test for this data.

6. A popular type of loaf sold by a regional chain of bakeries has been previously observed to have a mean mass of 500g, with a standard deviation of 20g. Following a change in the recipe used for the loaves, a random sample of 25 loaves yielded a mean mass of 483g.

   (a) State the condition for a sample size which implies that a sample mean will be approximately normally distributed even whilst the underlying distribution is not normally distributed.

   (b) State one assumption required for the valid use of a z-test for the population mean loaf mass.

   (c) Test at the 5% significance level the claim that the mean mass of a loaf has decreased.

7. The mass of a pine marten in Scotland is historically known to be normally distributed, with a mean of 1.31kg and a standard deviation of 0.23kg. A sample of 22 pine martens captured, weighed and released in Scotland in 2023 gives a mean mass of 1.38kg. Stating two assumptions required, test at the 5% level of significance the claim that the mean mass of a pine marten in Scotland has increased.

8. The time taken by runners to complete a 5km run in a weekly race in a local park is normally distributed with a mean time of 26.3 minutes and a standard deviation of 1.4 minutes. Following some development works completed in the park, the route is unchanged but some of the running surfaces have changed. A random sample of 40 runners completing the course after the development work gives a mean of 25.9 minutes.

   Stating one assumption required, test the claim at the 0.1% significance level that the mean run time is no longer 26.3 minutes.

9. A telecommunications provider wishes to test whether the mean amount of data used by its customer is more than 8.6 gigabytes per month. The amount of data used by a customer is known to be normally distributed with a standard deviation of 1.7 gigabytes. A random sample of 32 customers gives a mean of 9.1 gigabytes.

   Perform a hypothesis test at the 5% level of significance to assess the evidence that the mean amount of data used by a customer is greater than 8.6 gigabytes.

10. A large school plans to test whether the mean amount its teachers spend on photocopying is different from the £12 per year they currently budget for. A sample of 23 teachers gives a mean of £10.90 and a standard deviation of £2.30.

    Assume that the sample standard deviation of spend per teacher is a reliable estimate of the population standard deviation.

    (a) State a further assumption required for a z-test to be performed.

    (b) Perform a hypothesis test at the 10% level of significance to assess the evidence that the mean spend per teacher is different from £12.

11. A kettle manufacturer wishes to test whether the mean time for its kettles to boil one litre of water is more than the one minute and 5 seconds stated on the packaging. The standard deviation in the boiling time is 4 seconds, and the time taken to boil is normally distributed.

    A sample of 32 kettles gives a mean time of one minute and 7 seconds.

    Assuming that the standard deviation remains unchanged, perform a hypothesis test at the 5% level of significance to assess the evidence that the mean time taken to boil one litre is more than the packaging states.

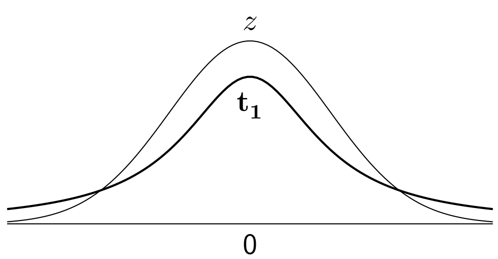
12. The volume of petrol a driver uses driving from Glasgow to Blackburn is normally distributed, with previous studies showing the mean to be 15.7 litres and the standard deviation to be 0.3 litres. The driver believes their new driving style may have caused the mean volume of petrol used to change. Subsequently, recording the fuel consumption for 5 journeys gives a mean volume of petrol for a journey of 15.4 litres.

    (a) State two assumptions required for the valid use of a z-test.

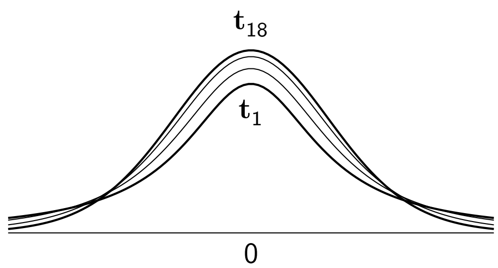
    (b) Perform a z-test at the 1% level of significance to assess whether there is evidence to support their claim.

10.4 One-Sample *t*-Test for the Population Mean

In the early 20th century William Sealy Gosset, statistician and Head Experimental Brewer at the Guinness factory in Dublin, described the distribution of the sample mean when the **population standard deviation is unknown**. This was necessary since he was working with **small sample sizes**, for which the sample standard deviation could not be used as a reliable estimator for the population standard deviation. Since Guinness did not want rival companies to know that they were employing a statistician to improve their brewing, he published his work under the pen name *Student*, and the distribution he described became known as **Student's *t* -distribution**, or simply *the t-distribution*.



t-distribution, t_1, and normal distribution, z



t-distributions with $\nu = 1, 2, 5, 18$

Not knowing the population standard deviation, σ, introduces additional uncertainty about the distribution of the sample mean, with the **sample standard deviation**, s, used instead. This results in a t-distribution that resembles a normal distribution but with *fatter tails*. The t-distribution takes one parameter called **degrees of freedom**, denoted using the Greek letter ν, which is the number of values that are free to vary once the sample mean has been fixed. For the one-sample t-test for population mean, the degrees of freedom can be calculated for a sample size n as $\nu = n - 1$.

As the sample size increases, and hence ν increases, the t-distribution approaches the normal distribution. This is why the Data Booklet provides a set of critical values for use when $\nu > 40$, which match critical values for z-tests.

The one-sample t-test for population mean is a **parametric test** that should be used in place of the equivalent z-test when the population standard deviation, σ, is not known, and the sample standard deviation, s, is to be used instead.

Conditions for the one-sample t-test for the population mean:

- The *underlying population* is **normally distributed**.
- The observations are **independent** (typically assured through *random sampling*).

The t-test statistic is identical to that of the equivalent z-test, except the use of the sample standard deviation, s, instead of the population standard deviation, σ.

For a one-sample t-test for the population mean:

Test statistic: $t = \dfrac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$ with $\nu = n - 1$ degrees of freedom

Critical values for the t-test can be found on *page 13* of the Data Booklet, referring to the significance level, whether the test is one-tailed or two-tailed and the degrees of freedom; constructing a sketch of the distribution is often useful.

Example

Problem: A supermarket sells sharing-sized bags of a particular brand of crisps. A consumer watchdog is asked to investigate a claim that the mean mass of crisps contained in a bag is less than the stated contents of 150 grams. A random sample of bags gives the following results, in grams:

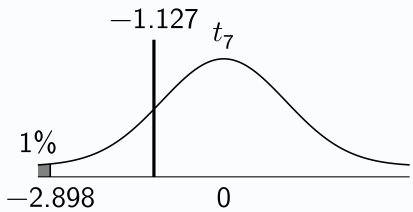
148 151 149 152 145 150 146 151

Stating a required assumption, perform a parametric test to assess the claim at the $\alpha = 0.01$ level.

Solution: $\mu = 150$, $\bar{x} = 149$, $n = 8$, $s = 2.51$ (calculating $\bar{x}$ and s from the sample)

Assume that the mass of crisps in a bag is normally distributed.

$\text{H}_0: \quad \mu = 150$
$\text{H}_1: \quad \mu < 150$



Test statistic: $t = \dfrac{149 - 150}{\frac{2.51}{\sqrt{8}}} = -1.127$

Degrees of freedom: $\nu = n - 1 = 8 - 1 = 7$

Critical value: $t = -2.898$

Since $-1.127 > -2.898$, do not reject H_0 at the 1% level of significance. There is insufficient evidence to suggest that the mean mass of crisps in a bag is less than the stated value of 150g.

As a reminder, assumptions and other comments should be written carefully and with reference to context wherever it is possible to. For example:

Missing context: "Assume the data is normally distributed."
With context: "Assume that the **mass of crisps in a bag** is normally distributed."

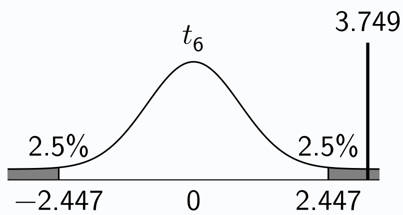
Example

Problem: Marks scored in an exam are known to be approximately normally distributed, and historically the mean mark achieved by students is 72. After some minor tweaks to the exam, marks gained by a random sample of 7 students generated the summary data below. Perform a hypothesis test at the 5% level of significance to determine whether there is evidence that the difficulty of the exam has changed for students.

$\Sigma x = 571 \qquad \Sigma x^2 = 46851 \qquad n = 7$

Solution: $\mu = 72$, $\bar{x} = 81.57$, $n = 7$, $s = 6.754$

$\text{H}_0: \quad \mu = 72$
$\text{H}_1: \quad \mu \neq 72$



Test statistic: $t = \dfrac{81.57 - 72}{\frac{6.754}{\sqrt{7}}} = 3.749$

Degrees of freedom: $\nu = n - 1 = 7 - 1 = 6$

Critical value: $t = 2.447$

Since $3.749 > 2.447$, reject H_0 at the 5% level of significance. There is sufficient evidence to suggest that the mean mark achieved by students is no longer 72, and that the difficulty of the exam has changed.

Exercise 10.4

1. The amount of sugar contained in frozen pizzas, per 100g, is normally distributed. A random sample of 8 brands of frozen pizza gives the following sugar content per 100g, measuring in grams:

   3.5 2.1 5.3 4.4 3.4 2.2 4.1 5.0

   Test at the 5% significance level the claim that the mean sugar content per 100g in a frozen pizza is less than 5 grams.

2. In a game of football, the amount of time between a goal being scored and the game resuming is normally distributed. The time taken to resume, in seconds, is recorded for a random sample of 6 goals, with results:

   49 25 31 23 38 29

   Test at the 10% level of significance the claim that the mean time taken for a game to resume after a goal is greater than the 30 seconds that is generally considered to be typical.

3. A camera tripod can supposedly safely hold masses of up to 4.5kg. A photography shop disputes this figure, and randomly selects 10 tripods. Each is loaded progressively with heavier masses until it fails. The results, in kg are summarised as follows:

   $$n = 10 \quad \Sigma x = 40.5 \quad \Sigma x^2 = 165.49$$

   Stating an assumption required, test at the 1% level the claim that the mean mass the tripods can withstand before failure is actually less than 4.5kg.

4. *Vinho Verde* is a type of Portuguese wine that originated in the Minho Province in the far north of the country. In 2009, the University of Minho supported a study which examined the extent to which wine quality can be judged through measuring its physiochemical properties. As part of the study, the acidity level of a random sample of 11 white *Vinho Verde* wines all made using the *alvarinho* grape were measured. The summary data obtained from this sample was as follows (measured in pH):

   $$\Sigma x = 34.89 \qquad \Sigma x^2 = 110.81$$

   Worldwide, the mean acidity level of white wines is 3.4 pH.

   Stating one assumption required, perform a parametric test at the 5% level of significance to determine whether the white *Vinho Verde* wines made from the *alvarinho* grape differ in acidity level from those made worldwide.

5. A *soporific drug* is one which has been shown to induce sleep, and increase drowsiness. An early 20th century study into possibly soporific drugs involved giving ten patients a dose of one such drug and recording whether they reported sleeping *more* than usual that night (recorded as a positive number) or *less* than usual (recorded as a negative number). The results recorded, in hours, were:

   0.7 -1.6 -0.2 -1.2 -0.1 3.4 3.7 0.8 0.0 2.0

   The study wished to determine whether there was evidence from the study of the drug leading to increased sleep. Perform a hypothesis test to assess the evidence at the 2.5% level of significance.

6. A particular model of bass guitar made by a company is known to have historically been produced with a mean mass of 9.4 pounds, where the mass of a bass guitar is normally distributed. Now under new ownership, a bass guitar magazine is looking into claims that the bass guitars of that model now being produced are heavier, possibly due to an undeclared change in the wood used. A random sample of 14 such bass guitars gives the following masses, in pounds:

9.4	9.6	9.3	9.5	9.7	9.6	9.6
9.5	9.4	9.7	9.6	9.3	9.8	9.7

Perform a hypothesis test at the 1% level of significance to assess evidence for the claims.

7. An experiment was conducted to measure and compare the effectiveness of various feed supplements on the growth rate of chickens. As part of this, 14 newly hatched chicks were randomly selected to be given a soybean-based feed supplement. Their masses in grams after six weeks were recorded, and the following summary statistics were obtained:

variable	n	min	max	mean	sd
weight (grams)	14	158.0	329.0	246.4	54.1

It has historically been observed that the mean mass of a chick after six weeks is 260 grams. Stating one assumption required, perform a hypothesis test to determine whether there is evidence at the 5% level of significance that giving a soybean-based supplement affects the mass a chick will reach by six weeks.

8. The second-most common species of tree in the United States is the *loblolly pine*, regarded as one of the most commercially important species of tree in the Southeastern US for its timber. A survey based on a sample of 15 of the loblolly trees in one region revealed a sample mean height of 32.4 feet and a standard deviation of 2.07 feet. The heights of the trees are known to be normally distributed. Stating one assumption required, test at the 1% significance level whether the data provides evidence to suggest the mean height of loblolly trees in the region is greater than 30 feet.

9. A school is looking to figure out a course for the annual school cross-country, with the goal of choosing a course such that the mean time it takes a student at the school to finish is 45 minutes. The first ten students to respond to a request by the PE department for volunteers to test the course have their time taken to complete the course recorded. The results of the t-test used to assess whether the course meets their requirements is as follows:

```
t-test for the population mean
n = 10    sample mean = 53.1    sample SD = 11.2492
alternative hypothesis: true mean is not equal to 45
t = ****    p-value = 0.0488
```

   (a) State the null and alternative hypotheses for the test conducted.
   (b) Determine the missing value of the t-test statistic, indicated by **** in the computer output.
   (c) Find the critical value for the hypothesis test, at the 5% level of significance.
   (d) Use the p-value to write a conclusion to the hypothesis test.
   (e) Name the sampling method used and comment on the appropriateness.

10. An experiment into the cold tolerance of a species of grass plant looked at the effect of low temperature on its carbon dioxide (CO2) uptake. It is known that, under normal conditions, the CO2 uptake rate for the species is known to be 33 $\mu mol/m^2$, whilst the sample of 7 areas exposed to chilling resulted in a sample mean uptake of 29.97 $\mu mol/m^2$ and a standard deviation of 8.335 $\mu mol/m^2$. Perform a parametric test to assess the evidence for CO2 uptake being affected by low temperature for the grass plant, at the 10% level of significance.

10.5 One-Sample z-Test for the Population Proportion

If X elements from a random sample of size n from a population of interest satisfy some condition, then the **sample proportion**, $\hat{P}$, can be calculated as:

$$\hat{P} = \frac{X}{n}$$

For example, if a survey of registered voters reveals 114 out of 250 intend to vote for the incumbent candidate, the *sample proportion*, $\hat{p}$, is $\frac{114}{250} = 0.456$. The true value of the **population proportion**, ***p***, may differ from this value. The **one-sample z-test for proportion** assesses whether an *observed sample proportion* provides statistically significant evidence against a *null-hypothesised population proportion*. The test statistic can be derived from the distributions of the random variables X and hence $\hat{P}$:

$$X \sim \mathsf{B}(n, p)$$

When $np > 5$ and $nq > 5$, X is *approximately* normally distributed:

$$\begin{aligned} \mathsf{E}(X) &= np \\ \mathsf{V}(X) &= npq \\ X &\approx \mathsf{N}(np, npq) \end{aligned}$$

The distribution of $\hat{P}$ can now be obtained since $\hat{P} = \frac{X}{n}$:

$$\mathsf{E}\left(\hat{P}\right) = \mathsf{E}\left(\frac{X}{n}\right) = \frac{1}{n}\mathsf{E}(X) = \frac{1}{n} \times np = p$$

$$\mathsf{V}\left(\hat{P}\right) = \mathsf{V}\left(\frac{X}{n}\right) = \left(\frac{1}{n}\right)^2 \mathsf{V}(X) = \frac{1}{n^2} \times npq = \frac{pq}{n}$$

Therefore:

$$\hat{P} \approx N\left(p, \frac{pq}{n}\right)$$

The z-transformation leads to the required test statistic:

For a one-sample z-test for the population proportion:

$$\text{Test statistic: } z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

Note that $\sqrt{\frac{pq}{n}}$ is called the **standard error of the sample proportion**.

Conditions for the one-sample z-test for the population proportion:

- ***np*** > 5 **and** ***nq*** > 5
- The observations are **independent** (typically assured through *random sampling*).

Critical values for the z-test for population proportion are determined in the same way as for the z-test for the population mean.

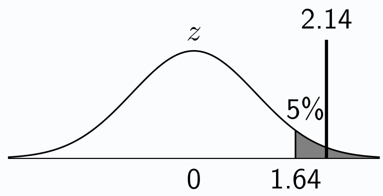
Example

Problem: A data storage company knows that 10% of the hard drives they use will fail within the first year of their use. They have decided to trial the use of a cheaper model of hard drive, but are concerned that it may have a higher failure rate. During this trial, 60 of the new hard drives are randomly selected to be monitored for a year once installed. Of those 60, 11 fail.

Use a parametric hypothesis test to assess the evidence, at the 5% level of significance, that the new drives have a higher first-year failure rate than the previous model.

Solution: $p = 0.1$, $q = 0.9$, $n = 60$, $\hat{p} = \frac{11}{60} = 0.183$

$H_0: \ p = 0.1$
$H_1: \ p > 0.1$



Test statistic: $z = \dfrac{0.183 - 0.1}{\sqrt{\dfrac{0.1 \times 0.9}{60}}} = 2.14$

Critical value: $z = 1.64$

Since $2.14 > 1.64$, reject H_0 at the 5% level of significance. There is sufficient evidence to suggest that the proportion of the new hard drives that fail within their first year of use is higher than 10%.

Since the alternative hypothesis for this test refers to the population *proportion*, conclusions should always aim to refer to a 'proportion' in context (and never the *mean*).

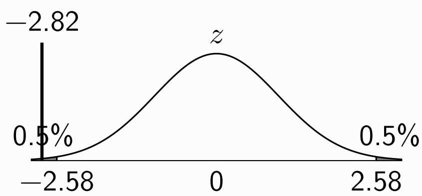
Example

Problem: A candidate in a mayoral election claims that 85% of voters intend to vote for them. Out of a random sample of 80 people eligible to vote, 59 say that they intend to vote for that candidate. Perform a hypothesis test to assess the evidence at the 1% level of significance that their claim may be untrue, justifying the use of a z test for this data.

Solution: $p = 0.85$, $q = 0.15$, $n = 80$, $\hat{p} = \frac{59}{80} = 0.7375$

$np = 80 \times 0.85 = 68$ and $nq = 80 \times 0.15 = 12$.
Since $np > 5$ and $nq > 5$, a normal approximation is appropriate.

$H_0: \ p = 0.85$
$H_1: \ p \neq 0.85$



Test statistic: $z = \dfrac{0.7375 - 0.85}{\sqrt{\dfrac{0.85 \times 0.15}{80}}} = -2.82$

Critical value: $z = -2.58$

Since $-2.82 < -2.58$, reject H_0 at the 1% level of significance. There is sufficient evidence to suggest that the true proportion of voters intending to vote for that candidate is not 85%, and that the candidate's claim may be untrue.

Exercise 10.5

For each question, show clearly that the conditions for the use of a normal approximation are satisfied.

1. A coin, suspected to be biased, is thrown 500 times and 267 heads are obtained. Perform a z-test to assess whether there is evidence to suggest that the coin is biased, using a 10% significance level.

2. A newspaper claims that over 60% of its readers are car owners. In a random sample of 312 readers there are 208 car owners. Suitable hypotheses are proposed of:

$$\begin{aligned} \mathsf{H}_0 &: \quad p = 0.6 \\ \mathsf{H}_1 &: \quad p > 0.6 \end{aligned}$$

   Test whether there is evidence at the 1% level of significance to support the newspaper's claim.

3. In a public opinion poll, 1000 randomly chosen teenagers were asked whether they would support a ban on mobile phones in schools, and 597 replied 'yes'. Test at the 5% significance level whether the data supports an article's claim that 'a majority of teenagers support a ban on mobile phones in schools'.

4. The proportion of left-handed people in the population is 10%. A random sample of 400 surgeons found that 29 were left-handed. Perform a hypothesis test at the 5% level of significance to assess whether there is evidence to suggest that the proportion of left-handed surgeons does not match the 10% seen in the wider population.

5. A seed company sells seeds in mixed packets and states that over 20% of the resulting plants will have red flowers. A packet of seeds is sown by a gardener who finds that only 9 out of 82 plants have red flowers.

   Perform a hypothesis test to assess the claim that fewer than 20% of the plants will actually have red flowers, using a 1% significance level.

6. A teacher says that the majority of pupils in a school prefer cats to dogs. A sample of 25 pupils is asked which they prefer, and 17 prefer cats. Perform a parametric test at the 5% level of significance to assess whether the data provides evidence to support the teacher's claim, stating an assumption required.

7. Scotland's 2022 census revealed that of those in employment, 7% were working in construction. Last year, a random sample of 200 adults living in Scotland revealed that, of the 189 who were in some form of employment, only 8 were working in construction. Perform a parametric test to assess whether the data gives evidence at the 10% level of significance that the proportion of working adults employed in construction has changed since the census.

8. A football manager claims that the *Video Assistant Referee* (VAR) system used in all games played in the top league leads to the on-field referee being instructed to watch a replay on a pitchside monitor in at least three-quarters of games. A journalist looks back at the 60 games played up to that point in the season and finds that the VAR instructed the referee to watch a replay in only 38 of the games.

   Assess at the 1% significance level whether the data provides evidence to support the journalist's claim that the football manager was incorrect in their statement.

9. In a factory, ready-sliced potatoes are cooked to make crisps, which are then placed into bags before flavouring is added and the bags sealed. Occasionally a packet of crisps is sealed without flavouring having been added. The crisp manufacturers have decided that they are content with this process provided no more than 1.8% of bags are flavourless.

   (a) A random sample of 50 bags is taken. Explain why performing a z-test for the proportion of flavourless crisps obtained from this sample would not be appropriate.

   (b) Determine the minimum sample size required such that a z-test would be appropriate.

   (c) From a much larger sample of 500 bags of crisps, 14 are found to be flavourless. Test at the 5% level of significance whether the manufacturers should be concerned with the process.

10.6 Choosing the Right Hypothesis Test

Selecting the appropriate hypothesis test to use in any given context is an important skill for a statistician and warrants careful practice. The following questions contain a mix of *z-tests for the population mean, t-tests for the population mean and z-tests for the population proportion.*

Exercise 10.6

1. A keen cyclist often takes a route which consists of a circuit, travelling through a forest and around a hill before returning to her starting point, timing herself on each occasion. The time it takes her to complete a circuit is normally distributed, with mean 42.7 minutes and standard deviation 3.8 minutes. After buying a new bike, she records the time it takes her to complete a circuit on 7 randomly selected occasions, obtaining a sample mean of 41.3 minutes. Assuming the standard deviation in her time taken remains unchanged, perform a hypothesis test to assess her data for evidence that her mean time taken to complete the route has reduced.

2. A popular chain of fast food restaurants states on its website that a vegan alternative to its most popular menu item is available in over 80% of its restaurants. A researcher for a website that publishes articles on veganism and vegetarianism selects 60 of its restaurants at random and asks one reader from nearby each restaurant to visit and see whether the vegan alternative is available. Out of the 60 restaurants visited, the vegan alternative was only available in 42 of them. Perform a hypothesis test at the 5% level of significance to assess whether the data suggests the chain is exaggerating the availability of the vegan alternative.

3. The mass of beef mince contained within packs that state they contain 1kg of mince is measured, for a random sample of 8 packs. The results obtained, in kg, are:

   0.97 1.01 0.99 0.97 1.00 0.98 0.99 0.96

   Perform a parametric test to assess the data for evidence that the mean mass of mince in a pack is less than that stated on the packaging.

4. A report by the *Energy Saving Trust* states that households in the UK use a mean of 349 litres of water each day. A family living in an area in which water usage is metered and billed for monitors its daily volume of water used on 8 randomly selected days. They find that the sample yields a mean volume of water used of 357 litres and a standard deviation of 12 litres. Stating one assumption required, perform a test at the 10% level of significance to assess whether the family has a higher mean daily water consumption than the figure quoted by the report.

5. *Screen time* refers to the amount of time a person spends in front of some kind of digital screen, such as a smartphone, tablet or computer monitor. Researchers exploring screen time in the UK reported that the mean daily screen time for an adult in the UK is 6 hours and 9 minutes, or 6.15 hours, with a standard deviation of 1 hour and 18 minutes, or 1.3 hours. Interested in regional variations, researchers located in Scotland contacted 20 randomly selected adults living in Scotland and asked them to report their screen time for the previous day. 16 of them responded, with a sample mean of 4 hours and 51 minutes, or 4.85 hours, obtained.

   (a) Assuming the standard deviation for adults in Scotland is the same as that for the UK as a whole, perform a hypothesis test at the 5% level of significance to assess whether the data supports any suggestion that mean daily screen time differs between adults in Scotland and the UK as a whole.

   (b) Give one reason why the method of gathering data used may call into question the reliability of the hypothesis test.

   A separate report by another group of researchers wishes to test the claim made by a newspaper that "a majority of adults in Scotland are taking steps to reduce their screen time". A survey of 120 randomly selected adults are asked whether they are taking steps to reduce their screen time, with 71 stating that they are, and the remaining 49 that they are not.

   (c) Perform a parametric test to assess, at the 5% significance level, whether there is evidence to support the newspaper's claim.

Review Exercise

1. A pharmaceutical company has developed a drug designed to reduce the recovery time needed for a specific type of operation from the well-established mean time of 11 days. Following trials involving treating 24 randomly selected patients admitted to hospital for the operation with the drug, it is found that the mean recovery time for these patients is 9.6 days, with a standard deviation of 2.7 days.

   (a) Explain why a z-test can be used for this data despite the population standard deviation, σ, not being known.

   (b) Perform a hypothesis test at the 1% level of significance to assess the evidence that the drug may reduce the mean recovery time for the operation.

2. A car company wishes to check that a new machine at their factory producing front bumpers is still maintaining the mean mass of 2.8kg for the part as the old machine it replaced. Eight bumpers produced by the machine are randomly selected and weighed. The results, in kilograms, are:

   $$2.9 \quad 2.6 \quad 3.3 \quad 3.1 \quad 3.0 \quad 2.9 \quad 3.1 \quad 2.8$$

   Perform a parametric test at the 5% level of significance to determine whether there is sufficient evidence to suggest that the mean mass of the part produced by the new machine has changed, stating an assumption required.

3. The speed of cars passing through a village is being studied in an effort to improve safety for pedestrians, and is known to be normally distributed. The mean speed has previously been monitored over a number of years and found to be consistently at 33 mph, with a standard deviation of 4 mph. Following the installation of speed bumps, the speed of 15 randomly selected cars is measured at the same point as before, with a sample mean of 31 mph observed.

   (a) Perform a parametric test to determine whether the data provides evidence to support the council's claim that the speed bumps have been successful in reducing the mean speed of cars passing through the village.

   (b) State an assumption required for the test conducted in part (a).

4. A driving instructor posts an advert online stating nine out of ten of their students pass their driving test first time. A random sample of 60 students taught by the instructor reveals that only 47 of them passed their driving test first time. Perform a hypothesis test at the 1% level of significance to assess the evidence for the claim that the proportion of students who pass first time is less than the instructor claims.

5. A coach working with a junior athletics club has observed that for many years the mean reaction time in 100m races for the sprinters at the club has been consistently stable at 185 milliseconds. Claiming that this figure may no longer be valid, she randomly selects 8 sprinters and records their reaction time in their next races. None of the athletes raced together in their next race. The summary data, where times were measured in milliseconds, is below:

   $$n = 8 \quad \Sigma x = 1372 \quad \Sigma x^2 = 237466$$

   (a) Stating an assumption required, perform a t-test to test the coach's claim at the 10% level of significance.

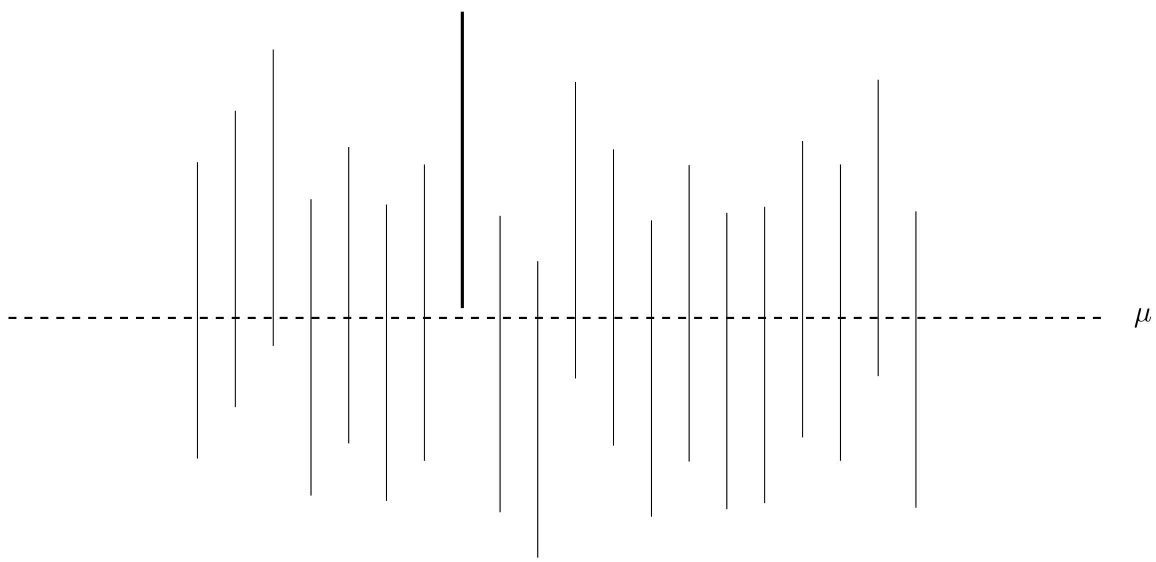
   (b) The coach initially debated recording the reaction times of the eight athletes competing in the gold medal race of club's annual 100m championships. Suggest one reason why she was correct to reject this approach.

11
Confidence Intervals

From a random sample drawn from a population, the sample mean can be used as a *point estimate* for the true, unknown value of the population mean. If the sample size is small, or the variability of the data is very high, the *precision* of this estimate may be low. To avoid the misleading sense of precision that point estimates on their own can give, many institutions will require *confidence intervals* to routinely be stated as part of any statistical analysis. Statistical software will often also report a confidence interval whenever a hypothesis test is performed.

"A 95% confidence interval for the mean weight of an adult pine marten, in kg, is (1.42 , 1.58)"

A common misconception is that such a confidence interval can be interpreted as *"There is a 95% probability that the true mean weight is between 1.42 and 1.58."* Care should be taken to *avoid* interpreting the confidence level used as a probability relating to any particular generated confidence interval and the parameter it relates to. Instead, *"95% confidence"* refers to the **long-run success rate of the procedure** used to generate the interval. A 95% confidence level means that if the sampling procedure were repeated many times, approximately 95% of the intervals produced would contain the true mean μ. The figure below shows twenty 95% confidence intervals generated using simulated data, one of which does not contain the true value of the population mean, μ, in the interval.



11.1 A z-Confidence Interval for the Population Mean

Given a *random sample* from a distribution that is *normally distributed* with *known population standard deviation* σ, a $100(1-\alpha)\%$ *z-confidence interval* for the population mean, μ, can be calculated:

A z-confidence interval for the population mean:

A $100(1-\alpha)\%$ CI for μ is: $\bar{x} \pm z_{1-\alpha/2} \times \dfrac{\sigma}{\sqrt{n}}$

As a reminder, the quantity $\dfrac{\sigma}{\sqrt{n}}$ is called the *standard error of the sample mean*.

Conditions for a z-confidence interval for the population mean:

- The population is **normally distributed**, or the sample size is sufficiently large $(n > 20)$ such that the Central Limit Theorem applies.
- The **population standard deviation**, σ, is known (and has not changed).
- The observed data are **independent** (obtained through a random sample).

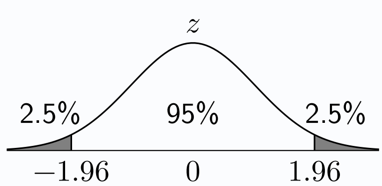
For large sample sizes, where the population standard deviation σ is not known, it may be decided that the sample standard deviation is a reasonable estimate $(\sigma \approx s)$. In practice, the t-distribution is generally preferred when only the sample standard deviation is known, and hence t-confidence intervals for the population mean, introduced in the next section, are far more commonly encountered.

Whilst the notation contained in the formula for a confidence interval can look complicated, it should be noted that the z-value for a two-tailed 95% confidence interval corresponds with that for the critical value of a two-tailed z test at the 5% level of significance. A sketch can be used to help find the required z-value, just as in the previous chapter. Although *one-tailed* confidence intervals exist, in this course only *two-tailed* confidence intervals will be calculated.

Example

Problem: The masses of ceramic dishes sold by a kitchenware shop are known to be normally distributed with a standard deviation of 12g. A random sample of 15 such ceramic dishes are weighed and the sample mean is 430 grams. Obtain a 95% confidence interval for the mean mass of a ceramic dish.

Solution: $\bar{x} = 430$, $n = 15$, $\sigma = 12$



A 95% CI for μ is: $\quad 430 \pm 1.96 \times \dfrac{12}{\sqrt{15}} \quad = \quad 430 \pm 6.07$

Therefore a 95% confidence interval for the population mean dish mass, in grams, is $(423.93, 436.07)$.

It should be observed that the z-confidence interval is symmetrical with the **sample mean at its centre**.

The width of the interval is determined by the standard error and the confidence level required. For the same standard deviation and confidence level, a larger sample size will produce a narrower interval; this is intuitive, as more data collected should lead to a more precise estimate of the population mean.

Exercise 11.1

1. The height of a red panda is known to be normally distributed with a standard deviation of 2cm. A random sample of 8 red pandas have their heights measured, with a sample mean of 57cm obtained. Obtain a 99% confidence interval for the mean height of a red panda.

2. An airline knows from historical data that the mass of the hold baggage for a passenger is normally distributed with a standard deviation of 2.1kg. The mass of the hold baggage is measured for a random sample of 11 passengers, with a sample mean of 13.8kg calculated. Assuming that the standard deviation remains unchanged, obtain a 90% confidence interval for the mean mass of the hold luggage for the airline's passengers.

3. The time taken for a chef to make a particular dish from their restaurant's main course menu is normally distributed, with the standard deviation known to be 0.9 minutes. On 7 randomly chosen occasions, the time it takes to make the dish is measured, with a sample mean of 15.2 minutes obtained. Calculate a 95% confidence interval for the mean amount of time it takes for the chef to make the dish.

4. The speeds of heavy goods vehicles (HGVs) crossing an old, rural bridge are normally distributed, with a historically observed standard deviation of 2.6 mph. Following improvements to the surface of the road on the bridge, the local council wishes to know what the mean speed of HGVs crossing the bridge is. The speeds of 17 randomly selected HGVs are recorded, with a sample mean of 39 mph. Obtain a 99% confidence interval for the mean speed of HGVs crossing the bridge, stating one assumption required.

5. The masses of the eggs laid by the chickens on a farm are normally distributed. The farmer weighs 25 randomly chosen eggs, in grams, obtaining the following summary statistics:

$$n = 25 \quad \Sigma x = 1254 \quad \Sigma x^2 = 63186$$

   It is assumed that the sample standard deviation is a reasonable estimate for the population standard deviation. Obtain a 90% confidence interval for the mean mass of the chicken eggs laid by the chickens at the farm.

6. A restaurant manager is concerned about the time a table has to wait to have a drinks order taken once they have been seated. On 30 random occasions throughout a week, the time taken is recorded, in minutes. The manager puts the data into some statistical software and obtains the following summary data:

```
x -> drinks order wait data
n = 30    sum of x = 171    x bar = 5.7    SD of x = 0.8
```

   Obtain a 95% z-confidence interval for the mean wait time, stating one assumption required.

7. A workers' rights organisation wishes to understand the number of hours worked per week by full-time employees of the local council. A random sample of 24 employees gives a sample mean of 38.2 hours and a standard deviation of 1.4 hours. It is decided that the standard deviation obtained is a reasonable estimate for the population standard deviation.
   (a) Obtain a 95% confidence interval for the mean number of hours worked by full-time employees of the local council.
   (b) The organisation would like a narrower confidence interval, and one organisation member suggests that obtaining instead a 99% confidence interval will provide this. Calculate the width of a 99% confidence interval for the mean hours worked and comment on the member's suggestion.
   (c) The organisation finally decides to stick with a 95% confidence interval, but to extend the sample size to obtain a confidence interval with width no greater than 1. Determine the minimum number of **additional** full-time workers that must be included in the sample in order to obtain this, assuming the sample standard deviation remains at 1.4 hours.

8. For a 90% z-confidence interval for the population mean to have a width of 12.4, based on a sample of size 7, calculate the value of σ.

11.2 Confidence Intervals and Hypothesis Testing

Consider the following hypotheses, where μ_0 is a possible value for a population mean of interest:

$$\begin{aligned} \mathrm{H}_0 : &\quad \mu = \mu_0 \\ \mathrm{H}_1 : &\quad \mu \neq \mu_0 \end{aligned}$$

The calculations involved in constructing a *two-tailed* z-confidence interval for the population mean correspond with those for a *two-tailed* z-test for the population mean. Such a confidence interval could therefore allow inferences to be made about the population mean.

Inference using confidence intervals:

If μ_0 is **outwith a calculated 95% confidence interval** then there is evidence at the 5% significance level to suggest that $\mu \neq \mu_0$.

If μ_0 is **within a calculated 95% confidence interval** then there is insufficient evidence at the 5% significance level to suggest that $\mu \neq \mu_0$.

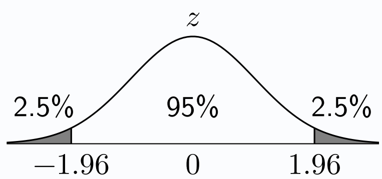
It should be noted that, in this course, an instruction to perform a *test* requires a hypothesis test to be conducted using either a p-value or a test statistic, rather than a confidence interval.

Example

Problem: The mass of food eaten each day by a family's cat is normally distributed with a mean of 280 grams and a standard deviation of 13 grams. Recently, they notice that their cat doesn't seem to eat the same amount of food as it used to. On 14 randomly chosen days they measure the mass of food that their cat eats, obtaining a mean of 273 grams per day. Stating an assumption required, obtain a 95% confidence interval for the mass of food eaten daily by their cat, and comment on their concern that their cat doesn't eat the same amount of the food as it did previously.

Solution: $\bar{x} = 273$, $n = 14$, $\sigma = 13$

Assume that the standard deviation of mass of food eaten per day is unchanged from 13 grams.



A 95% CI for μ is: $\quad 273 \pm 1.96 \times \dfrac{13}{\sqrt{14}} \quad = \quad 273 \pm 6.81$

Therefore a 95% confidence interval for the mean mass of food now being eaten daily by their cat, in grams, is $(266.19, 279.81)$.

Since 280 lies outwith the confidence interval, this provides evidence to suggest that the mean daily mass eaten has changed from 280g.

Since any confidence intervals encountered in this course will be two-tailed, it is only appropriate to comment on *two-tailed* conclusions (i.e. 'difference...') when using language such as 'significance' and 'evidence'. Making *firm one-tailed conclusions* should be avoided.

Exercise 11.2

1. A packing process packages coffee beans into bags. The mass of each bag is normally distributed with standard deviation 15g. A random sample of 25 bags is taken and the sample mean is found to be 493g. Obtain a 95% confidence interval for the mean mass of coffee beans in a bag, and comment on the claim that the mean mass being dispensed is not the 500g that it is supposed to be.

2. The lengths of metal bars produced by a particular machine are normally distributed with standard deviation 12cm. The machine is serviced, after which a random sample of 80 bars gives a mean length of 423cm.
   (a) Obtain a 99% confidence interval for the mean length of the bars now produced, assuming the standard deviation remains the same.
   (b) Comment on the concern that the mean bar length produced by the machine is no longer the 420cm that it was prior to it being serviced.

3. A large school is studying the amount of money its teachers spend on photocopying as it prepares its budget for the next academic year. A random sample of 23 teachers gives a mean spend of £10.90 per year and a standard deviation of £2.30. Assuming that the sample standard deviation is a reasonable estimate of the population standard deviation, construct a 90% z-confidence interval for the mean spend per teacher each year, and comment on the suggestion that the figure of £12 used previously is still accurate.

4. A sports statistician is analysing the number of goals scored in games of football. A random sample of 35 games gives a sample mean of 2.7 goals and a sample standard deviation of 0.6 goals. Stating an assumption required, obtain a 99% z-confidence interval for the mean number of goals scored in a game, and comment on their claim that the number of goals scored has changed from the historically observed mean of 2.5 goals per game.

5. Extensive records of the heights of mature oak trees throughout the 1970s and 1980s in some woodland show that they are normally distributed with a mean of 21 metres and a standard deviation of 1.5 metres. Recently, a random sample of 30 mature oak trees in the woodland trees yielded a sample mean of 20.4 metres.
   (a) Stating one assumption required, obtain a 90% confidence interval for the mean height of mature oak trees in the woodland now.
   (b) Comment on the suggestion that the mean height of mature oak trees in the woodland has changed.

6. A researcher is examining a statistical report on the salaries of delivery drivers in a city. She sees that a random sample of full-time delivery drivers in the city were asked their annual salaries. The report includes computer output related to a z-test on the hypothesis that the mean salaries for the delivery drivers is different to the national mean of £30 000. The report concludes that there is not sufficient evidence to suggest that the mean salary of delivery drivers in the city is different from that seen nationally.

```
z-test for the population mean
n = 22    sample mean = 29300    sample SD = 1720
alternative hypothesis: true mean is not equal to 30000
z = -1.91    p-value = 0.0563
```

   Assume the sample standard deviation is a reasonable estimate of the population standard deviation.

   (a) Obtain a 95% confidence interval for the mean annual salary of delivery drivers in the city.
   (b) With reference to the confidence interval, comment on whether this confidence interval supports the conclusion made by the report.

11.3 A t-Confidence Interval for the Population Mean

Given a random sample of data from an underlying population that is *normally distributed*, a t-confidence interval can be calculated using the *sample standard deviation*, s.

A t-confidence interval for the population mean:

A $100(1-\alpha)\%$ CI for μ is: $\bar{x} \pm t_{n-1,1-\alpha/2} \times \dfrac{s}{\sqrt{n}}$

A t-confidence interval is used when the population standard deviation is unknown and must be estimated using the sample standard deviation. The conditions for valid use of a t-confidence interval match those for a one-sample t-test, with the degrees of freedom calculated as $\nu = n - 1$.

Conditions for a t-confidence interval for the population mean:

- The population is **normally distributed**.
- The observed data are **independent** (obtained through a random sample).

As with a z-confidence interval, a *t-confidence interval* can be used to make inferences equivalent to those from a two-tailed one-sample t-test.

Example

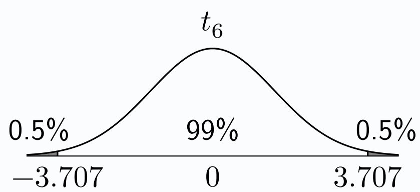
Problem: Seven mid-terrace houses in Fife are randomly chosen to have their electricity usage for 24 hours recorded. The results, in kWh, are:

7.8 10.1 8.4 9.8 8.3 9.4 7.3

Stating any assumption required, determine a 99% confidence interval for the mean daily electricity usage of mid-terrace houses in Fife, and comment on the claim that their electricity consumption is not the same as the mean of 9.7 kWh used by *semi-detached* houses in Fife.

Solution: $\bar{x} = 8.729$, $n = 7$, $s = 1.055$

Assume the underlying population of daily energy usage is normally distributed.



A 99% CI for μ is: $\quad 8.729 \pm 3.707 \times \dfrac{1.055}{\sqrt{7}} \quad = \quad 8.729 \pm 1.478$

Therefore a 99% CI for the mean daily electricity usage of mid-terrace houses in Fife, in kWh, is $(7.251, 10.207)$.

Since 9.7 kWh is within the confidence interval, there is insufficient evidence to support the claim that mid-terrace houses and semi-detached houses in Fife use different mean amounts of electricity.

Exercise 11.3

1. As part of a project to better understand acidity levels in lakes in England, a random sample of 8 lakes are selected and the pH level of each measured. The pH levels observed were:

   6.7 7.1 8.3 7.4 7.6 6.8 7.0 8.1

   Obtain a 95% confidence interval for the mean pH level of lakes in England.

2. EnergyPower makes replacement phone batteries. The capacity of the batteries is normally distributed with a mean of 4500 mAH. A phone repair store is concerned that a recent large delivery of EnergyPower batteries may not be genuine, and measures the capacity of a random sample of 7 batteries. The results, in mAH, are:

   4780 4120 4340 4360 4110 4570 4390

   Obtain a 99% confidence interval for the mean capacity of the batteries delivered and comment on their concern that they are not genuine EnergyPower batteries.

3. The lifespan of a type of light bulb made by a brand is normally distributed. A consumer watchdog records the lifespan of a random sample of ten such bulbs and obtains the following summary statistics, in hours:

   $$n = 10 \quad \Sigma x = 9920 \quad \Sigma x^2 = 9868100$$

   Obtain a 90% confidence interval for the mean lifespan of the bulbs.

4. A camera tripod can supposedly safely hold masses of up to 4.5kg. A photography shop disputes this figure, and randomly selects 10 tripods. Each is loaded progressively with heavier masses until it fails. The results, in kg are summarised in the following computer output:

   ```
   summary(tripod mass data)
   n = 10  median = 4.05  mean = 4.05
   sum of x = 40.5  sum of x squared = 165.49
   min = 3.40  max = 4.70
   ```

   (a) State an assumption required in order to calculate a t-confidence interval for the mean mass a tripod can withstand before failure.

   (b) Obtain a 99% confidence interval for the mean mass the tripods can support, and comment on the claim that the 4.5kg figure stated is not correct.

5. The amount of sugar contained in frozen pizzas, per 100g, is normally distributed. A random sample of 8 brands of frozen pizza gives the following sugar content per 100g, measuring in grams:

   3.5 2.1 5.3 4.4 3.4 2.2 4.1 5.0

   (a) Obtain a 95% confidence interval for the mean sugar content of frozen pizzas, and comment on a statement on a nutrition website that the mean amount is 5 grams.

   (b) Determine the minimum sample size required for an interval of width 0.5 grams, if the sample standard deviation remains the same.

11.4 A *z*-Confidence Interval for the Population Proportion

The sampling distribution of the sample proportion, $\hat{P}$, has the standard error: $\sqrt{\dfrac{pq}{n}}$.

When calculating a z-confidence interval for the population proportion, it is necessary to instead use the *sample proportion*, $\hat{p}$, as an *estimate* for p, leading to an **estimated standard error**.

Estimated standard error of the sample proportion: $\sqrt{\dfrac{\hat{p}\hat{q}}{n}}$.

A z-confidence interval for the population proportion:

A $100(1-\alpha)\%$ CI for p is: $\hat{p} \pm z_{1-\alpha/2} \times \sqrt{\dfrac{\hat{p}\hat{q}}{n}}$

This is only an *approximate confidence interval*, both because the sampling distribution of $\hat{p}$ is only *approximately normal* (and only for sufficiently large samples), and since the standard error is also an estimate.

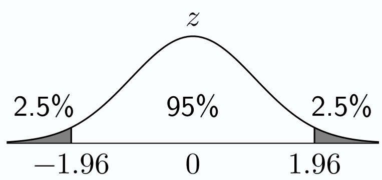
Conditions for a z-confidence interval for the population proportion:

- Both $n\hat{p} > 5$ and $n\hat{q} > 5$.
- The observed data are **independent** (obtained through a random sample).

Example

Problem: A tech website surveys a random selection of its subscribers and finds that 41 out of 50 of them have changed the password for their WiFi router from the default one as recommended by experts. Obtain an approximate 95% confidence interval for the proportion of its subscribers that have changed their router's password.

Solution: $\hat{p} = \frac{41}{50} = 0.82$, $\hat{q} = 0.18$, $n = 50$



A 95% CI for p is: $\quad 0.82 \pm 1.96 \times \sqrt{\dfrac{0.82 \times 0.18}{50}} \quad = \quad 0.82 \pm 0.060$

Therefore an approximate 95% confidence interval for the proportion of the website's readers that have changed their router's password is $(0.760, 0.880)$

Given the potential for confusion between the population proportion p and the sample proportion $\hat{p}$, as well as q and $\hat{q}$, it is recommended to check all working carefully to ensure the correct notation is being used.

Exercise 11.4

1. A newspaper conducts a survey to explore the proportion of its readers that are car owners. In a random sample of 312 readers there are 208 car owners. Obtain an approximate 99% confidence interval for the proportion of car owners amongst the newspaper's readers.

2. A sample of 25 pupils in a school are asked whether they prefer cats or dogs, and only 7 prefer cats.
   (a) State an assumption required about the sample of pupils.
   (b) Obtain an approximate 90% confidence interval for the proportion of pupils in the school that prefer cats.
   (c) Show the calculations required to verify that this data is suitable for a z-confidence interval.

3. In an opinion poll undertaken by the Scottish government, 1000 randomly chosen adults living in Scotland were asked whether they would support a proposed ban on advertising junk food on TV, and 643 replied 'yes'. Obtain an approximate 95% confidence interval for the true proportion of adults in Scotland who would support the ban.

4. The proportion of left-handed people in the general population is 10%. A random sample of 400 professional artists found that 47 were left-handed. Obtain a 95% confidence interval for the proportion of professional artists that are left-handed, and comment on the claim that the proportion of left-handed artists is not the same as that seen in the general population.

5. A seed company sells pansy seeds in mixed packets and claims that 20% of the resulting plants will have red flowers. A packet of seeds is sown by a gardener who finds that only 9 out of 82 plants have red flowers. Obtain an approximate 99% confidence interval for the proportion of plants that have red flowers, and comment on whether this supports the seed company's claim.

6. A particular gene is historically thought to be present in the DNA of 4% of people. Taking advantage of the increased affordability and availability of genetic testing, a charity randomly selects 240 people to be tested for the gene. Of the 240 people in the sample, 13 are found to have the gene. Obtain an approximate 95% confidence interval for the proportion of people that have the gene, and comment on the suggestion that the 4% figure is not correct.

7. The proportion of students at a university that subscribe to a music streaming service is thought to be between 45% and 75%. A random sample of n students gives a sample proportion of 58%. Determine the minimum sample size n such that a 95% confidence interval may be argued to support a claim that a majority of students at the university subscribe to a music streaming service.

8. A guitar website hosts a forum in which its users can advertise their guitars as for sale. The website claims that one in every three of the guitars listed on the forum are vintage instruments from the 1950s, 1960s and 1970s. A forum user selects at random one of the ten most recently listed guitars, then every tenth guitar after that until they have looked back over the last three months of for-sale guitars. The decade of manufacture of every guitar in the sample is recorded. The results are:

1950s	1960s	1970s	1980s	1990s	2000s	2010s	2020s
7	14	17	9	7	15	31	7

   (a) State the type of sampling used by the forum user.
   (b) Obtain an approximate 95% confidence interval for the proportion of guitars for sale that are vintage, manufactured in the 1950s, 1960s and 1970s.
   (c) Comment on whether the confidence interval supports the website's claim.

Review Exercise

1. A road safety researcher randomly selects eight cars passing by on a motorway and records their speed, in mph. The following results are obtained:

   63 74 57 84 80 68 70 91

   Obtain a 95% confidence interval for the mean speed of the cars passing by on the motorway, stating an assumption required.

2. A random sample of 50 residents of a town are asked whether or not they are aware that a proposal for a large retail park on the outskirts of the town is currently being considered by the local council. 14 of the 50 residents were aware of the proposals, whilst the rest were not. Obtain an approximate 99% confidence interval for the proportion of the town's residents that are aware of the proposals.

3. The volume of wine served when a restaurant's customer orders a small glass of wine is normally distributed with a standard deviation of 1.7ml. A random sample of 32 small glasses of wine served by the restaurant gives a mean of 126.3ml per glass.

   (a) Obtain a 90% confidence interval for the mean volume of wine served when a small glass of wine is ordered by a customer in the restaurant, stating an assumption required.

   (b) A repeat customer claims that the mean volume served is not the correct measure of 125ml, as stated on the restaurant's wine list. Comment on this claim with reference to the confidence interval.

4. A pharmaceutical company has developed a drug designed to reduce the recovery time needed for a specific type of operation. Following trials involving treating 25 randomly selected patients admitted to hospital for the operation with the drug, it is found that the mean recovery time for these patients is 11.4 days. Computer output summarising this data, measured in days, is as follows:

```
summary(recovery time in days)
n = 25  mean = 11.4
A 95% z-confidence interval for the mean is:
(10.3 , ******)
```

   (a) State the value from the computer output that has been replaced by ******.

   The sample standard deviation was considered a reliable estimator for the population standard deviation of recovery time, since the sample was sufficiently large.

   (a) Determine the value of the standard deviation used to calculate the confidence interval.

5. A driving instructor posts an advert online stating nine out of ten of their students pass their driving test first time. For a random sample of students taught by the instructor, the table below shows how many of them passed their driving test first time, and how many did not pass the first time.

Result of first driving test	**Frequency**
Pass	47
Fail	13

   Obtain an approximate 95% confidence interval for the proportion of the instructor's students that pass first time, and comment on the claim that the true proportion is not the nine out of ten claimed by the instructor.

12
Chi-Squared Goodness-Of-Fit Test

The **chi-squared distribution**, χ^2_ν, describes the distribution of *sums* of *independent squared and scaled random variables*. χ^2 hypothesis tests offer a way to analyse categorical data, and are a type of **non-parametric** test.

12.1 χ^2 Goodness-of-Fit Test

A *goodness-of-fit* test assesses how well observed data fits a hypothesised distribution by comparing observed and expected frequencies. It is applied to **categorical** data (which may arise from numerical variables grouped into categories for counting). Suppose a statistics teacher's cubic die is suspected of being *biased* and their students roll the die 48 times to check for any bias. The number of times each outcome occurs, or the **observed frequencies**, are recorded in a frequency table as follows:

Outcome	1	2	3	4	5	6
Observed Frequencies (O_i)	7	8	5	10	7	11

With the null hypothesis being that the die *is* fair, a third row of **expected frequencies** can be added to the table by dividing the total count of 48 *observations* by the 6 equally-likely outcomes:

Outcome	1	2	3	4	5	6
Observed Frequencies (O_i)	7	8	5	10	7	11
Expected Frequencies (E_i)	8	8	8	8	8	8

The χ^2 goodness-of-fit test asks whether the observed frequencies are so *significantly different* from the expected frequencies, under the model described in the null hypothesis, that there is evidence to suggest *the model is in fact incorrect*.

H_0 : The score on the die follows a uniform distribution (the die is fair).
H_1 : The score on the die follows some other distribution (the die is not fair).

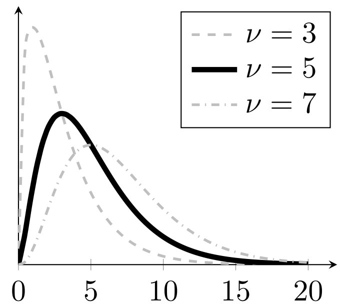
The theory underpinning the test statistic is beyond the scope of the Advanced Higher Statistics course. However, a glimpse into the rationale may be found by noting that the test statistic, on the next page, takes the *sum* of *squared differences* between the observed and expected frequencies, *scaled* by the expected frequencies.

For a χ^2 goodness-of-fit test:

Test statistic: $\chi^2 = \dfrac{(O_1 - E_1)^2}{E_1} + \dfrac{(O_2 - E_2)^2}{E_2} + \ldots + \dfrac{(O_n - E_n)^2}{E_n} = \sum \dfrac{(O_i - E_i)^2}{E_i}$

The test statistic follows a χ^2_ν distribution with degrees of freedom $\nu = c - 1 - m$, where:

- c is the number of *categories*.
- m is the number of *model parameters* estimated using the observed data (Such as estimating λ for a $\text{Po}(\lambda)$ distribution).



χ^2_ν distributions with ν degrees of freedom

The decision rule for χ^2 tests:

If **test statistic** $>$ **critical value**, *reject* H_0.

If **test statistic** $\leqslant$ **critical value**, *do not reject* H_0.

For the *"fair"* die, the test statistic can now be calculated:

$$\chi^2 = \frac{(7-8)^2}{8} + \frac{(8-8)^2}{8} + \frac{(5-8)^2}{8} + \frac{(10-8)^2}{8} + \frac{(7-8)^2}{8} + \frac{(11-8)^2}{8} = 3$$

There are six categories, so $c = 6$, and no parameters of the model were calculated using the data, so $m = 0$, leading to $\nu = 6 - 1 - 0 = 5$ degrees of freedom. This corresponds to the number of observed frequencies that can *vary independently* once the total and expected distribution are fixed, since the final observed frequency is determined once all others are known.

The test statistic can be compared against the χ^2 critical values from **page 14** of the data booklet. With $\nu = 5$ and taking $\alpha = 0.05$, the critical value of $\chi^2_{5,0.95} = 11.070$ can be found. The smallest possible observed test statistic is 0, representing data that perfectly matches the model, and for $\nu = 5$ approximately 5% of test statistics would exceed 11.070 by chance alone. Note that since the test statistic cannot be negative, and can only move further from 0 in the positive direction, the chi-squared tests covered in this course are always *one-tailed*.

Since $3 < 11.070$ the students should *not* reject the null hypothesis at the 5% level of significance. There is *insufficient evidence* to suggest that the die is not fair.

Note that whilst the χ^2 test statistic is a measure of how well data fits a model, the goodness-of-fit test does not show evidence *for* the model - it only assesses whether the data provides evidence *against* it.

Conditions for valid use of a goodness-of-fit test:

- No **expected** values should be *less than 1*.
- At least 80% of the **expected** values should be *at least 5*.
- The observations are **independent** (typically assured through *random sampling*).

The data for the teacher's die met the criteria for the use of a chi-squared test, given on **page 6** of the Data Booklet. Both conditions need to be checked before test statistics, degrees of freedom and critical values can be calculated.

Example

Problem: A chocolatier says that their boxes are randomly filled with caramel, coffee and strawberry flavoured chocolates in the ratio $2:2:1$. Suspecting that this is not true, a chocolate fan counts the number of chocolates of each flavour in a box and finds 8 are caramel, 13 are coffee and 11 are strawberry. Perform a non-parametric hypothesis test to assess the evidence at the 10% significance level that the flavours are not distributed as claimed by the chocolatier.

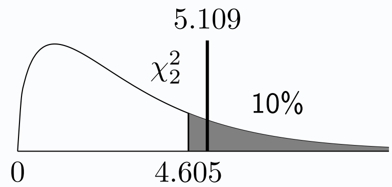
Solution:
H_0 : the chocolates' flavours are distributed in the ratio 2:2:1
H_1 : the chocolates' flavours are distributed differently

Flavour	caramel	coffee	strawberry
Observed Frequencies (O_i)	8	13	11
Expected Frequencies (E_i)	12.8	12.8	6.4

$\nu = 3 - 1 - 0 = 2$

Test statistic: $\chi^2 = \frac{(8-12.8)^2}{12.8} + \frac{(13-12.8)^2}{12.8} + \frac{(11-6.4)^2}{6.4} = 5.109$

Critical value: $\chi^2 = 4.605$



Since $5.109 > 4.605$, reject H_0 at the 10% level of significance. There is sufficient evidence to suggest that the flavours are not distributed in the ratio stated by the chocolatier.

Note that the χ^2 distributions displayed in the example solutions are illustrative of their typical characteristic shape, rather than precise graphs of each distribution matching the degrees of freedom.

Exercise 12.1

1. A fair, tetrahedral die with faces numbered 1 to 4 is rolled 24 times. The results obtained are:

Outcome	1	2	3	4
Observed Frequencies	6	1	8	9

   Perform a goodness-of-fit test at the 5% level of significance to assess whether there is evidence to support a claim that the die is biased.

2. A question in an old textbook states that a person is equally likely to be born on any day of the week. A student asks a random sample of 49 people which day of the week they were born on, and records the results.

Day	Mon	Tue	Wed	Thurs	Fri	Sat	Sun
Frequency	11	9	4	9	7	3	6

   Perform a non-parametric test at the 10% significance level to assess whether the data supports the textbook's statement.

3. A local council has previously monitored the numbers of cars entering a city, classifying each as either internal combustion engine (ICE), hybrid or fully electric. The ratio of cars in each category was found to be 5:2:1 respectively. Interested in whether this has since changed, a random sample is taken of 80 cars entering the city, with the results: 43 ICE, 16 hybrid and the remaining cars fully electric. Perform a hypothesis test at the 1% level of significance to test the claim that the ratio has changed from what it was previously.

12.2 Combining Columns

For each goodness-of-fit test encountered so far, the conditions for the **expected frequencies**, E_i, have been satisfied:

"At least 80% of the E_i should be at least 5 and none should be less than 1."

If the expected categories do not meet the conditions for a chi-squared test then two or more categories should be *combined* until the conditions are satisfied. The degrees of freedom for the test should be calculated based on the *final combined frequency table* rather than the original table. Whilst it is recommended to combine those categories with the smallest expected values, it may at times be more intuitive to instead combine categories that can be grouped more naturally.

Example

Problem: A video game allows players to buy in-game "loot boxes" with real money, each containing one in-game item. The game's publisher states that 60% of loot boxes contain a "Regular" item, 25% contain a "Premium" item, 10% contain a "Rare' item and the remaining 5% contain an "Ultra Rare" item. A player records the types of item contained in each of a series of loot boxes and records the results in the frequency table below.

Type	Regular	Premium	Rare	Ultra Rare
Observed Frequencies (O_i)	33	16	1	0

Perform a χ^2 goodness-of-fit test to determine whether there is evidence at the 1% significance level to suggest the publisher's claim is not correct.

Solution:

H$_0$: the item types are distributed as claimed
H$_1$: they are distributed differently

Type	Regular	Premium	Rare	Ultra Rare
Observed Frequencies (O_i)	33	16	1	0
Expected Frequencies (E_i)	30	12.5	5	2.5

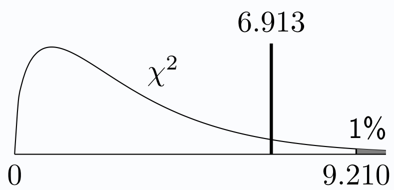
(Since only 75% of ***expected*** *values are 5 or greater, which is less than the required 80%, the last two columns must be combined.)*

Type	Regular	Premium	Rare/Ultra Rare
Observed Frequencies (O_i)	33	16	1
Expected Frequencies (E_i)	30	12.5	7.5

$\nu = 3 - 1 - 0 = 2$

Test statistic: $\chi^2 = \frac{(33-30)^2}{30} + \frac{(16-12.5)^2}{12.5} + \frac{(1-7.5)^2}{7.5} = 6.913$

Critical value: $\chi^2 = 9.210$



Since $6.913 < 9.210$, do not reject H$_0$ at the 1% level of significance. There is insufficient evidence to suggest that the item types are not distributed as stated by the game's publisher.

Exercise 12.2

1. Historically, grades attained (*A*, *B*, *C*, *D* and *No Award*) by pupils in a school course's end of year exam have been distributed following the ratio $4:5:3:2:1$. Following a change in the teaching resources used, the results of 40 students are recorded as follows:

Grade	*A*	*B*	*C*	*D*	*No Award*
Frequency	13	19	6	0	2

   Perform a chi-squared test to assess, at the 10% level of significance, whether the data provides evidence to suggest the distribution of grades attained has changed.

2. The proportion of people in the USA with each eye colour is as follows:

Eye Colour	*Brown*	*Blue*	*Hazel*	*Green*	*Other*
Proportion	0.45	0.27	0.18	0.09	0.01

   A random sample of 58 people living in San Francisco have their eye colour recorded. The results are shown in the table below:

Eye Colour	*Brown*	*Blue*	*Hazel*	*Green*	*Other*
Frequency	29	13	10	2	4

   Perform a chi-squared goodness-of-fit test to assess at the 1% significance level whether there is evidence of a difference in distribution of eye colours in San Francisco.

3. A coffee shop sells five of types of coffee. Historically, the five types of coffee have been sold in the following proportions:

   - 34% cappuccinos
   - 27% lattes
   - 18% espressos
   - 11% flat whites
   - 10% americanos

   Suspecting that these proportions may have changed, the coffee shop records the type of coffee sold for a random sample of 40 individual orders:

Cappuccinos	Lattes	Espressos	Flat whites	Americanos
7	8	7	9	9

   Perform a non-parametric test to assess at the 5% level of significance the claim that the historical proportions of each type of coffee sold are no longer accurate.

4. A sample of categorical data is to be collected to assess whether there is evidence to suggest the population is no longer distributed in the $4:3:2:1$ ratio for the four categories as previously was recorded. Determine a range of values for the sample size n such that:

   (a) A chi-squared goodness-of-fit test can be performed with exactly 3 degrees of freedom.

   (b) A chi-squared goodness-of-fit test can be performed with exactly 2 degrees of freedom.

12.3 Goodness-of-fit and Binomial Distributions

A goodness-of-fit test can also be performed to assess whether observed data fits a *binomial* distribution, in which the **total observed frequencies** will be split in proportion to the *calculated binomial probabilities*.

Example

Problem: A supermarket randomly samples boxes of duck eggs delivered by a supplier, each containing four eggs, and counts how many of the four eggs are broken for each box. The results of this are:

Eggs broken	0	1	2	3	4
Observed Frequencies (O_i)	122	10	0	4	8

The supermarket decides that the number of eggs broken in each box of four follows a binomial distribution, with $p = 0.1$. Perform a hypothesis test to assess this claim and comment on the findings with reference to the assumptions required for a binomial model.

Solution:

H$_0$: No. of eggs broken per box follows a B$(4, 0.1)$ distribution.
H$_1$: No. of eggs broken per box follows a different distribution.

Total observations $= \mathbf{144}$. Letting random variable X represent the number of eggs broken in a box:

$$\mathrm{P}(X=0) = 0.6561 \qquad \mathrm{P}(X=1) = 0.2916 \qquad \mathrm{P}(X=2) = 0.0486 \quad \cdots$$

$$0.6561 \times \mathbf{144} = 94.48 \quad 0.2916 \times \mathbf{144} = 41.99 \quad 0.0486 \times \mathbf{144} = 7.00 \quad \cdots$$

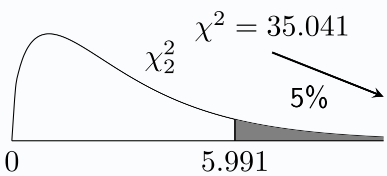
Eggs broken	0	1	2	3	4
Observed Frequencies (O_i)	122	10	0	4	8
Expected Frequencies (E_i)	94.48	41.99	7.00	0.52	0.01

(Combining the final three columns is necessary here to satisfy the conditions for expected frequencies.)

Eggs broken	0	1	2 or more
Observed Frequencies (O_i)	122	10	12
Expected Frequencies (E_i)	94.48	41.99	7.53

$\nu = 3 - 1 - 0 = 2$

Test statistic: $\chi^2 = \frac{(122-94.48)^2}{94.48} + \frac{(10-41.99)^2}{41.99} + \frac{(12-7.53)^2}{7.53} = 35.041$



Critical value: $\chi^2_{2,0.95} = 5.991$

Since $35.041 > 5.991$, reject H$_0$ at the 5% level of significance. There is sufficient evidence to suggest that the number of eggs broken per box does not follow a B$(4, 0.1)$ distribution.

A condition for a binomial model is that each trial is independent. Since broken eggs most likely occur together, this condition may not be satisfied and a binomial model is unlikely to be suitable.

Note that estimating the parameter p from *the data itself* would require an additional degree of freedom to be subtracted (since $m = 1$). For a binomial distribution, that is *estimating* the probability of success for each trial, p, by dividing the total number of successes (broken eggs) by the total number of trials (total eggs). Here:

$$\hat{p} = \frac{\text{broken eggs}}{\text{total eggs}} = \frac{1 \times 10 + 3 \times 4 + 4 \times 8}{4 \times (122 + 10 + 4 + 8)} = \frac{54}{576} = 0.09375$$

This could be used to check if the number of broken eggs per box fits *any* possible B$(4, p)$ distribution.

Exercise 12.3

1. A manufacturer of festive products sells Christmas crackers, in boxes of four. Since 20% of all crackers they produce are known to be faulty, failing to produce the required "crack" when pulled, they wish to test whether the number of faulty crackers in a box could be modelled using a binomial distribution.

   (a) State the distribution of X, representing the number of faulty crackers in a box.

   (b) Calculate:

   i. $\mathrm{P}(X=0)$ ii. $\mathrm{P}(X=1)$ iii. $\mathrm{P}(X=2)$ iv. $\mathrm{P}(X=3)$ v. $\mathrm{P}(X=4)$

   The manufacturer tests 60 boxes of crackers, and records the number of faulty crackers in each box, with the results shown in the table below:

No of Faulty Crackers	0	1	2	3	4
Frequency	34	17	4	2	3

   (c) Use a suitable hypothesis test to assess at the 10% level of significance whether the use of a binomial distribution is supported by the data.

2. A school's mathematics department runs a short multiple choice festive revision quiz, which requires students to choose the correct answer from one of four options. The quiz contains five questions in total.

   Teachers in the department suspect that, instead of carefully working using their knowledge to work out the answers to the mathematical questions, the students are simply guessing each question at random.

   (a) State the distribution of X, the number of correct answers out of 5, assuming that a student is guessing at random for every question.

   To investigate these, the number of correct answers out of five for each of the 80 students is first set out in a table:

No. of Correct Answers	0	1	2	3	4	5
Frequency	9	1	29	25	10	6

   (b) Perform a chi-squared test to determine if the teachers' suspicions are backed by the data.

3. A car mechanic has established that the proportion of tyres with tread that is below the legal minimum, for cars coming in for a service, is 15%. They are interested in whether the number of the four tyres on a car brought in for a service that have less than the legal minimum tread, and thus need to be replaced, may be modelled using a binomial distribution: $X \sim \mathrm{B}(4, 0.15)$.

   The number of tyres needing replaced for low tread for a random sample of cars brought in for a service are recorded:

No. of Tyres	0	1	2	3	4
Frequency	41	20	8	1	2

   (a) Perform a chi-squared test to assess whether the binomial distribution may be suitable, at the 5% significance level.

   (b) If the proportion of faulty tyres had not been known, and an estimate had instead been obtained using the data, explain how this would affect the calculation of the degrees of freedom, ν , for the test.

12.4 Goodness-of-fit and Poisson Distributions

Just as with a binomial distribution, using a Poisson distribution as a model for a goodness-of-fit test requires the total observations to be split according to calculated probabilities. If sample data is to be tested against a fully specified Poisson distribution (e.g. Po(4.5)) then no model parameter has been estimated using the data $(m = 0)$.

Should the only parameter for a Poisson distribution, λ, be required to be *estimated from the data*, an additional degree of freedom will be *subtracted*, since $m = 1$. This would arise when looking to compare the observed data against *any* Poisson distribution Po(λ).

In the following example, a value for the model parameter λ is not provided, and is instead estimated from the data, so here $m = 1$.

Example

Problem: It is generally said that the number of goals scored in a game of football follows a Poisson distribution. A football fan decides to see whether this statement is still reasonable, and records the number of goals scored in every game across a weekend in each of the top four tiers of English football:

Goals	0	1	2	3	4	5	6	7 or more
Observed Frequencies (O_i)	4	8	13	9	8	2	2	0

Perform a non-parametric test at the 1% level of significance to assess whether there is evidence to suggest the model is no longer valid.

Solution: $\lambda = \frac{\text{total no. of goals}}{\text{total no. of games}} = \frac{115}{46} = 2.5$

H$_0$: No. of goals per game follows a Poisson distribution (with $\lambda = 2.5$)
H$_1$: No. of goals per game follows a different distribution.

Total observations$= 46$.
P$(X = 0) = 0.0821$
Expected frequency for 0 goals $= 0.0821 \times 46 = 3.777$

Goals	0	1	2	3	4	5	6	7 or more
Observed Frequencies (O_i)	4	8	13	9	8	2	2	0
Expected Frequencies (E_i)	3.777	9.439	11.799	9.835	6.146	3.073	1.279	0.652

(Combining the final three columns is necessary here to satisfy the conditions for expected frequencies.)

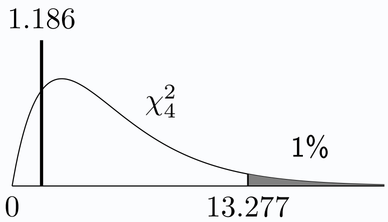
Goals	0	1	2	3	4	5 or more
Observed Frequencies (O_i)	4	8	13	9	8	4
Expected Frequencies (E_i)	3.777	9.439	11.799	9.835	6.146	5.004

(Since λ *was calculated using the data, subtract an additional degree of freedom.)*

$\nu = 6 - 1 - 1 = 4$

Test statistic: $\chi^2 = \frac{(4-3.777)^2}{3.777} + \ldots + \frac{(4-5.004)^2}{5.004} = 1.186$

Critical value: $\chi^2 = 13.277$



Since $1.186 < 13.277$, do not reject H$_0$ at the 1% level of significance.
There is insufficient evidence to suggest that the number of goals scored per football game does not follow a Poisson distribution, so the model still seems reasonable.

Exercise 12.4

1. A keen amateur astronomer recalls hearing during a statistics class that shooting stars can be seen in a local section of night sky at a mean rate of 1 per hour, and the number visible in an hour is Poisson distributed. Over the course of a year they carefully watch the night sky on clear nights, and record the number of shooting stars they see within each hour. The data they obtain for the total of 70 hours spent watching for shooting stars is shown in the table below:

Number of Shooting Stars	0	1	2	3 or more
Frequency	23	20	15	12

   (a) If the number of shooting stars visible in a section of sky in an hour follows the distribution described in the statistics class, calculate the probability of seeing, in any given hour:
      i. No shooting stars.
      ii. Exactly one shooting star.
      iii. Exactly two shooting stars.
      iv. Three or more shooting stars.

   (b) Use a non-parametric test to determine whether there is evidence at the 5% significance level that the distribution stated during the statistics class is not correct.

2. A restaurant had generally understood that the number of bottles of wine sold on any given night could be modelled by a Poisson distribution with parameter λ. Suspecting this is no longer the case, the restaurant recorded the number of bottles sold on 50 randomly selected nights. The table below shows the data obtained:

Number of Bottles	0	1	2	3	4	5 or more
Frequency	2	13	19	15	1	0

   (a) Calculate a suitable estimate for the value of λ.

   (b) Use a hypothesis test to show that there is evidence at the 1% level of significance to suggest that a Poisson distribution is no longer an appropriate model for the number of bottles of wine sold.

3. It is often stated that the number of goals scored in a game of football follows a Poisson distribution with a parameter of 2.5. To explore this, a sports statistician obtains the relevant data for all 380 games of the 2023-24 English Premier League season:

Number of Goals	0	1	2	3	4	5	6	7	8	9 or more
Frequency	11	42	81	80	81	48	24	10	3	0

   (a) Perform a chi-squared test to assess whether the data suggests a Poisson model with a parameter of 2.5 is suitable.

   It is subsequently claimed that a Poisson model is still appropriate, but that the value of the parameter has changed.

   (b) Calculate an estimated value for λ using the data, to 2 decimal places.

Review Exercise

1. The population of the United Kingdom can be approximately split between those living in England, Scotland, Wales and Northern Ireland in the ratio $30 : 3 : 2 : 1$. Out of a random sample of 117 people who applied to take part in a UK-wide television show, 102 lived in England, 4 lived in Scotland, and 1 lived in Northern Ireland.

   Perform a non-parametric test to assess whether there is evidence to suggest the distribution of people applying for the TV show differs from that of the wider UK population, at the 5% level of significance.

2. A football team practises for the possibility of a penalty shootout each training session by having their five chosen penalty takers take one penalty each. They know that 15% of all penalties taken as part of this training will result in a failure to score. Over the course of a season, the number of *failed attempts* for each of the 60 sets of five penalties is recorded:

Number of Failed Attempts	0	1	2	3	4	5
Frequency	33	12	7	5	3	0

   (a) Perform a test at the 10% significance level to assess the evidence that the data fits a binomial distribution where $p = 0.15$.

   (b) A coach comments that some of the five penalty takers are clearly better at taking penalties than others. Explain why this would mean that a binomial would be inappropriate.

3. A supermarket chain is monitoring concerns regarding shoplifting in one of its stores. It is proposed that a Poisson distribution may be used to model the number of incidents of shoplifting recorded in any given day.

   The table below shows *observed* frequencies of the numbers of incidents recorded per day across a period of time, with some *expected* frequencies calculated using a Poisson distribution estimating its parameter from the observed data.

Incidents	0	1	2	3	4	5	6 or more
Observed	19	37	27	10	6	1	0
Expected	22.31	33.47		12.55	4.71	1.41	

   (a) Calculate the mean of the Poisson distribution used.

   (b) Determine the missing expected values.

   (c) Carry out a non-parametric test at the 1% significance level to determine whether a Poisson distribution is an appropriate model for the data.

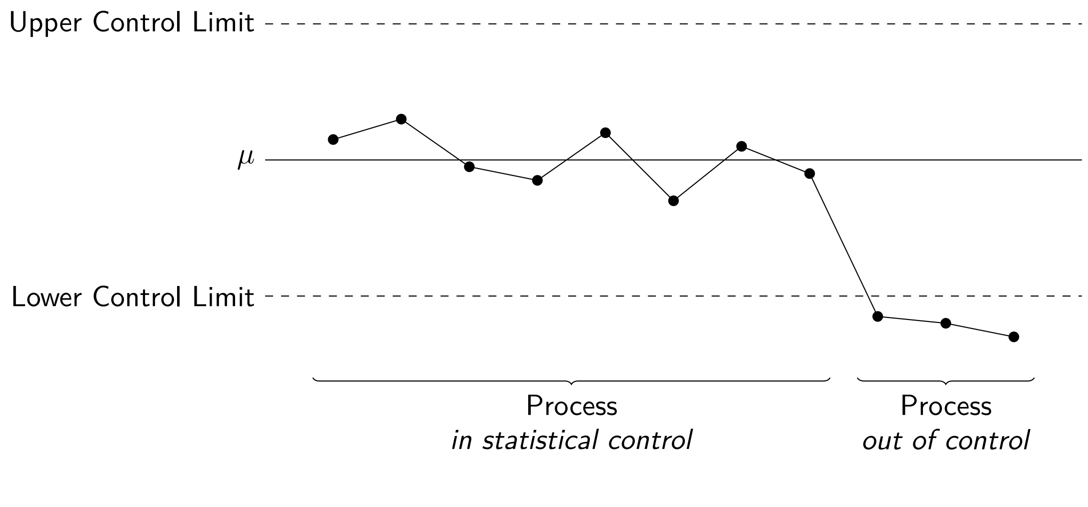
   It is suggested that the same mean number of incidents of shoplifting per day obtained in part (a) was already known to be the appropriate value to use for the Poisson distribution.

   (d) Had this been known before the test was conducted, state what the test statistic and critical value would have been.

13
Control Charts

In any manufacturing process, performance must be monitored to ensure that the products being manufactured are not faulty or that the process remains stable. A common tool used to monitor performance is a *control chart*, in which a series of samples are taken over time, measured and plotted against calculated limits to detect unusual variation.

A simplified control chart:



Variation naturally occurs as a result of random causes in most processes, referred to as **common cause variation**, and often there is no need for this to be investigated or corrected. This could be, for example, because the variation is within tolerable limits, or the cost of reducing the variation would outweigh the potential benefit of doing so. A process performing in this manner is said to be *in control*.

However, if something occurs within the process that results in variation that is not consistent with the typical random variation of the process, a system, such as a control chart, is used to signal that the process may be *out of control*. The process can then be reviewed and any possible **special cause variation** investigated, such as machine wear and tear, poor raw materials, poorly trained operatives, or an accident or malfunction.

The Principle of a Control Chart

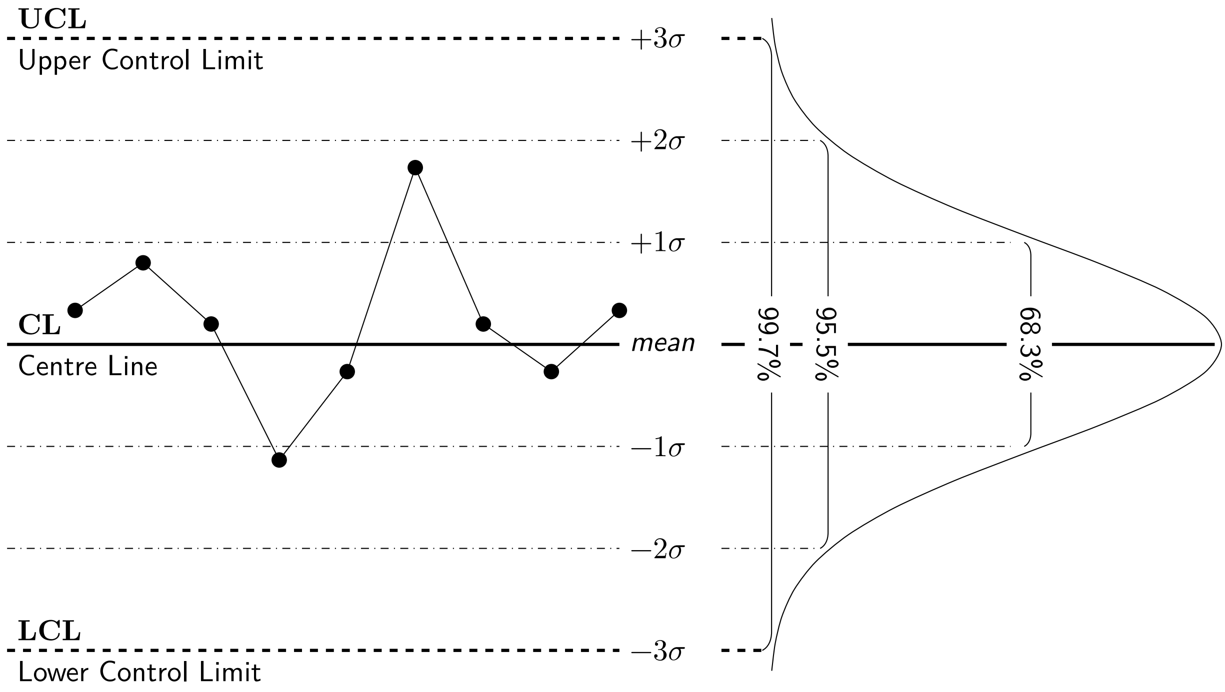
The control chart was invented in the 1920s by Walter Shewhart, who was working for a communications company whose engineers wanted to improve the reliability of their telephone transmission systems. His theory was based around the bell-shaped curve of the normal distribution.

The Advanced Higher Statistics course covers two types of control charts:

- A Shewhart x-bar chart for the sample **mean**.
- A Shewhart p-chart for the sample **proportion**.

In either case, random samples from the process are taken at regular intervals and the sample means or sample proportions are plotted onto the control chart. A line on the chart is drawn to indicate the target value for the population mean or population proportion, as well as 1-sigma , 2-sigma and 3-sigma lines either side of the target value where σ represents the population standard deviation of the process.

The 3-sigma (3σ) limits are often referred to as the *upper* and *lower control limits*, with the other limits called *warning lines*. The diagram below indicates the probability that a sample statistic falls within each set of limits, assuming the process is *in control* and certain conditions are met.



The Western Electric Company Rules (*WECO Rules*) are commonly used to determine whether a process is *in or out of statistical control*, and are provided on Page 4 of the Data Booklet.

A process may be considered out of statistical control using the WECO Rules when:

- *Any* single data point falls outside a 3σ limit.
- *Two out of three* consecutive points fall beyond the *same* 2σ limit.
- *Four out of five* consecutive points fall beyond the *same* 1σ limit.
- *Eight consecutive* points fall on the *same side of the centre line*.

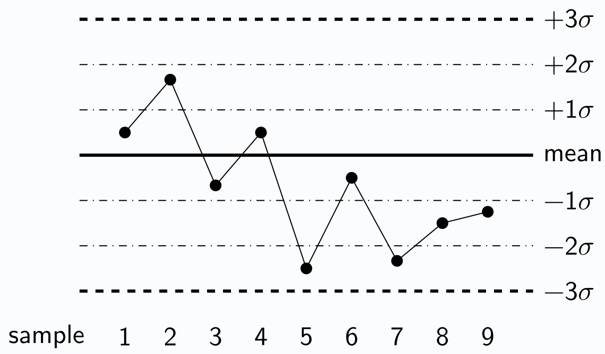
These rules help detect patterns that are unlikely to occur due to random variation alone.

13.1 Applying the WECO Rules

Each rule must be carefully checked for to determine whether a process is *in statistical control*. If it is *not* in statistical control, it is often necessary to identify the relevant WECO rule and at which point this could be declared. The following examples focus on context-free control charts for the purpose of practising *applying the WECO rules*.

Example

Problem: For the control chart below, determine whether the process is in control. If it is out of control, identify at which sample this could be determined, with justification.

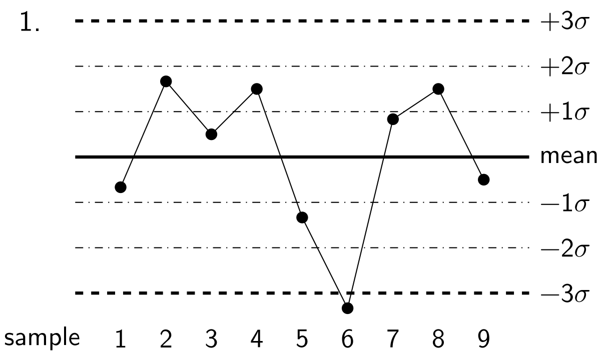


Solution: The process is out of control, as two out of the three consecutive points (from samples 5 to 7) lie below the lower 2-sigma limit. The process could have been identified as out of control at sample 7.

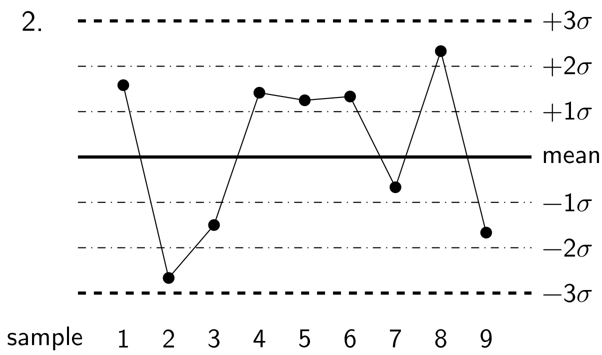
Exercise 13.1

For each control charts below, determine whether the process is in control, with justification.

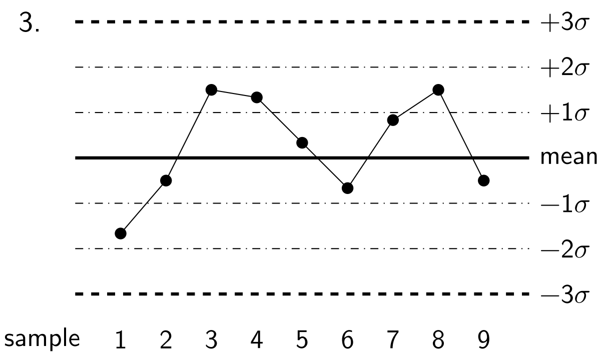
1.



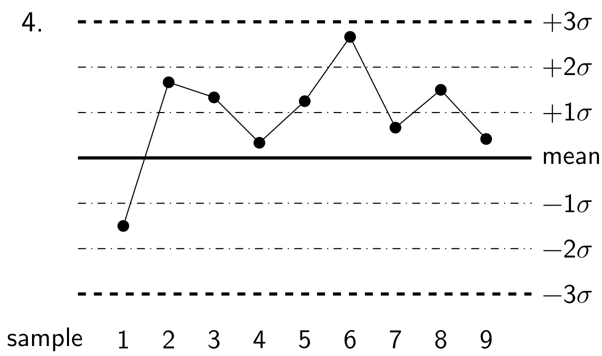
2.



3.



4.



13.2 Shewhart *x*-bar Charts for the Mean

Suppose that a manufacturing process is working properly and the output of the process is normally distributed, with mean μ and standard deviation σ. If samples of size n are taken regularly from the product, and the mean obtained for each sample, then those sample means will themselves also be normally distributed:

$$\overline{X} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

The standard error of each sample mean obtained is given by $\dfrac{\sigma}{\sqrt{n}}$, and the sigma limits are given by:

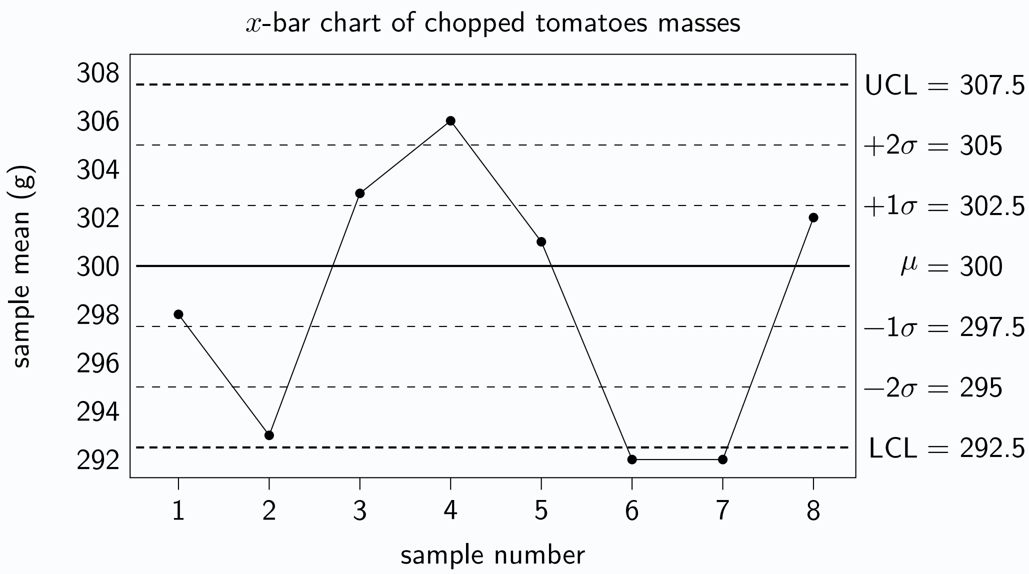
$$\underbrace{1\sigma \text{ limits: } \mu \pm \frac{\sigma}{\sqrt{n}} \qquad 2\sigma \text{ limits: } \mu \pm 2\frac{\sigma}{\sqrt{n}}}_{\textit{Warning lines}} \qquad \underbrace{3\sigma \text{ limits: } \mu \pm 3\frac{\sigma}{\sqrt{n}}}_{\textit{UCL and LCL}}$$

A control chart is created with horizontal lines indicating the population mean and each of the 6 sigma limits calculated. The sample mean for each of the samples drawn at regular intervals is plotted on the chart, and the *WECO rules* applied to determine if or when the production process is out of control.

Example

Problem: A machine fills cans with mean mass 300 grams and standard deviation 5 grams of chopped tomatoes. Every hour a sample of 4 cans is taken, the masses of their contents measured, and the sample mean calculated before being added to an x-bar chart.

Sample number	1	2	3	4	5	6	7	8
Sample mean	298	293	303	306	301	292	292	302



(a) Confirm the 2-sigma limits shown on the control chart.

(b) Determine when the process is out of control.

Solution:

(a) 2σ limits: $\mu \pm 2\dfrac{\sigma}{\sqrt{n}} = 300 \pm 2 \times \dfrac{5}{\sqrt{4}} = 300 \pm 5$, so 295 and 305.

(b) The process is out of statistical control, which can be identified at sample 6 when the sample mean first falls below the lower control limit (-3σ limit). The process should be stopped and investigated.

Exercise 13.2

1. Control charts are to be produced for sample means taken to monitor the following processes. If the samples are random, and it can be assumed that the processes can be suitably described by the normal distribution, copy and complete the following tables showing limits for the x-bar charts:

(a) Population mean = 80
Population standard deviation = 3
Sample size = 16

Upper Control Limit	
$+2\sigma$ Limit	
$+1\sigma$ Limit	
Target Value	
-1σ Limit	
-2σ Limit	
Lower Control Limit	

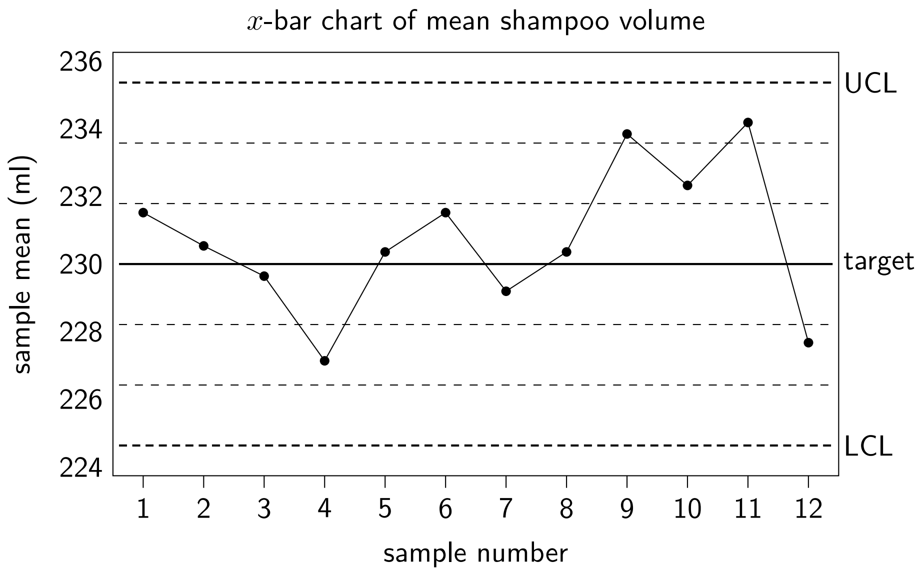
(b) Population mean = 486
Population standard deviation = 17
Sample size = 12

Upper Control Limit	
$+2\sigma$ Limit	
$+1\sigma$ Limit	
Target Value	
-1σ Limit	
-2σ Limit	
Lower Control Limit	

2. A factory produces plastic guitar plectrums which are sold with a labelled thickness of 1.4mm. The actual thickness of plectrums produced by the factory's equipment is normally distributed with a mean of 1.4mm and standard deviation 0.1mm. An x-bar chart is to be produced to monitor the performance of the process. Calculate the 2-sigma limits for the control chart.

3. A factory produces plastic guitar plectrums which are sold with a labelled thickness of 1.4mm. The actual thickness of plectrums produced by the factory's equipment is normally distributed with a mean of 1.4mm and standard deviation 0.1mm. An x-bar chart is to be produced to monitor the performance of the process, using random samples of size 18. Calculate the 2-sigma limits for the control chart.

4. A cosmetic company has a machine which dispenses shampoo into bottles. The machine has been calibrated to dispense a mean volume of 230ml into each bottle, and the amount dispensed has been found to be normally distributed with a standard deviation of 4ml. To monitor the performance of the machine, a sample of five bottles is taken each hour, the volume of shampoo measured and an x-bar chart created.



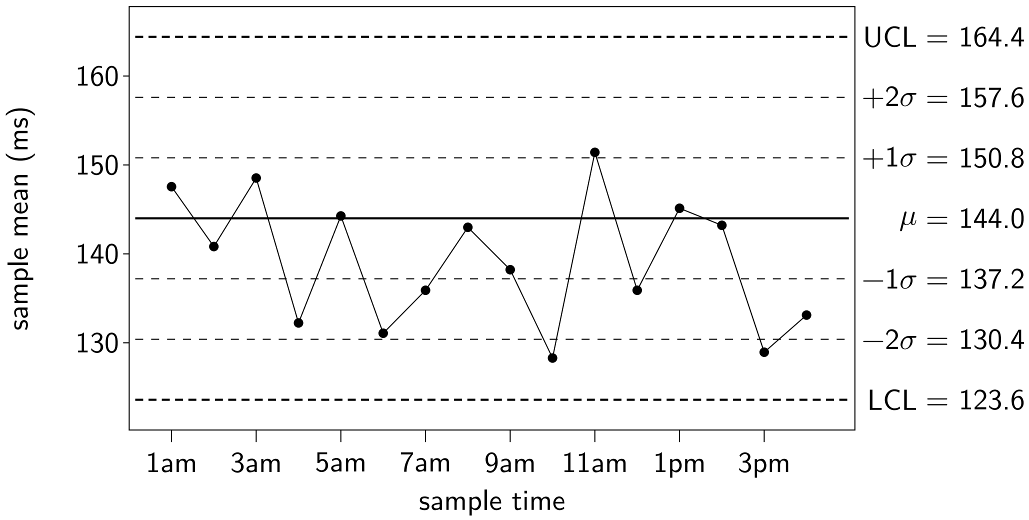
(a) Calculate the 1-sigma limits for the x-bar chart.

(b) Determine if the machine is under statistical control during this period.

(c) State an assumption required for the described use of the x-bar chart to be appropriate.

5. A video game producer monitors the responsiveness of the connections it provides to online gamers by measuring the amount of delay they experience, called their *ping*. This is measured in milliseconds and it is known that the amount of *ping* experienced by gamers is normally distributed with a mean of 144 milliseconds and a standard deviation of 18 milliseconds. Every hour, the ping is recorded for a random sample of 7 gamers, and the results added to an x-bar control chart, shown below.

control chart for ping



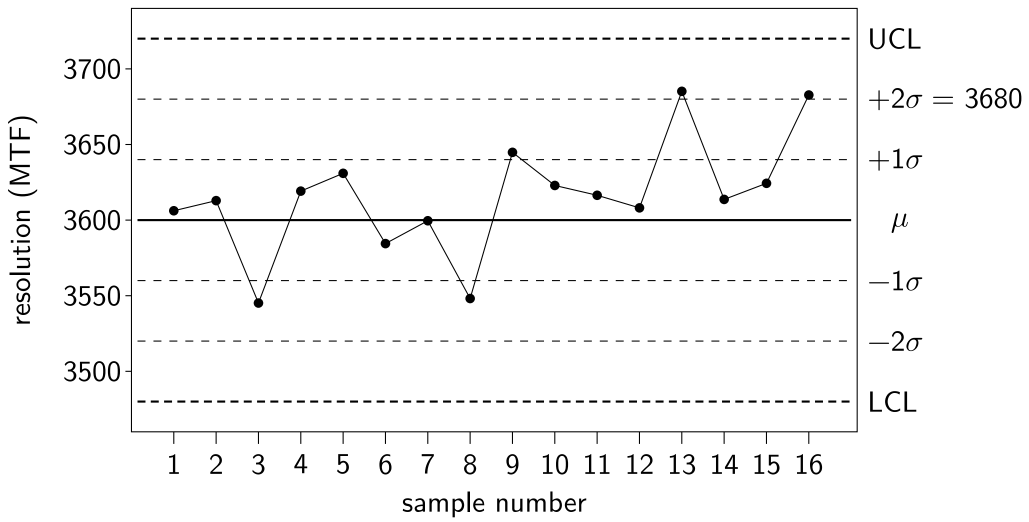
(a) Show the calculation required to obtain the lower control limit shown on the chart.

It can be determined from the control chart that the process is under statistical control for the duration of the times shown, since none of the WECO Rules have been broken.

(b) State the range of possible values for the next sample mean, $\bar{x}$, to be taken at 5pm, such that the process would remain in statistical control.

6. A camera lens manufacturer uses an x-bar control chart to monitor the quality of copies of a particular model of lens which it produces. The resolution of the lenses are measured using *MTF testing*, where a higher score indicates a higher resolution, which is desired. The manufacturer knows that the resolution of copies of the lens it produces varies, with a mean of 3600 and standard deviation of 120. Samples are taken at regular intervals, and their sample means plotted on the chart are below.

x-bar chart for lens resolution



(a) Describe the feature of the x-bar chart which means the process is out of statistical control, and explain why the lens manufacturers may not be concerned about this in practice.

(b) Use the 2σ-limit shown to determine the size of each sample of lenses taken.

(c) State two assumptions required for the use of the x-bar chart by the manufacturers.

7. A coffee company sells canned, chilled 'espresso' drinks in supermarkets. The amount of caffeine contained in each can is known to be normally distributed with a mean of 115mg and a standard deviation of 8mg. To comply with legal requirements, the company uses an x-bar control chart to monitor the manufacturing process, taking a random sample of 16 cans each hour and measuring the amount of caffeine they contain, recording the sample mean each time. The results of a series of 10 consecutive samples is shown in the table below.

Sample	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Caffeine (mg)	113.7	119.7	118.6	120.9	114.6	113.9	110.4	113.1	116.5	114.2

(a) Calculate each of the control limits and hence explain why it can be identified that the process is out of statistical control at sample 4.

The coffee company has created a new 'chilled cappuccino' drink, also sold in cans, and wishes to use an x-bar control chart to monitor caffeine content of the cans produced by the manufacturing process. Since at this stage the population mean caffeine content is unknown, its value must be estimated from the *mean of the sample means* of the first batch of sample measurements, known as 'x bar bar' ($\bar{\bar{x}}$), as is common when a control chart for a manufacturing process is being set up for the first time.
The results of the first 20 samples are shown in the table below.

Sample	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Caffeine (mg)	96.3	92.1	97.4	89.8	98.0	92.4	90.5	90.2	90.6	95.2
Sample	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
Caffeine (mg)	92.1	90.3	94.4	91.7	88.9	91.4	90.4	88.3	90.7	89.3

(b) Determine $\bar{\bar{x}}$, an estimate for the population mean caffeine content of the production process.

(c) Given that the population standard deviation is subsequently determined to be 6mg, determine the values of the upper and lower control limits for samples of size 4.

8. Control charts are to be produced for sample means taken to monitor the following processes. If the samples are random, and it can be assumed that the processes can be suitably described by the normal distribution, copy and complete the following tables for the x-bar charts:

(a)

Population mean	
Standard deviation	
Sample size	25
Upper Control Limit	
$+2\sigma$ Limit	78
$+1\sigma$ Limit	
Target Value	
-1σ Limit	
-2σ Limit	66
Lower Control Limit	

(b)

Population mean	2800
Standard deviation	60
Sample size	
Upper Control Limit	
$+2\sigma$ Limit	
$+1\sigma$ Limit	
Target Value	
-1σ Limit	
-2σ Limit	2760
Lower Control Limit	

13.3 Shewhart *p* Charts for a Proportion

For some quality control processes the attribute of interest will be a *quality* rather than a numerical value, hence calculating a mean is not possible. For such cases, a control chart for x-bar is not appropriate. Such a *quality* typically takes the form of a binary outcome, such as *defective* or *not defective*. The *proportion* of products within each random sample (of size n) that take a particular outcome (such as *defective*) may instead be calculated, and a control chart for proportion, or p-chart, created.

For $np > 5$ and $nq > 5$ the sampling distribution of the sample proportion may be described by $\hat{P} \approx \mathrm{N}\left(p, \frac{pq}{n}\right)$.

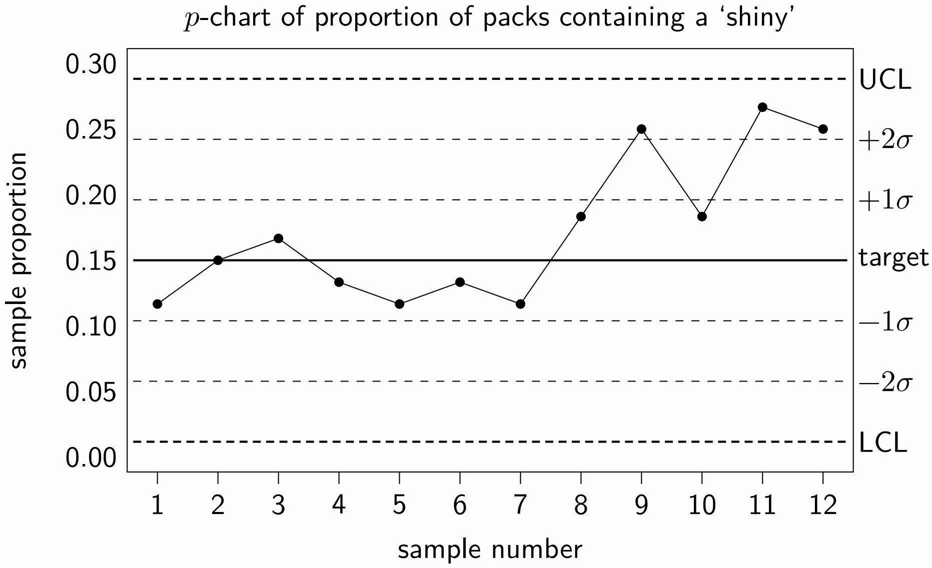
This leads to the 3-sigma control limits for a p-chart, for population proportion p, of $p \pm 3\sqrt{\frac{pq}{n}}$.

Similarly, the 1σ and 2σ limits can be calculated respectively as $p \pm \sqrt{\frac{pq}{n}}$ and $p \pm 2\sqrt{\frac{pq}{n}}$.

It should be noted that, depending on the value of p, it is not unusual for the lower control limit to be negative or the upper control limit to be greater than 1. When this happens the lower control limit should be given as 0 or the upper control limit given as 1.

Example

Problem: A machine produces packs of football stickers. 15% of the packs are supposed to contain a *'shiny'* sticker. Every 30 minutes a sample of 60 packs are taken and the number containing a 'shiny' is counted, with the proportion for each sample plotted on a p-chart.



Calculate the 3σ limits and determine if the p-chart is in statistical control for the samples given.

Solution:

3σ limits: $p \pm 3 \times \sqrt{\frac{pq}{n}} = 0.15 \pm 3 \times \sqrt{\frac{0.15 \times 0.85}{60}} = 0.15 \pm 0.1383$, so 0.0117 and 0.2883.

The process is out of statistical control at sample 11 as two out of three consecutive sample proportions (from 9 to 11) lie above the upper 2σ limit. The process needs to be stopped and investigated.

If no historical value of the population proportion has been suggested, then it may be appropriate to calculate an overall sample proportion $\widehat{p}$ from initial data, to be used as an estimate for the population proportion p.

Exercise 13.3

1. Control charts are to be produced for sample proportions taken to monitor the following processes. If the samples are random, copy and complete the following tables showing limits for the p-charts:

(a) Population proportion = 0.6
Sample size = 24

Upper Control Limit	
$+2\sigma$ Limit	
$+1\sigma$ Limit	
Target Value	
-1σ Limit	
-2σ Limit	
Lower Control Limit	

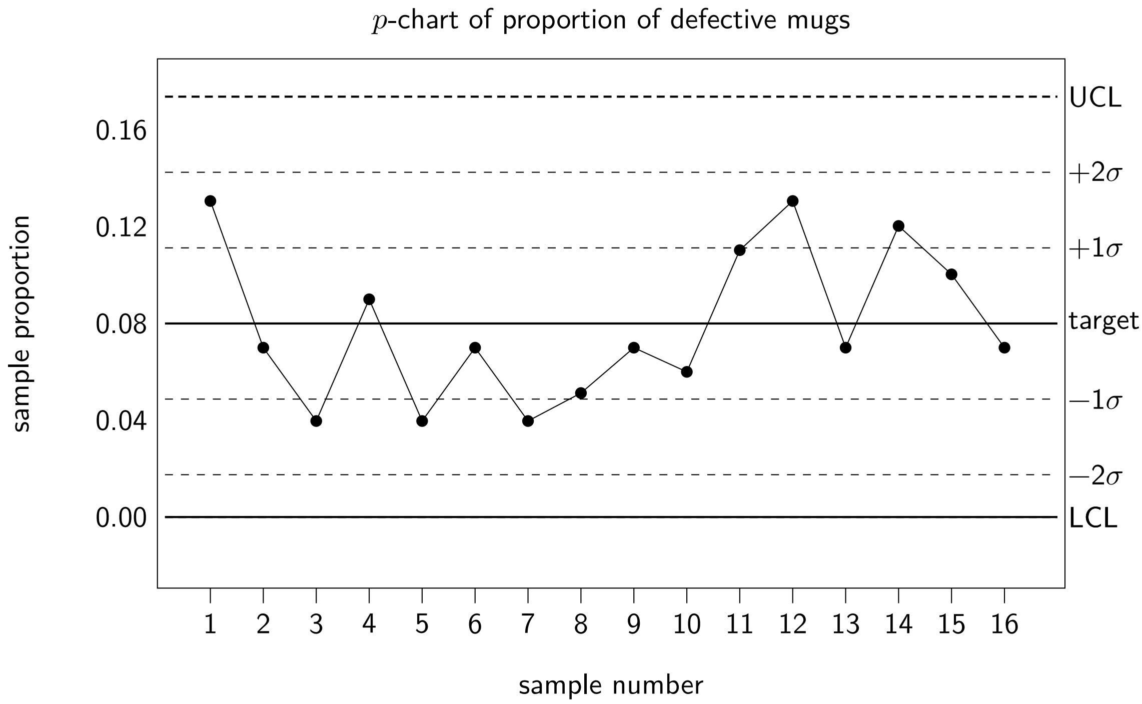
(b) Population proportion = 0.16
Sample size = 50

Upper Control Limit	
$+2\sigma$ Limit	
$+1\sigma$ Limit	
Target Value	
-1σ Limit	
-2σ Limit	
Lower Control Limit	

2. A manufacturer of kitchenware knows that 8% of the mugs produced by one of its factory lines do not meet the quality threshold. It has been determined that taking steps to reduce this proportion is not cost effective, but they want to monitor this proportion of 'defective' mugs using a control chart. One supervisor suggests checking the first 30 mugs each shift, calculating the proportion of 'defective' mugs each time and monitoring the results using a p-chart.

(a) Give two reasons why the supervisor's suggestion would not be suitable for creating a p-chart.

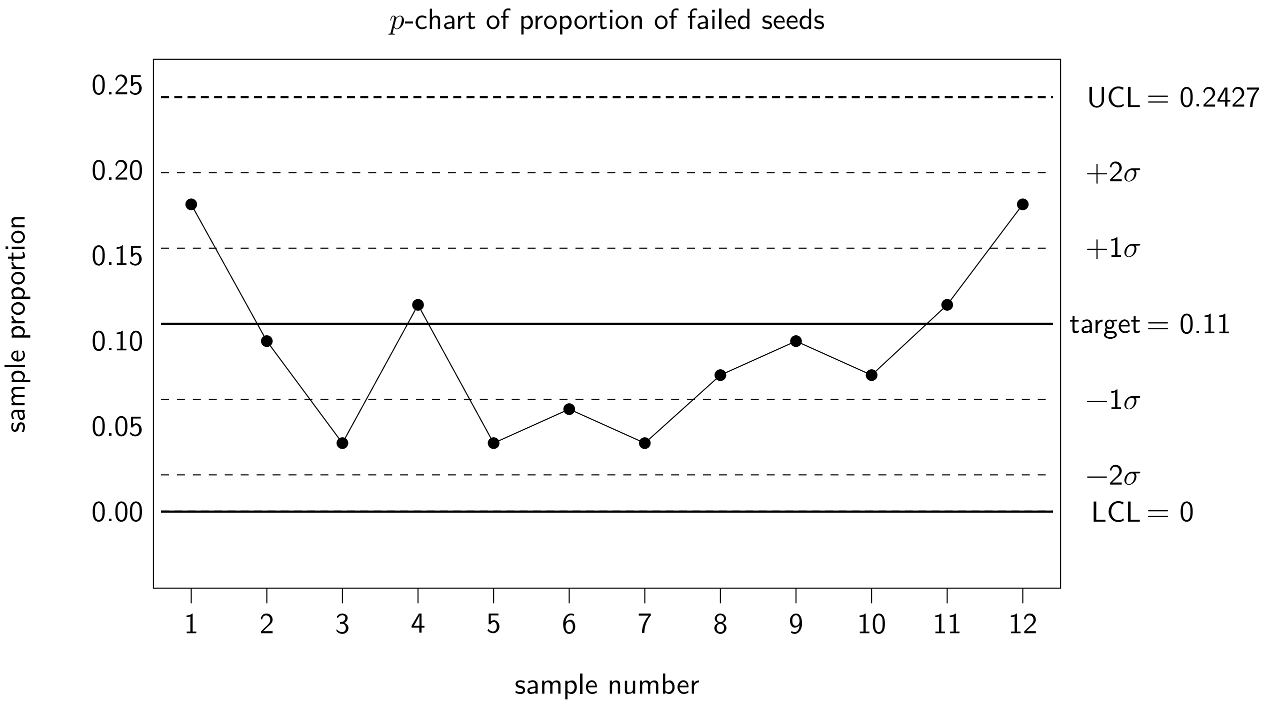
The manufacturer decides to take samples of 75 mugs each shift, and the results after the first week of monitoring are shown on the p-chart below.



(b) Calculate the 2-sigma limits for the p-chart.

(c) Determine whether the manufacturing process is in statistical control.

3. A company selling seeds for a commonly grown garden flower wishes to use a control chart to monitor the quality of its process. It is known that 11% of the seeds will fail to germinate even under ideal conditions. To monitor the rate of seeds that fail to germinate, the company randomly selects 50 seeds from each weekly batch and sows them individually in ideal conditions, recording the proportion of seeds which fail to germinate for each sample on a p-chart. The results of 12 such consecutive samples are shown below.



(a) Show the calculation required to determine that the upper control limit is 0.2427.

(b) Explain the use of the lower control limit of 0.

(c) State how it can be observed that the process is out of statistical control.

(d) Explain why the company will likely not be concerned about this.

4. A car insurance provider sets a target for 80% of claims to be processed and resolved within two weeks. To monitor this, each month 60 claims are randomly selected to be tracked, and the proportion of these claims resolved within two weeks recorded on a p-chart. Below is the data for 2024.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
$\hat{p}$	0.817	0.783	0.75	0.683	0.767	0.717	0.733	0.783	0.7	0.833	0.8	0.767

(a) Determine the value of the upper and lower control limits for the p-chart.

(b) Without performing any further calculations, explain why the data for 2024 shows that the processing and resolving of claims is out of statistical control.

After working on staff training and the efficiency of its systems, the car insurance provider is able to bring its proportion of claims resolved within two weeks up to 83%, where it remains steady throughout the year. For 2026, they wish to reduce the number of claims to sample each month, whilst still having a sufficient size for appropriate use of a p-chart.

(c) Determine the minimum sample size required for a p-chart with a population proportion of 83%.

5. A factory uses a machine to insert rubber corks to seal bottles of wine. When this process fails, the bottle is diverted back into the line of bottles waiting to be sealed, slowing down the process. The factory wishes to set up a p-chart to monitor the proportion of bottles which fail to be sealed on the first attempt, especially since an increase may suggest the machine is in need of additional maintenance. Each day over the course of two weeks, a random sample of 50 bottles are tagged and monitored, and the number of bottles for each sample which fail to be sealed on the first attempt is recorded. A table showing the results is shown below.

Week 1	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Failures	8	5	7	5	6	6	4

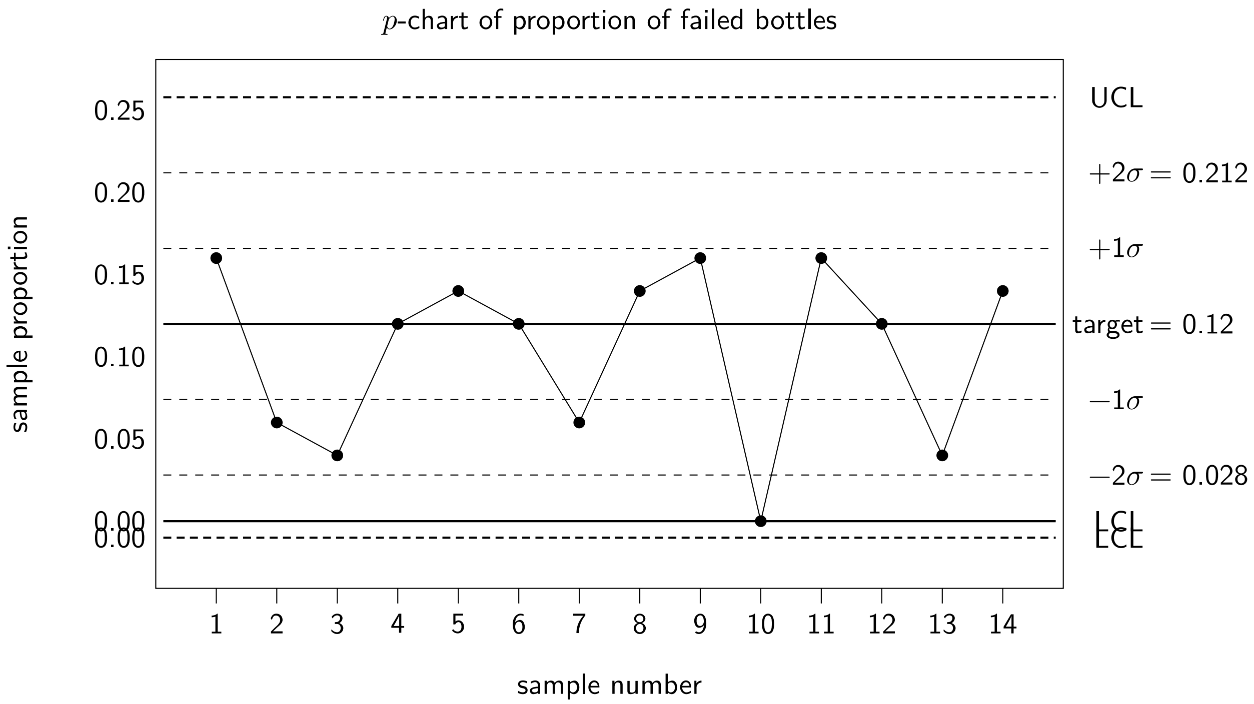
Week 2	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Failures	7	8	4	5	7	6	6

It is decided that this data is sufficient to allow a p-chart to be created to monitor the proportion of bottles which fail to be sealed by the machine, taking the total sample proportion $\hat{p}$ for the two weeks as a reliable estimator for the population proportion, p.

(a) Calculate the value of $\hat{p}$ from the table above, to be used as an estimator for p.

(b) Show the calculations required to obtain 2-sigma limits for the p-chart of 0.028 and 0.212.

A p-chart is constructed and data from the next two weeks is plotted on the chart.



(c) Determine whether the process is in statistical control during the two week period.

6. A p-chart is created using regular random samples.
The 2-sigma limits are calculated as 0.4 and 0.8.
Determine:

7. A p-chart is created using regular, random samples. The 2-sigma limits are calculated as 0.4 and 0.8.

(a) State the value of the population proportion, p.

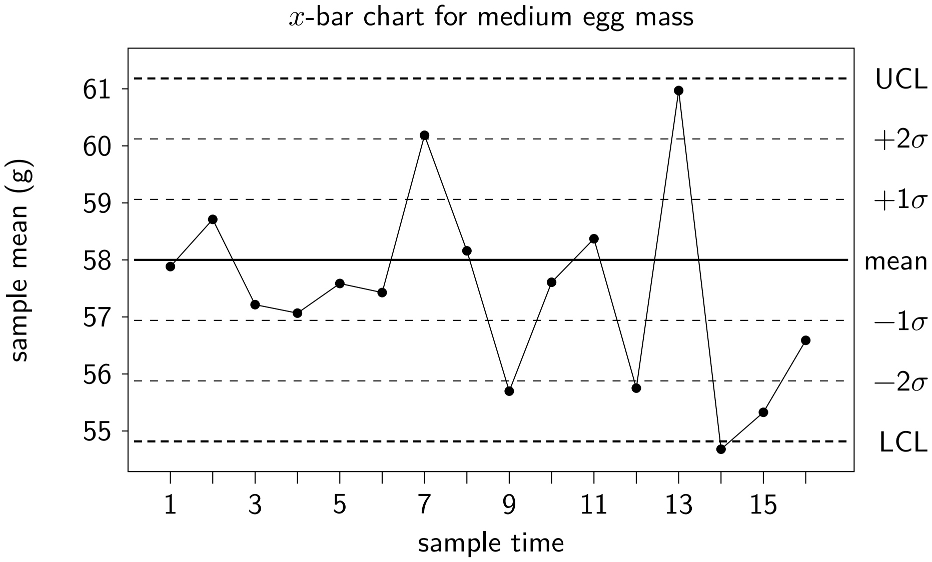
(b) Calculate the upper and lower control limits.

(c) Determine the size of each sample.

(d) Show that the sample size and the proportion meet the requirements for constructing a p-chart.

Review Exercise

1. Eggs are grouped according to size by a machine, and the eggs sorted into the *medium* category are known to have a mean mass of 58 grams with standard deviation 3 grams. To monitor the performance of the machine, eight eggs are randomly selected every fifteen minutes and the mean mass for each sample recorded on an x-bar chart, shown below.



   (a) Calculate the 2-sigma limits for the x-bar chart

   (b) Determine whether the process is under statistical control for the 16 samples shown.

2. A large company which employs tens of thousands of people uses p-charts to monitor its employee satisfaction. Each month, 50 employees are selected at random and asked to complete a short questionnaire, including being asked to rate their current satisfaction with their job. It is known that historically 88% of employees respond that they are either *satisfied* or *very satisfied*. The results of the last 14 months data is below, which is found to be *under statistical control*.

   (a) Calculate the 1-sigma limits for the control chart.

   (b) Given that the process remains to be in statistical control after the next month's data, state the range of possible values for the sample proportion $\hat{p}$ obtained from sample 15.

14

Chi-Squared Test for Association

Categorical variables can be described as *independent* if knowledge about one of the variables reveals nothing about the other. Another way of saying this is that there is **no association** between the variables. For example, knowing a person's *favourite colour* reveals nothing about whether or not they are *left-handed or right-handed*, and so there is no association between someone's favourite colour and their handedness.

On the other hand, knowing whether or not someone *enjoys camping* is likely to help in predicting whether or not they *own a tent*, and so these variables are *not independent*. Another way of saying this is that there **is an association** between enjoying camping and owning a tent.

Suppose a researcher wishes to investigate whether there is a link between someone's *handedness* and whether they have colour vision deficiency (*CVD*), often known as *colour-blindness*. Data is obtained for a random sample of 800 people, and the data is displayed in a *contingency table*:

Observed	CVD	No CVD
Left-handed	4	76
Right-handed	58	662

The contingency table shows that 5% (4 out of 80) of left-handed people had colour-blindness, whilst approximately 8.1% (58 out of 720) of right-handed people had colour-blindness. However, this difference in proportions may have occurred due to an association between colour-blindness and handedness, or through *random chance*.

A hypothesis test is required to assess whether the data in the contingency table gives sufficient evidence to suggest that there is in fact an association between handedness and colour-blindness.

The hypotheses for this test will be as follows:

H_0 : There is **no association** between *handedness* and *colour-blindness*.

H_1 : There is an **association** between *handedness* and *colour-blindness*.

14.1 χ^2 Test for Association

The χ^2 test for association is used to explore a possible *association* between *two categorical variables*, the data for which are typically displayed in a **contingency table**.

The hypotheses for this **non-parametric** test should be stated *in context* with reference to the *categorical variables*.

χ^2 Test for Association hypotheses:

H_0 : There is **no association** between the variables...

H_1 : There is an **association** between the variables...

The test uses the *same test statistic and conditions for validity* as the χ^2 goodness-of-fit test, given on **page 6** of the Data Booklet. The decision rule also matches that for the goodness-of-fit test (stated in chapter 12). The calculation for degrees of freedom, however, is different.

Returning to the data on handedness and colour-blindness, a χ^2 test for association can now be performed. The first step is to include *marginal totals* for each row and column of the contingency table, as well as the total number of observations. The expected value for each cell can then be calculated by multiplying marginal totals and dividing by the total frequency, as shown below:

Observed	CVD	No CVD	
Left-handed	4	76	*80*
Right-handed	58	662	*720*
	62	*738*	*800*

$$\frac{80 \times 738}{800}$$

Expected	CVD	No CVD	
Left-handed	6.2	73.8	*80*
Right-handed	55.8	664.2	*720*
	62	*738*	*800*

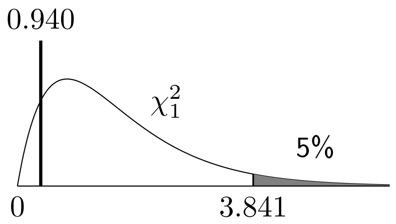
The expected values produced meet the required conditions, in that all are at least 1, and at least 80% are at least 5. Note that these expected values can also be produced quickly using some graphical calculators, as can the χ^2 test statistic. However, such tools will rarely first check that the expected frequencies meet the required conditions.

$$\chi^2 = \frac{(4-6.2)^2}{6.2} + \frac{(76-73.8)^2}{73.8} + \frac{(58-55.8)^2}{55.8} + \frac{(662-664.2)^2}{664.2} = 0.940$$

The number of *degrees of freedom* for the test statistic is calculated as $\nu = (r-1)(c-1)$, where r is the number of rows and c is the number of columns of data in the contingency table, *not including marginal rows and columns*.

Here there are *two rows* ($r = 2$) and *two columns* ($c = 2$), so $\nu = (2-1)(2-1) = 1$. The χ^2_1 critical value, taking a 5% level of significance, can be obtained from **page 14** of the Data Booklet:

Critical value: $\chi^2 = 3.841$



Since $0.940 < 3.841$, H_0 is not rejected at the 5% level of significance.
There is insufficient evidence to suggest that there is an association between handedness and colour-blindness.

Note that whilst the χ^2 test statistic is a measure of how well data fits a model of *no association* between the variables, the test for association does not show *evidence* of no association - it only assesses whether the data provides evidence *for* an association.

Example

Problem: Researchers working for an educational organisation wish to explore a possible connection between playing a musical instrument and learning a foreign language in school-age teenagers. They randomly select three schools within the region and then, within each school, select a number of S4 pupils to interview.

During the interview, each pupil is asked whether or not they play at least one musical instrument, and whether or not they are studying at least one modern language at National 5 level. The results are summarised in the contingency table below:

	Modern language	No modern language
Plays an instrument	32	18
Doesn't play an instrument	10	20

Perform a χ^2 test at the 1% level of significance to assess whether there is evidence to suggest that there is an association between playing a musical instrument and learning a language.

Solution:

H_0 : There is no association between playing a musical instrument and learning a language.

H_1 : There is an association between playing a musical instrument and learning a language.

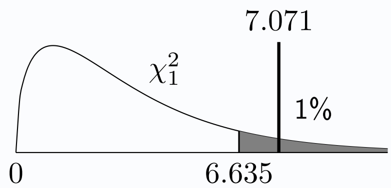
Observed	Language	No language	
Instrument	32	18	*50*
No instrument	10	20	*30*
	42	*38*	*80*

Expected	Language	No language	
Instrument	26.25	23.75	*50*
No instrument	15.75	14.25	*30*
	42	*38*	*80*

$\nu = (r-1)(c-1) = (2-1)(2-1) = 1$

Test statistic: $\chi^2 = \dfrac{(32-26.25)^2}{26.25} + \ldots + \dfrac{(20-14.25)^2}{14.25} = 7.071$

Critical value: $\chi^2 = 6.635$



Since $7.071 > 6.635$, reject H_0 at the 1% level of significance. There is evidence to suggest that there is an association between playing a musical instrument and learning a language.

A vital check carried out before calculating degrees of freedom and the test statistic was that the conditions for validity were met.

"At least 80% of the E_i should be at least 5 and none should be less than 1."

Here, none of the expected values were less than 1 and at least 80% ($\frac{4}{4} = 100\%$) were at least 5.

It should be noted that since the test's purpose of looking for evidence for an *association* is the same thing as looking for a *lack of independence*. The following hypotheses are statistically equivalent to those used in the example below:

H_0 : There is **no association** between *test centre used* and *test outcome*.

H_1 : There is an **association** between *test centre used* and *test outcome*.

It is not uncommon (outside of this course) to see a null hypothesis of **independence** between variables and an alternative hypothesis of **an association** between variables, combining the two choices of wording.

Example 2

Problem: The outcomes for a random sample of 80 driving tests sat in three different test centres located across Lancashire are recorded. The organisation responsible for overseeing driving test standards is seeking to explore a claim made that the centres do not use the same standards, and that the outcome of a driver's test is not independent of the location chosen.

		Test Centre		
		Blackburn	Preston	Chorley
Outcome	Pass	19	15	2
	Fail	13	21	10

Use a suitable non-parametric test to assess the data in the contingency table at the 10% level of significance, and hence comment on whether the data supports the claim.

Solution:

H_0 : The test outcome and test centre used are independent.

H_1 : The test outcome and test centre used are not independent.

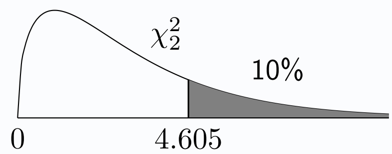
Observed	Blackburn	Preston	Chorley
Pass	19	15	2
Fail	13	21	10

Expected	Blackburn	Preston	Chorley
Pass	14.4	16.2	5.4
Fail	17.6	19.8	6.6

$\nu = (r-1)(c-1) = (2-1)(3-1) = 2$

Test statistic: $\chi^2 = \dfrac{(19-14.4)^2}{14.4} + ... + \dfrac{(10-6.6)^2}{6.6} = 6.726$

Critical value: $\chi^2 = 4.605$



Since $6.726 > 4.605$, reject H_0 at the 10% level of significance.
There is evidence to suggest the driving centre used and the outcome of the test are not independent. This suggests the claim may be true (although there could be other reasons apart from a difference in standards applied by the centres).

Note that an equivalent conclusion would be that there is evidence to suggest that there is an *association* between driving centre used and the outcome of the test. However, it is generally preferable for a solution to mirror the language used in the question.

Exercise 14.1

1. For all the games in the last 12 months, an ice hockey team records whether they won or lost, as well as whether the game was played at the weekend or on a weekday. The results were collated using a contingency table:

	Won	Lost
Weekend	13	8
Weekday	7	14

   Perform a χ^2 test, at the 10% level of significance, to assess whether there is evidence of an association between when a game is played and the result for the team.

2. A police service gives motorists caught speeding a choice between attending a speed awareness course and a fine together with points added to their licence. A random sample of such motorists has their records checked to see whether they committed a further offence in the following 12 months. The data is recorded in a contingency table:

	Awareness course	Fine and points
Further offence	12	16
No further offence	7	5

   Carry out a non-parametric test, at the 5% level of significance, to determine if there is an association between the choice of speeding penalty and whether a further speeding offence is committed.

3. A sports scientist is studying the effects of caffeine on the performance of professional darts players. 40 players are each randomly allocated to be given either a *caffeinated* or *caffeine-free* drink, each identical in appearance and taste. They are then given one attempt to hit a bullseye on a dart board. The results are as follows:

	Caffeinated	Caffeine-free
Hit	7	15
Miss	17	11

   The sports scientist claims that this shows that caffeine has an effect on the performance of darts players. Perform a χ^2 test of association to determine whether the data supports the claim, at the 10% level of significance.

4. Some modern cameras allow greater control over the sound the shutter makes when taking a photograph; however, some photographers claim that for one particular model there is an association between the shutter mode used and autofocus performance. 90 photos are taken with this model using the three different shutter modes, and each photograph is carefully checked to see whether it was in focus or not.

	In focus	Not in focus
Normal mode	11	4
Quiet mode	13	5
Silent mode	19	8

   Assess whether the data in the contingency table shows evidence for the claim, at the 5% level of significance.

5. Over the course of a month, a number of employees working at a company's office are selected at random each day to have their arrival time and their mode of transport noted. No employee is asked more than once. The results are shown in the contingency table below:

	Car	Bus	Cycle	Walk
On time	48	37	12	23
Late	11	9	3	7

   Perform a chi-squared test at the 1% level of significance to test whether there is an association between the mode of transport an employee uses and whether they make it into work on time.

14.2 Combining Columns or Rows

Where the conditions for validity for a χ^2 test of association are not met, it may be still possible to carry out a test by *combining rows and/or columns*. It is generally advised to combine either:

- Those rows or columns with the lowest expected frequencies.
- Those rows or columns which most naturally fit together.

In the example below, it would feel counterproductive to combine those patients who experienced a *reduction* in reported pain with those who experienced an *increase*.

Example

Problem: A pharmaceutical company arranges for a random sample of 120 patients to take part in a trial of a new pain relief medication. Some will be given the new medication, whilst others will instead be given a placebo treatment that is identical in appearance and packaging. At the end of the week, each patient will be asked whether they experienced a reduction, an increase in pain, or no change. The data collected is shown in the table below:

		Pain reported		
		Reduction	No change	Increase
Treatment	Medication	45	17	4
	Placebo	28	23	3

Assess at the 5% level of significance whether there is evidence of an association between the treatment given and the pain reported by the patient.

Solution:

H_0 : There is no association between the treatment given and pain reported.

H_1 : There is an association between the treatment given and pain reported.

Observed	Reduction	No change	Increase
Medication	45	17	4
Placebo	28	23	3

Expected	Reduction	No change	Increase
Medication	40.15	22	3.85
Placebo	32.85	18	3.15

Since less than 80% of expected values are at least 5, the last two columns should be combined.

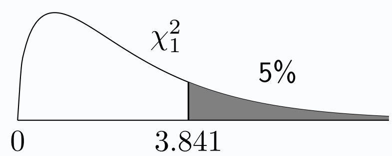
Observed	Reduction	No reduction
Medication	45	21
Placebo	28	26

Expected	Reduction	No reduction
Medication	40.15	25.85
Placebo	32.85	21.15

$\nu = (r-1)(c-1) = (2-1)(2-1) = 1$

Test statistic: $\chi^2 = \dfrac{(45-40.15)^2}{40.15} + \ldots + \dfrac{(26-21.15)^2}{21.15} = 3.324$

Critical value: $\chi^2 = 3.841$



Since $3.324 < 3.841$, do not reject H_0 at the 5% level of significance.
There is insufficient evidence to suggest that there is an association between the treatment given and the pain reported by patients.

Exercise 14.2

1. A website invited visitors via a pop-up survey to indicate whether their experience of using the site's new layout was *positive* or *negative*. The website logged the device type (*desktop* or *mobile*) used by each visitor, as well as the outcome of the survey. Some users ignored the survey, and this data was also captured as *NR* (no response). The results are shown in the table below:

	Positive	Negative	NR
Desktop	15	3	20
Mobile	35	7	23

Perform a chi-squared test to determine whether there is evidence of an association between the device a visitor uses to access a website, and their response to the survey, at the 5% level of significance.

2. A random sample of residents listed on the electoral register in four Scottish towns are selected and contacted to find out whether they voted in the most recent UK General Election, and the results shown in a contingency table. A *spoiled vote* means they deliberately left their ballot paper blank or otherwise voided it, often seen as a 'protest vote'.

Observed	Renfrew	Crieff	Moffat	Paisley
Voted	51	43	27	40
Didn't vote	17	11	18	22
Spoiled	3	7	2	1

Expected	Renfrew	Crieff	Moffat	Paisley
Voted	47.24	40.58	31.27	41.91
Didn't vote	19.95	17.14	13.21	17.7
Spoiled	3.81	3.28	2.52	3.38

Perform a hypothesis test to assess whether there is an association between voting habit and town of residence.

3. Data is collected from a random sample of people living across the United Kingdom on how many teaspoons (tsp) of sugar they typically add to a cup of tea. The contingency table below shows the results:

Observed	0 tsp	1 tsp	2 tsp	3+ tsp
England	132	34	11	7
Scotland	52	27	13	4
Wales	36	11	10	3
N. Ireland	27	16	5	0

Expected	0 tsp	1 tsp	2 tsp	3+ tsp
England	117.13	41.73	18.49	6.64
Scotland	61.11	21.77	9.65	3.46
Wales	38.20	13.61	6.03	2.16
N. Ireland	30.56	10.89	4.82	1.73

Use a chi-squared test to assess, at the 5% level of significance, whether there is a difference in the amount of sugar the residents of the various parts of the United Kingdom prefer in their tea.

4. A researcher studying a medical condition claims there is a link between having the condition and someone's blood type. A random sample of 620 patients registered with a local doctor have their blood tested, with the test revealing both which of the four major blood types they belong to and whether they test positive or negative for the condition.

	O	A	B	AB
Positive	7	24	10	2
Negative	318	188	50	21

Perform a non-parametric test at the 10% level of significance to assess whether there is evidence to support a claim that there is an association between a patient's blood type and testing positive for the medical condition.

Review Exercise

1. In statistician Ronald Fisher's 1935 book, '*A Design of Experiments*', he details setting up a test to judge a claim made by the scientist Muriel Bristol: that she could tell whether milk was added to a cup *before* tea, or *after*. A statistics student, whose friend claims to only like tea when the milk is added first, decides to set up a similar experiment for herself, disbelieving that her friend can really tell the difference.

   Over the course of a term, she makes 24 cups of tea for her friend, each time randomly selecting whether the milk is added before or after. Her friend then states whether he likes it or not, without knowing which method was used. The results are shown in a contingency table:

		Verdict	
		Likes the tea	Dislikes the tea
Method	Milk first	9	4
	Milk second	2	9

   Perform a non-parametric hypothesis test to assess at the 10% level of significance the evidence for an association between the order the milk is added in and whether her friend likes the tea.

2. Two bird feeders are placed in a garden, with a different type of bird food in each: *sunflower hearts* and *mealworms*. The garden's owner observes the number of each species of bird that visit each of the two feeders over the course of a week:

	Robins	Blue tits	Goldfinches	Bullfinches
Sunflower hearts	5	34	7	1
Mealworms	23	11	2	3

   Perform a non-parametric hypothesis test to assess at the 2.5% level of significance the evidence for an association between the species of the bird and the bird feeder it visits.

3. Two types of nutritional supplement, A and B, are given to newly hatched chicks. The growth over the next six months for each chick is categorised *high*, *medium* or *low*, with the results displayed in a contingency table set up as below:

	High	Medium	Low
Feed A	17	11	7
Feed B	13	18	5

   Computer output for a hypothesis test conducted on this data is shown below:

```
chi-squared test for association
n = 71        df = 2
alternative hypothesis: there is an association between the variables
test statistic = 2.543    critical value = 4.605    p-value = 0.2804
```

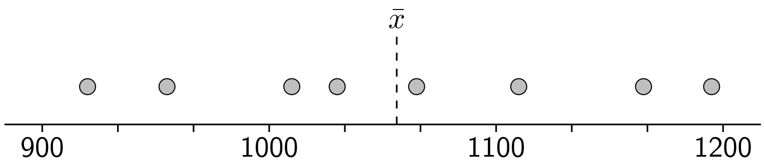
   (a) Use the degrees of freedom (*df*) to explain whether any columns or rows were combined.
   (b) State the significance level used.
   (c) State a suitable null hypothesis for the test.
   (d) Use the p-value to write a suitable conclusion to the test.

15

Two-Sample Parametric Tests

Chapter 10 introduced three hypothesis tests which use a single sample of data to assess the evidence for claims relating to a population parameter. For example, a study on the longevity of a brand's light bulbs might check a random sample of such bulbs, to assess whether there is sufficient evidence to suggest the population mean lifespan is greater than 1000 hours. The data from the sample is shown below, with the sample mean $\bar{x}$ indicated:

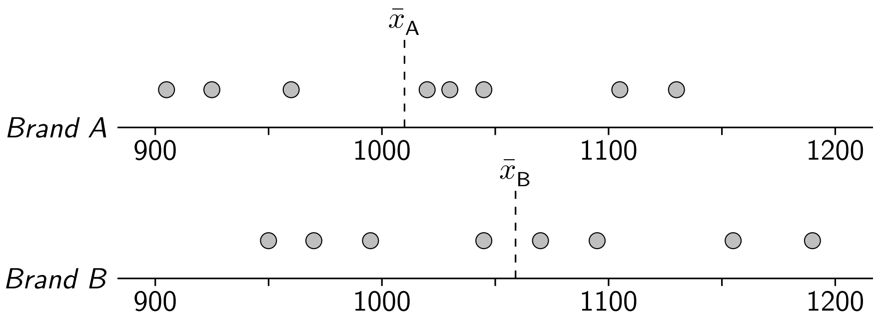
$$\begin{aligned} \mathsf{H}_0 &: \quad \mu = 1000 \\ \mathsf{H}_1 &: \quad \mu > 1000 \end{aligned}$$



This use of a *one* sample of data, investigating *one* population's parameter, requires a **one-sample test**.

Should a study instead wish to compare the longevity of *two different brands' light bulbs* they would instead take *two samples* of data: one sample from *Brand A* and one sample from *Brand B*. In such a case, a typical set of hypotheses might instead ask whether there is sufficient evidence to suggest that there is a difference between the population mean lifespan for Brand A, μ_{A}, and the population mean lifespan for Brand B, μ_{B}. The data from the samples are shown below, with the sample means $\bar{x}_{\mathsf{A}}$ and $\bar{x}_{\mathsf{B}}$ indicated:

$$\begin{aligned} \mathsf{H}_0 &: \quad \mu_{\mathsf{A}} = \mu_{\mathsf{B}} \\ \mathsf{H}_1 &: \quad \mu_{\mathsf{A}} \neq \mu_{\mathsf{B}} \end{aligned}$$



Such a test, comparing the parameters of *two* populations by using *two samples*, requires a **two-sample test**. Either one-tailed or two-tailed tests can be performed:

$$\begin{array}{lll} \mathsf{H}_0 : \mu_{\mathsf{A}} = \mu_{\mathsf{B}} & \quad \mathsf{H}_0 : \mu_{\mathsf{A}} = \mu_{\mathsf{B}} & \quad \mathsf{H}_0 : \mu_{\mathsf{A}} = \mu_{\mathsf{B}} \\ \mathsf{H}_1 : \mu_{\mathsf{A}} \neq \mu_{\mathsf{B}} & \quad \mathsf{H}_1 : \mu_{\mathsf{A}} > \mu_{\mathsf{B}} & \quad \mathsf{H}_1 : \mu_{\mathsf{A}} < \mu_{\mathsf{B}} \end{array}$$

Within calculations and hypotheses, populations can be numbered (e.g. μ_1 and μ_2), given letters (e.g. μ_{A} and μ_{B}), or named with words to aid readability (e.g. μ_{cat} and μ_{dog}).

15.1 Two-Sample *z*-test for Population Means

If **independent** random samples of sizes n_1 and n_2 are obtained from each of two normally distributed populations which have population means μ_1 and μ_2 respectively, and population standard deviations σ_1 and σ_2 respectively, then the difference in sample means $\bar{X}_1$ and $\bar{X}_2$ is also normally distributed, as below:

$$\overline{X}_1 \sim \mathrm{N}\left(\mu_1, \frac{\sigma_1^2}{n_1}\right) \quad \text{and} \quad \overline{X}_2 \sim \mathrm{N}\left(\mu_2, \frac{\sigma_2^2}{n_2}\right) \quad \Longrightarrow \quad \overline{X}_1 - \overline{X}_2 \sim \mathrm{N}\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

This leads to the test statistic for the *two-sample z-test for population means* given in the data booklet. Under the typical null hypothesis of no difference between the population means, or $\mu_1 - \mu_2 = 0$, the test statistic in the box may be used. From the Data Booklet:

$$Z \sim \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$$

For a two-sample z-test for means:

Test statistic: $z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$

Conditions for valid use of the two-sample z-test for population means:

- The *populations* are **normally distributed**.
- The *population standard deviations*, σ_1 and σ_2, are **known**.
- The data was obtained through **random** and **independent samples**.

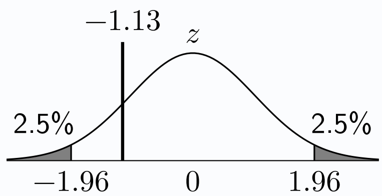
Note that when sample sizes n_1 and n_2 are greater than 20 the *Central Limit Theorem*, can be invoked. For large sample sizes, the sample standard deviations s_1 and s_2 may be used as estimates of σ_1 and σ_2, allowing an approximate z-test to be performed.

Example

Problem: To compare the battery life of two brands' mobile phones, a random sample of nine phones from *Apfel* and seven phones from *Ubhal* are tested. The sample of Apfel phones had a mean battery life of 34 hours, with a mean battery life of 36 hours for Ubhal. Assuming that the battery life for both populations of phones is known to be normally distributed, with a standard deviation of 3.5 hours, use a parametric test to determine whether there is evidence for a difference in battery life between the brands.

Solution: $\bar{x}_{\text{apfel}} = 34$, $\bar{x}_{\text{ubhal}} = 36$, $\sigma_{\text{apfel}} = 3.5$, $\sigma_{\text{ubhal}} = 3.5$, $n_{\text{apfel}} = 9$, $n_{\text{ubhal}} = 7$

$H_0: \quad \mu_{\text{apfel}} = \mu_{\text{ubhal}}$
$H_1: \quad \mu_{\text{apfel}} \neq \mu_{\text{ubhal}}$



Test statistic: $z = \dfrac{34 - 36}{\sqrt{\dfrac{3.5^2}{9} + \dfrac{3.5^2}{7}}} = -1.13$

Critical value: $z = -1.96$

Since $-1.13 > -1.96$, do not reject H_0 at the 5% level of significance. There is insufficient evidence to suggest a difference in the mean battery life of the two brands' mobile phones.

Exercise 15.1

1. A two-sample z-test for a difference in population means is to be carried out based on two independent random samples of data taken from normally distributed populations (A and B). The sample mean for A is 87 based on a sample size of 11, whilst the sample mean for B is 92 based on a sample size of 13. Populations A and B have standard deviations 7 and 5 respectively.
   (a) State suitable null and alternative hypotheses.
   (b) Show the calculation required to obtain a test statistic of $z = -1.98$.
   (c) Conclude the test at the 5% level of significance.

2. The population standard deviations of customer spend between two supermarkets, *The Food Maze* and *Thyme to Shop*, have been previously recorded as £9.50 and £13 respectively. Recent random samples of 21 The Food Maze customers and 7 Thyme to Shop customers produced sample mean customer spends of £34.80 and £42.13 respectively. Stating an assumption required, perform a parametric test to assess whether the data supports a claim that customers at Thyme to Shop spend more on average than those at The Food Maze.

3. The fuel efficiency in miles per gallon (mpg) is being studied by a car manufacturer for two of their hybrid car models, the *Ugo* and the *Theiris*. Across a month, a randomly selected employee each day drives one of the models from Dundee to Aberdeen and back. It has been previously recorded that the standard deviation of fuel efficiency for similar models of hybrid cars driving the same route is 2.7 mpg. The results from these journeys are shown below:

Model name	Ugo	Theiris
Sample mean	52	55
Sample size	16	14

   It is assumed that the population standard deviation in fuel efficiency is 2.7 mpg for both models.
   (a) State a further assumption required for the use of a two-sample z-test of a difference in mean fuel efficiency between the two models.
   (b) Perform a two-sample z-test at the 10% level of significance.

4. A restaurant chain has a branch in Edinburgh and a branch in Glasgow. Waiting times for food are known to be normally distributed for both branches, with standard deviations previously known to be 4 minutes for the Edinburgh branch and 3 minutes for the Glasgow branch.

   This year, waiting times were recorded for random samples of 17 tables at the Edinburgh branch, giving a mean of 23 minutes, and 19 tables at the Glasgow branch, giving a mean of 21 minutes.

   Perform a parametric test to assess whether the data provides evidence to suggest that there is a difference in mean waiting times for food between the two branches, at the 5% level of significance.

5. Researchers studying the effects of evening screen time on sleep quality for teenagers gather two random samples: one of teenagers who use their mobile phone in the hour leading up to the time they go to bed, and one of teenagers who do not. Fitness trackers are used to measure the quality of their sleep, giving a score from 0 (very poor) to 10 (excellent). The results are as follows:

Group	Phone use	No phone use
Sample mean	7.9	8.4
Sample standard deviation	0.7	0.4
Sample size	34	41

   It is assumed that the sample size is sufficiently large that the sample standard deviations in sleep quality score are reliable estimates for the population standard deviations.
   (a) Explain, with reference to the sample sizes, why the researchers do not need to assume that sleep quality scores in the populations are normally distributed.
   (b) Conduct a z-test to assess, at the 1% level of significance, whether the data supports a claim that screen use in the evening has a negative impact on sleep quality amongst teenagers.

15.2 Two-Sample z-test for Population Proportions

If random, independent samples of data are obtained from each of two populations, and the two **sample proportions** $\widehat{p}_1$ and $\widehat{p}_2$ are obtained, then a *two-sample z-test for population proportions* may be used to explore evidence for a difference in the **population proportions**, p_1 and p_2.

The test statistic, provided in the Data Booklet, uses an *estimate of the pooled population proportion*, $\widehat{p}$:

For a two-sample z-test for population proportions:

Test statistic: $z = \dfrac{\widehat{p}_1 - \widehat{p}_2}{\sqrt{\widehat{p}\widehat{q}\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$ where $\widehat{p} = \dfrac{n_1\widehat{p}_1 + n_2\widehat{p}_2}{n_1 + n_2}$

The test statistic may be derived from the use of the normal approximation to a binomial distribution to describe the approximate distribution of the sample proportion (see section 10.5) and the distribution of the difference of two *independent* sample proportions (similar to section 15.1). The conditions required for the use of this hypothesis test follow.

Conditions for valid use of the two-sample z-test for population proportions:

- Each of $n_1\widehat{p}_1, n_1\widehat{q}_1, n_2\widehat{p}_2$ and $n_2\widehat{q}_2$ must be **greater than 5**.
- The data was obtained through **random** and **independent samples**.

This parametric test can take one-tailed or two-tailed hypotheses which relate to the population proportions, p_1 and p_2:

$$\begin{array}{lll} \mathrm{H}_0 : p_1 = p_2 & \mathrm{H}_0 : p_1 = p_2 & \mathrm{H}_0 : p_1 = p_2 \\ \mathrm{H}_1 : p_1 \neq p_2 & \mathrm{H}_1 : p_1 > p_2 & \mathrm{H}_1 : p_1 < p_2 \end{array}$$

Example

Problem: Tyre pressures were checked for a random sample of 80 cars parked on streets in a town centre, and 15% were found to have at least one tyre with low pressure. For another sample of 75 vans in the town, 8% were found to have at least one tyre with low pressure. Perform a two-sample z-test for proportions, at the 1% level of significance, to assess whether there is a difference in the proportions of cars and vans that have at least one tyre with low pressure.

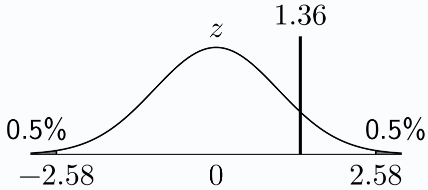
Solution: $\widehat{p}_{\text{car}} = 0.15$, $\widehat{p}_{\text{van}} = 0.08$, $n_{\text{car}} = 80$, $n_{\text{van}} = 75$

$$\begin{array}{ll} H_0 : & p_{\text{car}} = p_{\text{van}} \\ H_1 : & p_{\text{car}} \neq p_{\text{van}} \end{array}$$

Estimate of pooled population proportion: $\widehat{p} = \dfrac{80 \times 0.15 + 75 \times 0.08}{80 + 75} = 0.116$ (to 3dp)

Test statistic: $z = \dfrac{0.15 - 0.08}{\sqrt{0.116 \times 0.884\left(\dfrac{1}{80} + \dfrac{1}{75}\right)}} = 1.36$

Critical value: $z = 2.58$



Since $1.36 < 2.58$, do not reject H_0 at the 1% level of significance. There is insufficient evidence to suggest that the proportions of vans and cars with at least one tyre with low pressure parked in the town centre are different.

Exercise 15.2

1. Two publishing companies' Maths textbooks are known to contain occasional errors in the printed answers. Students randomly select questions from Brand A's textbook and from Brand B's textbook, checking the numbers of correct and incorrect answers for each sample. The results are:

Textbook	Brand A	Brand B
Correct	58	43
Incorrect	7	9

   Test at the 5% significance level the hypothesis that the Brand B's printed answers contain a higher proportion of incorrect answers.

2. An online travel magazine is investigating whether the younger or older holidaymakers are more likely to read books whilst on holiday. Random samples of those holidaymakers aged under 40 and those aged over 40 are selected and asked whether they read books whilst on holiday. The results are:

   - 27 of the 66 younger holidaymakers read on holiday.
   - 34 of the 59 older holidaymakers read on holiday.

   Perform a parametric hypothesis test at the 5% significance level to assess the hypothesis that there is a difference in the proportion of holidaymakers from each age group who read books whilst on holiday.

3. Out of a random sample of 60 customers using the *North East Train Service*, 42 rated their experience as positive, whilst 37 out of a random sample of 50 customers using the *South West Intercity Metro* service rated their experience as positive. Use a parametric test, at the 10% level of significance, to assess whether the data provides evidence of a difference between the two train services in the proportion of customers rating their experience as positive.

4. In a random sample of 140 customers in an airport branch of a coffee chain, 52 used a loyalty card. In a random sample of 153 customers in a city centre branch of a coffee chain, 68 used a loyalty card.
   (a) Show that the data meets the conditions required for the valid use of a z-test.
   (b) Perform a z-test to determine whether there is evidence of a difference in the proportion of people using a loyalty card between the airport and the city centre branches.

5. A recent survey of people living in Dollar showed that in a sample of 80 people, 35% had visited Islay. In another sample of 70 people living in Dollar, 20% had visited Orkney. Test at the 5% significance level the hypothesis that a higher proportion of people from Dollar have visited Islay than Orkney, stating two assumptions required.

6. An extensive study of two species of trees (one *native* and one *non-native*) commonly found in the Cairngorm National Park recorded the numbers of randomly selected trees from each species observed to be affected by a common disease. The aim of the study was to investigate whether the non-native species was more likely to be affected by the disease than the native species. For the native species, 35 out of 351 trees were observed to be affected, whilst 46 out of 289 non-native trees were affected. The results of a two-sample z-test for a difference in population proportions are shown below:

```
two sample z-test for equality of proportions
data: prop1: native, prop2: nonnative
alternative hypothesis: nonnative - native is greater than zero
z = 2.13        p-value = 0.0165
sample estimates: prop native = 0.0997, prop nonnative = ****
```

   (a) Find the value missing from the output, marked as ****.
   (b) State suitable hypotheses for the test.
   (c) Show the calculations required to obtain the test statistic shown.
   (d) Suggest a suitable conclusion to the test, at the 5% level of significance, using the p-value given.

15.3 Two-Sample t-test for Population Means

If random, independent samples of data are obtained from each of two *normally distributed* populations with **unknown population standard deviations**, then a *two-sample t-test for population means* may be used to explore evidence for a difference in the **population means**, μ_1 and μ_2, using the **sample standard deviations**.

Conditions for valid use of the two-sample t-test for population means:

- The *populations* are **normally distributed**.
- The populations are assumed to have **equal variance**.
- The data was obtained through **random** and **independent samples**.

The test statistic is provided in the Data Booklet, and follows a t-distribution with $\nu = n_1 + n_2 - 2$ degrees of freedom. It requires calculation of a *pooled estimate of the variance*, s^2, and as with the two-sample z-test, a more streamlined version of the test statistic under a null hypothesis of $\mu_1 = \mu_2$ is as follows:

For a two-sample t-test for population means:

Test statistic: $t = \dfrac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}$ where $s^2 = \dfrac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}$

with $\nu = n_1 + n_2 - 2$ degrees of freedom

While t-tests are especially valuable when dealing with small sample sizes, they remain the statistically *correct* test to use even when dealing with very large samples. Modern computing power reduces the need to switch to a z-test for very large samples where critical values for the t-distribution may not have been so easily obtained.

Example

Problem: The time taken for runners to complete 5km at each *Parkrun* location is known to be normally distributed. For a random sample of 8 runners at Dundee Parkrun the sample mean 5km time was 30.1 minutes, with a sample standard deviation of 3 minutes. At the Perth Parkrun the sample mean 5km time for 7 runners was 28.6 minutes, with a sample standard deviation of 2 minutes. Perform a t-test at the 10% level of significance to investigate whether the data suggests there is a difference in mean 5km running times between the two Parkrun locations.

Solution: $\bar{x}_{\text{Dundee}} = 30.1$, $\bar{x}_{\text{Perth}} = 28.6$, $s_{\text{Dundee}} = 3$, $s_{\text{Perth}} = 2$, $n_{\text{Dundee}} = 8$, $n_{\text{Perth}} = 7$

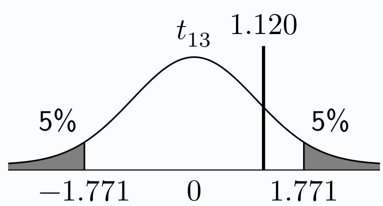
$$H_0: \ \mu_{\text{Dundee}} = \mu_{\text{Perth}}$$
$$H_1: \ \mu_{\text{Dundee}} \neq \mu_{\text{Perth}}$$

Estimate of pooled variance: $s^2 = \dfrac{(8-1)\times 3^2 + (7-1)\times 2^2}{8+7-2} = 6.692$

$s = \sqrt{6.692} = 2.587$

Test statistic: $t = \dfrac{30.1 - 28.6}{2.587 \times \sqrt{\dfrac{1}{8} + \dfrac{1}{7}}} = 1.120$

Critical value: $t = 1.771$



Since $1.120 < 1.771$, do not reject H_0 at the 10% level of significance. There is insufficient evidence to suggest a difference in the mean 5km running times between the two locations.

Exercise 15.3

1. A two-sample t-test for a difference in population means is to be carried out based on two independent random samples of data taken from normally distributed populations (A and B), which are assumed to have equal variances. A sample of 11 observations from A produces a mean of 87 and a standard deviation of 8, whilst a sample of 13 observations from B produces a mean of 92 and a standard deviation of 5.
   (a) State suitable null and alternative hypotheses.
   (b) Show the calculation required to obtain a test statistic of $t = -1.072$.
   (c) Conclude the test at the 5% level of significance.

2. A motorist and keen statistician tosses a coin before each of a series of drives to determine whether to use her car's air conditioning or not. The fuel efficiency (in miles per gallon) for those drives are:

With air conditioning (mpg)	47	48	51	52	52	53	55
Without air conditioning (mpg)	49	50	52	54	55	55	

   Assume that fuel efficiency is normally distributed for each population (with air conditioning and without), and that the population variances are equal. Perform a parametric test to determine whether the data supports her belief that fuel efficiency is higher when driving without air conditioning, at the 5% level of significance.

3. The contents of a random sample of six bags of coffee beans sold by *Brewhemian Rhapsody* were measured. The results, in grams, were:

   254 255 254 248 256 251

   The results for a random sample of eight bags sold by *The Bean Supremacy* were:

   240 249 244 246 245 250 240 253

   It is assumed that the mass of coffee in a bag is normally distributed, for each population. Use a parametric test to assess the evidence that the *Brewhemian Rhapsody* bags contain more coffee, stating one further assumption required.

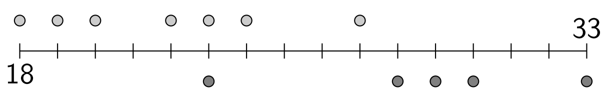
4. A medical trial wishes to compare the effectiveness of two new types of mild painkiller. Twenty volunteers are selected and randomly assigned either painkiller A or painkiller B, and an effectiveness score from 0 to 5 is obtained for each person through an industry-recognised pain tolerance test using ice-cold water. Computer output of summary statistics for the samples are shown below:

```
data          sample mean  sample sd  n
painkiller A  1.95         0.54       10
painkiller B  2.75         0.74       10
```

   Perform a t-test at the 2% level to assess whether there is evidence of a difference between the effectiveness of the two painkillers, stating two assumptions required.

5. Times taken to run 5km for a random sample of 7 hockey players and a random sample of 5 rugby players are recorded. The table below gives the results, to the nearest minute:

Hockey	18	19	20	22	23	24	27
Rugby	23	28	29	30	33		



   Assume that the time taken to run 5km is normally distributed for both the hockey and rugby players, and that the population variances in 5km time are equal. Perform a parametric test to assess the data for evidence of a difference in the mean time taken for hockey and rugby players to run 5km.

15.4 Paired t-test for the Population Mean Difference

If six randomly selected athletes have their 100 metre sprint times recorded, and then repeat the sprint a month later after some training, the two datasets (*time before* and *time after*) are *not independent*. Instead, each value from the first dataset pairs with a value from the second dataset; this is called *paired data*. The previous two-sample tests in this chapter would be inappropriate, since they require independent samples.

Instead, a sample of *differences*, d_i, for each pair should be calculated and the mean of the differences (also known as the *mean difference*) assessed using a *paired t-test*. Possible hypotheses would then relate to the population mean difference, μ_d:

$$\mathrm{H}_0 : \mu_d = 0 \qquad \mathrm{H}_0 : \mu_d = 0 \qquad \mathrm{H}_0 : \mu_d = 0$$
$$\mathrm{H}_1 : \mu_d \neq 0 \qquad \mathrm{H}_1 : \mu_d > 0 \qquad \mathrm{H}_1 : \mu_d < 0$$

The test statistic derives from the one-sample t-test covered in Chapter 10, with $\bar{d}$ (or $\bar{x}_d$) representing the sample mean difference and s_d the sample standard deviation in differences. The degrees of freedom are calculated based on the single sample of differences, as $\nu = n - 1$.

For a paired t-test:

Test statistic: $t = \dfrac{\bar{x}_d - \mu_d}{\frac{s_d}{\sqrt{n}}}$

with $\nu = n - 1$

Conditions for valid use of a paired t-test:

- The *population of differences* is **normally distributed**.
- The data was obtained through **random sampling**.

Careful attention must be given to how the sample of differences is defined, particularly for one-tailed hypotheses.

Example

Problem: Six randomly selected teachers have their pulse rates (in beats per minute) checked in at 10am and then four hours later at 2pm.

Pulse at 10am	58	51	68	83	71	78
Pulse at 2pm	65	57	63	98	71	82

Assuming that the sample of differences in pulse rate were independent, and that the population of differences is normally distributed, assess whether there is evidence that teachers' pulse rates are higher at 2pm than 10am, at the 5% level of significance.

Solution:

Differences (2pm—10am)	7	6	-5	15	0	4

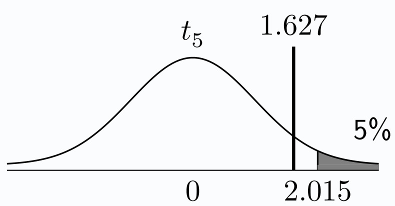
$\bar{d} = 4.5$, $s_d = 6.775$, $n = 6$

$$H_0 : \quad \mu_d = 0$$
$$H_1 : \quad \mu_d > 0$$

$\nu = 6 - 1 = 5$

Test statistic: $t = \dfrac{4.5 - 0}{\frac{6.775}{\sqrt{6}}} = 1.627$

Critical value: $t = 2.015$



Since $1.627 < 2.015$, do not reject H_0 at the 5% level of significance. There is insufficient evidence to suggest that teachers' pulse rates are higher at 2pm than 10am.

Exercise 15.4

1. A paired t-test for a population mean difference not equal to zero is to be carried out based on pairs of observations from a sample of six individuals. The population of differences is assumed to be normally distributed. The table below shows the results:

Individual	A	B	C	D	E	F
Observation 1	35	65	43	66	58	61
Observation 2	31	57	39	71	58	60

   (a) State suitable null and alternative hypotheses.
   (b) Show the calculation required to obtain a test statistic of $t = 0.251$.
   (c) Conclude the test at the 5% level of significance.

2. A school is studying whether a digital homework platform is helping pupils improve their times tables knowledge. Seven pupils are randomly selected to answer as many times tables questions correctly as they can in one minute. Six months later, after regular use of the homework platform, the same seven pupils try again to answer as many times tables questions correctly as they can in one minute. The results are:

Pupil	1	2	3	4	5	6	7
Before	31	13	27	19	14	25	24
After	42	15	26	19	18	22	26

   Perform a t-test at the 10% level of significance to assess whether the data suggests that there has been an improvement in the pupils' times tables ability.

3. Eight amateur runners are selected to study the effects of consuming energy gels before running on runners' performances at middle-distances. Each has their finish time recorded for a 5000m run without consuming energy gels beforehand, then a month later their time is recorded again for a 5000m run before which each consumed two energy gels. The data is shown in the table below, with times recorded to the nearest minute.

Runner	A	B	C	D	E	F	G	H
Without gels	24	19	21	17	26	32	18	20
With gels	22	19	20	18	23	28	15	20

   Perform a hypothesis test at the 5% level of significance to assess whether consuming energy gels leads to a difference in running performance

4. Seven red kites from a region's population have their masses measured using specially adapted feeding posts fitted with electronic scales. A year later, the masses of the same seven red kites are recorded, in kg. For each bird, the **differences** in mass between the first and second measurement are shown below.

differences in kg (second − first)	0.3	-0.2	0.4	0.0	0.1	0.2	0.1

   It is assumed that the population of differences in masses is normally distributed, and that the differences in mass for the sample are independent. A hypothesis test is conducted to assess the data for evidence of an increase in mass among the region's red kites:

```
paired t-test
alternative hypothesis: true difference in means is greater than 0
t = 1.722    df=6    p-value = 0.06798
sample estimates:
mean of differences = 0.1286    standard deviation of differences = 0.1976
```

   Show the calculation required to obtain the test statistic given, and give an appropriate conclusion to the test at the 10% level of significance.

Review Exercise

1. A commercial timber-selling company is concerned that the pine fence posts they are receiving from their new supplier have a lower density than those received from their old supplier. They measure the density (in kg/m^3) of a random sample of eight of the old fence posts and seven of the new fence posts. The results are:

old	563	576	534	580	556	519	540	548
new	550	571	527	506	524	549	571	

   Perform a parametric hypothesis test at the 5% level of significance to assess whether the company's concerns are supported by the data.

2. A number of fifth year and sixth year pupils studying maths at a school were randomly selected from the school's register by a maths teacher and asked whether or not they had attended at least one support session during this academic year. The results are shown in the table below:

Year	Fifth	Sixth
Attended at least one	14	11
Has not attended any	17	16

   (a) Show clearly that the data satisfies the conditions for the use of a z-test.

   (b) Perform a z-test at the 1% level of significance to assess whether there is evidence of a difference in the proportions of pupils who attend maths support sessions between the two year groups.

3. A random sample of nine students from a sports science course completed a 10km course whilst wearing *regular* running shoes. The same nine students then completed the same course again one week later, this time wearing *carbon plates* running shoes. The results are shown below, rounded to the nearest tenth of a minute.

Student	1	2	3	4	5	6	7	8	9
Regular	46.1	41.0	53.2	42.9	40.2	38.7	48.7	50.2	46.5
Carbon plate	45.6	38.7	53.3	41.8	40.2	35.6	47.5	50.0	46.1

   Assuming the population of differences in times to be normally distributed, test at the 5% level of significance whether the data gives evidence to suggest *carbon plate* running shoes improve 10km running times.

4. Last year a road safety campaigner recorded the speeds of randomly selected cars as they drove past a sign in their village which displays the car's speed. This year, after the council claimed average speeds of cars through the village had decreased, the campaigner recorded the speeds of another random sample of cars. The results are shown in the table below, measured in miles per hour:

Last year	This year
$\bar{x}_1 = 32.4$	$\bar{x}_2 = 29.7$
$s_1 = 4.3$	$s_2 = 1.3$
$n_1 = 29$	$n_2 = 34$

   Assume that the sample standard deviations in speed are reliable estimates of the population standard deviations.

   (a) Perform a z-test at the 10% level of significance to determine whether the data provides evidence to suggest that the council's claim is valid.

   (b) By considering the sample standard deviations, explain why a two-sample t-test may not be appropriate for this data.

16

Wilcoxon Signed Rank Test

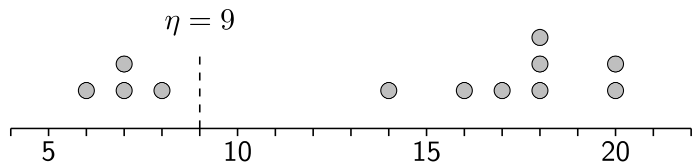
The z-tests and t-tests from chapters 10 and 15 which explore the *locations* of distributions have all focused on the *population mean*, μ. Those *parametric* tests each rely on either normally distributed populations or sample sizes greater than 20 such that the sample mean is approximately normally distributed (through the Central Limit Theorem).

Given small samples ($n < 20$) from populations whose distributions may not be assumed to be normally distributed, a number of **non-parametric** tests may be used which explore the locations of distributions by considering the **population median**, η (*'eta'*).

Sign Test

The Sign Test is a simple non-parametric test that, whilst *not required for the Advanced Higher Statistics course*, helps to lay the groundwork for the tests covered in this chapter. It uses a useful property of the population median, η, namely that 50% of observations sampled from a population can be expected to lie above η and 50% can be expected to lie below η (assuming no values are equal to the median). This means that the number of observations on either side of the population median in a sample of size n should follow a $\mathsf{B}(n, 0.5)$ distribution. For example, consider the following random sample of values ($n = 12$) and a claim that the population median is less than 9:

$$\mathsf{H}_0: \quad \eta = 9$$
$$\mathsf{H}_1: \quad \eta < 9$$



With 4 out of the 12 observed values being less than 9, the p-value for this test can be calculated as the probability of observing 4 or fewer values less than $\eta = 9$. Letting random variable X represent the number of such values, which follows a $\mathsf{B}(12, 0.5)$ distribution under the null hypothesis:

$$p\text{-value} = \mathsf{P}(X \leqslant 4) = 0.1938$$

Since $0.1938 > 0.05$, do not reject H_0 at the 5% level of significance.

There is insufficient evidence to suggest that the population median is less than 9.

16.1 Wilcoxon Signed Rank Test

Whilst the Sign Test is only concerned with the *number* of values that fall on either side of the hypothesised population median, η, the Wilcoxon Signed Rank test also considers how *far* from the median these values lie. More specifically, it applies weighting to the values by assigning *rank* 1 to the value closest to η, *rank* 2 to the second closest, and so on.

Zeros: Any values that are exactly equal to the median, η, are ignored and not given a rank, with the sample size n reduced to reflect the number of *remaining values to be ranked*.

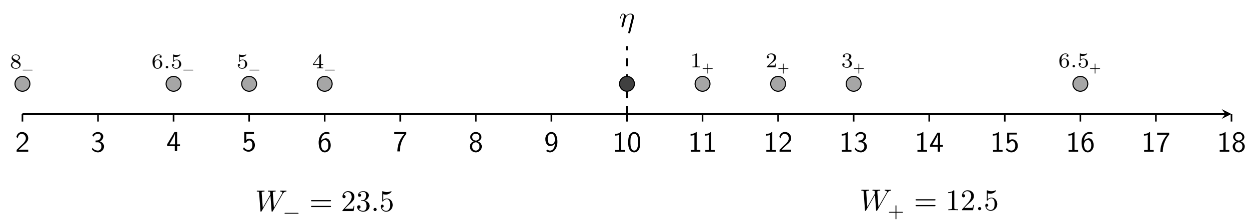
Tied ranks: Values that are equally distant from the median, η, share ranks. For example, if two values are equally far from η when looking to assign rank 4, then they are each ranked 4.5, since $\frac{4+5}{2} = 4.5$. If three values are equally distant from η when assigning rank 7, then they are each ranked 8 since $\frac{7+8+9}{3} = 8$.

- Ranks of values greater than the median are denoted as *positive ranks*, and their total as W_+.
- Ranks of values less than the median are denoted as *negative ranks*, and their total as W_-.

Consider a random sample of nine values from a distribution with claimed median $\eta = 10$:

2 4 5 6 10 11 12 13 16

$$\begin{aligned} \text{H}_0 &: \quad \eta = 10 \\ \text{H}_1 &: \quad \eta \neq 10 \end{aligned}$$



Note that the sum of W_+ and W_- is $12.5 + 23.5 = 36$, which will always be the total when ranks are assigned to $n = 8$ values, since $1 + 2 + \cdots + 8 = 36$. It is recommended to check that the sum of W_+ and W_- does give the appropriate total for the sample size, which for a sample of size n is given by $\frac{1}{2}n(n+1)$.

For a Wilcoxon Signed Rank test:

- The **test statistic**, denoted W, is the *lower* of W_+ and W_-, or $\min(W_+, W_-)$.
- The **critical value** can be obtained from page 15 of the Data Booklet.
- H_0 is **rejected** if the test statistic W is **less than or equal to** the critical value.

Test statistic: $W = \min(12.5, 23.5) = 12.5$\

Critical value: $W = 3$; ($\alpha = 0.05$, $n = 8$ and two-tailed.)\

Since $12.5 > 3$, do not reject H_0 at the 5% level of significance.
There is insufficient evidence to suggest that the population **median** is not 10.

The Wilcoxon Signed Rank test is often used for small samples for which the Central Limit Theorem cannot be invoked, from populations which cannot be reasonably assumed to be normally distributed in order to use a t-test. The table of critical values only extends to samples up to size $n = 20$, once zeros have been discarded.

Conditions for valid use:

- The underlying distribution is **symmetrical**.
- The observations are **independent** (typically assured through *random sampling*).

A suggested methodical approach to obtaining the signed ranks required without a diagram is:

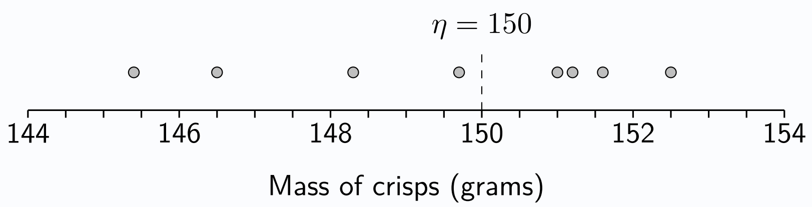
- **Subtract** η from every value in the sample.
- Ignoring any **zeros**, assign ranks to the remaining n values in ascending order by **absolute** value.

The following example should be compared to Example from Section 10.7, Chapter 10, in which a t-test was used.

Example

Problem: A supermarket sells sharing-sized bags of a particular brand of crisps. A consumer watchdog is asked to investigate a claim that the median mass of crisps contained in a bag is less than the stated contents of 150 grams. A random sample of bags gives the following results, in grams:

148.3 151.2 149.7 152.5 145.4 151.0 146.5 151.6



Stating a necessary assumption, perform a non-parametric test to assess the claim at the 1% level of significance.

Solution:
Assume that the mass of crisps in a bag is symmetrically distributed.

$$\text{H}_0: \quad \eta = 150$$
$$\text{H}_1: \quad \eta < 150$$

148.3	151.2	149.7	152.5	145.4	151.0	146.5	151.6	
-1.7	1.2	-0.3	2.5	-4.6	1.0	-3.5	1.6	(*subtract* $\eta = 150$)
5_-	3_+	1_-	6_+	8_-	2_+	7_-	4_+	(*assign signed ranks*)

$$W_+ = 15 \qquad \tfrac{1}{2} \times 8 \times 9 = 36$$
$$W_- = 21 \qquad \textit{Check: } 15 + 21 = 36$$

Test statistic: $W = \min(15, 21) = 15$

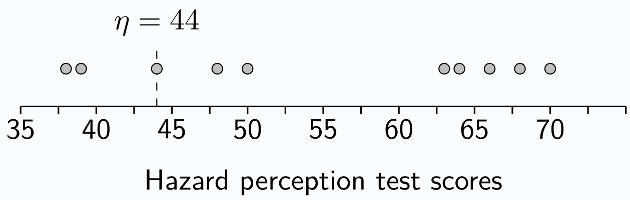
Critical value: $W = 1 \quad (\alpha = 0.01,\ n = 8$ and one-tailed.)

Since $15 > 1$, do not reject H_0 at the 1% level of significance. There is insufficient evidence to suggest that the median mass of crisps in the bags is less than the stated value of 150g.

Example

Problem: Before learner drivers can sit their practical test, they must sit a theory test. Part of this is a hazard perception test, in which a maximum of 75 points are available. Prior to some changes to the hazard perception test, the median score was 44. The points scored by ten randomly selected learner drivers in the updated hazard perception test are recorded:

63 38 48 44 50 70 68 39 64 66



Test whether there is sufficient evidence, at the 10% level of significance, that the median score has changed from 44.

Solution:

$H_0: \quad \eta = 44$

$H_1: \quad \eta \neq 44$

63	38	48	44	50	70	68	39	64	66	
19	-6	4	0	6	26	24	-5	20	22	(*subtract* $\eta = 44$)
5_+	3.5_-	1_+		3.5_+	9_+	8_+	2_-	6_+	7_+	(*assign signed ranks*)

$$W_+ = 39.5 \qquad \tfrac{1}{2} \times 9 \times 10 = 45$$

$$W_- = 5.5 \qquad \textit{Check: } 39.5 + 5.5 = 45$$

Test statistic: $W = \min(39.5, 5.5) = 5.5$

Critical value: $W = 8 \quad (\alpha = 0.1,\ n = 9$ and two-tailed.)

Since $5.5 < 8$, reject H_0 at the 10% level of significance. There is sufficient evidence to suggest that the median score achieved in the hazard perception tests has changed from 44 points.

Exercise 16.1

1. It is known that a certain tree produces apples with a median mass of 150g. Eight apples are chosen at random from a crate containing apples picked from the tree. The masses, in grams, are:

   147 138 171 142 152 145 141 143

   Use a non-parametric test, at the 5% significance level, to test whether the median mass of the apples is different from 150g.

2. The number of social media posts made in the last week is recorded for a random sample of eight UK teenagers, with the results:

   5 2 18 11 0 6 1 0

   Test at the 5% significance level the claim that the median number of social media posts made per week by a UK teenager is 10.

3. A recycling centre allows residents to deposit their metal waste into a skip set aside for it. The decision as to how often the skip will need to be emptied is based on an understanding that the median daily mass of metal waste deposited into the skip is 15kg, based on previous records. It has been suggested that this figure is no longer accurate. The recycling centre recorded the amount of metal waste deposited on eight randomly selected days. The results, in kg, were:

   11.6 14.7 15.1 14.3 16.2 9.7 15.2 10.7

   Perform a non-parametric test to assess, at the 10% level of significance, whether this sample provides evidence to support the suggestion that the median daily mass of metal waste deposited has changed.

4. The lifetimes of a random sample of a particular brand of candle, measured in minutes, are:

   372 352 335 364 345 360 354 358 348 341

   Determine if there is evidence at the 5% significance level that the median lifetime of this brand of candles is different from 6 hours.

5. A company's complaints department has a policy that states that the median length of time its customers should have to wait before receiving a response to any complaint should not exceed 14 days. A sample of ten customers who have received responses to complaints were asked how long they had to wait. The results, in days, were:

   15 10 17 13 18 12 23 20 19 21

   (a) State two assumptions required in order for a Wilcoxon Signed Rank Test to be performed.
   (b) Assess whether there is evidence to support a claim that the complaint department's policy is being breached, at the 5% level of significance.

6. A company is concerned about the length of time customers have to wait on a helpline before being connected to an agent. Previous records show that the median waiting time was 34 minutes, and following additional training for the agents the company wishes to gauge the impact it may have had. Customer waiting times, in minutes, for a random sample of ten calls are recorded:

   55 20 31 12 18 32 28 16 14 33

   Stating an assumption required, perform a non-parametric test at the 5% level of significance to assess whether there is evidence the median waiting time has reduced from 34 minutes.

7. A medical centre schedules doctors appointments for patients on the basis of a median appointment duration of 8 minutes. Wishing to determine whether this figure is reasonable, the length of ten randomly chosen appointments are recorded, and displayed below (in minutes).

   7 8 6 5 9 5 7 5 10 4

   Assess at the 5% level whether the data suggests the median appointment duration is not 8 minutes.

16.2 Wilcoxon Signed Rank Test for Paired Data

In Chapter 15, the *paired t-test* was introduced. It is used for assessing the *mean of the differences between paired values* from small samples, and a key assumption for the test is that the *population of differences is normally distributed*. Where this assumption cannot reasonably be made, a **Wilcoxon Signed Rank Test for Paired Data** can instead be used to assesses the **median of the differences**, η_d.

Conditions for valid use of the Wilcoxon Signed Rank Test for Paired Data:

- The **population of differences is symmetrical**.
- The observed differences are **independent** (typically assured through *random sampling*).

Just as with the paired t-test, the paired samples are used to produce a *single sample of differences*, allowing a similar process to the one-sample Wilcoxon Signed Rank Test.

With a null hypothesis of $\eta_{\text{difference}} = 0$, the step of subtracting the median from each value can be omitted for a paired Wilcoxon test, as subtracting zero will have no effect.

Example

Problem: An athletics coach wishes to assess the value to athletes of an intensive period of weight training. Twelve 400-metre runners are selected at random and their times to complete this distance, in seconds, are recorded. Following a programme of weight training they run the distance again, with their times in seconds again recorded. The table below summarises the results.

Athlete	A	B	C	D	E	F	G	H	I	J	K	L
Before	51.0	49.8	49.5	50.1	51.6	48.9	52.4	50.6	53.1	48.6	52.9	53.4
After	50.6	50.4	48.9	49.1	51.6	47.6	53.5	49.9	51.0	48.5	50.6	51.7

Stating an assumption required, perform a non-parametric test to assess whether there is sufficient evidence that the training programme improves athletes' times for the 400 metre distance.

Solution:
(Note that d = *Before* − *After* will give differences such that a positive number represents an improvement in running time, and η_d represents the population median of differences).

$\text{H}_0: \quad \eta_d = 0$
$\text{H}_1: \quad \eta_d > 0$

0.4	-0.6	0.6	1.0	0.0	1.3	-1.1	0.7	2.1	0.1	2.3	1.7	
2_+	3.5_-	3.5_+	6_+		8_+	7_-	5_+	10_+	1_+	11_+	9_+	(*assign signed ranks*)

$$W_+ = 55.5 \qquad \tfrac{1}{2} \times 11 \times 12 = 66$$
$$W_- = 10.5 \qquad \textit{Check: } 55.5 + 10.5 = 66$$

Test statistic: $W = \min(55.5, 10.5) = 10.5$

Critical value: $W = 13$ $(\alpha = 0.05$, $n = 11$ and one-tailed.)

Since $10.5 < 13$, reject H_0 at the 5% level of significance. There is sufficient evidence to suggest that the median difference in 400 metres times is greater than zero, meaning times have improved after the weight training programme.

Exercise 16.2

1. To measure the effectiveness of a drug for asthmatic relief, twelve subjects, all susceptible to asthma, were each randomly administered either the drug or a placebo during two separate asthma attacks. After one hour an asthmatic index was obtained on each subject with the following results:

Subject	1	2	3	4	5	6	7	8	9	10	11	12
Drug	28	31	17	18	31	12	33	24	18	25	19	17
Placebo	32	33	23	26	34	17	30	24	19	23	21	24

Investigate the claim that the drug reduces the asthmatic index.

2. A group of 8 patients with a particular illness were given a special diet and it was desired to assess the impact at the end of a 2-week period. Their masses, measured before and after the diet, are shown below, in kg.

Patient	A	B	C	D	E	F	G	H
Before	82.27	78.18	86.36	85.00	95.45	75.45	83.18	83.64
After	82.87	79.54	87.36	86.10	94.99	75.48	83.54	82.15

Use a non-parametric test to determine if there is evidence at the 5% level that the diet increases patient mass, stating any assumptions made.

3. As part of her research into the behaviour of the human memory, a psychologist asked 15 randomly selected school pupils to talk for five minutes on 'my day at school'. Each pupil was then asked to estimate how many times they thought that they had used the word "like" during the five minutes. The table below gives their estimates together with the true values.

Pupil	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
True value	12	20	1	8	0	12	12	17	6	5	24	23	10	18	16
Estimated value	9	19	3	14	4	12	16	14	5	9	20	16	11	17	19

Use a non-parametric test to investigate whether pupils can remember accurately the frequency with which they use a particular word in a verbal description.

4. Eight amateur runners are selected to study the effects of consuming energy gels before running on runners' performances at middle-distances. Each has their finish time recorded for a 5000m run without consuming energy gels beforehand, then a month later their time is recorded again for a 5000m run before which each consumed two energy gels. The data is shown in the table below, with times recorded to the nearest minute.

Runners	A	B	C	D	E	F	G	H
Without gels	24	19	21	17	26	32	18	20
With gels	22	19	20	18	23	28	15	20

Perform a hypothesis test at the 5% level of significance to assess whether the data supports a conclusion that consuming energy gels improves performances for runners.

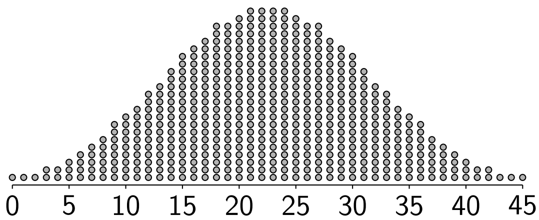
5. The delay in sound transmission through an audio interface is called *latency*. A music technology website wishes to investigate whether a company's new audio interface, really offers lower latency than its older one. A random sample of ten computers each have their audio latency measured, first with the older interface and then with the new interface. The results are as follows, measured in milliseconds:

Computer	1	2	3	4	5	6	7	8	9	10
Old interface	5.7	3.6	7.3	6.1	2.8	4.8	5.3	2.9	6.7	11.7
New interface	5.4	3.5	7.5	5.4	2.7	4.0	5.3	2.6	6.3	11.6

Stating an assumption required, perform a non-parametric test to assess whether the newer interface offers lower latency, at the 5% level of significance.

16.3 Normal Approximation to the Wilcoxon Signed Rank Test

Although large samples may allow data from *not normally-distributed* populations to be assessed using a parametric test due to the Central Limit Theorem, non-parametric tests are typically more robust when handling data from *strongly non-normal* populations. This means the Wilcoxon test is not only used for small samples.



Discrete distribution of W for $n = 9$

The discrete distribution of W under the null hypothesis is symmetrical and, as the sample size n increases, can be increasingly accurately *approximated* by the continuous normal distribution.

The Data Booklet only provides Wilcoxon critical values for $n \leqslant 20$. When dealing with a larger sample, a Wilcoxon Signed Rank Test may still be performed using a **normal approximation** to the distribution of W. **Page 15** of the Data Booklet provides the required properties of this distribution:

$$\mathsf{E}(W) = \frac{1}{4}n(n+1) \quad \text{and} \quad \mathsf{V}(W) = \frac{1}{24}n(n+1)(2n+1)$$

Since here a *discrete* distribution is being approximated by the *continuous* normal distribution, a **continuity correction** is required. When using the smaller rank sum $\min(W_+, W_-)$, a continuity correction of $+0.5$ should be applied.

Example

Problem: It is claimed by a local resident that more than 50% of all the vehicles on an urban road exceed the 30 mph speed limit. The speed of each of a random sample of 24 vehicles is recorded with the following results:

22	24	26	28	29	30	30	32	33	34	35	35
37	38	38	39	40	41	42	45	48	56	62	72

Sums of signed ranks for this data were calculated as: $W_+ = 223$ and $W_- = 30$.

Use a non-parametric test to investigate the resident's claim.

Solution:

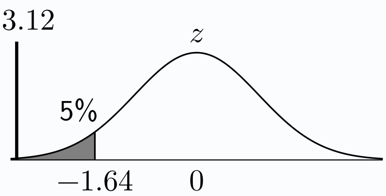
$\mathsf{H}_0: \ \eta = 30$
$\mathsf{H}_1: \ \eta > 30$

$W = \min(223, 30) = 30$

$$\mathsf{E}(W) = \frac{1}{4} \times 22 \times 23 = 126.5 \quad \text{and} \quad \mathsf{V}(W) = \frac{1}{24} \times 22 \times 23 \times 45 = 948.75$$

Test statistic: $z = \dfrac{30.5 - 126.5}{\sqrt{948.75}} = -3.12$

Critical value: $z = -1.64$



Since $-3.12 < -1.64$, reject H_0 at the 5% level of significance. There is sufficient evidence to suggest that the median speed of the vehicles on the road is greater than 30 mph, and so more than 50% of the vehicles are speeding.

Exercise 16.3

Note: sums of signed ranks are often provided in questions that require a normal approximation to the Wilcoxon test. The data must still be inspected to check whether any values have been discarded, to determine the effective sample size

1. Twenty six apples are chosen at random from a crate containing a large number of apples. The masses (to the nearest gram) are:

136	137	138	139	141	141	142	142	143	145	146	146	147
148	149	150	152	152	156	157	159	161	162	167	171	173

$$W_+ = 149.5 \qquad W_- = 175.5$$

   Use a Wilcoxon Signed Rank test, at the 5% significance level, to test whether the median mass of the apples is different than 150 grams, stating any assumptions made.

2. The lifetimes of a random sample of a particular type of candle, measured to the nearest minute, are:

268	335	339	341	341	345	346	348	352	354	355	356
357	358	359	361	362	363	364	367	371	373	374	374

$$W_+ = 91 \qquad W_- = 209$$

   The manufacturer claims that the average lifetime is at least 6 hours. Use a non-parametric test to assess whether the sample gives evidence to suggest the manufacturer's claim is not correct.

3. A local politician stated that the median hourly wage paid to young people aged 16-17 in the area is £9.20. Believing that the true median hourly wage paid is lower than this, a newspaper randomly selected 28 young people currently doing paid work and recorded their hourly wage (in £):

8.65	9.10	8.71	9.80	8.60	8.95	10.35	8.60	8.86	9.25	8.60	8.70	9.20	8.92
8.82	8.60	9.83	13.95	9.00	8.60	8.63	9.35	8.90	9.70	8.70	9.00	10.10	9.50

$$W_+ = 152 \qquad W_- = 173$$

   Use a non-parametric test to determine whether the sample supports the newspaper's suspicions.

4. A hillwalking website publishes details of popular hiking routes around Scotland. One particular route states that it takes 1 hour and 50 minutes to complete. Following upgrades to some of the paths which the route follows, the time taken to complete the route is recorded for a random sample of 32 hikers. The results of a non-parametric test conducted on the resulting data is shown below, with times recorded in minutes:

```
Wilcoxon test for the population median
data: hike times
alternative hypothesis: true location is less than 110
n = 32    sample median = 102
W = 148    p-value = 0.0154
```

   (a) State the null and alternative hypotheses for the test conducted.
   (b) Use the value of W provided in the computer output to perform a test of the above hypotheses at the 5% level of significance.
   (c) State an assumption required for the test.

Review Exercise

1. A random sample of nine primary school pupils are selected to test the number of times tables questions they can answer correctly in one minute. The results are as follows:

   24 33 36 19 37 21 16 47 20

   Perform a test to determine whether the data supports a claim that the median number of times tables questions a primary school pupil can answer in a minute is greater than 20.

2. To test whether displaying the speeds of cars travelling up a school drive on a digital display is effective in reducing driving speeds, ten cars that regularly drive up the school drive are randomly selected. For each, a speed is recorded on an occasion in which their speed is not displayed on a digital sign, and a speed is recorded when their speed is displayed. The results are as follows, in mph:

Vehicle	A	B	C	D	E	F	G	H	I	J
Speed not displayed	8	9	11	9	11	13	9	16	7	21
Speed is displayed	7	10	10	9	10	14	8	10	9	14

   Use a non-parametric test to assess whether this suggests displaying the driving speeds of cars travelling up the school drive is effective in reducing speeds, stating an assumption required.

3. A national newspaper states that the median amount of fuel motorists put into their vehicles' tanks when stopping at a petrol station is 20 litres. To investigate this claim, a motoring magazine stops thirty people at random whilst leaving a petrol station and asks how much fuel they just bought. The results, to the nearest litre, are:

32	45	17	54	20	9	15	34	38	26
40	24	61	6	18	21	34	48	28	25
9	10	13	10	25	22	38	41	23	29

$$W_+ = 106 \qquad W_- = 329$$

   Perform a non-parametric test to assess whether there is evidence to suggest that the figure claimed by the national newspaper is incorrect, at the 5% level of significance.

4. A car company wishes to check that a new machine at their factory producing front bumpers is still maintaining the median mass of 2.8kg for the part as the old machine it replaced. Eight bumpers produced by the machine are randomly selected and weighed, in kilograms. Computer output from a test performed on the data is as follows:

```
Wilcoxon test for the population median
data: bumper mass
alternative hypothesis: true location is not equal to 2.8
W = 25    n = 7 ranks assigned    sample median = 2.95
```

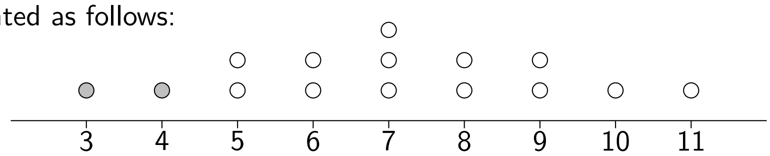
   By first calculating the required test statistic, perform a non-parametric test at the 5% level of significance to determine whether there is evidence to suggest that the median mass of the part produced by the new machine has changed, stating an assumption required.

17 Mann-Whitney Rank Sum Test

Consider a random sample consisting of two mice from species A and four from species B. The six observations are ordered from lightest to heaviest, and ranks from 1 to 6 are assigned accordingly. Under a null hypothesis that the distributions of mass for the two species are *identical*, each of the 15 possible allocations of ranks for the two mice from species A is *equally likely*. These 15 arrangements are shown below alongside the corresponding **rank sum** for species A:

Arrangement	Rank Sum	Arrangement	Rank Sum	Arrangement	Rank Sum
AABBBB	$1+2=3$	ABBBAB	$1+5=6$	BBABAB	$3+5=8$
ABABBB	$1+3=4$	ABBBBA	$1+6=7$	BBABBA	$3+6=9$
ABBABB	$1+4=5$	BABBAB	$2+5=7$	BBBAAB	$4+5=9$
BAABBB	$2+3=5$	BBAABB	$3+4=7$	BBBABA	$4+6=10$
BABABB	$2+4=6$	BABBBA	$2+6=8$	BBBBAA	$5+6=11$

Let W denote the rank sum for species A. The possible values of W and their frequencies (number of arrangements) can be represented as follows:



For example, the probability, *under the null hypothesis*, of obtaining a rank sum less than or equal to 4 is:

$$\mathsf{P}(W \leqslant 4) = \frac{2}{15} = 0.1333$$

since only two arrangements (with rank sums 3 and 4) produce such values.

If it is assumed that the population distributions of mass for the two species of mice have the *same shape and variance* then any difference between them corresponds to a shift in their medians. In this case, the Mann–Whitney test can be interpreted as a test comparing population medians, which may be *two-tailed* or, as below, *one-tailed*:

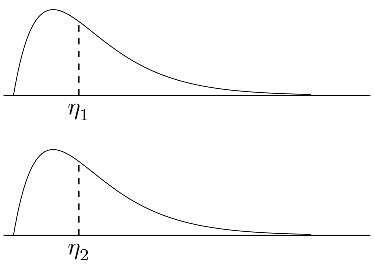
$$\begin{aligned} \mathsf{H}_0 &: \quad \eta_{\mathsf{A}} = \eta_{\mathsf{B}} \\ \mathsf{H}_1 &: \quad \eta_{\mathsf{A}} < \eta_{\mathsf{B}} \end{aligned}$$

Returning to the example rank sum of $W = 4$, a p-value of 0.1333 would mean there is insufficient evidence, at the 5% level of significance, to suggest that the median mass of a mouse of species A is less than the median mass of a mouse of species B (since $0.1333 > 0.05$).

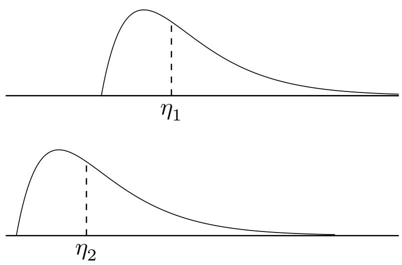
Since listing all possible arrangements quickly becomes impractical for larger samples, in practice Mann-Whitney tests are typically carried out using a rank sum test statistic W and *critical values* (from the Data Booklet).

17.1 Mann-Whitney Rank Sum Test using Critical Values

Given two independent samples, the *non-parametric* Mann-Whitney Rank Sum test can be used to assess whether the two population distributions from which the samples were taken have different *medians*. It is typically used in place of a two-sample t-test where the two distributions cannot be assumed to be *normal* and sample sizes are small. Instead a required assumption is that the population distributions have the **same shape and variance**, to test whether there is evidence of a difference in *locations* (medians):



Same locations: $\eta_1 = \eta_2$



Different locations: $\eta_1 \neq \eta_2$

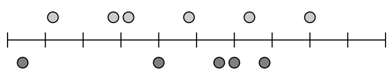
Conditions for valid use of the Mann-Whitney Rank Sum Test:

- The two populations have **the same shape and variance**.
- The observations are **independent** (typically assured through *random sampling*).

Consider two independent samples taken from populations A and B with the following hypotheses:

$$\left.\begin{array}{l} \mathsf{H}_0 : \eta_\mathsf{A} = \eta_\mathsf{B} \\ \mathsf{H}_1 : \eta_\mathsf{A} \neq \eta_\mathsf{B} \end{array}\right\} \begin{array}{l} \alpha = 0.05 \\ \text{Two-tailed} \end{array}$$

A	2.8	3.2	3.3	3.7	4.1	4.5
B	2.6	3.5	3.9	4.0	4.2	



To perform the test using a test statistic and critical value approach, first the samples are combined, ordered and ranked, with **rank sums** W_A and W_B obtained:

Values	2.6	2.8	3.2	3.3	3.5	3.7	3.9	4.0	4.1	4.2	4.5
Sample	B	A	A	A	B	A	B	B	A	B	A
Ranks	1	2	3	4	5	6	7	8	9	10	11

$W_\mathsf{A} = 35$
$W_\mathsf{B} = 31$

In this course, the test statistic W is defined as the *smallest possible rank sum that can be obtained for the smaller sample*, here B.
One option would be to reorder the values and rerank them to see if this produces a smaller rank sum W_B.

Values	4.5	4.2	4.1	4.0	3.9	3.7	3.5	3.3	3.2	2.8	2.6
Sample	A	B	A	B	B	A	B	A	A	A	B
Ranks	1	2	3	4	5	6	7	8	9	10	11

$W_\mathsf{A} = 37$
$W_\mathsf{B} = 29$

However this value can be obtained without reordering using the formula $m(m+n+1) - W_m$, where m and n represent the sizes of the smaller and larger samples respectively, and W_m is a rank sum for the smaller sample:

$$m(m+n+1) - W_m = 5(5+6+1) - 31 = 29$$

The test statistic can now be confirmed as $W = \min(31, 29) = 29$.

For a Mann-Whitney Rank Sum test:

- The **test statistic**, denoted W, is the *smaller* of W_m and $m(m+n+1) - W_m$, or:

$$W = \min\left(W_m,\ m(m+n+1) - W_m\right)$$

- The **critical value** can be obtained from the Data Booklet.
- H_0 is rejected if the test statistic W is **less than or equal to** the critical value.

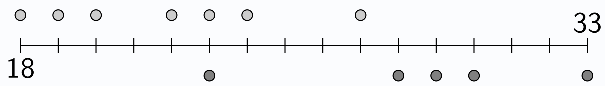
Taking as the test statistic the value $W = 29$, the test can now be concluded. **Page 16** of the Data Booklet contains critical values for the Mann-Whitney Rank Sum test, which are based on the smallest possible rank sum for the smaller sample. Using this approach, *small values* for the test statistic W offer support towards rejecting H_0 in favour of H_1.

The critical value of $W = 18$ can be obtained, using $m = 5$, $n = 6$ and a two-tailed test at the 5% significance level. Since $29 > 18$, H_0 cannot be rejected at the 5% level of significance. There is insufficient evidence to suggest that the two distributions have different population medians.

Example

Problem: Times taken to run 5km for a random sample of 7 hockey players and a random sample of 5 rugby players are recorded. The table below gives the results, to the nearest minute:

Hockey	18	19	20	22	23	24	27
Rugby	23	28	29	30	33		



Stating an assumption required, perform a non-parametric test to assess the data for evidence of a difference in the median time taken for hockey and rugby players to run 5km.

Solution: Assume the distributions of time taken to run 5km for hockey players and rugby players have the same shape and variance.

$H_0: \quad \eta_{\text{hockey}} = \eta_{\text{rugby}}$

$H_1: \quad \eta_{\text{hockey}} \neq \eta_{\text{rugby}}$

Values	18	19	20	22	23	23	24	27	28	29	30	33
Sample	H	H	H	H	R	H	H	H	R	R	R	R
Ranks	1	2	3	4	5.5	5.5	7	8	9	10	11	12

(Tied values are assigned the average of the ranks they occupy: $\frac{5+6}{2} = 5.5$)

$n = 7$, $W_{\text{hockey}} = 30.5$

$m = 5$, $W_{\text{rugby}} = 47.5$

Check for a smaller possible rank sum for W_{rugby}: $5 \times (1 + 5 + 7) - 47.5 = 17.5$

Test statistic: $W = \min(47.5, 17.5) = 17.5$

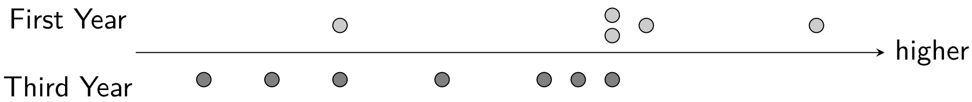
Critical value: $W = 20$ ($\alpha = 0.05$, $m = 5, n = 7$ and two-tailed)

Since $17.5 < 20$, reject H_0 at the 5% level of significance. There is evidence to suggest that the median time taken for rugby players to run 5km is different than the median time taken for hockey players.

Exercise 17.1

1. One year on World Maths Day (14th March) pupils across all the year groups in a school attempt a times tables speed challenge. A teacher checks the scores obtained by five randomly selected First Year pupils and seven randomly selected Third Year pupils. Their results are shown below.

First Year	14	6	14	15	20		
Third Year	12	9	6	13	2	14	4



A Mann-Whitney test is performed to assess whether the median score for First Year pupils is greater than that of Third Year pupils.

(a) Use the data provided in the table to verify the test statistic of 20.5.

(b) Conclude the test at the 5% level of significance.

2. A shopper regularly buys boxes of *medium-sized* eggs from two different supermarkets, *WaitThistle* and *Largl*. Over a period of time, they randomly select a single egg from each box they buy and measure their masses, in grams. The results are shown in the table below.

WaitThistle	59.7	56.9	53.9	54.4	50.0	51.9
Largl	55.8	49.3	49.4	56.3	51.2	

Test at the 10% level of significance whether the data indicates that there is a difference in median egg mass between the two supermarkets, for medium eggs.

3. A medical trial wishes to compare the effectiveness of two new types of mild painkiller. Twenty volunteers are selected and randomly assigned either painkiller A or painkiller B, and an effectiveness score from 0 to 5 is obtained for each person through an industry-recognised pain tolerance test using ice-cold water.

Painkiller A	2.5	1.4	1.6	1.9	2.2	1.4	1.5	2.9	1.6	2.5
Painkiller B	2.2	1.6	2.0	2.3	3.7	3.4	2.8	3.0	2.7	3.8

Perform a Mann-Whitney test at the 2% level, using the test statistic of $W = 18.5$, to assess whether there is a difference between the effectiveness of the two painkillers.

4. The contents of a random sample of six bags of coffee beans sold by *Brewhemian Rhapsody* were measured. The results, in grams, were:

254 255 254 248 256 251

The results for a random sample of eight bags sold by *The Bean Supremacy* were:

240 249 244 246 245 250 240 253

Use a non-parametric test to assess the evidence that the *Brewhemian Rhapsody* bags contain more coffee.

5. A gardener tests a new bag of compost against their old stored compost by sowing twenty seeds for the *Monstera deliciosa* plant in a large seed tray with individual compartments to keep each separate. Ten compartments were randomly selected to be filled with the new compost, and ten with the old compost. Only nine seeds from each group germinated, and their heights after one month were recorded, and are displayed in the stem-and-leaf diagram below.

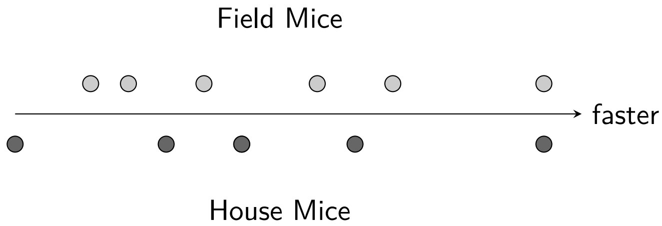
Old Compost		New Compost
7	1	
9 4	2	8
8 6 4 2	3	4 7 9
0	4	3 4 5 6
1	5	0

5 | 4 represents 5.4 cm

Use a Mann-Whitney test to determine whether the data suggests that the new compost supports faster growth.

6. The times (in minutes) it took six randomly selected field mice to learn to run a simple maze and the times it took six randomly selected house mice to learn to run the same maze are given.

Field Mice	House Mice
18	24
24	16
12	10
20	19
13	14
15	16



Stating an assumption required, use a non-parametric test to determine if there is a difference in learning time between the two types of mice.

7. A motoring magazine tested the range of electric car models from two manufacturers. A looping route was devised with a mix of road conditions, and each car was driven around this route until they ran out of charge. Twelve cars from *Ohm-G Motors* and nine from *Watts-Up Motors* were tested, and their observed driving range in miles for the route recorded in the table below.

Ohm-G	314	327	364	352	373	385	305	384	326	387	352	330
Watts-Up	355	320	295	314	329	291	332	305	336			

A non-parametric test for a difference in population medians was performed at the 5% level of significance.

The computer software used is known to return a test statistic which may or may not be the *smallest possible* rank sum.

```
Mann-Whitney rank sum test
data: Ohm-G and Watts-Up
alternative hypothesis: true location shift is not equal to 0
W for Watts-Up = 128    p-value = 0.04262
```

(a) State the suitable null and alternative hypotheses the test conducted.

(b) By performing the usual check and the rank sum for *Watts-Up* given in the computer output, obtain the required test statistic.

(c) Use the answer from (b) to complete the hypothesis test and interpret the result.

17.2 Normal Approximation to the Mann-Whitney Rank Sum Test

Although large samples may allow data from *not normally-distributed* populations to be assessed using a parametric test due to the Central Limit Theorem, non-parametric tests are typically more robust when handling data from *strongly non-normal* populations. This means the Mann-Whitney test is not only used for small samples.

The Data Booklet only provides Mann-Whitney critical values for sample sizes where $n \leqslant 20$ and $m \leqslant 20$. When dealing with larger samples, where $m, n > 20$, a Mann-Whitney Rank Sum test can still be performed using a **normal approximation** to the distribution of W (under H_0). **Page 16** of the Data Booklet provides the required properties of the this distribution:

$$\mathsf{E}(W) = \frac{1}{2}m(m+n+1) \quad \text{and} \quad \mathsf{V}(W) = \frac{1}{12}mn(m+n+1)$$

Since here a *discrete* distribution is being approximated by the *continuous* normal distribution, a **continuity correction** is required. When using the smaller rank sum W_m, a continuity correction of $+0.5$ should be applied.

Example

Problem: 26 Atlantic herrings and 24 North Sea herrings were caught and their lengths measured in cm. The 50 lengths were ordered from smallest to largest and the ranks summed:

$$W_{\text{Atlantic}} = 503 \qquad W_{\text{North Sea}} = 772$$

Assess at the 5% significance level whether Atlantic and North Sea herrings differ in median length.

Solution:

$\mathrm{H}_0: \ \eta_{\text{Atlantic}} = \eta_{\text{North Sea}}$
$\mathrm{H}_1: \ \eta_{\text{Atlantic}} \neq \eta_{\text{North Sea}}$

Find the smallest possible rank sum for the smaller sample (to obtain the required value of W):

$m = 24, n = 26, W_{\text{North Sea}} = 772$

Check for a smaller possible rank sum for $W_{\text{North Sea}}$: $24 \times (24 + 26 + 1) - 772 = 452$

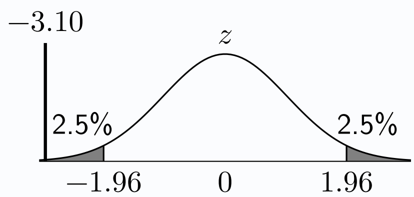
$W = \min(772, 452) = 452$

$\mathsf{E}(W) = \dfrac{1}{2} \times 24 \times (24 + 26 + 1) = 612$

$\mathsf{V}(W) = \dfrac{1}{12} \times 24 \times 26 \times (24 + 26 + 1) = 2652$

$W \approx \mathsf{N}(612, 2652)$

Test statistic: $z = \dfrac{452 + 0.5 - 612}{\sqrt{2652}} = -3.10$



Critical value: $z = -1.96$

Since $-3.10 < -1.96$, reject H_0 at the 5% level of significance. There is evidence to suggest that the median length of an Atlantic herring differs from the median length of a North Sea herring.

Exercise 17.2

1. A professional street photographer with an interest in dramatic scenes believes she takes more photographs which she wants to save, which she calls *keepers*, on rainy days than on dry days. She randomly selects 48 of the days from the previous year during which she was out taking photographs and uses her dated database of photographs to note for each day whether it was dry or rainy, and the number of *keepers* she took.

Dry	20	Dry	30	Dry	35	Dry	39	Rainy	41	Rainy	50
Dry	24	Rainy	31	Rainy	36	Rainy	39	Rainy	42	Rainy	52
Rainy	25	Rainy	31	Dry	36	Rainy	39	Rainy	43	Rainy	53
Dry	25	Dry	32	Dry	36	Dry	39	Dry	43	Dry	54
Dry	27	Dry	32	Dry	37	Rainy	40	Rainy	43	Rainy	57
Rainy	27	Rainy	34	Rainy	37	Dry	41	Rainy	48	Rainy	57
Dry	28	Rainy	34	Rainy	38	Rainy	41	Rainy	48	Dry	62
Dry	30	Dry	35	Rainy	38	Dry	41	Rainy	50	Rainy	66

The data was ranked from smallest to largest, and the rank sum for the 21 *dry* days was 396.5. Perform a Mann-Whitney test using $\alpha = 0.05$ to determine whether or not her belief is supported.

2. An agricultural scientist measures the soil pH level of thirty randomly selected fields in the North West of England and thirty in the South East. The data was ranked and rank sums were obtained:

$$\text{North West rank sum} = 1015 \qquad \text{South East rank sum} = 815$$

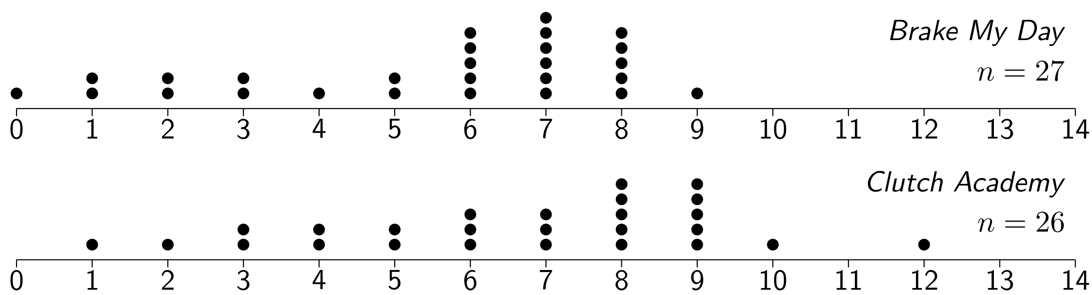
Carry out a hypothesis test to assess the evidence for a difference in median soil pH levels for fields in the two regions, at the 1% level.

3. A technology magazine tested the cooling power of random samples of 21 fans from brand A and 25 from brand B. When ranked from lowest cooling power to highest, the brand A fans had the higher rank sum. The results were ranked and rank sums were:

$$W_{\mathsf{A}} = 583 \qquad W_{\mathsf{B}} = 498$$

Perform a non-parametric hypothesis test to assess whether the data provides evidence that brand A fans have more cooling power.

4. A driving test centre notes, for a random selection of students who sit and pass their test, how many *minor faults* they made and which of the town's two driving schools they used, between *Brake My Day* and *Clutch Academy*. The results are shown on dotplots below.



(a) Explain why the dot plots suggest a Mann-Whitney test would be more appropriate for this data than a t-test.

The rank sum for *Clutch Academy* is found to be 606.5.

(b) Carry out a Mann-Whitney test to determine if there is evidence at the 0.1% significance level that, on average, students from *Brake My Day* make fewer errors when they pass their test.

(c) With reference to the target population and sampling frame, explain why this data may not be helpful to students looking to choose a driving school to help them pass.

17.3 Mann-Whitney from First Principles

In Chapter 10, hypothesis tests were initially introduced using p-values, which can be compared directly to the chosen significance level to conclude a test. Exact p-values can also be calculated directly for Mann-Whitney tests using an approach introduced on the first page of this chapter. The conditions required for a Mann-Whitney test using the *first principles* method are the same as when using critical values, but with the addition that the data must contain *no tied values*, and hence no shared ranks.

The standard steps involved in performing a one-tailed test in this manner are, for samples of size m and n (where $m \leqslant n$):

- Use ${}^{m+n}C_m$ to calculate the number of possible arrangements of the $m+n$ values.
- Use the observed data to obtain the rank sum for the smaller sample (of size m) as usual.
- Typically by listing all relevant *subsets* of ranks for the smaller sample, determine the number of possible such subsets that result in a rank sum equal to or less than the rank sum obtained from the data.
- The p-value for the test can be calculated using the values from the previous steps as $\dfrac{\text{relevant arrangements}}{\text{possible arrangements}}$.

Example

Problem: Three randomly selected cameras from Brand A and seven from Brand B are tested for reliability, with the number of *shutter actuations* before failure recorded. When ranked with 10 as the most reliable (greatest number of actuations) to 1 as the least reliable (least number of actuations), a rank sum of 9 is obtained for Brand A.

By considering possible arrangements of rankings, and stating any assumptions required, perform a Mann-Whitney test at the 10% significance level to assess whether the data suggests Brand A's cameras are less reliable.

Solution:

Assume that the population distributions of number of actuations for each brand's cameras have the same shape and variance.

$\text{H}_0: \quad \eta_\text{A} = \eta_\text{B}$
$\text{H}_1: \quad \eta_\text{A} < \eta_\text{B}$

Number of possible arrangements $= {}^{10}C_3 = 120$.

Subsets of relevant arrangements of ranks with a rank sum for A$\leqslant 9$:

Ranks for A	Rank Sum
1,2,3	6
1,2,4	7
1,2,5	8
1,2,6	9
1,3,4	8
1,3,5	9
2,3,4	9

p-value $= \text{P}(\text{rank sum} \leqslant 9) = \dfrac{7}{120} = 0.0583$

Since $0.0583 < 0.1$, reject H_0 at the 10% level of significance. There is evidence to suggest that the median number of shutter actuations before failure for Brand A is less than that for Brand B, and so Brand A's cameras are less reliable.

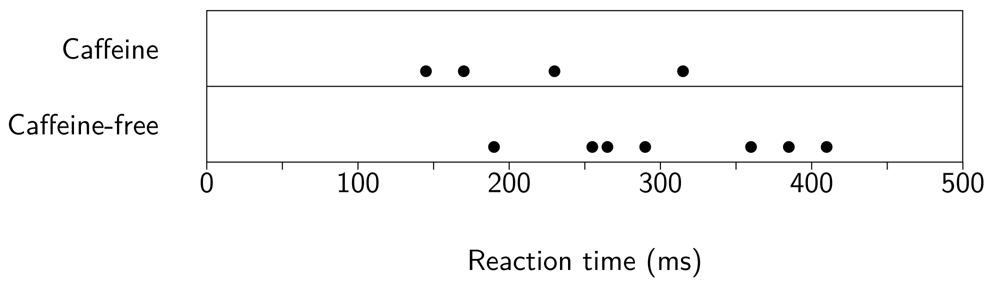
Exercise 17.3

1. A Mann-Whitney test is to be performed using two independent samples of sizes 3 and 5. Once data for the samples have been collected and ordered, a rank sum will be obtained for the smaller sample.
   (a) Determine the number of possible arrangements of the 8 values.
   (b) List all the possible subsets of ranks for the smaller sample which would result in a rank sum of 8 or lower.
   (c) Hence determine P(rank sum $\leqslant 8$).

2. A Mann-Whitney test is to be performed using two independent samples of sizes 4 and 5. Once data for the samples have been collected and ordered, a rank sum will be obtained for the smaller sample.
   (a) Determine the number of possible arrangements of the 9 values.
   (b) List all the possible subsets of ranks for the smaller sample which would result in a rank sum of 12 or lower.
   (c) Hence determine P(rank sum $\leqslant 12$).

3. To investigate the effects of caffeine on reaction times, four students from a group of eleven are randomly chosen to be given a soft drink containing caffeine, whilst the remaining seven are given the caffeine-free version of the soft drink. The students will be unaware which they have consumed.

   The reaction time of each student is then tested using an online test, with the results shown on the dot plot below.



   (a) Show the calculation required to obtain the rank sum of 15 for the caffeine group.
   (b) List all of the subsets of ranks for the caffeine group which would produce a rank sum equal to or less than 15.

   A non-parametric hypothesis test is to be conducted under the following hypotheses:

$$\mathrm{H}_0 : \eta_{\text{caffeine}} = \eta_{\text{caffeine-free}}$$
$$\mathrm{H}_1 : \eta_{\text{caffeine}} < \eta_{\text{caffeine-free}}$$

   (c) State an assumption required for the test.
   (d) Under the null hypothesis, calculate the probability of the rank sum for the caffeine group being equal to or less than 15.
   (e) Use your answer to part (b) to conclude the test at the 5% level of significance.

4. A 800m race between four hockey players and eight rugby players results in the hockey players finishing first, second, fourth and sixth. Assume that the population distributions of times for each group have the same shape and are equal in variance.
   (a) By considering possible arrangements and relevant subsets, calculate the p-value for a Mann-Whitney test with an alternative hypothesis that hockey players are faster over 800m than rugby players.
   (b) Use your answer from (a) to conclude the test at the 10% level of significance.
   (c) State an assumption required for the test to be valid.

Review Exercise

1. A wildlife enthusiast sets up a remote camera which is configured to take a photo when movement is detected. Sometimes it is placed in some local woods, whilst at other times it is positioned beside a pond. The photos are later reviewed and the number of animals spotted each day by the camera is recorded. The number of animal sightings at each location over the last two weeks is shown below.

Woods	7	3	6	2	10	8		
Pond	6	4	1	7	3	1	4	3

   Perform a non-parametric hypothesis test at the 5% level of significance to determine whether there is a difference in the median number of wildlife sightings between the two camera locations.

2. 23 red wines and 23 white wines in stock at a large supermarket are randomly chosen to have their alcohol content (%ABV) recorded. The results are displayed in the stem-and-leaf diagram below.

white		red
	15	0 0
0	14	0 2 5 5 8
5 5 3 0 0	13	0 0 5 5 5 5 5 8
9 8 5 5 5 5 0 0	12	0 5 5 8 8
8 5 5 3 0 0	11	0 5
5 5 2	10	5

   $12 \mid 8$ represents 12.8%

   (a) Comment on what the stem-and-leaf diagram shows.

   The data are ranked from lowest to highest, and a rank sum for the white wines of 401.5 is obtained.

   (b) Perform a Mann-Whitney test to assess at the 10% level of significance whether there is evidence to suggest the median alcohol content differs between white and red wines.

   A t-test was also considered for a comparison of the alcohol content of the two types of wine.

   (c) Comment on what the stem-and-leaf diagram may suggest about the suitability of both a t-test and a Mann-Whitney test.

3. Independent samples, of sizes 4 and 9 respectively, are taken from populations A and B. A non-parametric hypothesis test is performed at the 2% significance level, with truncated computer output from the test as follows:

```
Mann-Whitney test
data: sample A and sample B
alternative hypothesis: true location shift is not equal to zero
W = 12
```

   (a) State the null and alternative hypotheses used for the test.

   (b) Determine the critical value, and hence make a suitable conclusion.

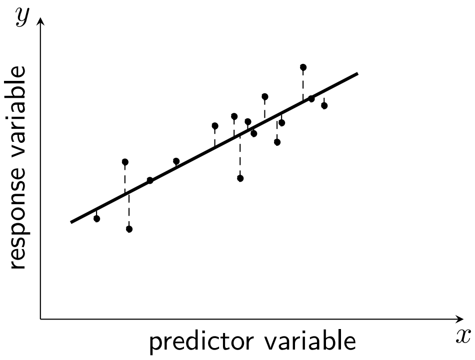
   (c) State two assumptions required for the test performed to be valid.

   (d) Explain the steps required in order to obtain the test statistic of `W=12`.

18

Further Linear Regression

Given bivariate data, one variable is selected as the **predictor** x_i, whilst the other variable is selected as the **response** Y_i (modelled as a random variable). It is common to seek to estimate an unknown Y_i value given a known x_i value.



The linear regression model assumes a straight line describing the best fit for a relationship between variables. However, observed data will not lie exactly along this line due to random variation.

The Linear Regression Model

$$Y_i = \alpha + \beta x_i + \varepsilon_i$$

The **errors** ε_i (*"epsilon"*) represent the random deviation of the response variable from the true regression line.

The true values of the **intercept parameter** α and the **slope parameter** β are unknown, as they relate to the full *population* of bivariate data. However, the sample observations (x_i, y_i) can be used to give estimates, a and b, of α and β.

Since *errors* are the vertical distances between observations (x_i, y_i) and the *true* regression line defined by α and β, they cannot be known precisely. Instead, vertical distances to the *estimated* regression line (using a and b) are considered, called **residuals** (e_i). Residuals provide observable estimates of the underlying errors.

The method of least squares produces formulae for a and b derived by minimising the sum of squared residuals, $\sum e_i^2$. These rely on several assumptions about the population of errors given on **page 5** of the Data Booklet, along with the formulae:

$$b = \frac{S_{xy}}{S_{xx}} \qquad \text{and} \qquad a = \bar{y} - b\bar{x}$$

Assumptions for the Errors

- ε_i are independent.
- $\mathrm{E}(\varepsilon_i) = 0$
- $\mathrm{V}(\varepsilon_i) = \sigma^2$

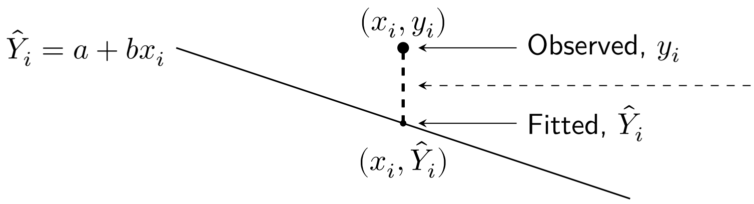
This leads to the regression line $\hat{Y}_i = a + bx_i$.

Whilst the terms ‘errors’ and ‘residuals’ are often casually used interchangeably, recognising the distinction is helpful. Assumptions about a population of *errors* ε_i play an important role in linear regression, and learning about these errors (which are treated as random variables) comes from observing a sample of *residuals* e_i.

18.1 Calculating the Value of a Residual

For any given x_i, the equation of the least squares regression line $\hat{Y}_i = a + bx_i$ gives a *predicted* value $\hat{Y}_i$, also referred to as a **fitted** value. For any data point (x_i, y_i), the value of y_i is called the **observed** value.

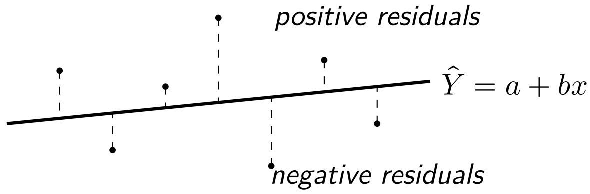
The difference between an observed value y_i and a corresponding fitted value $\hat{Y}_i$ is called a **residual**:



Calculating a Residual

Residual = Observed − Fitted

$e_i = y_i - \hat{Y}_i$



Data points *above* the regression line have a *positive* residual: the observed y_i was *greater* than that predicted by the regression line for the value of x_i.

Data points *below* the regression line have a *negative* residual: the observed y_i was *less* than that predicted by the regression line for the value of x_i.

To calculate a residual the fitted value, $\hat{Y}_i$, needs to first be found using the equation of the regression line.

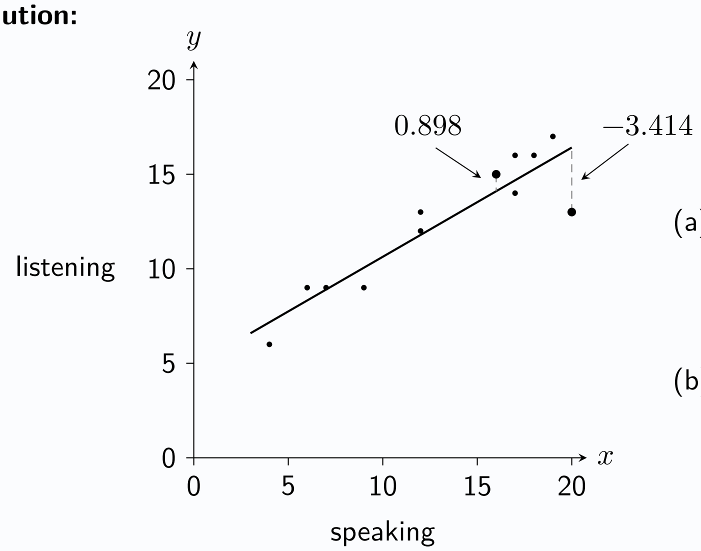
Example

Problem:
Students sitting a French exam are given a mark for each of the two components: speaking (x) and listening (y). From a random sample of 12 students, the y on x regression line is calculated as $\hat{Y} = 4.854 + 0.578x$. Calculate the residual for a student who gains:

(a) 20 in the speaking test and 13 in the listening test.

(b) 16 in the speaking test and 15 in the listening test.

Solution:



(a) $\hat{Y} = 4.854 + 0.578 \times 20 = 16.414$
Residual $= 13 - 16.414 = -3.414$

(b) $\hat{Y} = 4.854 + 0.578 \times 16 = 14.102$
Residual $= 15 - 14.102 = 0.898$

Exercise 18.1

1. Copy and complete the table below:

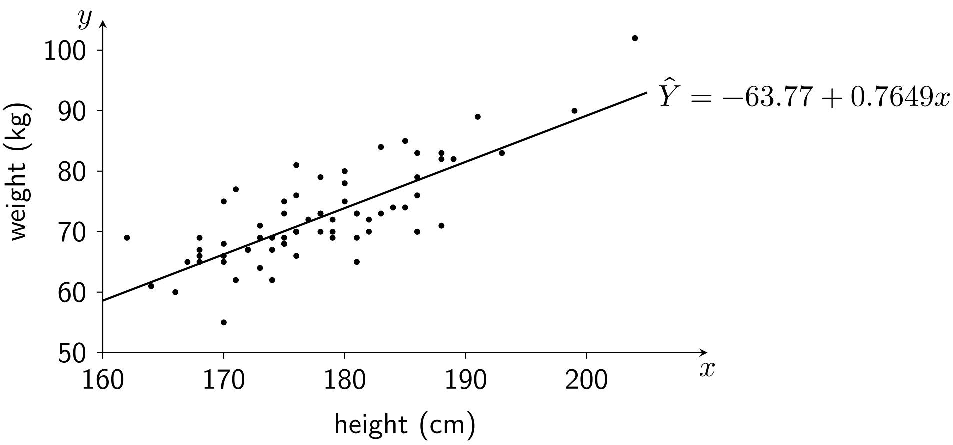
Regression Line $\hat{Y} = a + bx$	Observed x	Observed y	Fitted $\hat{Y}$	Residual ε
$\hat{Y} = 5.3 + 0.1x$	7	8		
$\hat{Y} = -0.7 + 0.3x$	6.1	1.04		
$\hat{Y} = 180 - 2.5x$	40	178		
$\hat{Y} = -45 + 3.5x$	200			-30
$\hat{Y} = 100 + 20x$	-15			50

2. The relationship between the number of calories (x) and the carbohydrate content (y grams) of different types of drinks available in a city's coffee shops can be described using the least squares regression line:

$$\hat{Y} = 5.5572 + 0.6361x$$

   (a) Calculate the expected carbohydrate content for a small cappuccino, which contains 50 calories.
   (b) Determine the value of the residuals for each of the following:
      i. An extra large white chocolate mocha (450 calories and 310 grams of carbohydrates).
      ii. A large caramel macchiato (240 calories and 150 grams of carbohydrates).
      iii. A double espresso (10 calories and 1 gram of carbohydrates).

3. The heights (x cm) and weights (y kg) are recorded for a sample of male professional footballers.



A linear model is fit to the data as shown with a regression line equation of:

$$\texttt{weight = -63.77 + 0.7649} \times \texttt{height}$$

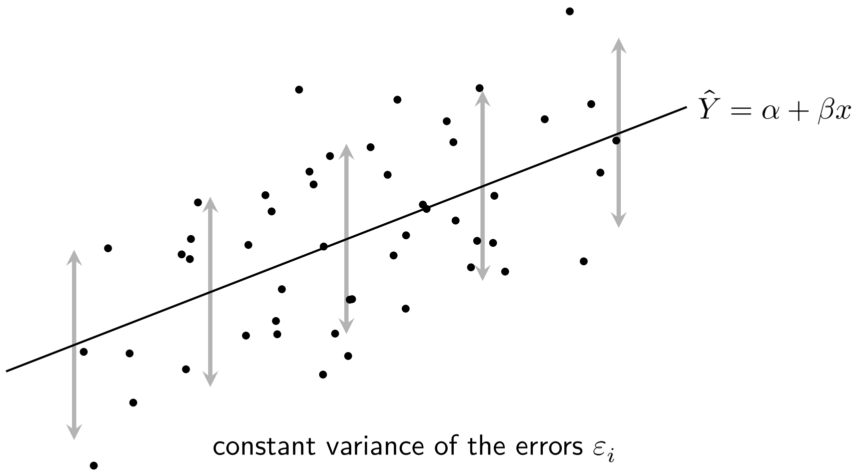
   (a) State whether each of the following have positive or negative residuals:
      i. The tallest footballer shown.
      ii. The lightest footballer shown.
   (b) Find the value of the residual for a footballer with:
      i. Height 186cm and weight 70kg.
      ii. Height 191cm and weight 89kg.

18.2 Estimating the Error Variance

The least squares regression model assumes that the variance of the errors is a constant for all x_i (and Y_i):

$$\mathsf{V}(\varepsilon_i) = \sigma^2$$

The error variance represents the variability of the response variable around the regression line, loosely indicated below by the series of arrows below.



The error variance σ^2 is estimated using the sample of *residuals* as:

Estimating the Error Variance

$$s^2 = \frac{\mathsf{SSR}}{n-2}$$

Where the (always non-negative) *sum of squared residuals*, SSR, is calculated as: $\mathsf{SSR} = S_{yy} - \dfrac{S^2{}_{xy}}{S_{xx}}$

Both the formula for estimating the error variance σ^2 and the more convenient second formula for the sum of squared residuals SSR are given on page 5 of the Data Booklet.

Example

Problem:
Students sitting a French exam are given a mark for each of the two components: speaking (x) and listening (y). From a random sample of 12 students: $S_{xx} = 344.9$, $S_{yy} = 132.9$, $S_{xy} = 193.6$.

(a) Calculate the sum of squared residuals for this data set.

(b) Obtain an estimate of the error variance.

Solution:

(a) $\mathsf{SSR} = S_{yy} - \dfrac{S^2{}_{xy}}{S_{xx}} = 132.9 - \dfrac{193.6^2}{344.9} = 24.23$

(b) $s^2 = \dfrac{\mathsf{SSR}}{n-2} = \dfrac{24.23}{12-2} = 2.423$

Exercise 18.2

1. A bivariate random sample has summary statistics:

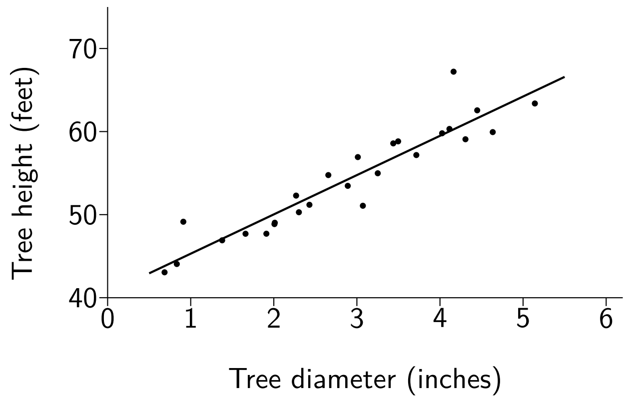
$$S_{xx} = 28.92 \qquad S_{yy} = 495.0 \qquad S_{xy} = 105.3 \qquad n = 14$$

It is determined that a linear relationship appears to exist between the variables.

(a) Obtain an estimate s^2 for the error variance, σ^2.

(b) Hence determine the value of s for this bivariate data.

2. 26 randomly selected trees in a forest have their diameter (x) at a set distance from the ground measured, in inches, and the height (y) of each tree also measured, in feet. The equation of the least squares regression line for y on x is obtained, and shown on the scatterplot below.



Summary statistics:

$\sum x = 74.76$

$\sum x^2 = 253.76$

$\sum y = 1408.3$

$\sum y^2 = 77277$

$\sum xy = 4233.1$

$n = 26$

(a) Obtain an estimate s^2 for the error variance, σ^2.

(b) Hence determine the value of s for this bivariate data.

3. A running website randomly selected eleven amateur runners who recently completed a marathon and asked them to complete a survey which included their finishing time in the marathon and information about their training routine. The scatterplot below shows, for each runner, their finishing time (x, in hours) and the total distance they ran in the final four weeks of their training (y, in miles).

Summary statistics:

$\sum x = 44.5$

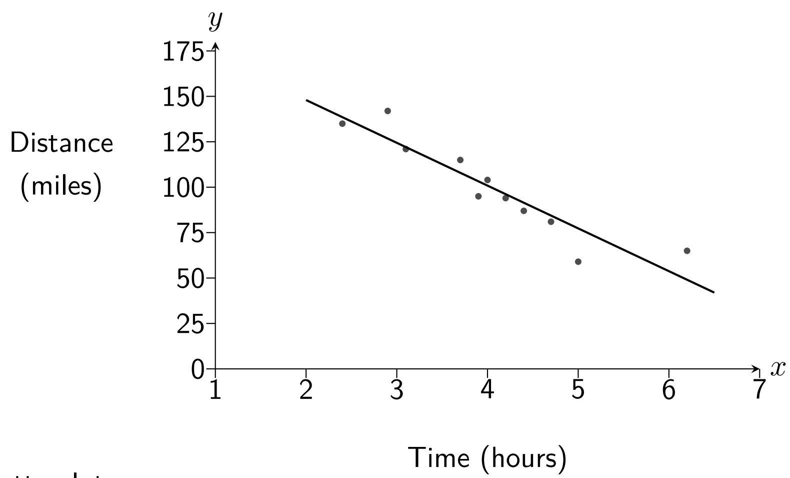
$\sum x^2 = 191.21$

$\sum y = 1098$

$\sum y^2 = 116768$

$\sum xy = 4179.2$

$n = 11$



(a) Make a comment on the scatterplot.

(b) Obtain an estimate s^2 for the error variance, σ^2.

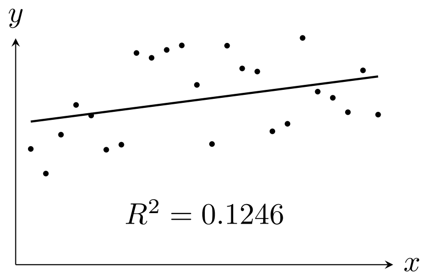
(c) Determine the equation of the least squares regression line.

(d) Calculate the correlation coefficient for the data and comment on its value.

(e) A runner, not shown, completed the marathon in 3 hours and 30 minutes, after running a total of 110 miles in the four weeks before the race. Calculate the residual for this runner.

18.3 The Coefficient of Determination

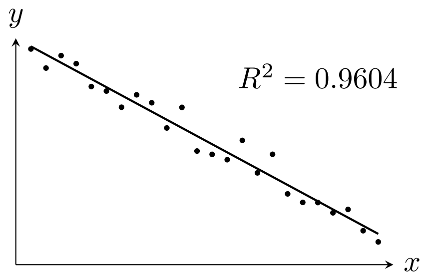
The *coefficient of determination*, R^2, is the proportion of the total variation in the response variable, Y_i, that is explained by the linear regression model. The higher the value of R^2, the less unexplained variation remains in the response variable Y_i. This means that the coefficient of determination is a very useful measure of how well a model explains variation in the data. (However, it guarantee that a linear model is appropriate, nor does it imply *causation*).



12.5% of the variation in Y is explained by the linear model.

Coefficient of Determination

$$R^2 = \frac{\text{explained variation}}{\text{total variation}}$$



96.0% of the variation in Y is explained by the linear model.

For the linear regression models covered in this course, the coefficient of determination is connected to the *correlation coefficient*, r, by a simple relationship:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \qquad \Longrightarrow \qquad \boxed{R^2 = r^2}$$

Both the formula above and the interpretation of the coefficient of determination, as a proportion, implies that $0 \leqslant R^2 \leqslant 1$. Note that R^2 does not indicate the direction of the relationship; the sign of r must be used for this.

Example

Problem:
Students sitting a French exam are given a mark for each of the two components: speaking (x) and listening (y). From a random sample of 12 students, the following summary statistics are obtained:

$$S_{xx} = 334.9 \quad S_{yy} = 132.9 \quad S_{xy} = 193.6 \quad n = 12$$

A linear regression model is proposed, and an equation for the regression line of y on x is obtained:

$$\hat{Y} = 4.854 + 0.578x$$

Calculate the coefficient of determination and interpret this value in context.

Solution:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{193.6}{\sqrt{334.9 \times 132.9}} = 0.9177$$

$$R^2 = r^2 = 0.9177^2 = 0.842$$

This indicates that 84.2% of the variation in listening marks can be explained by the linear model.

The conclusion from the example above could also be phrased as:

"*84.2% of the variation in listening marks can be explained by the speaking marks.*"

Exercise 18.3

1. A bivariate random sample has summary statistics:

$$S_{xx} = 28.92 \qquad S_{yy} = 495.0 \qquad S_{xy} = 105.3 \qquad n = 14$$

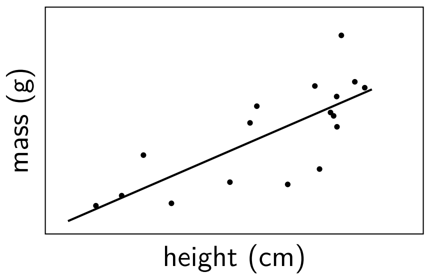
   Determine the value of the coefficient of determination and interpret it.

2. For a series of studies, computer output giving equations of least square regression lines and some summary statistics are provided. For each, determine the coefficient of determination and interpret it.

   (a) The salary and time employed is recorded for a sample of 70 employees at a company.


```
y on x regression line
salary = 20359 + 1506 × years_employed

sample estimate for r
0.857
```

   (b) For a random sample of 17 red squirrels, the mass and height are recorded.



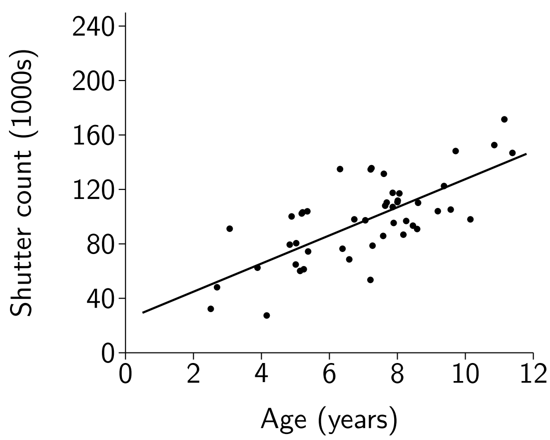

```
y on x regression line
mass = 45.5 + 6.50 × height

sample correlation coefficient
0.757
```

   (c) The top speed reached (in mph) and the lap times (in seconds) are recorded for a sample of drivers on a race track.

```
y on x regression line              n     r        s
top_speed = 269.2 - 1.72 × lap_time    34    -0.670   8.49
```

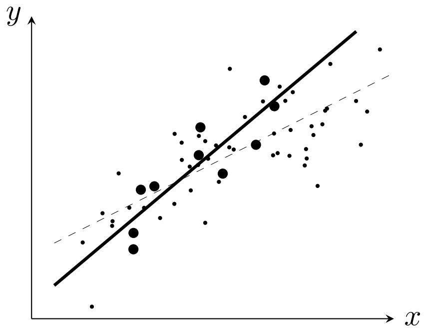
3. A seller on an online auction site has a stock of used digital cameras, all aged somewhere between 2 and 12 years. For each camera, the total number of photos taken (the *shutter count*) is determined using special software. A y on x regression line is shown on the scatterplot of the data below. A correlation coefficient is calculated for the data of 0.718.

   (a) Make one comment on the scatterplot.
   (b) Interpret the correlation coefficient.
   (c) Calculate the coefficient of determination, and interpret it in context.
   (d) The seller is considering expanding their business to also sell a range of film cameras, produced between 1976 and 1984. Give two reasons why they should not use the regression line shown to make predictions about the shutter counts of the any film cameras they sell.



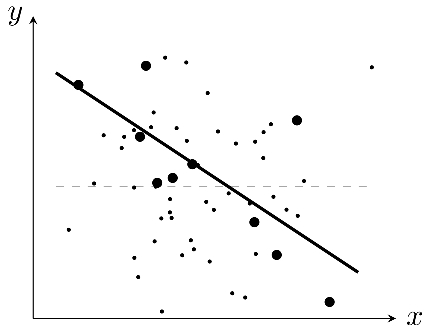
18.4 Test for the Slope Parameter β

The true value of the *slope parameter*, β, cannot be known without observing the full population. The *sample statistic* b is an estimator for β, and may differ slightly or substantially from it, as shown below.



Population:
$\beta = 0.5$

Sample:
$b = 0.84$



Population:
$\beta = 0$

Sample:
$b = -0.66$

The slope parameter β can be *positive*, *negative* or *zero*. Where $\beta = 0$ it can be said that knowing x_i has no value in predicting y_i. In other words, a linear regression model is only **useful for prediction** in cases where the slope parameter is *non-zero*, or $\beta \neq 0$. The *parametric* **Test for Beta** assesses whether data from a sample provide evidence to suggest that this is the case. The test statistic below follows a t-distribution with $n-2$ degrees of freedom, and in this course the test will always be *two-tailed* with hypotheses $\mathrm{H}_0 : \beta = 0$ and $\mathrm{H}_1 : \beta \neq 0$.

Test Statistic

$$t_{n-2} = \frac{b\sqrt{S_{xx}}}{s}$$

Conditions

- ε_i are independent and identically distributed as $\mathrm{N}(0, \sigma^2)$.
- The data was obtained through random sampling.

Example

Problem: A bivariate sample has summary statistics:

$$S_{xx} = 157.875 \qquad S_{yy} = 188 \qquad S_{xy} = -133.5 \qquad n = 8$$

Test at the 5% significance level whether a linear regression model would be useful for prediction.

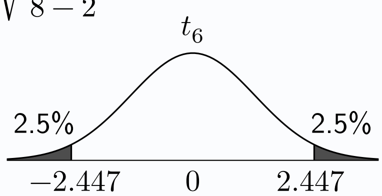
Solution:

$$H_0 : \quad \beta = 0$$
$$H_1 : \quad \beta \neq 0$$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{-133.5}{157.875} = -0.8456$$

$$\mathrm{SSR} = S_{yy} - \frac{(S_{xy})^2}{S_{xx}} = 188 - \frac{(-133.5)^2}{157.875} = 75.11, \quad s = \sqrt{\frac{\mathrm{SSR}}{n-2}} = \sqrt{\frac{75.11}{8-2}} = 3.538$$

Test statistic: $t = \dfrac{b\sqrt{S_{xx}}}{s} = \dfrac{-0.8456 \times \sqrt{157.875}}{3.538} = -3.003$



Critical value: $t = -2.447$ ($\alpha = 0.05$, two-tailed and $\nu = 6$ degrees of freedom)

Since $-3.003 < -2.447$, reject H_0 at the 5% level of significance. There is evidence to suggest that the slope parameter is not zero, so evidence that a linear model would be useful for prediction.

Exercise 18.4

1. A random sample of bivariate data, sample size 27, yields the following summary statistics:

$$S_{xx} = 87.8 \qquad S_{yy} = 415.5 \qquad S_{xy} = -129.6 \qquad \text{SSR} = 224.2$$

   A test for β is to be performed on the data.

   (a) State suitable null and alternative hypotheses.

   (b) Show the calculations required to reach a test statistic of -4.618.

   (c) State the degrees of freedom, ν, of the t-distribution required for the test.

   (d) Conclude the test at the 5% level of significance.

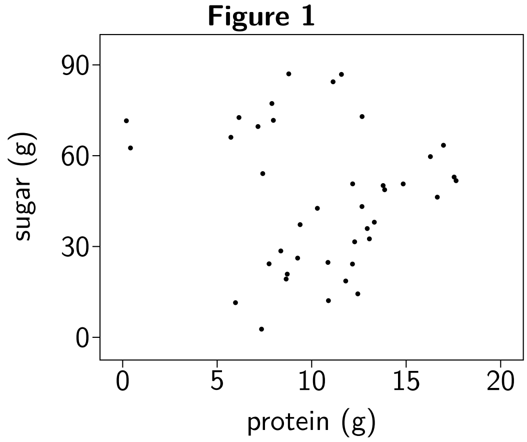
2. The top speed reached (in mph) and the lap times (in seconds) are recorded for a sample of drivers on a race track. Computer output related to a linear model constructed using this data is below.

```
y on x regression line              n     s       sum of squares for x
top_speed = 269.2 - 1.72 × lap_time  34    8.49    638.6
```

   Perform a test for the slope parameter at the 10% level of significance to assess whether there is evidence that a linear model is useful for prediction for this data.

3. A nutritionist purchases a sample of drinks sold from vending machines inside a selection of gyms and health centres in his area. The protein content in grams and the sugar content in grams is recorded for each drink. **Figure 1** shows a scatterplot of the data and **Output 1** shows results of a hypothesis test conducted (where $\alpha = 0.05$) to assess whether a linear model would be useful for prediction.

Figure 1



Output 1

```
model: sugar = 4.01 - 0.124 × protein

t = -0.42295
df = 38
p-value = 0.6747
alternative: true slope parameter is not
equal to zero
```

   (a) State suitable null and alternative hypotheses for the test conducted.

   (b) State the sample size.

   (c) Determine the critical value required, and hence write a conclusion for the hypothesis test.

   (d) State the required assumption about the population of errors for this test.

4. A sample of bivariate data with summary statistics below is proposed to be used for a linear model:

$$S_{xx} = 75.6 \qquad S_{xy} = -235.2 \qquad s^2 = 4.04 \qquad n = 70$$

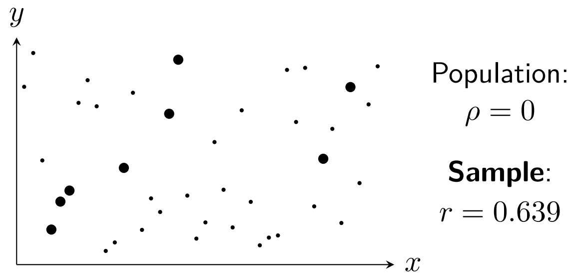
   Conduct a parametric test at the 1% level of significance to assess whether there is evidence that the slope parameter is non-zero.

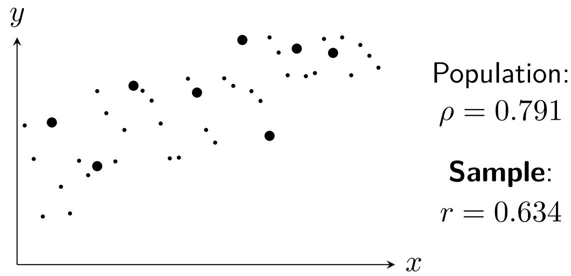
5. Test whether the sample data below provides evidence that a linear model would be useful for prediction.

x	38	27	19	45	8	23	11	34
y	34	53	42	13	19	30	29	45

18.5 Test for the Correlation Coefficient ρ

The previously introduced correlation coefficient, r, is a sample statistic which measures the strength of the linear association between two variables. It is an estimate of the population parameter ρ (*"rho"*), where $-1 \leqslant \rho \leqslant 1$. If $\rho = 0$ then there is no *linear* relationship between the variables. Given a sample of bivariate data for which there exists *no* linear relationship, it is common for the sample to give a *non-zero* value for r:





The *parametric* **Test for Rho** assesses a sample of bivariate data for evidence of a **linear association** between the variables, or $\rho \neq 0$. The test statistic below follows a t_{n-2} distribution, and in this course the test will always be *two-tailed* with hypotheses $\mathrm{H}_0 : \rho = 0$ and $\mathrm{H}_1 : \rho \neq 0$.

Test Statistic

$$t_{n-2} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Conditions

- The observations (x_i, y_i) are independent (through random sampling).
- The variables follow an approximately bivariate normal distribution.

Any conclusion should state whether there is sufficient evidence to suggest a linear association between the variables.

Example

Problem: A bivariate sample has summary statistics:

$$S_{xx} = 157.875 \qquad S_{yy} = 188 \qquad S_{xy} = -133.5 \qquad n = 8$$

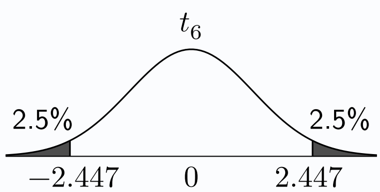
Test at the 5% significance level for evidence of a linear association between x and y.

Solution:

$$\begin{aligned} H_0 &: \quad \rho = 0 \\ H_1 &: \quad \rho \neq 0 \end{aligned}$$

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{-133.5}{\sqrt{157.875 \times 188}} = -0.7749$$

Test statistic: $t = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \dfrac{-0.7749\sqrt{8-2}}{\sqrt{1-(-0.7749)^2}} = -3.003$



Critical value: $t = -2.447$ ($\alpha = 0.05$, two-tailed with $\nu = 6$ degrees of freedom)

Since $-3.003 < -2.447$, reject H_0 at the 5% level of significance.
There is sufficient evidence to suggest that there is a linear association between the two variables.

Exercise 18.5

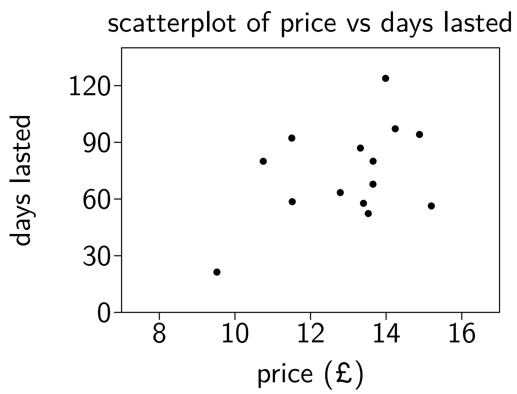
1. A random sample of bivariate data, sample size 27, yields a sample correlation coefficient of −0.678. A test for linear association is to be performed on the data.
   (a) State suitable null and alternative hypotheses.
   (b) Show the calculations required to reach a test statistic of −4.612.
   (c) State the degrees of freedom, ν, of the t-distribution required for the test.
   (d) Conclude the test at the 5% level of significance.

2. A bivariate random sample has summary statistics:

$$S_{xx} = 28.92 \qquad S_{yy} = 495.0 \qquad S_{xy} = 105.3 \qquad n = 14$$

   Perform a parametric test to assess at the 10% level of significance whether there is sufficient evidence to suggest a linear association exists between x and y.

3. A guitarist records over the course of several years the price of sets of strings that she buys and the number of days that they last without a string breaking. A scatterplot of the data collected is shown below alongside some summary statistics from computer software.




```
data:
guitar_strings

sample correlation coefficient:
0.437

sample size:
14
```

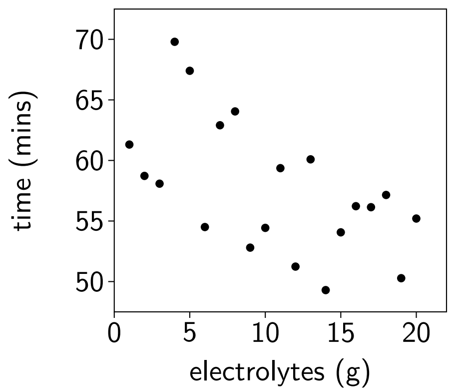
   Test at the 1% level of significance whether her data provides evidence of a linear association between the price of a set of strings and the time until a string breaks.

4. *Electrolyte powders* are often used to help improve hydration during exercise. A keen cyclist records the time (y minutes) it takes them to complete a familiar route during a series of rides over the summer, randomly choosing the mass (x grams) of electrolyte powder to add to their water bottle each time.

   Summary statistics for the rides are below:

S_{xx}	S_{yy}	S_{xy}	n
665	567.2	-346.4	20

   Test whether the data supports a claim from the cyclist that a linear association exists between the amount of electrolyte powder they use and their cycling performance, stating one assumption required.



5. Test whether the sample data below provides evidence of a linear association between x and y.

x	38	27	19	45	8	23	11	34
y	34	53	42	13	19	30	29	45

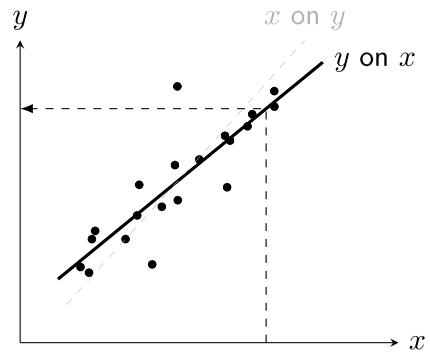
18.6 Regression Line of x on y

All least squares regression lines covered so far have been y *on* x regression lines, calculated with the aim of minimising errors when *predicting* y. However, a y *on* x regression line model should not be used to predict x values. This is because the y on x line minimises vertical distances, not horizontal distances.

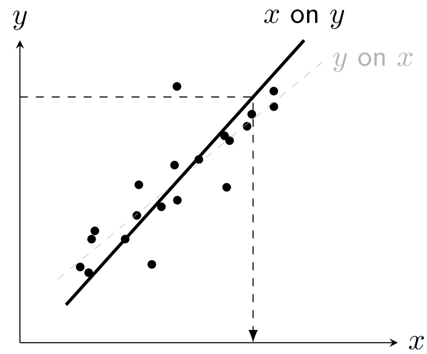
If the prediction of x values based on y values is of interest, it is necessary to obtain the equation of an x *on* y regression line. This takes a different form and uses different values of a and b as shown below:

Calculating an x on y Regression Line

$$\hat{X} = a + by \qquad \text{where} \qquad b = \frac{S_{xy}}{S_{yy}} \qquad \text{and} \qquad a = \bar{x} - b\bar{y}$$



Prediction with a y on x model
(Minimises squared vertical distances)



Prediction with an x on y model
(Minimises squared horizontal distances)

Example

Problem:
Students sitting a French exam are given a mark for each of the two components: speaking (x) and listening (y). From a random sample of 12 students, the following summary statistics are obtained:

$$\Sigma x = 157 \quad \Sigma x^2 = 2389 \quad \Sigma y = 149 \quad \Sigma y^2 = 1983 \quad \Sigma xy = 2143$$

$$S_{xx} = 334.9 \quad S_{yy} = 132.9 \quad S_{xy} = 193.6 \quad n = 12 \quad r = 0.9177$$

Determine the equation for the least squares regression line of x on y, and hence estimate the speaking mark for a student who has only completed the listening component and obtained a mark of 17 in it.

Solution:

$$b = \frac{S_{xy}}{S_{yy}} = \frac{193.6}{132.9} = 1.457$$

$$a = \bar{x} - b\bar{y} = \frac{157}{12} - 1.457 \times \frac{149}{12} = -5.008$$

$$\hat{X}_i = -5.008 + 1.457 y_i$$

For a student with a listening mark (y) of 17:

$$\hat{X} = -5.008 + 1.457 \times 17 = 19.76$$

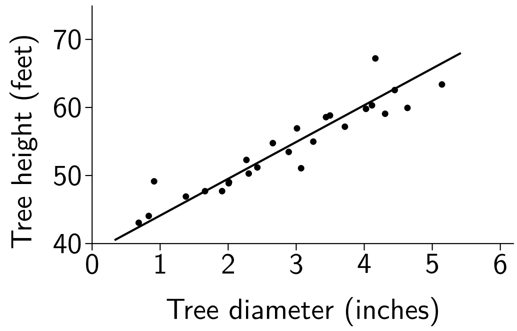
Exercise 18.6

1. A bivariate sample has summary statistics:

$$\sum x = 117 \qquad \sum x^2 = 1869 \qquad \sum y = 660 \qquad \sum y^2 = 54638 \qquad \sum xy = 9519 \qquad n = 8$$

   It is determined from inspecting a scatterplot that there appears to be a linear relationship between the variables. Obtain the equation of the least squares regression line of x on y.

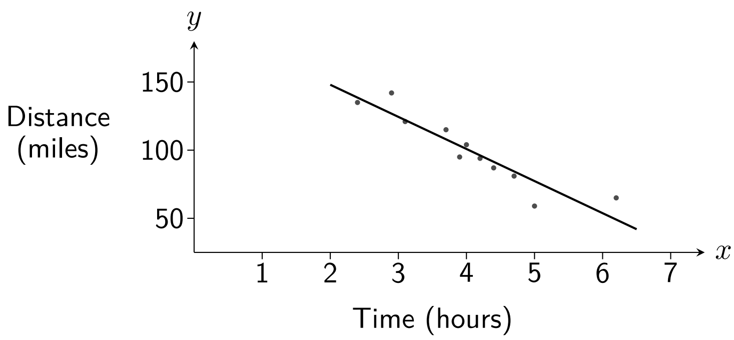
2. 26 randomly selected trees in a forest have their diameter at a set distance from the ground measured, in inches, and the height of each tree also measured, in feet. The equation of the least squares regression line for x on y is obtained, and shown on the scatterplot below.



   Summary statistics:
   $\sum x = 74.76$
   $\sum x^2 = 253.76$
   $\sum y = 1408.3$
   $\sum y^2 = 77277$
   $\sum xy = 4233.1$
   $n = 26$

   (a) Calculate the equation of the least squares regression line of x on y.
   (b) Hence estimate the diameter of a tree which is 65 feet tall.

3. A running website randomly selected eleven amateur runners who recently completed a marathon and asked them to complete a survey which included their finishing time in the marathon and information about their training routine. The scatterplot below shows, for each runner, their finishing time (x, in hours) and the total distance they ran in the final four weeks of their training (y, in miles).



   Summary statistics:
   $\sum x = 44.5$
   $\sum x^2 = 191.21$
   $\sum y = 1098$
   $\sum y^2 = 116768$
   $\sum xy = 4179.2$
   $n = 11$

   A least squares regression line for y on x is shown on the graph, with equation $\widehat{Y} = 40.6 + 4.73x$.

   The website wishes to allow users the ability to input the number of miles in their training plan for the four weeks leading up to the race, and receive in response an estimate for their finishing time.

   (a) Explain why a *new* regression line will need to be determined for this tool on the website.
   (b) Calculate the equation of this new regression line.
   (c) Determine an estimate of the finishing time for a runner who completed 70 miles.
   (d) Explain why the new regression line obtained in part (b) is not suitable for predicting the finish time of a runner who completed 162 miles.

4. Determine the equation of an x on y regression line for the bivariate data below.

x	38	27	19	45	8	23	11	34
y	34	53	42	13	19	30	29	45

18.7 Residual Plots

As a reminder, a residual is the difference between an observed value y_i and its predicted value $\hat{Y}_i$:

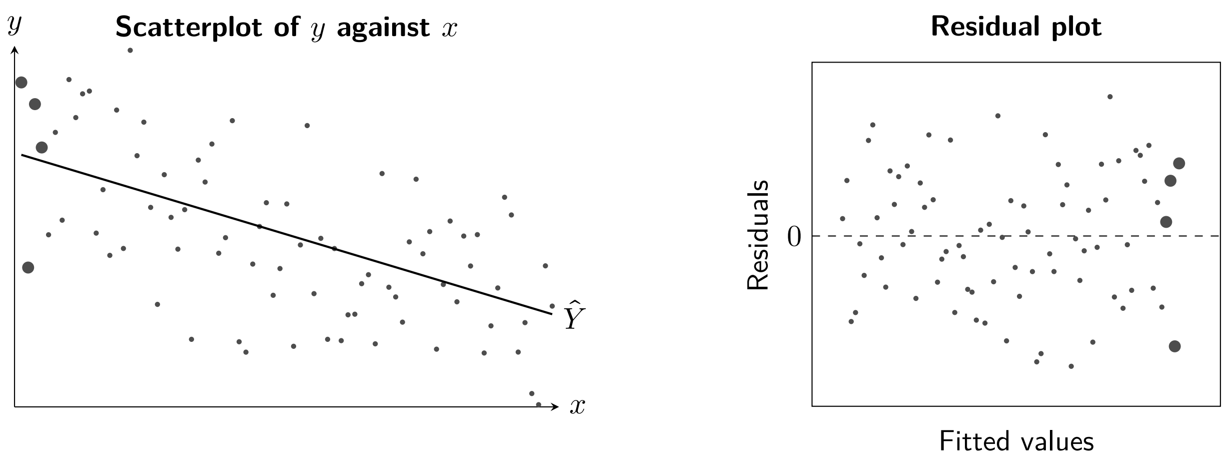
$$e_i = y_i - \hat{Y}_i$$

When considering the use of a linear regression model for bivariate data, it is standard practice to produce a plot of *residuals* (e_i) against *fitted values* $(\hat{Y}_i)$, which have been calculated using the sample regression line. The purpose of this is both to ascertain whether a linear regression model is *the most appropriate* model, as well as to use the observed **residuals** to assess whether key assumptions about the **errors** are reasonable, including those required to obtain the estimates for α and β:

$$\mathsf{E}(\varepsilon_i) = 0 \qquad \text{and} \qquad \mathsf{V}(\varepsilon_i) = \sigma^2$$

These correspond to residuals being centered around zero and having constant spread across the fitted values.

Below is an illustration of a standard scatterplot showing a sample of bivariate data on the left, with a y on x regression line included, and a residual plot for the same data on the right. The data points highlighted with larger circles on the y against x scatterplot have their residuals similarly highlighted on the residual plot.

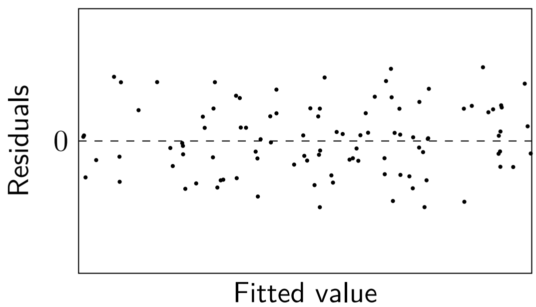


From the residual plot, the following can be observed:

Interpreting Residual Plots	
A random scatter of points...	
...centred on zero...	$\mathsf{E}(\varepsilon_i) = 0$
...with a constant variance.	$\mathsf{V}(\varepsilon_i) = \sigma^2$

This means that the residual plot would give no reason to doubt the assumptions made regarding the errors. This *ideal* residual plot would suggest that the linear regression model is *appropriate*.

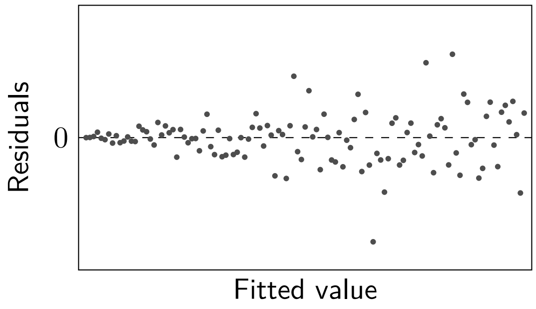
Any clear pattern in the residual plot suggests that the assumptions of the linear model may not be valid. The following residual plots are accompanied with relevant comments and suitable conclusions.



The residual plot appears to show a random scatter of points centred on zero, with a constant variance as the fitted value increases.

$$\mathsf{E}(\varepsilon_i) = 0$$
$$\mathsf{V}(\varepsilon_i) = \sigma^2$$

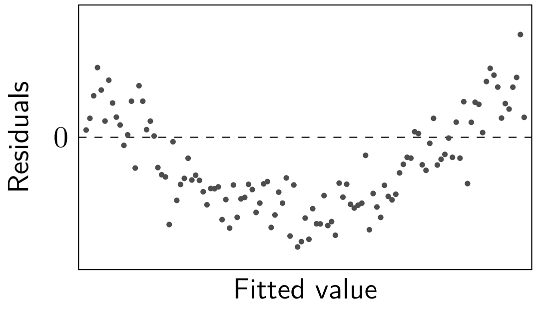
This suggests that the linear model may be **appropriate**.



The residual plot appears to show a 'funnel' shaped pattern of points centred on zero, with variance increasing as the fitted value increases.

$$\mathsf{E}(\varepsilon_i) = 0$$
$$\mathsf{V}(\varepsilon_i) \neq \sigma^2$$

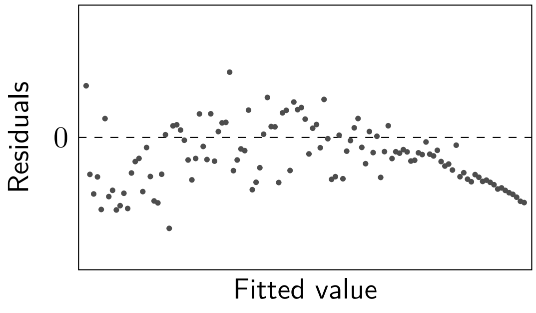
This suggests that the linear model may be **not appropriate**.



The residual plot appears to show a pattern of points following a curve, with constant variance.

$$\mathsf{E}(\varepsilon_i) \neq 0$$
$$\mathsf{V}(\varepsilon_i) = \sigma^2$$

This suggests that the linear model may be **not appropriate**.



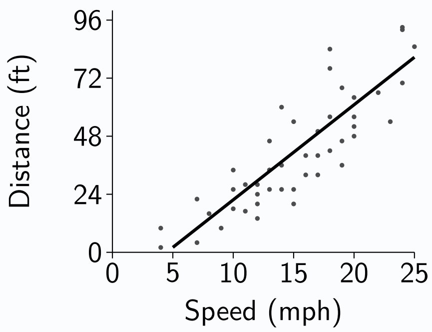
The residual plot appears to show a pattern of points following a curve, with variance decreasing as the fitted value increases.

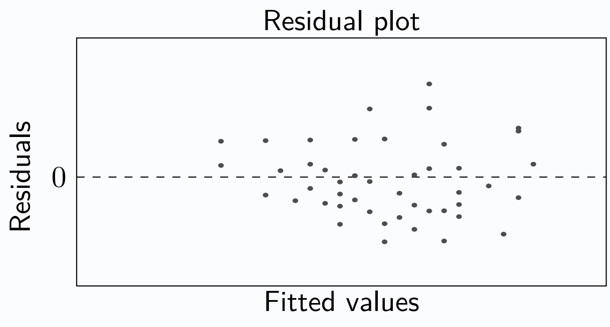
$$\mathsf{E}(\varepsilon_i) \neq 0$$
$$\mathsf{V}(\varepsilon_i) \neq \sigma^2$$

This suggests that the linear model may be **not appropriate**.

Example

Problem: A 1920s study on automobile safety involved recording the stopping distances (in feet) from selected speeds (in miles per hour) for a random sample of 47 cars. Below is a scatterplot of the results with a regression line obtained from a linear model, and a residual plot.





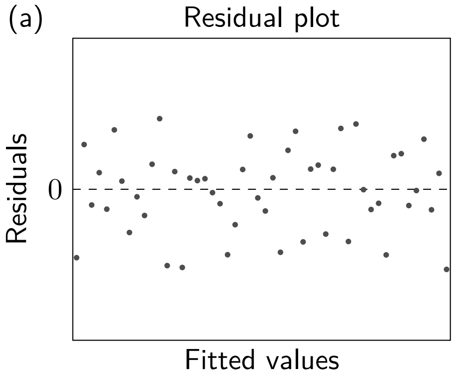
Use the residual plot to comment on the suitability of the linear model.

Solution: The residual plot appears to show a random scatter of points centred on zero, with a broadly constant variance as the fitted value increases. This suggests the linear model between stopping distance and speed is likely to be appropriate.

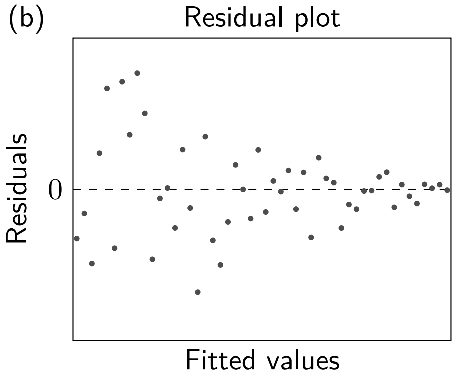
Exercise 18.7

1. For each, of the linear model from which they were produced.

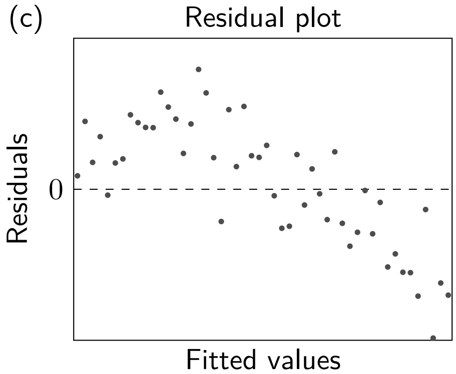
(a)



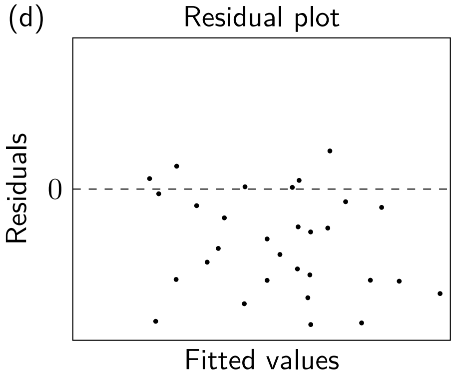
(b)



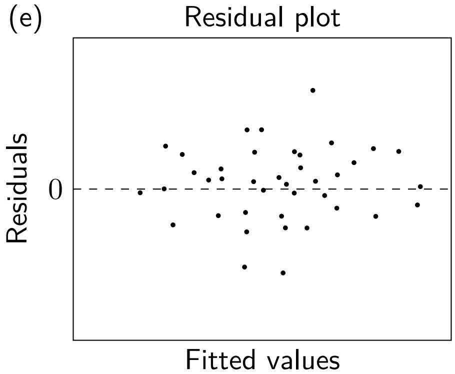
(c)



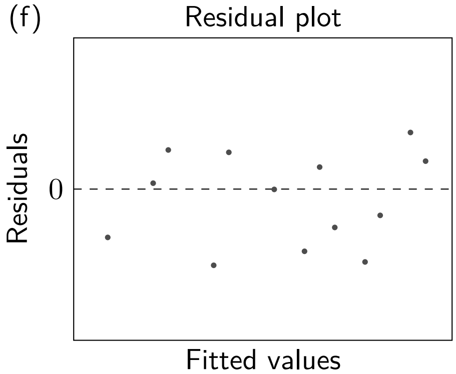
(d)



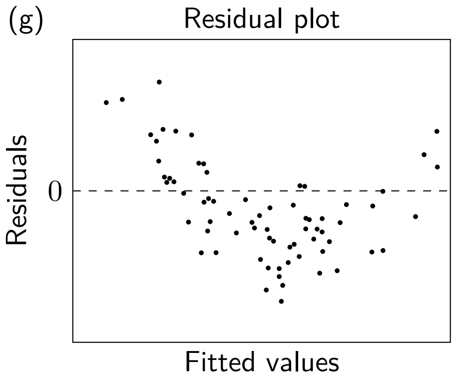
(e)



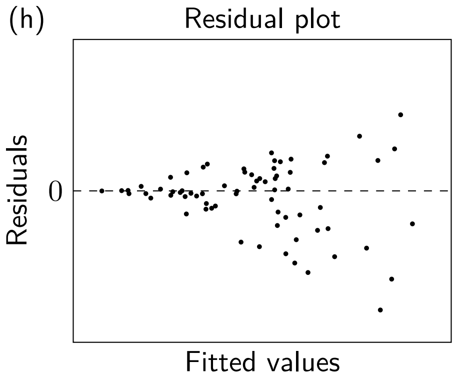
(f)



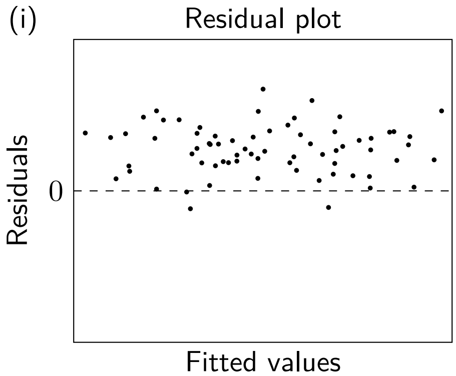
(g)



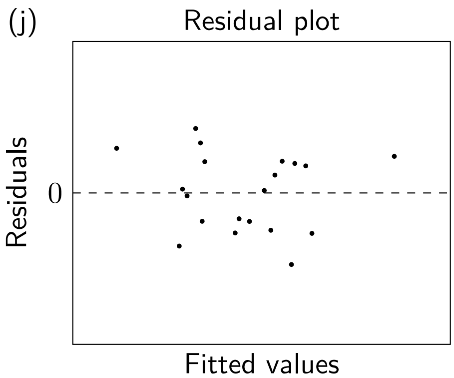
(h)



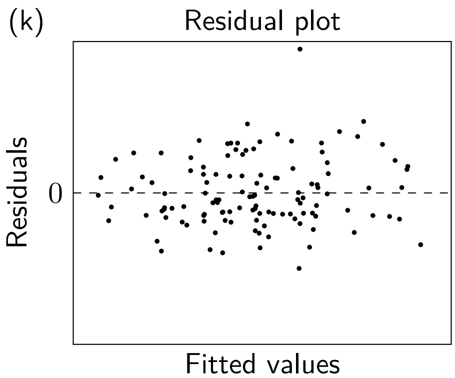
(i)



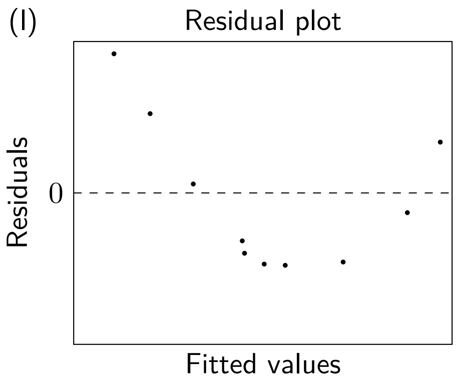
(j)



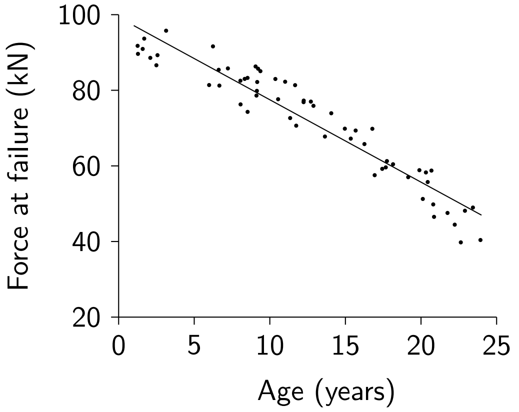
(k)

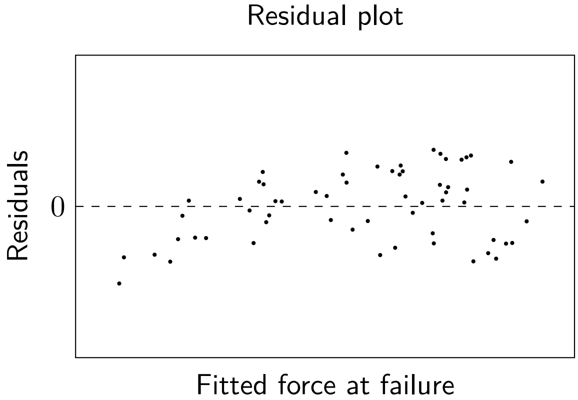


(l)



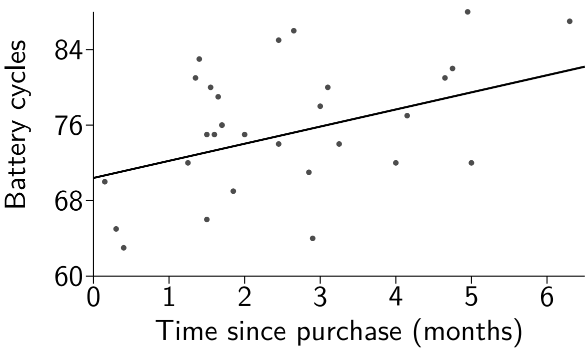
2. Suspension clamps are used to hold electricity cables in place on pylons. Over time these clamps are subject to wear, making them more likely to fail. A random sample of clamps are subject to an increasing force, until they fail. A proposed linear regression line is shown on the scatterplot of force at failure (kN) against age in years below, as well as a residual plot.

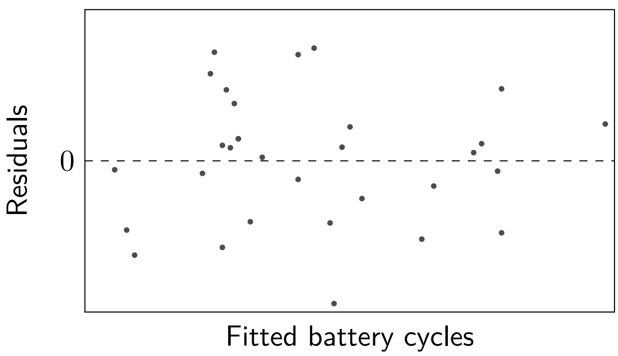




Comment on the suitability of the linear model fitted.

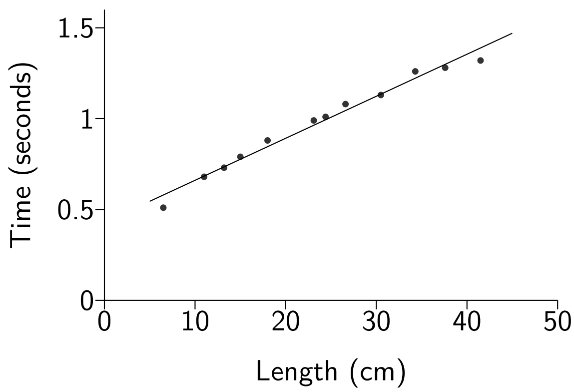
3. A number of randomly selected mobile phones returned as faulty within a six month warranty period each have their time since purchase (x) and number of completed battery cycles (y) recorded. The equation of the least squares regression line for y on x is obtained, and shown on the scatterplot below alongside a residual plot.

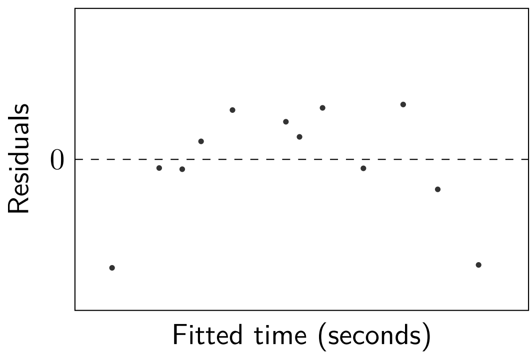




Explain why the residual plot suggests that a linear model would be suitable for the data.

4. A student investigating the relationship between the length of a pendulum and the time taken for one complete swing to occur made a simple pendulum from string and a metal disc, suspended from a clamp over the edge of a table. The time taken for one complete swing (y seconds) was recorded for 12 different string lengths (x cm).





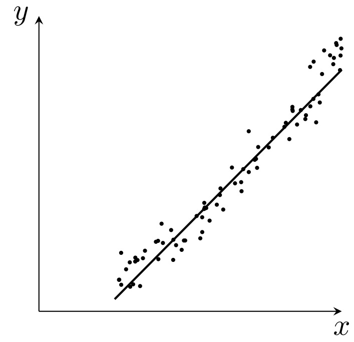
Comment on the suitability of the linear model fitted.

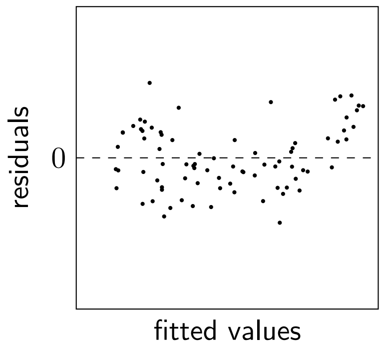
18.8 Transforming Data

Where a *non-linear* relationship seems to exist between two variables then it would not be appropriate to use a model which describes a *linear* relationship between them. The presence of a non-linear relationship may be immediately clear from an initial scatterplot, or it may only be revealed once a residual plot for a proposed linear model is created.

A linear model might initially seem to describe the relationship between x and y, as might the result of $R^2 = 0.9228$.

However, the residual plot suggests that a linear model between x and y is *not* appropriate, since the residuals follow a curved pattern.

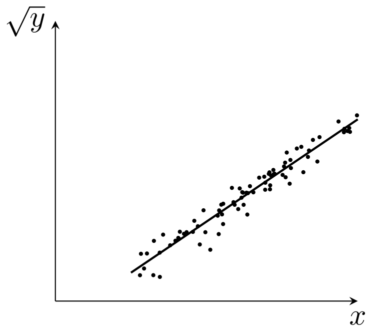


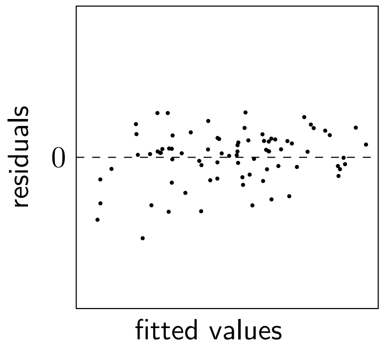


The simulated data above was actually generated using a *quadratic* relationship between x and y, which explains the curved pattern seen in the residuals.
A scatterplot of $\sqrt{y}$ against x for this data is shown below, alongside a residual plot:

Both the scatterplot and the residual plot for this new model of $\sqrt{y}$ on x suggest that it is appropriate for the data.

This new, better-fitting model also produces a higher value for the coefficient of determination, of $R^2 = 0.9402$.





Creating a model such as this, in which one or both variables have some operation performed on them (such as taking the *square root*) is called applying a *transformation* to the data. In practice, knowing which transformation(s) to try will generally be guided by a background understanding of the variables, or previous studies. In this course, any required transformations will be clearly specified.

Familiar statistics may still be calculated using transformed variables:

Statistic	For a y on x model	New model	New formula
Sum of Squares for x	$S_{xx} = \Sigma x^2 - \dfrac{(\Sigma x)^2}{n}$	y on $\sqrt{x}$	$S_{\sqrt{x}\sqrt{x}} = \Sigma(\sqrt{x})^2 - \dfrac{(\Sigma\sqrt{x})^2}{n}$
Slope parameter estimate	$b = \dfrac{S_{xy}}{S_{xx}}$	$\log_a y$ on x	$b = \dfrac{S_{x\log_a y}}{S_{xx}}$
Correlation coefficient	$r = \dfrac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$	y^3 on x	$r = \dfrac{S_{xy^3}}{\sqrt{S_{xx}S_{y^3y^3}}}$

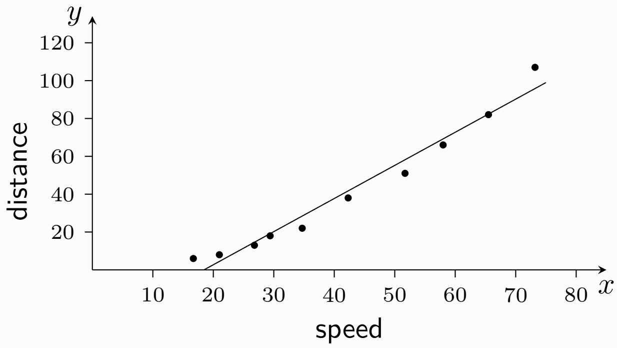
Care should be taken when interpreting these statistics and when calculating using regression line equations. The example opposite illustrates this for the coefficient of determination, and calculating a predicted value.

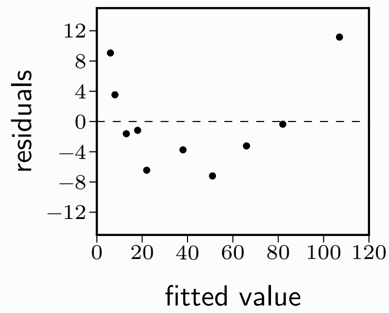
Example

Problem: Data was recorded for a sample of 10 cars of their stopping distance (y metres) when travelling at a range of speeds (x miles per hour). The table below shows the results.

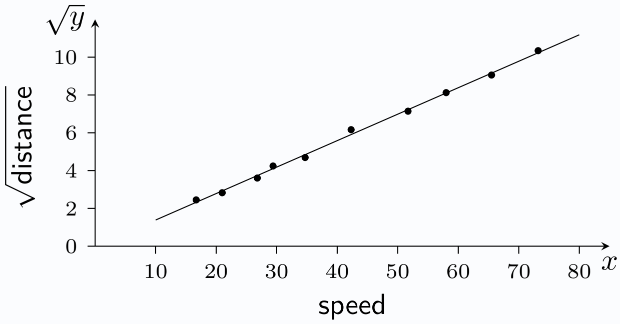
Speed (x)	16.7	21.0	26.8	29.4	34.7	42.3	51.7	58.0	65.5	73.2
Distance (y)	6	8	13	18	22	38	51	66	82	107

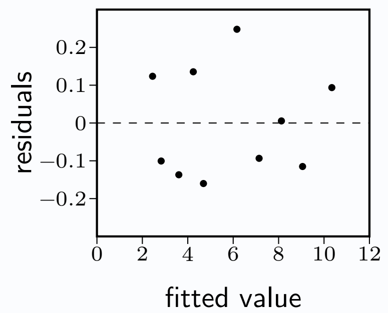
The line $\hat{Y} = -32.3 + 1.75x$ was fitted, and a scatterplot and a residual plot drawn.





A new linear model was created between speed (x) and the square root of the stopping distance ($\sqrt{y}$). For this new model, a scatterplot and residual plot were produced, and a least squares regression line was computed as: `square root of distance = -0.016 + 0.14 × speed`





(a) Comment on the appropriateness of the linear model of y on x.

(b) Comment on the appropriateness of the new model of $\sqrt{y}$ on x.

(c) By first creating a new table for x and $\sqrt{y}$ and finding summary statistics for this data, calculate the coefficient of determination for this new model and interpret its value.

(d) Estimate the stopping distance at a speed of 50 miles per hour.

Solution:

(a) The residual plot appears to show a curved pattern, so this linear model is not appropriate.

(b) The residual plot for the new model appears to show a random scatter of points centered on zero, with a constant variance as fitted values increase, so the new model seems appropriate.

(c) $r = \dfrac{S_{x\sqrt{y}}}{\sqrt{S_{xx}S_{\sqrt{y}\sqrt{y}}}} = \dfrac{3400.0}{\sqrt{10758.9 \times 5951.5}} = 0.9987$ and $R^2 = 0.9987^2 = 0.997$.

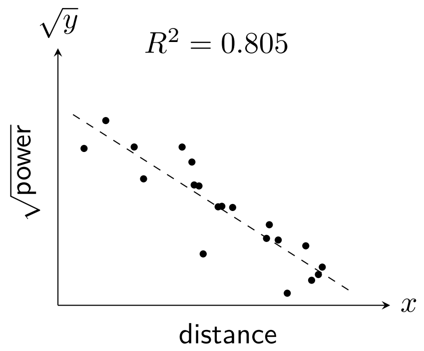
99.7% of the variation of $\sqrt{\text{distance}}$ is explained by speed.

(d) $\sqrt{\text{distance}} = -0.016 + 0.14 \times 50 = 6.984$, so distance $= 6.984^2 = 48.8$ metres

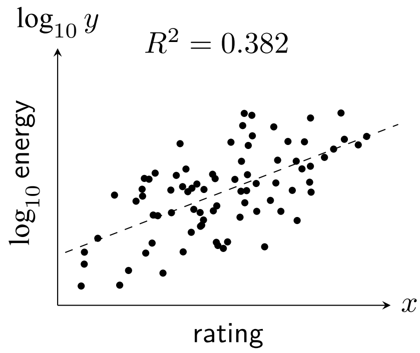
Exercise 18.8

1. For each of the following scatterplots, showing transformed data, in context:
   - Describe any relationship shown by the scatterplot of the transformed data.
   - Interpret the coefficient of determination of the proposed linear model shown by the regression line.

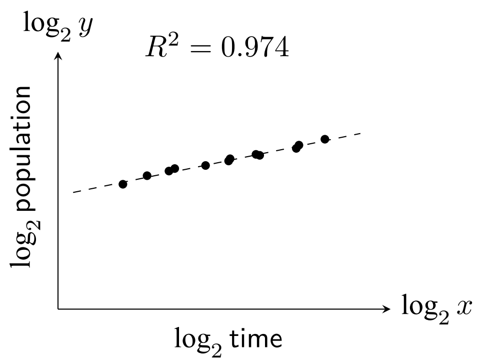
   (a) Data before transformation:
   $x =$ distance
   $y =$ power



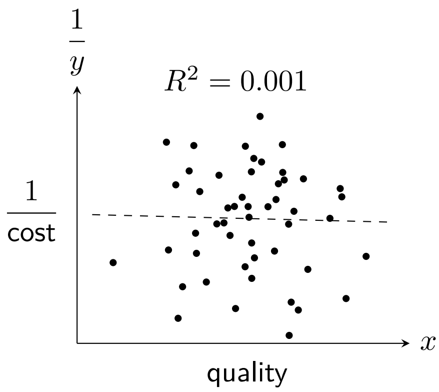
   (b) Data before transformation:
   $x =$ rating
   $y =$ energy



   (c) Data before transformation:
   $x =$ time
   $y =$ population size



   (d) Data before transformation:
   $x =$ quality
   $y =$ cost



2. It is determined that a linear relationship exists between x and $\frac{1}{y}$.
   Summary statistics for this model are as follows:

$$\Sigma x = 35.9 \qquad \Sigma\frac{1}{y} = 133.4 \qquad \Sigma x^2 = 142.3 \qquad \Sigma\frac{1}{y^2} = 1922.4 \qquad \Sigma x\frac{1}{y} = 517.8 \qquad n = 10$$

   (a) Determine the equation of the least squares regression line for the model of $\frac{1}{y}$ on x.

   (b) Calculate an estimate for the value of y when $x = 5$.

3. It is determined that a linear relationship exists between $\log x$ and $\log y$.
   Summary statistics for this model are as follows:

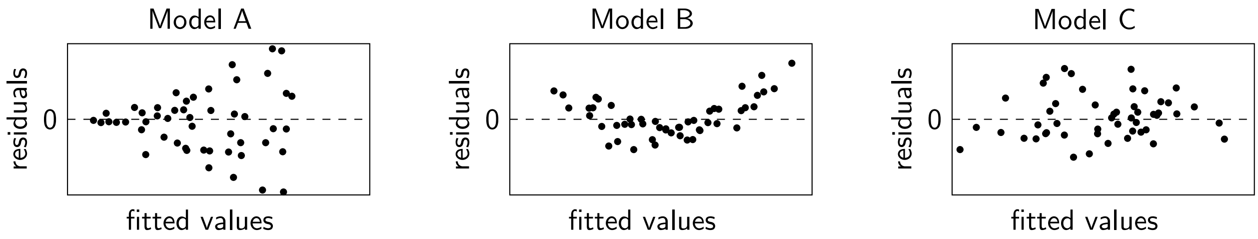
$$\Sigma \log x = 46.3 \qquad \Sigma \log y = 1870.4 \qquad \Sigma(\log x)^2 = 308.4$$
$$\Sigma(\log y)^2 = 350761.2 \qquad \Sigma(\log x)(\log y) = 8370.3 \qquad n = 10$$

   Calculate the correlation coefficient for the linear model and interpret it.

4. A linear model created by a statistician for variables x and y is used to produce a residual plot. After studying the residual plot, the statistician concludes that the assumption for the population of errors that $\mathsf{V}(\varepsilon_i) = 0$ is not supported.

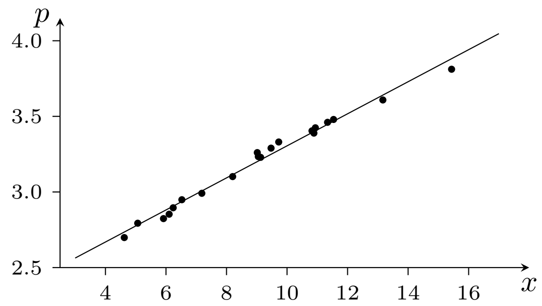
   (a) Suggest a feature of the residual plot which could have led to their conclusion.

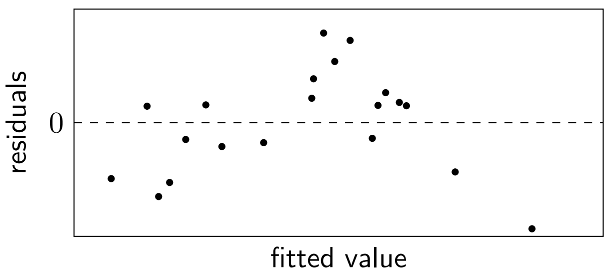
   The statistician tries three different transformations, producing three new models (A, B and C). Residual plots for each model are below:



   (b) State which model the statistician should use, with justification.

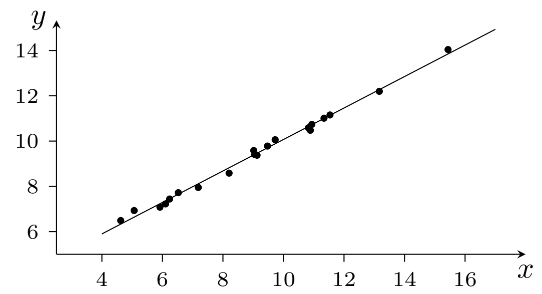
5. Data was collected on each weekday across four weeks of a university computer's peak processing speed (p GHz) and the amount of power consumed (x kWh). The regression line $\hat{U} = 2.245 + 0.106x$ is fitted to the data.

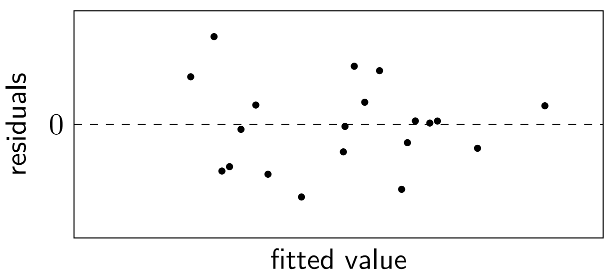




   (a) Comment on the suitability of the linear model of p on x.

   A transformation using $y = 2^p$ was applied, and a new least squares regression line of y on x found, with the following scatterplot and residual plot.





   (b) Comment on the suitability of this new model of y on x.

   The new model has summary statistics:

$$\Sigma x = 180.4 \qquad \Sigma y = 187.8 \qquad \Sigma x^2 = 1779 \qquad S_{xy} = 105.9 \qquad n = 20$$

   (c) Find the new least squares regression line of y on x.

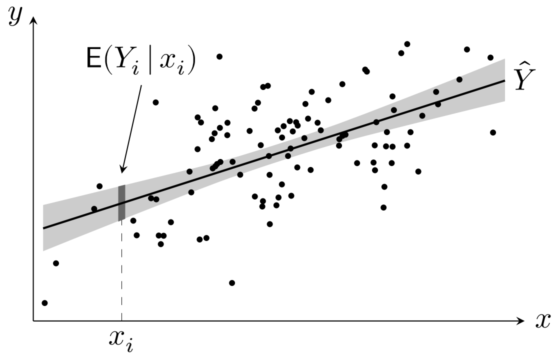
   (d) Estimate the peak processing speed corresponding to a power consumption of 145 kWh. (Note: $t = a^b$ can be written as $b = \log_a t$).

18.9 Confidence Intervals and Prediction Intervals

One purpose of adopting a regression approach to examine the relationship between two variables is the wish to make predictions related to the response variable (Y) for a particular value for the predictor variable (x).

The least squares regression line equation of $\hat{Y}_i = a + bx_i$ gives an *estimate* for the **mean** value of the response variable for any given x_i, whilst the *true* mean response value for a given x_i, denoted $\mathsf{E}(Y_i \,|\, x_i)$, is unknown.

It can be desirable to obtain a **confidence interval** for $\mathsf{E}(Y_i \,|\, x_i)$.



It can be shown that the *point estimate* for the mean response, $\mathsf{E}\,(Y_i \,|\, x_i)$, has a sample standard error of:

$$s\sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}}$$

If the errors, ε_i, are identically and independently distributed as $\mathsf{N}\,(0, \sigma^2)$ then the formula below, given on Page 5 of the Data Booklet, can be derived.

A Confidence Interval for the Mean Response

A $100(1-\alpha)\%$ CI for $\mathsf{E}(Y_i \,|\, x_i)$ is given by: $\hat{Y}_i \pm t_{n-2,1-\alpha/2} \times s\sqrt{\dfrac{1}{n} + \dfrac{(x_i - \bar{x})^2}{S_{xx}}}$

This interval estimates the mean value of Y for all observations with predictor value x_i, not the value for a single observation. As with all confidence intervals, the level of confidence relates to the success rate of the *procedure used to generate the interval*, rather than referring to the probability of a calculated interval containing the mean response value $\mathsf{E}(Y_i \,|\, x_i)$.

Example

Problem:
The lengths in centimetres (x) and masses in kilograms (y) of 12 salmon were measured, with the results:

$$\Sigma x = 30 \quad \Sigma y = 540 \quad S_{xx} = 12 \quad S_{yy} = 4800 \quad S_{xy} = 150$$

Construct a 95% confidence interval for the mean length of salmon with mass 3kg.

Solution:

$$b = \frac{S_{xy}}{S_{xx}} = \frac{150}{12} = 12.5, \quad a = \bar{y} - b\bar{x} = \frac{540}{12} - 12.5 \times \frac{30}{12} = 13.75 \quad \Longrightarrow \quad \hat{Y}_i = 13.75 + 12.5x_i$$

$$\hat{Y} = 13.75 + 12.5 \times 3 = 51.75$$

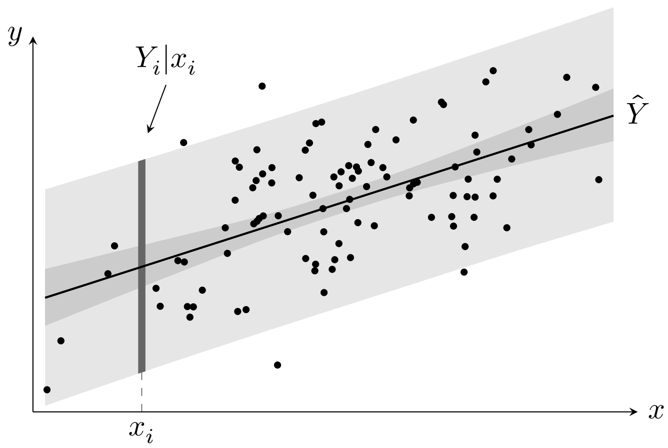
$$\mathsf{SSR} = S_{yy} - \frac{(S_{xy})^2}{S_{xx}} = 4800 - \frac{150^2}{12} = 2925, \quad s^2 = \frac{\mathsf{SSR}}{n-2} = \frac{2925}{10} = 292.5, \quad s = \sqrt{292.5} = 17.1$$

A 95% CI for $\mathsf{E}(Y \,|\, x = 3)$ is $51.75 \pm 2.228 \times 17.1 \times \sqrt{\dfrac{1}{12} + \dfrac{(3-2.5)^2}{12}} = 51.75 \pm 12.30$

So a 95% confidence interval for the mean length of a 3kg salmon is $(39.45\text{cm}\,,\, 64.05\text{cm})$.

Instead of an interval for the value of *mean* response, it may be desired to obtain an interval for **individual** response values for a given x_i, denoted as $Y_i \,|\, x_i$.

The desired interval is called a **prediction interval**, and it reflects both the variability of the regression line and the additional variability of individual observations about the line.



Prediction intervals are always wider than confidence intervals at the same confidence level, because they account for both uncertainty in the regression line *and* variability of individual observations.

Note the more darkly shaded region lying closer to the regression line, showing the narrower confidence interval for the same data.

A 95% prediction interval for $Y_i \,|\, x_i$, such as indicated in the diagram, will capture approximately 95% of individual response values in the long run.

A Prediction Interval for an Individual Response

A $100(1-\alpha)\%$ PI for $Y_i \,|\, x_i$ is given by: $\hat{Y}_i \pm t_{n-2,1-\alpha/2} \times s\sqrt{1+\frac{1}{n}+\frac{(x_i-\bar{x})^2}{S_{xx}}}$

This formula, also given on Page 5 of the Data Booklet, relies also on the errors, ε_i, being identically and independently distributed as $\mathrm{N}\left(0, \sigma^2\right)$

Example

Problem:
The lengths in centimetres (x) and masses in kilograms (y) of 12 salmon were measured, with the results:

$$\Sigma x = 30 \quad \Sigma y = 540 \quad S_{xx} = 12 \quad S_{yy} = 4800 \quad S_{xy} = 150$$

Construct a 95% prediction interval for the length of salmon with mass 3kg.

Solution:

$$b = \frac{S_{xy}}{S_{xx}} = \frac{150}{12} = 12.5, \quad a = \bar{y} - b\bar{x} = \frac{540}{12} - 12.5 \times \frac{30}{12} = 13.75 \quad \Longrightarrow \quad \hat{Y}_i = 13.75 + 12.5x_i$$

$$\hat{Y} = 13.75 + 12.5 \times 3 = 51.75$$

$$\text{SSR} = S_{yy} - \frac{(S_{xy})^2}{S_{xx}} = 4800 - \frac{150^2}{12} = 2925, \quad s^2 = \frac{\text{SSR}}{n-2} = \frac{2925}{10} = 292.5, \quad s = \sqrt{292.5} = 17.1$$

A 95% PI for $Y \,|\, x = 3$ is $51.75 \pm 2.228 \times 17.1 \times \sqrt{1+\frac{1}{12}+\frac{(3-2.5)^2}{12}} = 51.75 \pm 40.03$

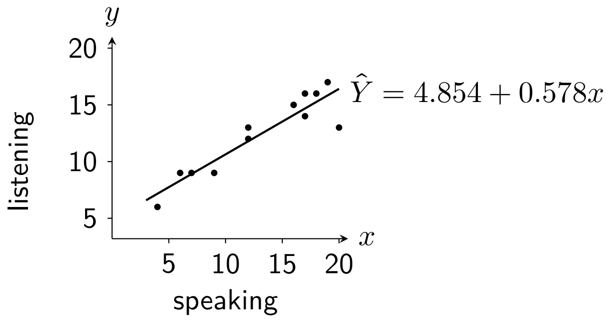
So a 95% prediction interval for the observed length of a 3kg salmon is $(11.72\text{cm}\,,\, 91.78\text{cm})$.

Exercise 18.9

1. Students sitting a French exam are given a mark for each of the two components: speaking (x) and listening (y). Data is gathered from a random sample of 12 students, and a regression line obtained.

   Summary statistics are calculated as:

$$\Sigma x = 157$$
$$S_{xx} = 334.9$$
$$s = 1.45$$
$$n = 12$$

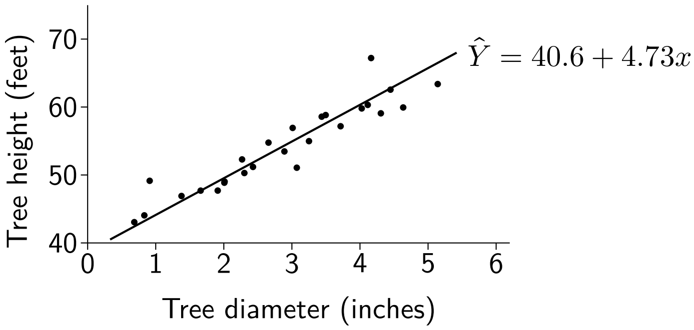


   A pupil achieves a speaking score of 15. They are yet to complete the listening component.

   (a) Obtain a 95% confidence interval for the mean listening score given a speaking score of 15.

   (b) Obtain a 95% prediction interval for the listening score given a speaking score of 15.

2. 26 randomly selected trees in a forest have their diameter at a set distance from the ground measured, in inches, and the height of each tree also measured, in feet. The equation of the least squares regression line for x on y is obtained, and shown on the scatterplot below.



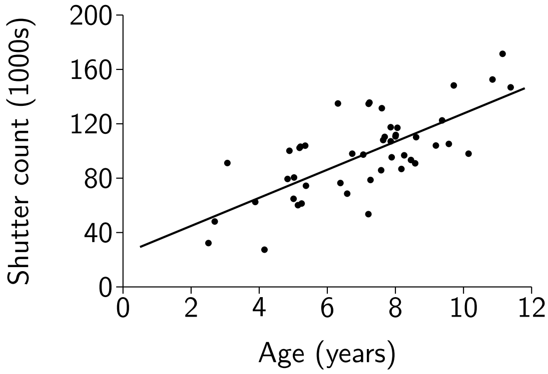
   Summary statistics:

$$\sum x = 74.76$$
$$S_{xx} = 38.8$$
$$s^2 = 5.09$$
$$n = 26$$

   (a) Obtain a 99% confidence interval for the mean height of a tree with diameter 4 inches.

   (b) Obtain a 99% prediction interval for the height of a tree with diameter 4 inches.

3. A seller on an online auction site has a stock of used digital cameras, all aged somewhere between 2 and 12 years. For a random sample of 48 cameras, the total number of photos taken (the *shutter count*) is determined using special software. A scatterplot with a least squares regression line is below.



   Summary statistics for the data are as follows:

$$\Sigma x = 337.7 \qquad S_{xx} = 221.8$$
$$S_{yy} = 44085.2 \qquad S_{xy} = 2192.8$$

   The equation for regression line was found:

```
shuttercount = 24.22 + 10.36 × age
```

   (a) Obtain a 90% prediction interval for the shutter count of a 6 year old camera.

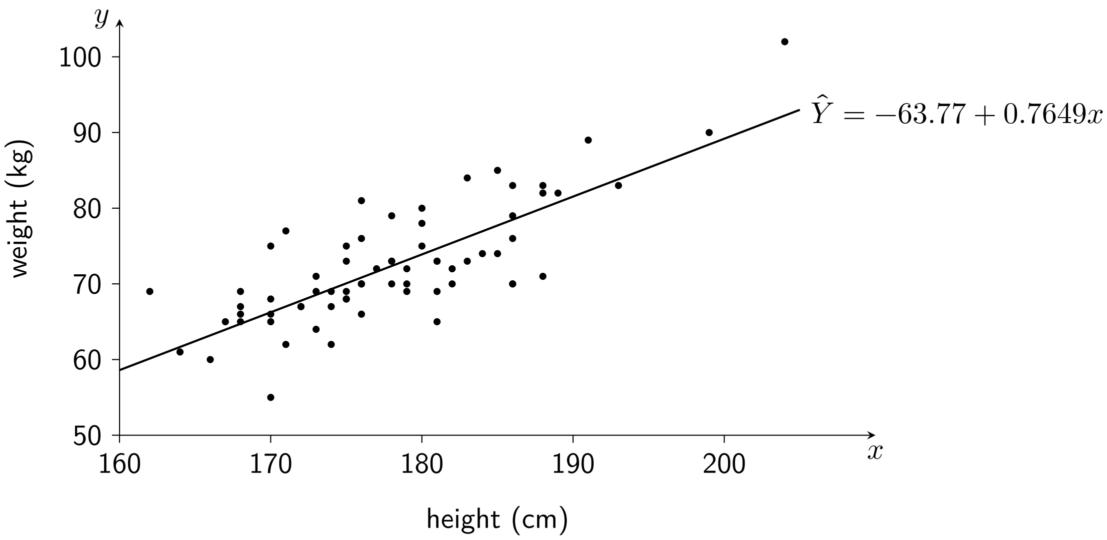
   (b) Obtain a 90% confidence interval for the mean shutter count of a 6 year old camera.

4. The top speed reached (in mph) and the lap times (in seconds) are recorded for a sample of drivers on a race track. Computer output related to a linear model constructed using this data is below.

y on x regression line	n	s	sum of squares for x	sigma x
speed = 269.2 - 1.72 $\times$ time	34	8.49	638.6	72.05

One driver completed their lap in a time of 52 seconds, but their top speed was not recorded. They wish to estimate a range of values for their top speed. Obtain a suitable interval at the 95% level.

5. The heights (x cm) and weights (y kg) are recorded for a random sample of professional footballers.



A linear model is fit to the data as shown with a regression line equation of:

```
weight = -63.77 + 0.7649 × height
```

A footballer, whose weight is unknown, has a height of 190 centimetres.

(a) Use the equation of the regression line to calculate an estimate for the footballer's weight.

Computer output is used to generate a 95% prediction interval for the weight of a footballer with a height of 190 centimetres.

```
95% PI
(71.59,****)
```

(b) Determine the missing upper value for the prediction interval above, indicated by '****'.

(c) State the assumptions about the population of errors required for the predication interval above.

Summary statistics are obtained for the data, as shown below:

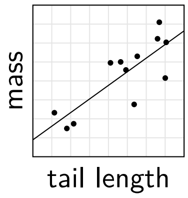
$$S_{xx} = 4319.0 \qquad S_{yy} = 4122.4 \qquad S_{xy} = 3303.6 \qquad n = 69$$

(d) Calculate a confidence interval for the weight of a footballer of height 172cm.

(e) Explain why the method used above may not be appropriate to help estimate the mean weight of a footballer with a height of 158 centimetres.

Review Exercise

1. Biologists studying populations of European hamsters record the tail length (x centimetres) and body mass (y grams) of a random sample of 12 wild hamsters found living in central Vienna. The linear regression equation was calculated for the data shown in the scatterplot as $\widehat{Y} = -15.24 + 71.88x$.

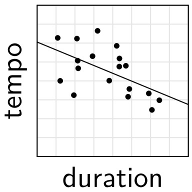


$$\Sigma x = 62.25, \quad \Sigma y = 4291.5, \quad \Sigma x^2 = 325.9, \quad \Sigma y^2 = 1558042, \quad \Sigma xy = 22475$$

(a) Calculate the residual of a 410 gram hamster with a tail length of 5.2cm.

(b) Use a suitable hypothesis test to assess the usefulness of the linear model for prediction at the 10% level of significance.

2. Data is collected on the tempo (T beats per minute) and duration (d minutes) of a random sample of pop-punk songs. The regression line $\widehat{T} = 256.65 - 20.65d$ is obtained for the data.

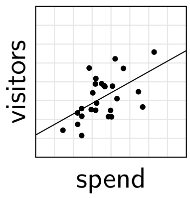


$$\Sigma d = 43.6 \qquad \Sigma T = 3719.5 \qquad \Sigma dT = 8947.2$$
$$\Sigma d^2 = 108.5 \qquad \Sigma T^2 = 772792 \qquad n = 18$$

(a) Calculate the coefficient of determination and interpret its value.

(b) Perform a hypothesis test at the 1% level of significance to check for evidence of a linear association between the tempo and duration of pop-punk songs.

3. The creators of a brand new sports website collect data on the annual social media marketing spend (£x thousand) and new daily visitors (y) of similar sites, to plan their own budget. The results are shown below, including a least squares regression line of equation $\widehat{Y} = 29.5 + 5.6x$.



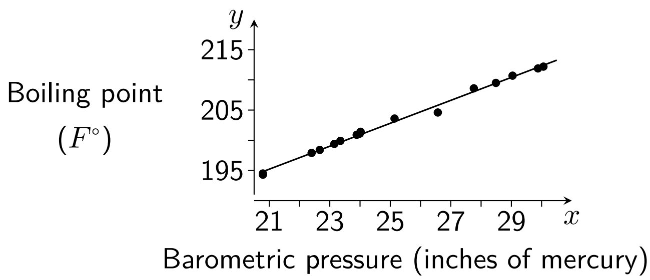
$$\Sigma x = 230.6 \qquad \Sigma x^2 = 2244.6 \qquad S_{xy} = 1118.6 \qquad S_{yy} = 20640.9 \qquad n = 26$$

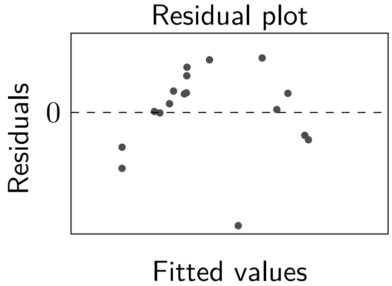
(a) Calculate an estimate for the error variance.

(b) Calculate a 95% prediction interval for the daily new visitors of a site with an annual social media marking spend of £15,000.

(c) Explain why a prediction interval is more likely to be of interest to the magazine's creators than a confidence interval.

4. The Scottish physicist James Forbes was interested in finding a way to estimate altitude from measurements of the boiling temperature of water. Some of his observations on boiling point (y, in F°) and barometric pressure (x, in inches of mercury) are summarised below. After plotting a scatterplot for this data, a least squares regression line was obtained with the equation $\widehat{Y} = 155.29 + 1.9018x$, from which a residual plot was also created.





(a) Comment on the suitability of the linear model.

(b) Suggest what next steps might be taken on the basis of the residual plot for the linear model.

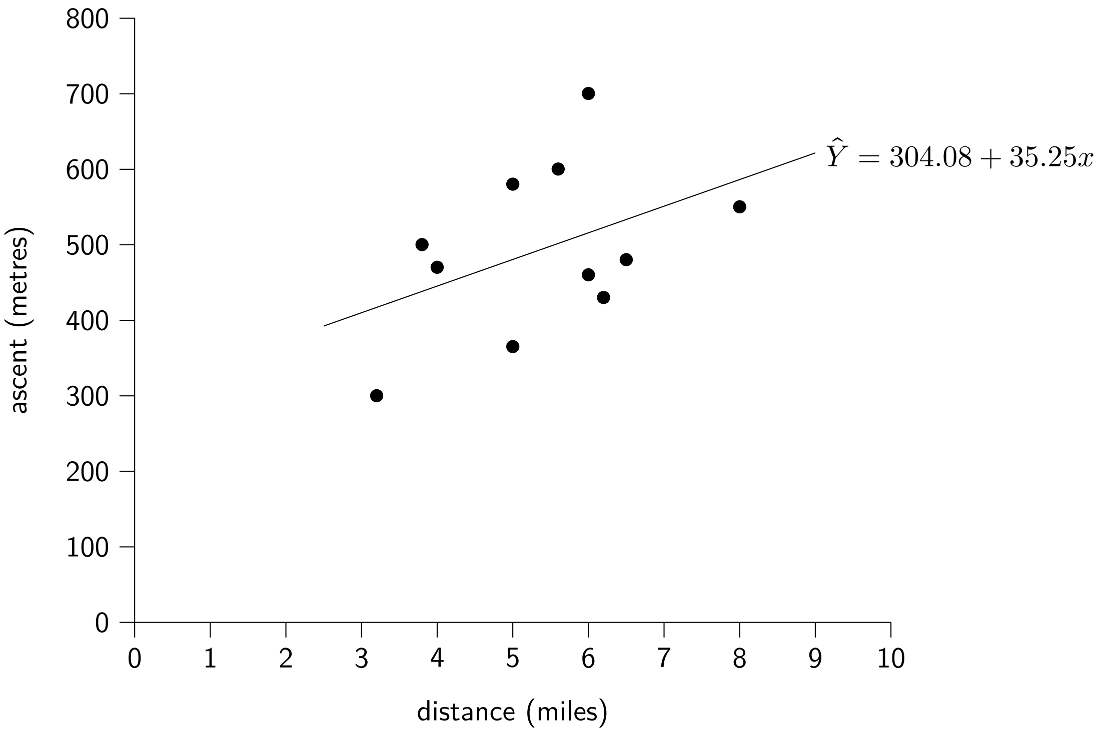
5. The website *Fife Walking* contains an extensive collection of walking routes grouped by area. The length (x miles) and total vertical ascent (y metres) is shown in **table 1** below for a random sample of eleven routes in the *Ochil Hills* area.

Table 1

Route Name	Distance (miles)	Ascent (metres)
Beld Hill and Craig Rossie	6	460
Dumyat	3.8	500
Myretoun Hill and Balquharn Glen	4	470
Colsnaur Circuit	5	580
Core Hill	6.5	480
Eastbow Hills and Steele's Knowe	6.2	430
King's Seat Circuit	5.6	600
Burn of Sorrow Circular	6	700
Hilfoot Hill and Castle Campbell	3.2	300
Innerdownie, Whitewisp and Glenquey Circuit	8	550
Ben Shee	5	365

The data is displayed below in a scatterplot below (**figure 1**), with the equation of a y on x least squares regression line calculated as $\hat{Y} = 304.08 + 35.25x$.

Figure 1



(a) Calculate the correlation coefficient and interpret it, with reference to the context.

(b) Explain why the regression line obtained would not be suitable for estimating the distance a route covers if it is known that the total ascent is 430 metres.

(c) Determine the equation of a more suitable regression line, and use it to estimate the distance covered for a route with an ascent of 430 metres.

(d) A walker completes a route in the Lomond Hills area, which also features on the *Fife Walking* website. Explain why it may not be appropriate for them to use either of the regression lines obtained to predict their total ascent if their distance covered is known.

Answers

Chapter 1 Answers - Exploratory Data Analysis

Exercise 1.1

1. (a) Quantitative (b) Qualitative (c) Quantitative (d) Qualitative (e) Quantitative (f) Qualitative

2. (a) Discrete (b) Continuous (c) Discrete (d) Discrete (e) Continuous (f) Continuous

3. Categorical: *activity level* and *smoker*. Continuous: *height* and *mass*. Discrete: *pulse*.

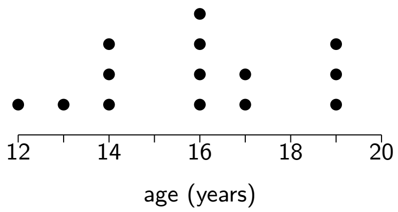
Exercise 1.2

1. Population mean μ
2. Sample mean $\bar{x}$
3. Sample standard deviation s
4. Population standard deviation σ
5. Sample mean $\bar{x}$

Exercise 1.3

1.

dot plot of golfer ages



2.

stem-and-leaf diagram

(ice cream cost)

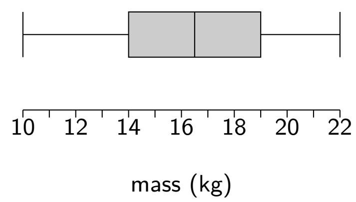
0	9					
1	4	5	5	7	8	9
2	0	1	2	5		
3	6					

$n = 12$

$2|5$ represents £2.50

3.

box plot of luggage masses



Exercise 1.4

1. $\bar{x} = 35.9, s = 15.74$ 2. $\bar{x} = 5.8, s = 2.145$ 3. $\bar{x} = 29.125, s = 7.586$ 4. median $= 40.5$, IQR $= 28$

Exercise 1.5

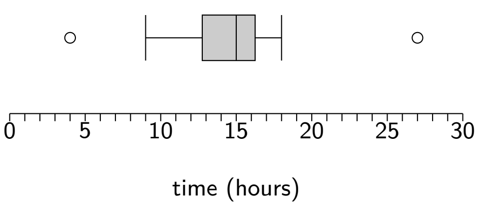
1. $\frac{23}{94}$ or 0.2447
2. A clear key/legend linked to the plot, numbers/scale on the y axis.
3. The proportions of pupils in year group taking the bus seem to be broadly similar.
4. Approximately 15 minutes.
5. 7

Exercise 1.6

1. Lower fence $= 7.5$ and upper fence $= 51.5$. Since all values lie within the interval $(7.5, 51.5)$ there are no outliers.
2. (a) Lower fence $= 6.5$ and upper fence $= 22.5$. Since $4 < 6.5$ and $27 > 22.5$ they are both outliers.

   (b)

   box plot of studying time



3. (a) Upper fence $= 26.5$. Since both circled points shown on the box plot (47 and what appears to be 28) are greater than 26.25, they are outliers.

   (b) If the lengths of their stays were incorrectly recorded in the hospital's records, it would be valid to remove them from the data set before conducting any further analysis. *Other valid reasons are possible.*

Exercise 1.7

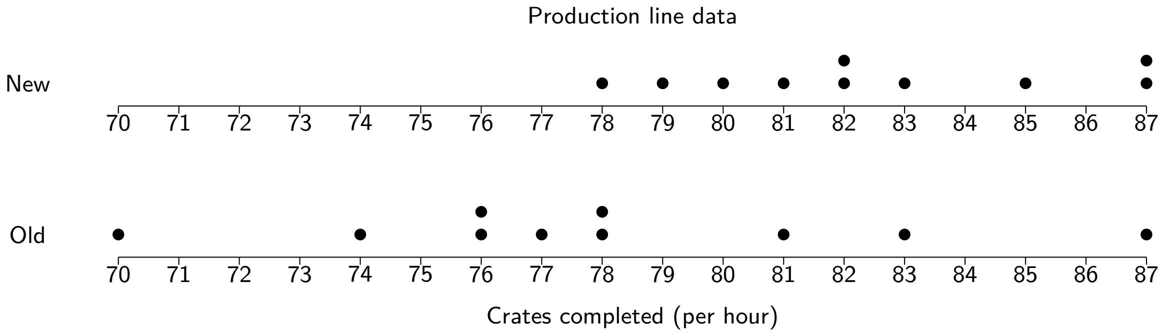
1. The population of hazelnuts masses appear to possibly follow a positively skewed distribution.
2. The population of tree heights appear to possibly follow a symmetrical distributed (such as a normally distribution).
3. The population of waiting times appear to be possibly uniformly distributed.
4. The population of number of successful passes appears to possibly follow a bimodal distribution.

Exercise 1.8

1. (a) $\bar{x}_A = 301.3, s_A = 4.57$
   $\bar{x}_B = 302.3, s_B = 12.3$

   (b) The mean volume of coffee dispensed by Model B is appears to be greater (larger sample mean).
   The amount of coffee dispensed by Model B is more varied (larger sample standard deviation).

2. The median score awarded by judge A was higher.
   The higher interquartile range for judge A shows that their scores were more varied.
3. (a) The daily 2pm temperature was more varied in January than in June, as shown by the slightly higher standard deviation. The median 2pm temperature in June was higher.
   (b) Consistent scales have not been used for the box plots, making the location and spread of each less easily comparable.
4. (a)



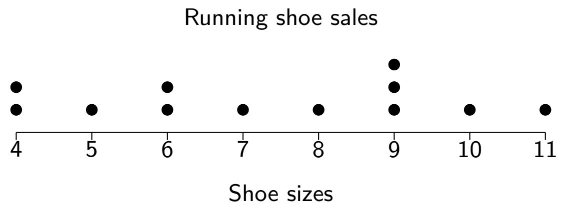
   (b) $\bar{x}_{\text{new}} = 82.4, s_{\text{new}} = 3.13$
   $\bar{x}_{\text{old}} = 78, s_{\text{old}} = 4.76$
   (c) The mean number of crates completed per hours was higher for the new production line.
   The number of crates completed per hour was more consistent for the new production line.

Chapter 1 Review Exercise

1. (a) i. Track; artist. ii. Release year; no. of streams. iii. Length.

   (b) The upper fence could be calculated to determine whether it is an outlier.

2. (a)



   (b) median $= 7.5$ and interquartile range $= 3.5$
   (c) mean $= 7.33$ and sd $= 2.35$
   (d) Shoe sizes are discrete data.

3. (a) A legend should be provided for the diagram, such as $3|2 = 3.2$ metres.
   (b) The stem-and-leaf diagram shows that trees in Woodland A were taller, on average.
4. (a) The fences are 1.13 and 1.85. Since both the minimum value of 1.00 and the maximum of 1.90 lie outwith the interval $(1.00 < 1.13$ and $(1.90 > 1.85)$ the *iris setosa* data contains at least two outliers.
   (b) The median length of *iris versicolor* petals appears to be greater than the length of *iris setosa* petals. The length of *iris versicolor* petals appears to be more varied than those of the *iris setosa* species, as seen by their interquartile ranges of 0.18 and 0.60 centimetres respectively.

Chapter 2 Answers - An Introduction to Probability Theory

Exercise 2.1

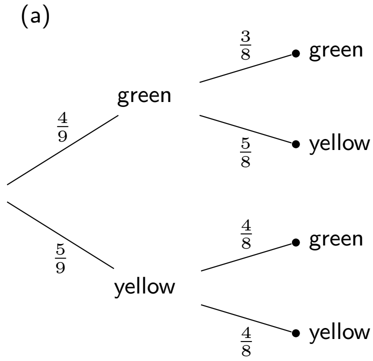
1. (a) P(vowel) $= \frac{5}{26}$
   (b) P(consonant) $= \frac{21}{26}$
   (c) P(statistics) $= \frac{5}{26}$
2. (a) P(blue) $= \frac{4}{15}$
   (b) P(green or red) $= \frac{11}{15}$
   (c) P(not red) $= \frac{3}{5}$
3. (a) P(orange) $= \frac{1}{3}$
   (b) P(orange and strawberry) $= 0$
   (c) P(not caramel) $= \frac{7}{12}$
4. (a) P(Scotland) $= \frac{2}{19}$
   (b) P(Wales or NI) $= \frac{13}{76}$
   (c) P(not England) $= \frac{21}{76}$
5. (a) P(blue and 5) $= \frac{1}{18}$
   (b) P(5) $= \frac{1}{9}$
   (c) P(blue or 5) $= \frac{7}{18}$
6. (a) P(G) $= \frac{9}{10}$
   (b) P(G and A) $= \frac{27}{80}$
   (c) P(not A) $= \frac{5}{8}$
7. (a) P(nurse) $= \frac{25}{64}$
   (b) P(not ward 2) $= \frac{39}{64}$
   (c) P(ward 2 and not doctor) $= \frac{21}{64}$

Exercise 2.2

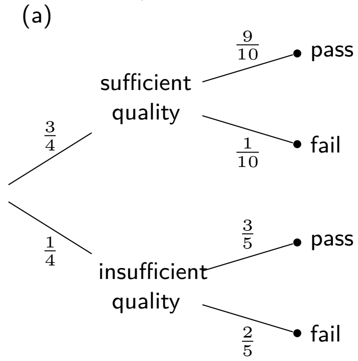
1. (a) {H1,H2,H3,H4,H5,H6 T1,T2,T3,T4,T5,T6}
   (b) P(even and tails) $= \frac{1}{4}$
2. (a) P(tea and toast) $= \frac{1}{12}$
   (b) P(museli and not coffee) $= \frac{1}{6}$
   (c) P(not porridge and not tea) $= \frac{1}{2}$
3. (a) P(tails twice in a row) $= \frac{3}{8}$
   (b) P(tails at least twice) $= \frac{1}{2}$
4. P(sum $= 3$) $= \frac{1}{20}$
5. P(product < 5) $= \frac{2}{9}$
6. P(at least one number $\leqslant 2$) $= \frac{2}{3}$
7. (a) P(20p first) $= \frac{1}{5}$
   (b) P(20p selected) $= \frac{2}{5}$
   (c) P(total $\geqslant 15$p) $= \frac{1}{2}$
   (d) P(total < 12p) $= \frac{2}{5}$
8. (a) 56
   (b) i. P(A board and C pens) $= \frac{1}{56}$
       ii. P(neither chosen) $= \frac{15}{28}$
       iii. P(at least one chosen) $= \frac{13}{28}$

Exercise 2.3

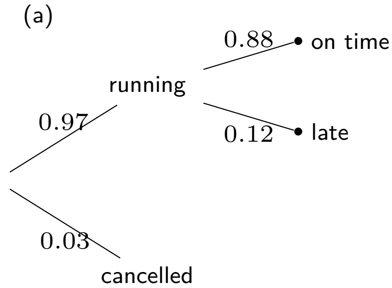
1. (a)



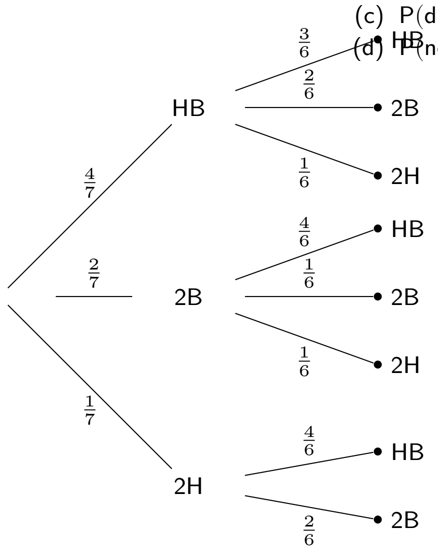
   (b) P(both green) $= \frac{1}{6}$
   (c) P(at least one green) $= \frac{13}{18}$
2. P(both same colour) $= \frac{9}{19}$
3. P(neither O_+) $= 0.4281$
4. (a)



   (b) P(both pass) $= 0.825$
5. (a)



   (b) P(on time) $= 0.8536$
6. (a)



   (b) P(both same) $= \frac{1}{3}$
   (c) P(different) $= \frac{2}{3}$
   (d) P(neither HB) $= \frac{1}{7}$
7. (a) P(no stops) $= 0.224$
   (b) P(at least one stop) $= 0.776$
   (c) P(exactly two stops) $= 0.252$

Exercise 2.4

1. (a) $P(A) = \frac{5}{9}$
   (b) $P(B) = \frac{4}{9}$
   (c) $P(\bar{A}) = \frac{4}{9}$
   (d) $P(\bar{B}) = \frac{5}{9}$
   (e) $P(A \cap B) = \frac{1}{3}$
   (f) $P(A \cup B) = \frac{2}{3}$
   (g) $P(\bar{A} \cap B) = \frac{1}{9}$
   (h) $P(A \cap \bar{B}) = \frac{2}{9}$
2. (a) $x = 0.35$
   (b) i. $P(N) = 0.55$
       ii. $P(\bar{M}) = 0.5$
       iii. $P(M \cup N) = 0.7$
3. (a) P(prime) $= \frac{3}{8}$
   (b) P(odd) $= \frac{1}{2}$
   (c) $P(\overline{\text{prime}}) = \frac{5}{8}$
   (d) $P(\overline{\text{odd}}) = \frac{1}{2}$
   (e) $P(\text{prime} \cap \text{odd}) = \frac{5}{16}$
   (f) $P(\text{prime} \cup \text{odd}) = \frac{9}{16}$
   (g) $P(\overline{\text{prime}} \cap \text{odd}) = \frac{3}{16}$
   (h) $P(\text{prime} \cap \overline{\text{odd}}) = \frac{1}{16}$
4. (a) $P(C \cap D) = 0.34$
   (b) $P(C \cup D) = 0.91$
   (c) $P(\bar{C} \cap \bar{D}) = 0.09$
   (d) $P(\bar{C} \cup \bar{D}) = 0.66$
   (e) $P(C \cap \bar{D}) = 0.45$
   (f) $P(\bar{C} \cap D) = 0.12$
   (g) $P(C \cup \bar{D}) = 0.88$
   (h) $P(\bar{C} \cup D) = 0.55$
5. P(biology) $= \frac{13}{24}$
6. $P(\text{PM} \cap \text{NO}_2) = \frac{1}{10}$

Exercise 2.5

1. $\mathsf{P}(X \cup Y) = 0.6$
2. $\mathsf{P}(A \cup B) = \frac{29}{30}$
3. $\mathsf{P}(E \cup F) = 0.61$
4. $\mathsf{P}(C \cup D) = \frac{11}{14}$
5. $\mathsf{P}(A \cap B) = 0.163$
6. $\mathsf{P}(X \cap Y) = \frac{9}{80}$
7. $\mathsf{P}(V \cap W) = 0.2$
8. $\mathsf{P}(\bar{J}) = \frac{7}{72}$
9. $\mathsf{P}(X \cup Y) = 0.9$
10. $\mathsf{P}(\bar{B}) = \frac{1}{6}$
11. $\mathsf{P}(\text{fish}) = 0.85$
12. $\mathsf{P}(\bar{N}) = \frac{3}{8}$
13. $\mathsf{P}(\bar{R}) = 0.19$
14. (a) $\mathsf{P}(X \cup Y) = 0.9 \neq 1$
    $\therefore X, Y$ not exhaustive
    (b) $\mathsf{P}(R \cup T) = 1$
    $\therefore R, T$ exhaustive
    (c) $\mathsf{P}(E \cup F) = 0.85 \neq 1$
    $\therefore E, F$ not exhaustive
15. (a) $\mathsf{P}(W \cup V) = 0.3 \neq 0$
    $\therefore W, V$ not mutually exclusive
    (b) $\mathsf{P}(X \cap Y) = 0$
    $\therefore X, Y$ mutually exclusive
    (c) $\mathsf{P}(A \cap B) = 0$
    $\therefore A, B$ mutually exclusive

Chapter 2 Review Exercise

1. (a) $\mathsf{P}(\overline{\text{yellow}} \cup 1) = \frac{2}{3}$
   (b) $\mathsf{P}(\text{neither yellow}) = \frac{5}{18}$
2. (a) $\mathsf{P}(\text{English} \cap \text{N5}) = \frac{21}{64}$
   (b) $\mathsf{P}(\text{History} \cup \text{Higher}) = \frac{43}{64}$
3. $\mathsf{P}(F) = 0.9$
4. $\mathsf{P}(\text{takes payment}) = 0.8926$
5. (a) $\mathsf{P}(\text{not a tie}) = \frac{5}{6}$
   (b) $\mathsf{P}(\text{attacker wins}) = \frac{7}{12}$
   (c) $\mathsf{P}(\text{bonus point}) = \frac{1}{8}$
6. $\mathsf{P}(N \cup \bar{A}) = \frac{7}{25}$
7. $\mathsf{P}(B \cap C) = 0.2$

Chapter 3 Answers - Sampling Theory

Note: Whilst the answers included here are intended to include an appropriate level of detail, variations in the amount and type of information given in exam questions will require a flexible and common-sense approach to be used in any exam. Past exam questions and marking schemes should be used as a guide, as well as course reports and relevant CPD events.

Exercise 3.1

1. (a) i. People who use his local supermarket.
   ii. All shoppers who use the self-checkout tills on that Sunday morning.
   (b) The sampling frame excludes people who prefer to shop at other times, such as during the evening or midweek. It also only includes those using self-checkout tills, likely excluding those who have a large trolly of shopping. Both of these issues are likely to make the sample unrepresentative of the target population, and may introduce bias.
   (c) The sampling methods is non-random and so will likely introduce bias to the data, making it unsuitable.
2. (a) There may be so many flowers that it is too time-consuming and impractical to make a sampling frame comprised of individual flowers.
   (b) A better choice of sampling unit for the sampling frame may be square metres of field, forming a numbered grid.
   (c) A random number generator will help avoid possible bias in the selection of square metres to include in the sample, by ensuring it is random and not influenced by the botanist. It should help produce a sample that is more representative of the population of wildflowers.

Exercise 3.2

1. (a) One-stage cluster sampling.
   (b) Stratified sampling.
   (c) Systematic sampling.
   (d) Simple random sampling.
   (e) Quota sampling.
   (f) Convenience Sampling.
2. (a) Each of the different types of room may have their own distinct characteristics in terms of ageing or the quality of maintenance. Stratified sampling will ensure that a proportionate number from each type of room will be included in the sample.
   (b) 30 out of 360 cabins means 1 in every 12 cabins should be sampled, so the sample should contain: 2 Deluxe cabins; 6 Premium cabins and 22 Standard cabins. To sample the Deluxe cabins, a numbered list of those cabins should be obtained as a suitable sampling frame, and then a random number generator used to produce 2 different random numbers. Those cabins corresponding to the numbers selected should be included in the sample. This process should then be repeated for the Premium cabins and the Standard cabins, using a numbered sampling frame for each and randomly selecting the appropriate number of cabins of each type. Combining all of the selected cabins will make a stratified sample.
3. (a) Convenience sampling.
   (b) He only samples coral reefs close to his town, ignoring that those near other towns or villages, or rural locations, may have a different level of health. This means the sample is unlikely to be representative of the the population of coral reefs along the West Coast of Scotland. (Alternatively, the non-random selection method may be discussed.)
4. (a) Using cluster sampling means less physical travelling to a larger number of petrol stations, which would cost more and take more time.
   (b) To sample 40 pumps, 5 petrol stations are needed if one-stage cluster sampling is used. A sampling frame of all the petrol stations numbered from 1 to 126 should be drawn up. A random number generator can then be used to select 5 numbers, and the petrol stations corresponding to those numbers should be included in the sample. From each of those petrol stations, all of the pumps should be sampled.
5. (a) Systematic sampling.

(b) She didn't use a random starting point, so it is not a true random sampling process that is being used.

6. (a) Quota sampling.
   (b) This is a non-random sampling method which is liable to produce a sample not representative of the target population, and may introduce bias into the data gathered.

Chapter 3 Review Exercise

1. (a) i. Convenience sampling.
       ii. This is a non-random sampling method which may produce a sample that is not representative of the target population, and may introduce bias to the data.
   (b) i. Simple random sampling.
       ii. With the 150 employees in an ordered list, the random cell selection tool could be used to select the first employee to be included in the sample. From there, every 5th employee on the list could be selected, looping back around to the start of the list if needed until the sample of size 30 has been obtained.
   (c) i. A numbered list of the restaurants in the chain could be obtained. A random number generator could be used to pick several numbers, and those restaurants corresponding to the numbers could be selected to be sampled. For a one-stage cluster sample, all employees at those restaurants should be sampled.
       ii. One advantage of this method is that it will require less restaurants to be visted for an equivalent sample size, reducing travel costs. One disadvantage is that visiting less restaurants may make it more likely than any restaurants in the chain experiencing problems have none of their employees included in the sample.
2. For an overall sample size of 150 out of 5000 (3%), the sample needs to include 3 people from Management, 12 from Sales and 135 from the Shop Floor. For each strata, an ordered sampling frame should be obtained, and numbers assigned. For the Management sample, a random number generator should be used to choose 3 numbers, and those staff corresponding to those numbers should be selected to be in the sample. The same process should be repeated to choose 12 from the Sales samling frame, and 135 from the Shop Floor sampling frame. Put together, this will make a sample of size 150 with proportions from each strata matching those of the target population.
3. The pizzas sampled will only be those from takeaway restaurants, and may not be representative of pizzas sold in the UK as a whole. Additionally, only those takeaways with a presence on social media will be included.
4. (a) It is unlikely to be possible for the researcher to obtain a list of all pupils in Scotland studying Higher Mathematics.
   (b) From an ordered list of all centres in Scotland which presents pupils for Higher Maths, simple random sampling could be used to select a number of schools. For each of those schools selected, an ordered list of all pupils taking Higher Maths should be obtained, and used to selected a random number of pupils from each selected school. Each of those pupils selected should be included in the overall sample.
5. A random letter selector and a random number generator could be used together to select a letter and a number, to correspond to grids, such as 'C' and '5' to select grid C5. This process could be repeated to create a list of grids to for the botanist to visit and study the health of flowers in those grids.

Chapter 4 Answers - Further Probability Theory

Exercise 4.1

1. (a) P$(X) = \frac{5}{9}$
   (b) P$(X|Y) = \frac{3}{4}$
   (c) P$(X \cap Y) = \frac{1}{3}$
   (d) P$(X \cup Y) = \frac{2}{3}$
   (e) P$(Y) = \frac{4}{9}$
   (f) P$(Y|X) = \frac{3}{5}$
   (g) P$(Y \cap X) = \frac{1}{3}$
   (h) P$(Y \cup X) = \frac{2}{3}$
2. (a) P(scored) $= \frac{73}{135}$
   (b) P(scored | 3-pointer) $= \frac{15}{41}$
   (c) P(3-pointer) $= \frac{41}{135}$
   (d) P(3-pointer | scored) $= \frac{15}{73}$
3. (a) P(rainy)$= 0.3$
   (b) P(not rainy)$= 0.7$
   (c) P(leaves | rainy)$= 0.4$
   (d) P(does not leave | rainy)$= 0.6$
   (e) P(leaves | not rainy)$= 0.9$
   (f) P(does not leave | not rainy)$= 0.1$
4. (a) P$(A) = \frac{2}{5}$
   (b) P$(B) = \frac{2}{5}$
   (c) P$(C) = \frac{1}{3}$
   (d) P$(A \cap B) = \frac{2}{15}$
   (e) P$(B \cup C) = \frac{3}{5}$
   (f) P$(A \cap B \cap C) = \frac{1}{15}$
   (g) P$(A|C) = \frac{3}{5}$
   (h) P$(C|B) = \frac{1}{3}$
   (i) P$(\overline{B}|C) = \frac{3}{5}$
   (j) P$(C|\overline{A}) = \frac{2}{9}$
   (k) P$((A \cap B)|C) = \frac{1}{5}$

Exercise 4.2

1. (a) P$(A|B) = 0.2$
   (b) P(red | blue) $= 0.4849$
   (c) P$(X|Y) = \frac{2}{3}$
   (d) P$(M|N) = 0.3233$
   (e) i. P$(\overline{A}|B) = 0.8$
   ii. P($\overline{\text{red}}$ | blue) $= 0.5151$
   iii. P$(\overline{X}|Y) = \frac{1}{3}$
   iv. P$(\overline{M}|N) = 0.6767$
2. P$(\overline{V}|W) = 0.3$
3. P(5 | yellow) $= 0.2$
4. P(lose | concede ≥ 2) $= \frac{5}{8}$
5. P$(M|N) = 0.25$
6. P$(V \cup W) = 0.725$

Exercise 4.3

1. (a) Independent.
   (b) Not independent.
2. (a) Not independent.
   (b) Independent.
   (c) Independent.
3. (a) Show e.g. P$(V) =$ P$(V|W)$
   (b) Show e.g. P$(T) \neq$ P$(T|W)$

Exercise 4.4

1. (a) P$(X \cap Y) = 0.072$
   (b) P(blue $\cap$ yellow) $= \frac{3}{8}$
   (c) P$(E \cap F) = 0.246$
2. (a) P$(A \cap B) = 0.392$
   (b) P$(A \cup B) = 0.708$
3. P$(X \cup Y) = 0.88$
4. P(working $\cap$ Calculo) $= 0.12$
5. P(out-of-date $\cap$ weak) $= 0.096$
6. P(blue $\cap$ pink) $= 0.38$

Exercise 4.5

1. (a) P$(X|Y) = \frac{7}{12}$
   (b) P$(V|W) = 0.625$
2. P(1 | red) $= 0.6$
3. P(gene | positive) $= 0.2186$
4. (a) P$(M|N) = \frac{8}{15}$
   (b) P$(\overline{M}|N) = \frac{7}{15}$
5. P(pass | car) $= 0.8446$
6. P(Pacific 525 | music) $= 0.7467$
7. P$(\overline{E}|F) = 0.1946$

Exercise 4.6

1. P(UK) $= 0.8708$
2. P(survives) $= 0.6745$
3. P(win) $= 0.695$
4. P(likes) $= 0.54$
5. P(not make second part) $= 0.28$
6. P(do not survive) $= 0.268$
7. P(win) $= 0.3882$

Exercise 4.7

1. (a) P(damaged) $= 0.0655$
   (b) P(express | damaged) $= 0.1069$
2. (a) P(flagged) $= 0.0308$
   (b) P(malicious | flagged) $= 0.3584$
3. P(guessed | wrong) $= 0.7273$
4. (a) P(visits) $= 0.033$
   (b) P(social media | visits) $= 0.8409$
5. P(lost | rest day) $= 0.0559$
6. P(flaw | doesn't leave) $= 0.8048$
7. (a) P(has condition | positive) $= 0.4580$
   (b) P(has condition | positive test) $= 0.4975$

Chapter 4 Review Exercise

1. $\mathsf{P}(E|F) = \frac{7}{9}$
2. (a) i. $\mathsf{P}(W \cup A) = \frac{17}{20}$
   ii. $\mathsf{P}(\overline{W} \cap B) = \frac{1}{8}$
   (b) $\mathsf{P}(\text{power}) = 0.825$
   (c) $\mathsf{P}(\text{Mac} \mid \overline{\text{power}}) = \frac{1}{7}$
3. (a) $\mathsf{P}(\text{wins}) = 0.3917$
   (b) $\mathsf{P}(\text{1st} \mid \text{lost}) = 0.1370$
4. $\mathsf{P}(M|B) = \frac{1}{5} = \mathsf{P}(M)$
   hence P, M are independent
5. (a) $\mathsf{P}(\text{incorrect}) = 0.2875$
   (b) $\mathsf{P}(\text{cat} \mid \text{incorrect}) = 0.2348$

Chapter 5 - An Introduction to Linear Regression

Exercise 5.1

Exercise 5.2

Exercise 5.3

Chapter 5 Review Exercise

Chapter 6 - Random Variables

Exercise 6.1

Exercise 6.2

Exercise 6.3

Exercise 6.4

Exercise 6.5

Exercise 6.6

Chapter 6 Review Exercise

Chapter 7 - Discrete Distributions

Exercise 7.1

1. $E(X) = 3$ and $V(X) = 2$
2. $E(X) = 8$ and $V(X) = \frac{56}{3}$
3. (a) $E(X) = 4.5$ and $V(X) = 5.25$
   (b) $E(2X+5) = 14$ and $V(2X+5) = 21$
4. (a) $X \sim U(20)$
   (b) $E(X) = 10.5$ and $V(X) = 33.25$
   (c) $E(1-3X) = -30.5$ and $V(1-3X) = 299.25$
5. (a) $E(X-Y) = -1$
   (b) $V(X-Y) = \frac{8}{3}$
6. (a) $k = 10$
   (b) $P(2 \leqslant X < 6) = 0.4$
   (c) $SD(X) = \frac{\sqrt{33}}{2}$
7. (a) $k = 8$
   (b) $E(1-2Y) = -8$
8. $X \sim U(11)$ and $Y \sim U(7)$

Exercise 7.2

1. 0.2503
2. 0.2787
3. 0.0579
4. 0.0988
5. (a) $X \sim B(4, \frac{1}{7})$
   (b) 0.3599
6. 0.0214

Exercise 7.3

1. (a) 0.9894
   (b) 0.0367
   (c) 0.9527
   (d) 0.0106
   (e) 0.0473
2. (a) 0.9527
   (b) 0.2235
   (c) 0.5756
3. (a) 0.7414
   (b) 0.0003
   (c) 0.4829
4. (a) 0.1468
   (b) 0.5033
   (c) 0.1927

Exercise 7.4

1. (a) $\mu = 0.8, \sigma^2 = 0.64$
   (b) $\mu = 2, \sigma^2 = \frac{4}{3}$
   (c) $\mu = 0.4, \sigma^2 = 0.36$
   (d) $\mu = 12, \sigma^2 = 7.2$
2. (a) $S \sim B(10, \frac{1}{4})$
   (b) $E(S) = 2.5$ and $V(S) = 1.875$
   (c) 0.0197
3. (a) $p = 0.2$
   (b) $SD(X) = 0.96$
4. (a) $p = \frac{2}{3}$
   (b) $E(Y) = 8$
5. (a) $n = 20$ and $p = \frac{4}{5}$
   (b) $P(5 < X < 10) = 0.0006$

Exercise 7.5

1. (a) 0.1336
   (b) 0.0821
   (c) 0.2873
   (d) 0.7127
2. (a) $X \sim Po(4)$
   (b) 0.0183

Exercise 7.6

1. (a) 0.01512
   (b) 0.5543
   (c) 0.0620
   (d) 0.3937
2. (a) 0.02231
   (b) 0.0611
   (c) 0.4412
   (d) 0.1912
3. (a) $X \sim Po(1)$
   (b) 0.0190

Exercise 7.7

1. 0.9473
2. 0.9389
3. 0.2661
4. 0.1353
5. 0.0144

Chapter 7 Review Exercise

1. (a) $\frac{5}{8}$
   (b) $E(X) = 4.5$ and $V(X) = 5.25$
2. (a) 0.1954
   (b) 0.2071
3. (a) 0.1294
   (b) 0.4769
4. 0.4335
5. (a) 0.8298
   (b) $\mu = 17$ and $\sigma = 1.597$
   (c) The germination of the seeds is independent.
6. (a) 0.6703
   (b) 0.1954
7. (a) $P(M|V) = 0.2$
   (b) $X \sim B(8, 0.3)$
   (c) 0.0580
8. (a) 0.2
   (b) 0.1528
   (c) 0.0267

Chapter 8 Answers - Continuous Distributions

Exercise 8.1

1. (a) $\frac{2}{3}$
   (b) 0.7667
   (c) 0.1
2. (a) $\frac{1}{3}$
   (b) 0
   (c) 0.6583
3. (a) $\frac{1}{6}$
   (b) $\frac{5}{6}$
   (c) $\frac{5}{12}$
4. (a) $E(X) = 1,\ SD(X) = 2.89$
   (b) $E(3X - 1) = 2$, $V(5 - 2X) = \frac{100}{3}$
   (c) 0.64
   (d) 0.4
5. (a) $\frac{5}{6}$
   (b) 7.5 mins
   (c) 4.33 mins
6. $\frac{4}{7}$
7. (a) 0.4
   (b) $X \sim B(20, 0.4)$,0.5841
8. (a) 8
   (b) 4.04
   (c) 0.5
   (d) 0.5771
9. (a) $X \sim U(2, 14)$
   (b) $\sqrt{12}$
   (c) $\frac{5}{12}$
10. (a) $Y \sim U(16, 22)$
    (b) 0.4082

Exercise 8.2

1. (a) $52, 60, 68$ marked
   (b) $237, 243, 249$ marked
   (c) $5.4, 8, 10.6$ marked
   (d) $12.6, 15, 17.4$ marked
2. $3, 7, 11$ marked
3. $5, 8, 11$ marked
4. $26, 32, 38$ marked

Exercise 8.3

1. (a) 0
   (b) 0.9573
   (c) 0.9573
   (d) 0.5
   (e) 0.6368
   (f) 0.9788
   (g) 0.0212
   (h) 0.1170
   (i) 0.1170
   (j) 0.7123
   (k) 0.9251
   (l) 0.2843
2. (a) 0.0740
   (b) 0.2888
   (c) 0.0339
   (d) 0.1618
   (e) 0.6612
   (f) 0.6826
   (g) 0.9544
   (h) 0.9500
   (i) 0.8990

Exercise 8.4

1. (a) 0.7475
   (b) 0.1587
   (c) 0.6306
   (d) 0.4719
2. (a) 0.1587
   (b) 0.4305
   (c) 0.9660
   (d) 0.3891
3. (a) $X \sim N(67.4, 6^2)$
   (b) i. 0.6676
   ii. 0.6554
   iii. 0.2647
4. (a) i. 0.8997
   ii. 0.9500
   (b) i. 143
   ii. 595
   iii. 536

Exercise 8.5

1. (a) 1.28
   (b) 1.64
   (c) 3.09
   (d) 2.33
   (e) 2.58
   (f) -1.96
   (g) -2.58
   (h) -1.28
   (i) -1.96
   (j) 3.09
   (k) 0.81
   (l) 2.75
2. (a) $X \sim N(2.4, 0.2^2)$
   (b) 0.0228
   (c) 2.866 litres
3. (a) 8.692 mins
   (b) 7.612 to 8.788 mins
4. $\sigma = 9.9$
5. $\mu = 64.1, \sigma = 14.8$

Exercise 8.6

1. (a) $N(34, 13.09)$
   (b) 0.6103
2. 0.1867
3. 0.1379
4. 0.0192

Exercise 8.7

1. (a) $np = 16, nq = 24$ $np, nq > 5$
   (b) i. 0.9265
   ii. 0.0031
   iii. ≈ 0
2. (a) $np = 54, nq = 27$ $np, nq > 5$
   (b) i. 0.3632
   ii. 0.0338
   iii. 0.6971
3. $X \sim B(600, 0.85)$ 0.1151
4. 0.2578
5. 0.0044

Exercise 8.8

1. (a) 0.6217
   (b) 0.0661
   (c) 0.5398
2. (a) 0.6736
   (b) 0.5640
   (c) 0.1357
3. (a) 0.2611
   (b) 0.2349
   (c) 0.0550
4. 0.8907
5. (a) 0.7384
   (b) 3.8
6. (a) 0.4562
   (b) 0.1718
   (c) 0.8315
7. (a) 0.2389
   (b) 0.4641
8. 0.8529

Chapter 8 Review Exercise

1. (a) 4
   (b) 3
   (c) $\frac{1}{3}$
2. (a) 0.7977
   (b) 0.3413
3. (a) $H \sim N(18.1, 0.7^2)$
   (b) 0.0033
   (c) 19.248 feet
4. (a) 0.1353
   (b) 0.0733
5. 22.99
6. (a) 0.3
   (b) 0.6172
   (c) 0.4562
7. (a) 0.0951

Chapter 9 - Distribution of the Sample Mean

Exercise 9.1

1. (a) i. 0.3821
   ii. 0.4404
   iii. 0.3974
   (b) i. $N(24, \frac{4^2}{3})$
   ii. 0.3015
2. 0.1922
3. (a) 0.3085
   (b) 0.1170
4. 0.0228
5. (a) 0.0082
   (b) 0.3974
6. 0.9992

Exercise 9.2

1. (a) $X \approx N(2.3, \frac{5^2}{40})$
   (b) $Y \sim N(83, \frac{9}{25})$
   (c) $D \sim N(126, \frac{10}{7})$
   (d) Comment.
2. 0.9857
3. 0.0071
4. 0.1038
5. 0.3758
6. (a) 0.0526

Chapter 9 Review Exercise

1. (a) $\bar{X} \sim N(15, \frac{3^2}{8})$
   (b) 0.9706
2. 0.9981
3. (a) Comment.
   (b) $\bar{X} \approx N(600, \frac{70^2}{40})$
   (c) 0.1489
4. (a) i. $X \sim N(60, 2^2)$
   ii. 0.3085
   (b) i. $\bar{X} \sim N(60, \frac{2^2}{8})$
   ii. 0.1271
   (c) Comment.
5. (a) $X \sim Po(4.5)$
   (b) 0.1736
   (c) $X \approx N(4.5, \frac{4.5}{30})$
   (d) 0.0985

Chapter 10 - An Introduction to Hypothesis Testing

Exercise 10.1

1. $H_0 : \mu = 8.6$
   $H_1 : \mu > 8.6$
2. $H_0 : \mu = 12$
   $H_1 : \mu > 12$
3. $H_0 : \mu = 4.2$
   $H_1 : \mu < 4.2$
4. $H_0 : \mu = 2.5$
   $H_1 : \mu \neq 2.5$
5. $H_0 : \mu = 65$
   $H_1 : \mu > 65$
6. $H_0 : \mu = 15.7$
   $H_1 : \mu \neq 15.7$

Exercise 10.2

}

1. $0.0162 < 0.05 \implies$ *reject* H_0
2. $0.1056 > 0.01 \implies$ *do not reject* H_0
3. $0.0485 < 0.1 \implies$ *reject* H_0
4. $0.0764 > 0.05 \implies$ *do not reject* H_0

Exercise 10.3

1. $0.0286 < 0.1 \implies$ *reject* H_0
2. $0.0702 > 0.001 \implies$ *do not reject* H_0
3. $0.0485 < 0.05 \implies$ *reject* H_0
4. $0.0220 < 0.1 \implies$ *reject* H_0
5. $0.0294 < 0.05 \implies$ *reject* H_0
6. $0.0488 > 0.01 \implies$ *do not reject* H_0
7. $0.0023 < 0.05 \implies$ *reject* H_0
8. $0.0250 > 0.01 \implies$ *do not reject* H_0

Exercise 10.4

1. $1.67 < 1.96 \implies$ *do not reject* H_0
2. $-2.5 < -1.64 \implies$ *reject* H_0
3. $2.24 < 2.33 \implies$ *do not reject* H_0
4. $-0.77 > -2.33 \implies$ *do not reject* H_0
5. $2.00 > 1.96 \implies$ *reject* H_0

Exercise 10.5

1. $-2.985 < -1.895 \implies$ *reject* H_0
2. $0.636 < 1.476 \implies$ *do not reject* H_0
3. $2.317 < 2.447 \implies$ *do not reject* H_0
4. $-3.531 < -2.821 \implies$ *reject* H_0

Exercise 10.6

1. $1.52 < 1.64 \implies$ *do not reject* H_0
2. $2.42 > 2.33 \implies$ *reject* H_0
3. $-2.78 < -1.64 \implies$ *reject* H_0
4. $1.17 < 1.96 \implies$ *do not reject* H_0
5. $-2.04 > -2.58 \implies$ *do not reject* H_0
6. $-2.2 < -1.64 \implies$ *reject* H_0

Chapter 10 Review Exercise

1. $-2.54 < -2.33 \implies$ *reject* H_0
2. $2.158 < 2.365 \implies$ *do not reject* H_0
3. $-1.94 < -1.64 \implies$ *reject* H_0
4. $-3.00 < -2.33 \implies$ *reject* H_0
5. $-2.170 < -1.895 \implies$ *reject* H_0

Chapter 11 - Confidence Intervals

Exercise 11.1

1. $(55.18, 58.82)$
2. $(12.76, 14.84)$
3. $(14.53, 15.87)$
4. $(37.38, 40.62)$
5. $(49.03, 51.29)$
6. $(5.41, 5.99)$
7. (a) $(37.64, 38.76)$
   (b) $(37.46, 38.94)$
   (c) 31
8. 10.0

Exercise 11.2

1. $(499.1, 510.9) \implies$ within, comment
2. $(419.5, 426.5) \implies$ within, comment
3. $(10.11, 11.69) \implies$ outwith, comment
4. $(2.439, 2.961) \implies$ within, comment
5. $(19.95, 20.85) \implies$ outwith, comment
6. $(28580, 30020 \implies$ within, comment$)$

Exercise 11.3

1. $(6.88, 7.87)$
2. $(4048.5, 4714.4) \implies$ within, comment
3. $(960.0, 1024.0)$
4. $(3.636, 4.464) \implies$ outwith, comment
5. $(2.760, 4.740) \implies$ outwith, comment

Exercise 11.4

1. $(0.598, 0.736)$
2. $(0.133, 0.427)$
3. $(0.327, 0.387)$
4. $(0.086, 0.149) \implies$ within, comment
5. $(0.021, 0.199) \implies$ outwith, comment
6. $(0.026, 0.083) \implies$ within, comment
7. $n = 147$
8. (a) Systematic sampling
   (b) $(0.264, 0.446) \implies$ within, comment

Chapter 11 Review Exercise

1. $(63.98, 82.77)$
2. $(0.116, 0.444)$
3. (a) $(125.8, 126.8)$
   (b) Outwith, comment
4. (a) 12.5
   (b) $s = 25.77$
5. $(0.679, 0.888) \implies$ outwith, comment

Chapter 12 - Chi-Squared Goodness-of-Fit Test

Exercise 12.1

1. $6.333 < 7.815$, do not reject H_0
2. $7.143 < 10.645$, do not reject H_0
3. $13.88 > 9.210$, reject H_0

Exercise 12.2

1. $7.779 < 9.488$, do not reject H_0
2. $3.049 < 11.345$, do not reject H_0
3. $14.906 > 7.815$, reject H_0

Exercise 12.3

1. (a) $X \sim \mathrm{B}(4, 0.2)$
   (b) i. 0.4096
   ii. 0.4096
   iii. 0.1536
   iv. 0.0256
   v. 0.0016
   (c) $6.266 > 4.605$, reject H_0
2. (a) $X \sim \mathrm{B}(5, 0.25)$
   (b) $257.862 > 9.488$, reject H_0 at the 5% level
3. (a) $3.144 < 5.991$, do not reject H_0
   (b) An additional 1 must be subtracted from the number of categories to obtain the degrees of freedom.

Exercise 12.4

1. (a) i. 0.3679
   ii. 0.3679
   iii. 0.1839
   iv. 0.0803
   (b) $9.173 > 7.815$, reject H_0
2. (a) $\lambda = 2$
   (b) $14.839 > 11.345$, reject H_0
3. (a) $98.443 > 14.067$, reject H_0 at the 5% level
   (b) $\lambda = 3.28$

Review Exercise

1. $3.759 < 5.991$, do not reject H_0
2. (a) $9.790 > 4.605$, reject H_0
   (b) Some players being better penalty takers than others would mean the probability of success for each trial (penalty attempt) would not be equal, and hence a binomial distribution would not be appropriate.
3. (a) $\lambda = 1.5$
   (b) $25.10, 0.45$
   (c) $1.553 < 11.345$, do not reject H_0
   (d) The test statistic would still be 1.553, whilst the critical value would instead be 13.277

Exercise 12.1

Exercise 12.2

Exercise 12.3

Exercise 12.4

Chapter 12 Review Exercise

Chapter 13 - Control Charts

Exercise 13.1

1. The process is out of statistical control, since sample 6 falls below the lower control line. The process should have been stopped after sample 6.
2. The process is out of statistical control, since from samples 4 to 8 at least four of the five samples lie above the upper 1-sigma line. The process should have been stopped after sample 8.
3. The process is under statistical control.
4. The process is out of statistical control, since from samples 2 to 9 eight consecutive samples lie above the centre line. The process should have been stopped after sample 9.

Exercise 13.2

1. (a) UCL = 82.5, $+2\sigma$ = 81.75, $+1\sigma$ = 80.75, target = 80, -1σ = 79.25, -2σ = 78.5, LCL = 77.75
   (b) UCL = 500.72, $+2\sigma$ = 495.81, $+1\sigma$ = 490.91, target = 486, -1σ = 481.09, -2σ = 476.19, LCL = 471.28
2. $+2\sigma = 1.447$, $-2\sigma = 1.353$
3. (a) $+1\sigma = 231.79$, $-1\sigma = 228.21$
   (b) The process is out of statistical control since, from samples 9 to 11, two of the three samples lie above the upper 2-sigma limit. The process should have been stopped after sample 11.
   (c) A required assumption is that the samples of bottles are random.
4. (a) $144 - 3 \times \dfrac{18}{\sqrt{7}} = 123.59$
   (b) $130.4 \leqslant \bar{x} \leqslant 164.4$
5. (a) The process is out of statistical control since eight consecutive points lie above the centre line, from samples 9 to 16. However, since these values suggest the lenses being produced have a higher resolution than the target value, which is a good thing, they may not be concerned.
   (b) Samples of size 9 are being taken.
   (c) Two required assumptions are that:
      - The samples of lenses taken are random.
      - The resolution scores of manufactured lenses are normally distributed.
6. (a) UCL = 121, $+2\sigma$ = 119, $+1\sigma$ = 117, target = 115, -1σ = 113, -2σ = 111, LCL = 109.
   The process is out of statistical control since, from samples 2 to 4, two out of three sample means lie above the upper 2-sigma limit.
   (b) $\bar{\bar{x}} = 92$
   (c) UCL = 101, LCL = 83
7. (a)

Population mean	72
Standard deviation	15
Sample size	25
Upper Control Limit	81
$+2\sigma$ Limit	78
$+1\sigma$ Limit	75
Target Value	72
-1σ Limit	69
-2σ Limit	66
Lower Control Limit	63

(b)

Population mean	2800
Standard deviation	60
Sample size	9
Upper Control Limit	2860
$+2\sigma$ Limit	2840
$+1\sigma$ Limit	2820
Target Value	2800
-1σ Limit	2780
-2σ Limit	2760
Lower Control Limit	2740

Exercise 13.3

1. (a) Population proportion $= 0.6$
Sample size $= 24$

Upper Control Limit	0.9
$+2\sigma$ Limit	0.8
$+1\sigma$ Limit	0.7
Target Value	0.6
-1σ Limit	0.5
-2σ Limit	0.4
Lower Control Limit	0.3

(b) Population proportion $= 0.16$
Sample size $= 50$

Upper Control Limit	0.316
$+2\sigma$ Limit	0.264
$+1\sigma$ Limit	0.212
Target Value	0.16
-1σ Limit	0.108
-2σ Limit	0.056
Lower Control Limit	0.004

2. (a) The samples of mugs would not be random. Also, for appropriate use of a p chart it is required that $np > 5$ and $nq > 5$. Here, $0.08 \times 30 = 2.4 < 5$.
(b) Upper 2-sigma limit = 0.14265 and lower 2-sigma limit = 0.01735.
(c) The process is under statistical control for the samples shown, as none of the WECO rules have been violated.
3. (a) $0.11 + 3 \times \sqrt{\dfrac{0.11 \times 0.89}{50}} = 0.2427$
(b) The calculated lower limit would be -0.0227. However, a negative sample proportion is not possible, so the lower control limit is set as 0.
(c) The process is out of statistical control since, from samples 3 to 7, four out of five samples lie below the lower 2-sigma limit.
(d) Since these sample proportions suggest less seeds aere failing to germinate than expected, which is a good thing, the company many not be concerned.
4. (a) UCL = 0.9549, LCL = 0.6451.
(b) The process is out of statistical control since, from February to September, eight consecutive sample proportions fall below the target value.
(c) The sample size needs to be at least 30 to ensure both $np > 5$ and $nq > 5$.
5. (a) $\hat{p} = 0.12$
(b) Upper 2-sigma limit $= 0.12 + 2 \times \sqrt{\dfrac{0.12 \times 0.88}{50}} = 0.212$ and lower 2-sigma limit
$= 0.12 - 2 \times \sqrt{\dfrac{0.12 \times 0.88}{50}} = 0.028$
(c) The process is under statistical control for the samples shown, as none of the WECO rules have been violated.
6. (a) The population proportion is $p = 0.6$.
(b) UCL = 0.9 and LCL = 0.3.
(c) The sample size is $n = 24$.
(d) The sample size and population proportion are appropriate since both of $24 \times 0.6 = 14.4$ and $24 \times 0.4 = 9.6$ are greater than 5.

Chapter 13 Review Exercise

1. (a) Upper 2-sigma limit $= 60.12$ and lower 2-sigma limit $= 55.88$.
(b) The process is out of statistical control since the sample mean is below the lower control limit at sample 14.
2. (a) Upper 1-sigma limit $= 0.926$ and lower 1-sigma limit $= 0.834$.
(b) $0.88 \leqslant \hat{p} \leqslant 1$

Chapter 14 - Chi-Squared Test for Association

Exercise 14A

1. $3.436 > 2.706$, reject H_0
2. $0.807 < 3.841$, do not reject H_0
3. $4.121 > 2.706$, reject H_0
4. $0.0465 < 5.991$, do not reject H_0
5. $0.282 < 11.345$, do not reject H_0

Exercise 14B

1. $3.328 > 7.815$, do not reject H_0
2. $15.737 > 12.592$, reject H_0
3. $25.106 > 4.605$, reject H_0

Review Exercise

1. $6.254 > 2.706$, reject H_0
2. $23.478 > 7.378$, reject H_0
3. (a) No, with justification
   (b) 10%
   (c) There is no association between the type of feed used and the growth of a chick.
   (d) $0.2804 > 0.1$, do not reject H_0, with context

Exercise 14.1

Exercise 14.2

Chapter 14 Review Exercise

Chapter 15 - Two-Sample Parametric Tests

Exercise 15.1

Exercise 15.2

Exercise 15.3

Exercise 15.4

Chapter 15 Review Exercise

Chapter 16 - Wilcoxon Signed Rank Test

Exercise 16.1

1. $9 > 3 \implies$ do not reject H_0
2. $5.5 < 3 \implies$ reject H_0
3. $8 > 5 \implies$ do not reject H_0
4. $7.5 > 5 \implies$ do not reject H_0
5. $10 \leqslant 10 \implies$ reject H_0
6. $9 < 10 \implies$ reject H_0
7. $6.5 > 5 \implies$ do not reject H_0

Exercise 16.2

1. $8.5 < 13 \implies$ reject H_0
2. $11 > 5 \implies$ do not reject H_0
3. $46 > 21 \implies$ do not reject H_0
4. $1.5 < 2 \implies$ reject H_0
5. $4 < 8 \implies$ reject H_0

Exercise 16.3

1. $-0.34 > -1.96 \implies$ do not reject H_0
2. $-1.67 < -1.64 \implies$ reject H_0
3. $-0.27 > -1.64 \implies$ do not reject H_0
4. (a) $H_0 : \eta = 110, \quad H_1 : \eta < 110$
   (b) $-2.16 > -1.64 \implies$ reject H_0
   (c) Assume that time taken to complete the hike is symmetrically distributed.

Chapter 16 Review Exercise

1. $5 \leqslant 5 \implies$ reject H_0
2. $14 > 8 \implies$ do not reject H_0
3. $-2.40 > -1.96 \implies$ reject H_0
4. $3 > 2 \implies$ do not reject H_0

Chapter 17 - Mann Whitney Rank Sum Test

Exercise 17.1

1. (a) Rank sum of 20.5, or rank sum of 44.5 leading to smaller value of 20.5.
   (b) $20.5 < 21 \implies$ reject H_0
2. $24 > 20 \implies$ do not reject H_0
3. $73.5 < 74 \implies$ reject H_0
4. $25 < 31 \implies$ reject H_0 at the 5% level
5. $66.5 > 66 \implies$ do not reject H_0 at the 5% level
6. Assume distributions of learning times for each type of mouse have the same shape and variance.
   $38.5 > 26 \implies$ do not reject H_0
7. (a) $\mathrm{H}_0 : \eta_{\text{ohm}} = \eta_{\text{watts}}$ and $\mathrm{H}_1 : \eta_{\text{ohm}} \neq \eta_{\text{watts}}$
   (b) $W = 70$
   (c) Since $70 > 71$ reject H_0 at the 5% level of significance. There is evidence to suggest that electric cars from Ohm-G Motors and Watts-Up Motors have a different mean driving range.

Exercise 17.2

1. (a) $-3.45 < -1.64 \implies$ reject H_0
   (b) Assume that the population distributions of number of keepers for each weather condition have the same shape and variance.
2. $-2.08 > -2.58 \implies$ do not reject H_0
3. $-2.78 < -1.64 \implies$ reject H_0 at the 5% level
4. (a) The distributions of number of minor faults for each school appear to have the same shape and variance, so a Mann-Whitney test is appropriate. The distributions do not appear to be normal, so a t-test would not be appropriate.
   (b) $-2.39 > -3.09 \implies$ do not reject H_0
   (c) This data only considers the number of minor faults for students who passed, and does not show the proportions of students who passed or failed from each school, which students are likely to want to know before choosing a driving school.

Exercise 17.3

1. (a) 56
   (b) $\{1,2,3\},\{1,2,4\},\{1,2,5\},\{1,3,4\}$
   (c) $\frac{4}{56} = \frac{1}{14}$ or 0.0714
   (a) 126
   (b) $\{1,2,3,4\},\{1,2,3,5\},\{1,2,3,6\},\{1,2,4,5\}$
   (c) $\frac{4}{126} = \frac{2}{63}$ or 0.0317
2. (a) $1+2+4+8 = 15$
   (b) $\{1,2,3,4\},\{1,2,3,5\},\{1,2,3,6\},\{1,2,3,7\},\{1,2,3,8\}$
   $\{1,2,3,9\},\{1,2,4,5\},\{1,2,4,6\},\{1,2,4,7\},\{1,2,4,8\}$
   $\{1,2,5,6\},\{1,2,5,7\},\{1,3,4,5\},\{1,3,4,6\},\{1,3,4,7\}$
   $\{1,3,5,6\},\{2,3,4,5\},\{2,3,4,6\}$
   (c) Assume that the population distributions of reactions times for each group have the same shape and variance.
   (d) $\frac{18}{330} = \frac{3}{55}$ or 0.0545
   (e) $0.0545 > 0.05 \implies$ do not reject H_0
3. (a) $\frac{7}{495} = 0.0141$
   (b) $0.0141 < 0.1 \implies$ reject H_0
   (c) Assume that the samples of hockey and rugby players are random.

Chapter 17 Review Exercise

1. $34 > 29 \implies$ do not reject H_0
2. (a) The stem-and-leaf diagram shows that red wines seem to contain more alcohol on average than white wines.
   (b) $-3.04 < -1.64 \implies$ reject H_0
   (c) Since the stem-and-leaf diagram seems to suggest that the distributions of alcohol content for the two groups have the same shape and variance, a Mann-Whitney test would be appropriate. Since the distributions both appear to be normal-like, a t-test would also be appropriate.
   (d) i. $H_0 : \eta_A = \eta_B$ and $H_0 : \eta_A \neq \eta_B$
   ii. CV= 13. Since $12 < 13$ reject H_0 at the 2% significance level, so there is evidence to suggest that the two populations do not have the same median.
   iii. Assuming that the samples are random, and assume that the populations have the same shape and variance.

iv. The data for samples A and B will have been combined and ordered, with ranks assigned to each value. The rank sum for sample A will have been found, W_A, and then the value $4 \times (4 + 9 + 1) - W_A$ will have also been calculated. `W=12` will have been the smaller of those two values.

Chapter 18 - Further Linear Regression

Exercise 18.1

1. Completed table:

Regression Line $\hat{Y} = ax + b$	Observed x	Observed y	Fitted $\hat{Y}$	Residual ϵ
$\hat{Y} = 5.3 + 0.1x$	7	8	6	2
$\hat{Y} = -0.7 + 0.3x$	6.1	1.04	1.13	−0.09
$\hat{Y} = 180 - 2.5x$	40	178	80	98
$\hat{Y} = -45 + 3.5x$	200	625	655	−30
$\hat{Y} = 100 + 20x$	-15	-150	-200	50

2. (a) 37.36 grams

   (b) i. Fitted $= 291.8$
   Residual $= 18.2$

   ii. Fitted $= 158.2$
   Residual $= -8.2$

   iii. Fitted $= 11.9$
   Residual $= -1.9$

3. (a) i. positive
   ii. negative

   (b) i. Fitted $= 78.5$
   Residual $= 8.5$

   ii. Fitted $= 82.3$
   Residual $= -6.7$

Exercise 18.2

1. (a) $s^2 = 9.3$ (b) $s = 3.05$
2. (a) $s^2 = 5.257$ (b) $s = 2.29$
3. (a) The scatterplot suggests there is a strong negative linear association between the distance a runner covers in training in the four weeks before a marathon and their finishing time.
   (b) $s^2 = 10.97$
   (c) $\hat{Y} = 194.8 - 23.48x$
   (d) $r = -0.928$
   The correlation coefficient also suggests there is a strong negative linear association between the distance a runner covers in training in the four weeks before a marathon and their finishing time.
   (e) Fitted $= 112.6$, residual $= -2.6$

Exercise 18.3

1. $R^2 = 0.775$. This suggests that 77.5% of the variation in Y is explained by the linear model.
2. (a) $R^2 = 0.734$. This suggests that 73.4% of the variation in the exmployees' salaries is explained by the linear model.
   (Or: *...by the number of years they have been employed for*).
   (b) $R^2 = 0.573$. This suggests that 57.3% of the variation in red squirrel mass is explained by the linear model.
   (Or: *...by the height of the red squirrels*).
   (c) $R^2 = 0.449$. This suggests that 44.9% of the variation in the top speed reached is explained by the linear model.
   (Or: *...by their lap time*).
3. (a) The scatterplot appears to show a moderate/strong, positive linear association between a camera's shutter count and age.
   (b) The correlation coefficient also suggests there is a moderate/strong, positive linear association between a camera's shutter count and its age.
   (c) $R^2 = 0.516$. This suggests that 51,6% of the variation in a camera's shutter count is explained by the linear model.
   (Or: *...by its age*).
   (d) Since the data collected only includes digital cameras from within the last 12 years, any results from it may not apply to much older film cameras.
   Also, the regression line is y on x, suitable only for predicting shutter count based on a camera's age.

Exercise 18.4

1. (a) H_0: $\beta = 0$
   H_1: $\beta \neq 0$
   (b) $s^2 = \dfrac{224.2}{27-2} = 8.968,\ s = \sqrt{8.987} = 2.995,\ b = \dfrac{-129.6}{87.8} = -1.476,\ t = \dfrac{-1.476 \times \sqrt{87.8}}{2.995} = -4.618$
   (c) $\nu = 27 - 2 = 25$
   (d) Critical value for $\nu = 25$ and a 5% two tailed test (and a negative test statistic) is $t_{25,0.975} = -2.060$.
   Since $-4.618 < -2.060$ reject H_0 at the 5% level of significance.
   There is evidence to suggest that the slope parameter $\beta \neq 0$, and so the linear model is useful for prediction.
2. Test statistic $t = -5.120$, critical value $t_{32,0.95} = -1.694$, reject H_0 since $-5.120 < -1.694$.
3. (a) H_0: $\beta = 0$
   H_1: $\beta \neq 0$
   (b) $n = 40$
   (c) Critical value $t_{0.975,38} = -2.024$. Since $-0.42295 > -2.024$, do not rejecrt H_0 at the 5% significance level. There is insufficient evidence to suggest that the slope parameter β is not equal to zero, and so no evidence that the linear model is useful for prediction.
   (d) Assume that the population of errors are independent and identically distributed as $\mathsf{N}(0, \sigma^2)$.
4. Test statistic $t = -13.458$, critical value $t_{68,0.995} = -2.576$, reject H_0 since $-13.458 < -2.576$.
5. Test statistic $t = 0.013$, critical value $t_{6,0.975} = 2.447$, do not reject H_0 at the 5% level since $0.013 < 2.447$.

Exercise 18.5

1. (a) H_0: $\rho = 0$
   H_1: $\rho \neq 0$
   (b)
   $$t = \frac{-0.678\sqrt{27-2}}{\sqrt{1-(-0.678)^2}} = -4.612$$
   (c) $\nu = 27 - 2 = 25$
   (d) Critical value for $\nu = 25$ and a 5% two tailed test (and a negative test statistic) is $t_{25,0.975} = -2.060$.
   Since $-4.612 < -2.060$ reject H_0 at the 5% level of significance.
   There is evidence to suggest that there is a linear association between the variables.
2. Test statistic $t = 6.418$, critical value $t_{12,0.95} = 1.782$, reject H_0 at the 10% level since $6.418 > 1.782$.
3. Test statistic $t = 1.683$, critical value $t_{12,0.995} = 3.055$, do not reject H_0 at the 1% level since $1.683 < 3.055$.
4. Test statistic $t = -2.898$, critical value $t_{18,0.975} = -2.101$, reject H_0 at the 5% level since $-2.898 < -2.101$.
   Assume that electrolytes amount and time are independent and follow approximately bivariate normal distributions.
5. Test statistic $t = 0.013$, critical value $t_{6,0.975} = 2.447$, do not reject H_0 at the 5% level since $0.013 < 2.447$.

Exercise 18.6

1. $\hat{X} = 73.2 - 0.719y$
2. (a) $\hat{X} = -7.09 + 0.184y$
   (b) 4.87 inches
3. (a) The existing regression line is a y on x regression line, suitable for predicting the distance covered based on their finish time. To predict a finish time based on the distance covered in training, an x on y regression model is needed.
   (b) $\hat{X} = 7.709 - 0.0367y$
   (c) 5.14 hours
   (d) This would be an example of extrapolation, which may not be considered reliable.
4. $\hat{X} = 25.46 + 0.0051y$

Exercise 18.7

1. (a) The residual plot shows a random scatter of points centred on zero ($\mathsf{E}(\epsilon) = 0$) with constant variance ($\mathsf{V}(\epsilon) = \sigma^2$), which suggests the linear regression model is suitable.
   (b) The residual plot appears to show the points centred on zero ($\mathsf{E}(\epsilon) = 0$) in a funnel pattern with decreasing variance as fitted values increase ($\mathsf{V}(\epsilon) \neq \sigma^2$), which suggests that the linear regression model is not suitable.
   (c) The residual plot appears to show the points in the pattern of a curve ($\mathsf{E}(\epsilon) \neq 0$) with constant variance ($\mathsf{V}(\epsilon) = \sigma^2$), which suggests that the linear regression model is not suitable.
   (d) The residual plot shows a random scatter of points not centred on zero ($\mathsf{E}(\epsilon) \neq 0$) with constant variance ($\mathsf{V}(\epsilon) = \sigma^2$), which suggests the linear regression model is not suitable.
   (e) The residual plot shows a random scatter of points centred on zero ($\mathsf{E}(\epsilon) = 0$) with constant variance ($\mathsf{V}(\epsilon) \neq \sigma^2$) which suggests the linear regression model is suitable.
   (f) The residual plot shows a random scatter of points centred on zero ($\mathsf{E}(\epsilon) = 0$) with constant variance ($\mathsf{V}(\epsilon) \neq \sigma^2$) which suggests the linear regression model is suitable.
   (g) The residual plot appears to show the points in the pattern of a U-shaped curve ($\mathsf{E}(\epsilon) \neq 0$) with constant variance ($\mathsf{V}(\epsilon) = \sigma^2$) which suggests that the linear regression model is not suitable.
   (h) The residual plot appears to show the points centred on zero ($\mathsf{E}(\epsilon) = 0$) in a funnel pattern with increasing variance as fitted values increase ($\mathsf{V}(\epsilon) \neq \sigma^2$), which suggests that the linear regression model is not suitable.
   (i) The residual plot shows a random scatter of points not centred on zero ($\mathsf{E}(\epsilon) \neq 0$) with constant variance ($\mathsf{V}(\epsilon) = \sigma^2$), which suggests the linear regression model is not suitable.
   (j) The residual plot shows a random scatter of points centred on zero ($\mathsf{E}(\epsilon) = 0$) with constant variance ($\mathsf{V}(\epsilon) = \sigma^2$), which suggests the linear regression model is suitable.
   (k) The residual plot shows a random scatter of points centred on zero ($\mathsf{E}(\epsilon) = 0$) with constant variance ($\mathsf{V}(\epsilon) = \sigma^2$), which suggests the linear regression model is suitable.
   (l) The residual plot appears to show the points in the pattern of a curve ($\mathsf{E}(\epsilon) \neq 0$), which suggests that the linear regression model is not suitable.
2. The residual plot appears to show the points form a curved pattern ($\mathsf{E}(\epsilon) = 0$) with increasing variance as fitted values increase ($\mathsf{V}(\epsilon) \neq \sigma^2$), which suggests that the linear regression model is not suitable.
3. The residual plot shows a random scatter of points centred on zero ($\mathsf{E}(\epsilon) = 0$) with constant variance ($\mathsf{V}(\epsilon) = \sigma^2$), which suggests the linear regression model is suitable.
4. The residual plot appears to show the points in the pattern of a curve ($\mathsf{E}(\epsilon) \neq 0$), which suggests that the linear regression model is not suitable.

Exercise 18.8

1. (a) There appears to be a strong negative linear association between 'distance' and '$\sqrt{\text{power}}$'.
   The coefficient of determination suggests 80.5% of the variation in '$\sqrt{\text{power}}$' is explained by the model.
   (b) There appears to be a weak positive linear association between 'rating' and '$\log_{10}$ energy'.
   The coefficient of determination suggests 38.2% of the variation in '$\log_2$ energy' is explained by the model.
   (c) There appears to be a very strong positive linear association between '$\log_2$ time' and '$\log_2$ population'.
   The coefficient of determination suggests 97.4% of the variation in '$\log_2$ population' is explained by the model.
   (d) There appears to be no linear association between 'quality' and '$\frac{1}{\text{cost}}$'.
   The coefficient of determination suggests 0.1% of the variation in '$\frac{1}{\text{cost}}$' is explained by the model.
2. (a) $\widehat{\frac{1}{Y}} = 2.936 + 2.898x$
   (b) 0.0574
3. $r = -0.984$. This suggests that there is a very strong negative linear correlation between $\log x$ and $\log y$.
4. (a) They would have observed the points not showing an equal variance as fitted values increased, such as a funnel shape.
   (b) They should use model C, as it is the only one whose residual plot appears to show a random scatter of points centred on zero, with a constant variance as fitted values increase. This suggests it is the only model of the three which is appropriate.
   (a) The residual plot appears to show a curved pattern, suggesting a linear model between p and x is not appropriate.
   (b) The residual plot for this new model appears to show a random scatter of points centred around zero, with a constant variance, so this model appears to be appropriate.
   (c) $\hat{Y} = 0.0598 + 8.85x$
   (d) 10.3

Exercise 18.9

1. (a) (12.53,14.52)
   (b) (10.14,16.90)
2. (a) (57.80,61.17)
   (b) (52.95,66.02)
3. (a) (49.69,123.0)
   (b) (80.52,92.17)
4. (161.35,198.62)
5. (a) 81.56 kg
   (b) 91.53

(c) Assume that the population of errors are independently and identically distributed as $\mathsf{N}(0, \sigma^2)$.
(d) (66.34,69.24)
(e) This is an example of extrapolation, which may not be reliable.

Chapter 18 Review Exercise

1. (a) 51.46
   (b) Since $4.380 > 1.812$ reject H_0.
2. (a) $R^2 = 0.300$. This suggests that 30.0% of the variation in the tempo of a pop-punk song is explained by the linear model.
   (b) Since $-2.620 < -2.921$ do not reject H_0.
3. (a) 598.9
   (b) (57.58,169.46)
   (c) They are interested in a range of values for the number of daily visitors they might receive, rather than the mean number of daily visitors received by websites spending the same amount.
4. (a) The residual plot shows a curved pattern of points, suggesting that the linear model of boiling point on barometric pressure is not appropriate.
   (b) A transformation could be taken on one or both of the variables and a new linear model created between those transformed variables.
5. (a) $r = 0.436$, suggesting that there is a moderate positive linear association between the distance a route covers and the ascent.
   (b) The regression line given is a y on x regression line, suitable for predicting the ascent of a route given its distance. To predict the distance given the ascent, an x on y regression line would be needed.
   (c) $\hat{X} = 2.73 + 0.00539y$
   predicted distance $= 5.05$ miles
   (d) This data has only been taken from the Ochil Hills, and any results from this data may not be relevant for routes in the Lomond Hills.